



THE HONG KONG POLYTECHNIC UNIVERSITY

Department of Mechanical Engineering

**BEng(Hons) Scheme in Mechanical Engineering
(43499)**

Awards Offered Through the Scheme:

BEng(Hons) in Intelligent Robotics Engineering (43499-IRE/SYI)
BEng(Hons) in Mechanical Engineering (43499-ME/SYM)

(4-Year & Senior Year undergraduate degree structure)

Programme Requirement Document

(For 2025/26 Cohort)

August 2025

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This Programme Requirement Document is subject to review and changes which the programme offering Department may decide to make from time to time. Students will be informed of the changes as and when appropriate.

PART A

PROGRAMME SCHEME

1. GENERAL INFORMATION

1.1 Programme Information

Programme Title	Bachelor of Engineering(Honours) Scheme in Mechanical Engineering (BEng Scheme in ME) 機械工程學(榮譽)工學士組合課程
Programme Code	43499 (JUPAS Code: JS3741)
Host Department	Department of Mechanical Engineering (ME)
Programme Structure	Credit-based
Mode of Attendance	Full-time
Duration of Study	4 Years (2 Years for Senior Year Intake)
Award Title	There are two awards operating under the BEng Scheme in ME: • Bachelor of Engineering (Honours) in Intelligent Robotics Engineering (BEngIRE) 智能機械人工程學（榮譽）工學士學位 <i>Programme Code: 43499-IRE (4yrs) / 43499-SYI (2yrs)</i> • Bachelor of Engineering (Honours) in Mechanical Engineering (BEngME) 機械工程學（榮譽）工學士學位 <i>Programme Code: 43499-ME (4yrs) / 43499-SYM (2yrs)</i> <i>Students admitted into the BEng Scheme in ME will study together during the first two years and then complete their preferred award (BEngIRE or BEngME) in the next two years until graduation.</i>
Credits Required for Graduation	<u>4-Year Curriculum</u> (a) Academic Credits: 120 (b) Training Credits: 10 (c) Work-Integrated Education Training: 1 <u>Senior Year Curriculum</u> (a) Academic Credits: 64 (BEngME) / 62 (BEngIRE) (b) Training Credits: 6 (c) Work-Integrated Education Training: 1

1.2 Minimum Entrance Requirements

In addition to the general requirements for admission to the honours degree programmes offered by the University, a candidate needs to satisfy one of the following requirements (a), (b), (c), (d) or (e):

(a) For entry with Hong Kong Diploma of Secondary Education Examination (HKDSE) Qualification

The general minimum entrance requirements are as follows:

HKDSE Subjects	Core Subjects				Electives (incl. M1/M2)	
	Chinese Language	English Language	Mathematics	Citizenship & Social Development*	1 st Elective	2 nd Elective
Level Requirement	3	3	2	Attained	3	3

* An attainment at "Attained" in "Citizenship and Social Development" is required for meeting the entrance requirement but NOT included in the admission score calculation.

There is no compulsory subject requirement. Preferred electives include:

Physics, Biology, Chemistry, Combined Science, Information and Communication Technology, Extended Modules in Mathematics and Design and Applied Technology

(b) For entry with A-Level Qualification

Pass in 3 A-Level subjects **AND** Satisfy the English Language Requirement

(c) For entry with International Baccalaureate (IB) Qualification

IB Diploma **AND** Satisfy the English Language Requirement

(d) For those with other Qualifications

A relevant Diploma passed with credit or a relevant Higher Certificate from a recognized institution **OR** a relevant Associate Degree / Higher Diploma from a recognized institution.

(e) Qualifications equivalent to (a), (b), (c) or (d)

Note 1: Credit transfer may be granted to applicants with A-Level / IB qualification / Higher Diploma / Associate Degree, or the equivalent.

Note 2: Holder of a Higher Diploma or Associate Degree in Mechanical Engineering/ Engineering Design or a related discipline with good academic results at graduation (as determined by the Department) may be considered for the Senior Year curriculum.

1.3 External Recognition

The BEng (Hons) in Mechanical Engineering programme has been granted full accreditation by the Hong Kong Institution of Engineers (HKIE). Accreditation for BEng (Hons) in Intelligent Robotics Engineering will be sought when the first cohort of graduates is produced.

To view the list of Accredited/Recognised Programmes, please visit the HKIE's website:

<https://www.hkie.org.hk/en/quali/program/>

1.4 Summer Semester Teaching

There will be no teaching during the summer semester. However, Industrial Centre (IC) training and CAR subjects may still be scheduled to take place during the summer.

1.5 Daytime and Evening Teaching

Subjects will primarily be offered on weekdays. However, some subjects may only be available in the evenings or on Saturdays/Sundays.

2. PREAMBLE

The Hong Kong Polytechnic University (PolyU) aspires to be a leading university that excels in professional education, applied research and partnership with the industry for the betterment of Hong Kong, the nation and the world. It is the mission of the University (a) *To nurture socially responsible professionals and leaders with a strong sense of national pride and a global perspective*; (b) *To pursue world-leading research and innovation for societal benefits*; and (c) *To foster a University community in which all members are united with a strong sense of belonging and pride, empowering the University to scale new heights*. Being one of the oldest departments in the University, the Department of Mechanical Engineering (ME Department) follows closely the progress and development of the University, and shares its vision and mission.

In line with the vision of the University, the ME Department aspires to achieve excellence in education and research in the discipline of mechanical engineering with global out-reach and impact. It is also our mission (a) To train future leaders, with creativity, broad vision, global outlook, and professional ethics for industry, academia, government and communities, who have sound knowledge in mechanical engineering with effective communication, analytical, and problem-solving skills; and (b) To create knowledge and technologies through fundamental research and its applications in mechanical engineering, in order to serve the societal needs.

Hong Kong is facing a fast-evolving and increasingly competitive world. In order to maintain economic growth in the face of globalization, its economy has to change from being efficiency-based to knowledge-based. The goal of the ME Department is to produce all-round graduates who can lead a changing society. This goal is accomplished by having forward looking course curricula. Thus, it is one of the objectives of the Department to provide a holistic education which is outcome-based, work-integrated, professionally and globally-oriented, and student centred to strengthen the holistic development of the students.

The ME Department is currently offering the following two undergraduate degree programmes:

- **Bachelor of Engineering (Honours) in Intelligent Robotics Engineering (BEngIRE); and**
- **Bachelor of Engineering (Honours) in Mechanical Engineering (BEngME)**

3. RATIONALE, AIMS AND OBJECTIVES

3.1 Rationale

One of the missions of the ME Department is to produce graduates with a good general education, a competent command of the English and Chinese languages, a broad knowledge of mechanical engineering, and a deeper understanding of one of its sub-fields, with an aim to endow our graduates with competence to meet and lead the changing technological challenges.

PolyU aspires to develop all its students as all-round graduates with professional competence, and has identified a set of highly valued graduate attributes as the learning goals for students. While many of these graduate attributes can be developed through the curricular activities of the BEng Scheme in ME, some (including global outlook, interest in local and international affairs, sense of social and national responsibility, cultural appreciation) are primarily addressed through co-curricular activities offered by faculties, departments and various teaching and learning support units of the University. Students are encouraged to make full use of such opportunities to develop these attributes.

Students admitted into the Scheme will study together during the first two years before choosing their preferred programme—either *Bachelor of Engineering (Honours) in Intelligent Robotics Engineering (BEngIRE)* or *Bachelor of Engineering (Honours) in Mechanical Engineering (BEngME)*—for the final two years leading to graduation. Upon successful completion of their chosen programme, students will receive their respective degrees.

Besides, students also have the opportunity to expand their expertise by pursuing a Secondary Major alongside their primary Major. This pathway offers a meaningful enhancement to their academic portfolios, striking a balance between challenge and feasibility. It allows students to develop complementary skills that can significantly improve their career prospects through a substantial yet manageable commitment. Please refer to [Section 8.2](#) for more detail about the Secondary Major.

The structure of the BEng Scheme in ME is illustrated in Figure 3.1(a).

3.2 Programme Aims and Objectives of BEngIRE

The BEngIRE programme is designed with the following objectives:

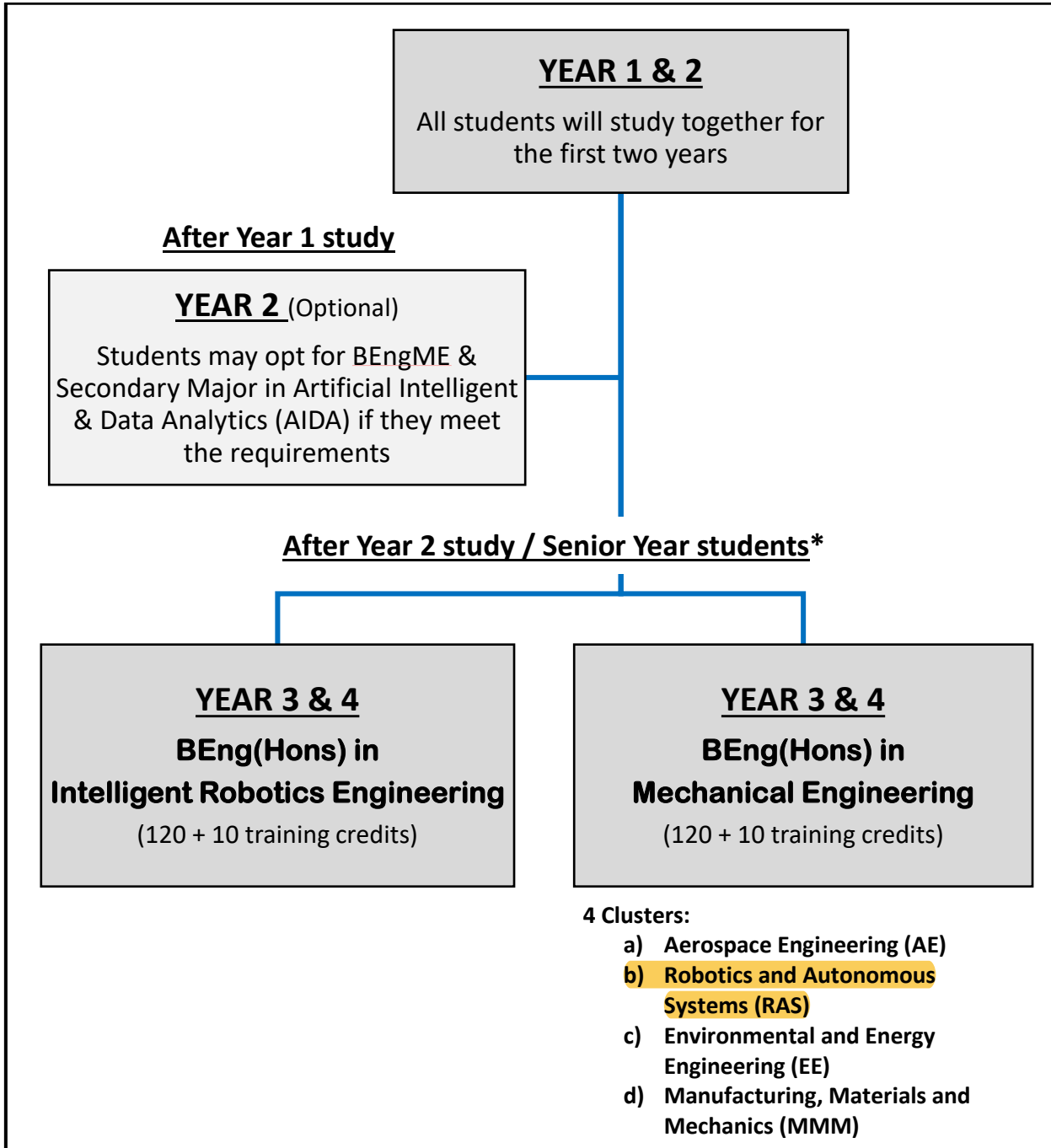
1. To provide students with a theoretical foundation and hands-on skills in intelligent robotics.
2. To help students develop the ability to engage in life-long learning and professional development and to acquire professional recognition from professional bodies including the Hong Kong Institution of Engineers
3. To produce graduates with an awareness of the global, national, societal, ethical and professional issues in the practice of engineering.

3.3 Programme Aims and Objectives of BEngME

The BEngME programme is designed with the following objectives:

1. To provide students with a broad base of knowledge in the fundamentals of mechanical engineering.
2. To help students develop the ability to engage in life-long learning and professional development.
3. To produce graduates that are aware of the global, national, societal, ethical and professional issues in the practice of engineering.

Figure 3.1(a) - Structure of BEng(Hons) Scheme in Mechanical Engineering



* Senior Year students will be admitted directly to either BEng (Hons) in Intelligent Robotics Engineering or BEng (Hons) in Mechanical Engineering.

4 LEARNING OUTCOMES (LOs)

The BEng(Hons) in Intelligent Robotics Engineering (BEngIRE) and BEng(Hons) in Mechanical Engineering (BEngME) programmes offered by the ME Department are designed to produce graduates that are broad-based and knowledgeable in mechanical engineering and design engineering respectively. It is expected that our graduates would accept responsibilities as professionals in academic, industrial and governmental organizations.

4.1 PolyU Institutional Learning Outcomes

It is PolyU's educational mission to nurture competent professionals who are also critical thinkers, effective communicators, innovative problem solvers, lifelong learners, and ethical leaders. The institutional learning outcomes for these attributes are provided as follows:

- (a) ***Socially responsible leaders with a strong sense of national pride and a global outlook:*** Care about and understand local, national and global issues, and be able to think globally, act responsibly, and lead with integrity and pride for the benefit of society and a sustainable future.
- (b) ***Future-ready professionals who possess technical acumen:*** Be able to integrate and apply in-depth discipline knowledge and specialised skills, leverage changing and emerging technologies for work, function in variable interdisciplinary contexts, and demonstrate professionalism and entrepreneurial spirit at work.
- (c) ***Critical thinkers and creative problem solvers:*** Be able to critically evaluate information and arguments, draw logical and informed conclusions, identify problems and formulate innovative solutions, in both professional and everyday contexts.
- (d) ***Effective communicators and collaborators:*** Be able to communicate effectively in English and Chinese in professional and everyday contexts*, collaborate with people from diverse backgrounds and different perspectives, and contribute to effective teamwork and positive group dynamics.
- (e) ***Adaptable and resilient lifelong learners:*** Committed to continual learning and self-improvement, engage in learning with a sense of purpose, manage their own learning, adapt to different learning situations, and deal effectively with the arising stress and challenges.

** The expectation to communicate in Chinese does not apply to foreign students.*

4.2 Desired Learning Outcomes of The Hong Kong Institution of Engineers (HKIE)

One of the main objectives of our undergraduate degree programmes is to obtain professional accreditation from The Hong Kong Institution of Engineers (HKIE). Thus, the desired learning outcomes achieved by graduates of accredited engineering programmes proposed by HKIE are used as reference for the Department to develop learning outcomes of our academic programmes. Desired learning outcomes proposed by HKIE are presented as below:

- (a) an ability to apply knowledge of mathematics, science, and engineering appropriate to the degree discipline;
- (b) an ability to design and conduct experiments, as well as to analyze and interpret data;
- (c) an ability to design a system, component or process to meet desired needs within realistic constraints, such as economic, environmental, social, political, ethical, health and safety, manufacturability and sustainability;
- (d) an ability to function on multi-disciplinary teams;
- (e) an ability to identify, formulate and solve engineering problems;
- (f) an ability to understand professional and ethical responsibility;
- (g) an ability to communicate effectively;
- (h) an ability to understand the impact of engineering solutions in a global and societal context, especially the importance of health, safety and environmental considerations to both workers and the general public;

- (i) an ability to stay abreast of contemporary issues;
- (j) an ability to recognize the need for, and to engage in life-long learning;
- (k) an ability to use the techniques, skills, and modern engineering tools necessary for engineering practice appropriate to the degree discipline; and
- (l) an ability to use the computer/IT tools relevant to the discipline along with an understanding of their processes and limitations

More information can be obtained from the HKIE's website: <https://www.hkie.org.hk/en/>.

4.3 Programme Learning Outcomes

A) **BEng(Hons) in Intelligent Robotics Engineering (BEngIRE)**

The BEngIRE programme aims to equip students with 12 learning outcomes. Each student is expected to achieve these outcomes, which are classified into the following two groups, before/at graduation:

Professional/academic knowledge and skills (PAK)

- (a) An ability to identify, formulate and solve engineering problems with innovative robot technology.
- (b) An ability to apply and incorporate their knowledge of mathematics, science and engineering into robot systems.
- (c) An ability to design and conduct experiments, as well as to analyze and interpret data.
- (d) An ability to design intelligent robot systems with creative thinking to meet desired needs.
- (e) An ability to use the techniques, skills and modern engineering tools, including computational tools necessary for engineering practice.
- (f) An ability to work professionally in the field of robotics engineering, with thorough considerations of technical and human aspects in design and implementation of robot systems.
- (g) An ability to appreciate the current state-of-the-art and to anticipate the future trends in robot systems.

Professional outlook and workplace skills (POW)

- (a) A knowledge of contemporary issues and the broad education necessary to understand the impact of robotics engineering in a local, national, global, and societal contexts.
- (b) An ability to function professionally in multidisciplinary teams.
- (c) An understanding of professional and ethical responsibility.
- (d) An ability to communicate effectively.
- (e) A recognition of the need for and an ability to engage in life-long learning.

The following table illustrates the relationship between the programme outcomes and the programme objectives of **BEngIRE**:

Programme Learning Outcomes	Programme Objectives		
	1	2	3
PAK (a)	√	√	
PAK (b)	√	√	
PAK (c)	√	√	
PAK (d)	√	√	
PAK (e)	√	√	
PAK (f)	√	√	
PAK (g)	√	√	
POW (a)		√	√

Programme Learning Outcomes	Programme Objectives		
	1	2	3
POW (b)		√	
POW (c)		√	√
POW (d)		√	
POW (e)		√	

The following table illustrates the relationship between **BEngIRE** Programme Learning Outcomes and PolyU Institutional Learning Outcomes:

Programme Learning Outcomes	PolyU Institutional Learning Outcomes				
	(a)	(b)	(c)	(d)	(e)
PAK (a)	√	√		√	
PAK (b)	√	√		√	
PAK (c)	√	√		√	
PAK (d)	√	√		√	
PAK (e)	√	√		√	
PAK (f)	√	√		√	
PAK (g)	√				
POW (a)	√	√		√	
POW (b)	√				
POW (c)	√				
POW (d)	√		√		
POW (e)	√				√

Relationship of BEngIRE Programme Learning Outcomes to the Desired Learning Outcomes of the Hong Kong Institution of Engineers (HKIE)

Since the development of the BEngIRE intended learning outcomes is initially referred to the desired learning outcomes proposed by HKIE, thus the HKIE desired learning outcomes can be fully covered.

B) BEng(Hons) in Mechanical Engineering (BEngME)

To fulfill PolyU's educational mission as well as BEngME's educational objective, the ME programme aims to equip students with the following 12 learning outcomes developed by referring to the desired learning outcomes proposed by HKIE. Each student is expected to achieve these learning outcomes, which are classified into two groups, before graduation:

Professional/academic knowledge and skills (PAK)

- (a) an ability to identify, formulate and solve engineering problems;
- (b) an ability to apply their knowledge of mathematics, science and engineering;
- (c) an ability to design and conduct experiments, as well as to analyze and interpret data;
- (d) an ability to design a system, component or process to meet desired needs;
- (e) an ability to use the techniques, skills and modern engineering tools, including computational tools necessary for engineering practice;

- (f) an ability to work professionally in general mechanical systems, including the design and realization of such systems;
- (g) a basic understanding of manufacturing methods and maintain awareness of contemporary issue.

Professional outlook and workplace skills (POW)

- (a) a knowledge of contemporary issues and the broad education necessary to understand the impact of engineering solutions in a local, national and global contexts;
- (b) an ability to function professionally in multidisciplinary teams;
- (c) an understanding of professional and ethical responsibility;
- (d) an ability to communicate effectively;
- (e) a recognition of the need for and an ability to engage in life-long learning.

The following table illustrates the relationship between the programme outcomes and the programme objectives of **BEngME**:

Programme Learning Outcomes	Programme Objectives		
	1	2	3
PAK (a)	√	√	
PAK (b)	√	√	
PAK (c)	√	√	
PAK (d)	√	√	
PAK (e)	√	√	
PAK (f)	√	√	
PAK (g)	√	√	
POW (a)		√	√
POW (b)		√	
POW (c)		√	√
POW (d)		√	
POW (e)		√	

The following table illustrates the relationship between **BEngME** Programme Learning Outcomes and PolyU Institutional Learning Outcomes:

Programme Learning Outcomes	PolyU Institutional Learning Outcomes				
	(a)	(b)	(c)	(d)	(e)
PAK (a)	√	√	√		
PAK (b)	√	√	√		
PAK (c)	√	√	√		
PAK (d)	√	√	√		
PAK (e)	√	√	√		
PAK (f)	√	√	√	√	
PAK (g)	√	√			
POW (a)	√	√	√	√	
POW (b)	√	√		√	
POW (c)	√	√		√	

Programme Learning Outcomes	PolyU Institutional Learning Outcomes				
	(a)	(b)	(c)	(d)	(e)
POW (d)	√	√		√	
POW (e)	√				√

Relationship of BEngME Programme Learning Outcomes to the Desired Learning Outcomes of the Hong Kong Institution of Engineers (HKIE)

Since the development of the BEngME intended learning outcomes is initially referred to the desired learning outcomes proposed by HKIE, thus the HKIE desired learning outcomes can be fully covered.

4.4 General Approach to Teaching, Learning and Assessment

To accomplish the learning outcomes of the programme, students are expected to achieve specific learning outcomes for each subject outlined in Part B. These learning outcomes are spelt out explicitly in the syllabus of each subject. They provide a motivation and a target for students who may use this information to formulate their study plan before the teaching. The students may also use the information to conduct a self-assessment after the teaching.

Generally speaking, a one-credit subject is allocated with a contact time of one hour per week. Hence, a typical PolyU subject offered by the Department normally requires 3 hours per week of class attendance. There are 13 weeks in each semester leading to a total of 39 hours of contact time for a three-credit subject. The structuring of those 39 contact hours varies from subject to subject, and the details are given in the syllabi.

The Department uses a wide variety of teaching methods, in a number of different settings including formal lectures, invited lectures by guest speakers, seminars, laboratory work, practical work, project work, case studies and student project presentations. In most of the classroom activities, the staff member will begin with a formal lecture that is designed to give students an overview of the topic on hand, which may also require their engagement through questioning or interactive hand-outs. Some of these hand-outs form a part of the assignments where the students are required to work after the class. The students are frequently required to contribute through presentations, through working on case studies and mini-projects, through experimental studies by laboratory classes. In many of these teaching/learning activities, students are asked to participate in small groups. These different teaching and learning approaches will be assessed with appropriate methods. In case of group activity, both the overall performance of the group as well as the individual effort/contribution of each team member will be assessed.

The prime purpose of assessment is to enable students to demonstrate that they have met the aims and objectives of the academic programme: in particular, they have fulfilled the requirement of each subject and have, at the end of their study, achieved the standard appropriate to the award.

Assessment also fulfils two major functions. It is used to evaluate whether the specific student-learning-outcomes of a subject have been achieved by the students, and distinguish their performance in achieving them.

Assessment will also serve as prompt and useful feedback to students. Students will be informed of their performance in the assessment so that they are aware of their progress and attainment to facilitate teaching and learning. Students' performance in a subject will be judged by continuous assessment, or final examination and continuous assessment as deemed appropriate. Where both methods are used, the weighting of each in the overall subject grade will be clearly stated in the relevant subject syllabi. Continuous assessment may include tests, assignments, project reports and oral presentations, laboratory work and other forms of classroom participation. As assessment should be a matter of judgment, the subject lecturer will have the discretion to assign a final grade which is considered to reflect more appropriately the overall performance of the student in a subject.

4.5 Alignment of Teaching, Learning and Assessment Methods with Programme Outcomes

There are compulsory and electives offered in the programme. The details for each individual subject are contained in the respective syllabus listed in Part B. These explain how the objectives, teaching/learning activities, and eventually student learning outcomes, can be matched together so that they are constructively aligned in the context of these subjects. Typical teaching methods include lectures, tutorials, laboratory work, case studies which are supplemented by mini-projects, and presentations by individual students.

The major forms of assessment used in the programme are written examinations (open or closed book) and continuous assessment. In assessing students' academic performance and attainment of teaching and learning outcomes, much emphasis is placed on their ability to analyze, synthesize, integrate and apply what they have learnt in the course of their studies.

Details of the alignment of teaching, learning and assessment methods with programme outcomes are shown in Section 5.4 and the individual subject syllabus.

5. PROGRAMME STRUCTURE, CURRICULUM AND STUDY PATTERN

5.1 General Structure and Curriculum

The number of credits required for graduation is **120 academic credits and 10 training credits for the 4-year curriculum**, and **62/64 academic credits and 6 training credits for the 2-year curriculum**. Furthermore, students are required to fulfill the **Work-Integrated Education (WIE) component**.

4-year Curriculum: The 120 academic credits consist of **27 credits of General University Requirements (GUR)**, **7 credits of Free Elective** and **86 credits of Major Subjects**. Students who do not have Level 2 or above in HKDSE Physics (or Combined Science with a component in Physics), or the equivalent qualifications, “AP10001 Introduction to Physics” will be assigned before proceeding to take “AP10005 Physics I” and “AP10006 Physics II”. Upon the completion of “AP10001 Introduction to Physics”, students can fulfill a Free Elective (3 credits).

2-year Curriculum: The 62/64 academic credits consist of **9 credits of General University Requirements (GUR)** and **53/55 credits of Major Subjects**. However, students who do not meet the equivalent standard of the undergraduate degree may need to complete additional credits.

Details of GUR, Free Electives and Major Subjects of BEngIRE and BEngME are shown in the following tables:

Table 5.1.1: General University Requirements (GUR) & Free Electives for both BEngIRE and BEngME

Table 5.1.2: Major Subjects for BEngIRE

Table 5.1.3: Major Subjects for BEngME

Table 5.1.1: General University Requirements (GUR) & Free Electives for both BEngIRE and BEngME

Subject Code	Subject Title	Credits	
		FY Entry	SY Entry
--	Cluster Areas Requirements (CAR A) - Human Nature, Relations and Development	3	
	Cluster Areas Requirements (CAR A) - Human Nature, Relations and Development – English language with English Reading and Writing (ER/EW)		3
--	Cluster Areas Requirements (CAR M) - Chinese History and Culture	3	3
--	Cluster Areas Requirements (CAR N) - Cultures, Organisations, Societies and Globalisation	3	
--	Language & Communication Requirements (LCR I) – English	3	*
--	Language & Communication Requirements (LCR II) – English	3	*
	Language & Communication Requirements (LCR III) – Chinese ⁽ⁱ⁾	3	*
APSS1L01	Tomorrow’s Leaders (Leadership Education and Development) (LEAD)	3	
EIE1005	Fundamental AI & Data Analytics (Artificial Intelligence and Data Analytics) (AIDA)	2	
MM1031	Introduction to Innovation and Entrepreneurship (Innovation and Entrepreneurship) (IE)	1	
--	Service-Learning (SL)	3	3
--	Healthy Lifestyle (HLS)	(required)	
--	Essential Components of General Education e-modules on: a) Academic Integrity (OTAI) b) Artificial Intelligence and Data Analytics (AIDA) c) Innovation and Entrepreneurship (IE) d) National Education (NE)		(required)
Total credits of GUR :		27	9
	Free Electives	7	
Total credits of Free Electives :		7	0

Table 5.1.2: Major Subjects for BEngIRE

Subject Code	Subject Title	Pre-requisites (if any)	Credits	
			FY Entry	SY Entry
Compulsory Subjects				
AMA1110	Basic Mathematics I – Calculus and Probability & Statistics		3	
AMA1120 ⁽ⁱⁱ⁾	Basic Mathematics II– Calculus and Linear Algebra	AMA1110	3	
AMA2111	Mathematics I	AMA1120	3	
AMA3708	Differential Equations	AMA2112	3	3
AP10005 ⁽ⁱⁱ⁾⁽ⁱⁱⁱ⁾	Physics I		3	
AP10006 ⁽ⁱⁱⁱ⁾	Physics II		3	
COMP2011	Data Structure	ENG2002	3	3
EE2901S	Basic Electricity and Electronics		3	
EIE3320	Object-Oriented Design and Programming	ENG2002	3	3
ELC3531	Professional Communication in English for Engineering Students	LCR-English	2	2
ENG1004	Engineering Professionals in Society I		1	1
ENG2001 ^(iv)	Fundamentals of Materials Science and Engineering		3	
ENG2002	Computer Programming		3	
ENG2003 ⁽ⁱⁱ⁾	Information Technology		3	
ENG3003	Engineering Management		3	3
ENG3006	Engineering Professionals in Society II		2	2
ME22003	Visualization and Communication in Design Engineering		3	
ME23001	Engineering Mechanics	AP10005	3	
ME31002	Linear Systems and Control	ME23001	3	3
ME32002	Engineering Design Fundamentals	ME22003	3	
ME33001	Mechanics of Materials	ME23001 & ENG2001	3	
ME35001	Robotics and Computer Integrated Manufacturing	Ref to syllabus in Part B	3	3
ME35002	Sensors and Actuators for Mechatronics	Ref to syllabus in Part B	3	3
ME42011	Fundamentals of Robotics	ME31002 or ME41004	3	3
ME42012	Advanced Robotics Studio	EIE3320	3	3
ME42013	Robot Mechanisms and Motion	ME23001	3	3
ME49007	Capstone Project	EIE3320, ME31002, ME35001, ME35002, ME33001 & ME39003	6	6
Electives				
Group A				
COMP3011	Design Analysis of Algorithms	COMP2011 or EIE3320	3 (any ONE elective subject from this group)	3 (any ONE elective subject from this group)
COMP3423	Human Computer Interaction	ENG2002		
EE3002	Electromechanical Energy Conversion	Ref to syllabus in Part B		
EIE3103	Digital Signals and Systems	Ref to syllabus in Part B		
EIE3373	Microcontroller Systems and Interface	Ref to syllabus in Part B		
ISE3018	Logistics Automation			
ISE3021	Warehousing Technologies and Operations			
Group B				
AAE4011	Artificial Intelligence in Unmanned Autonomous Systems	Ref to syllabus in Part B	3 (any ONE elective from this group)	9 (any THREE electives from this group)
ENG4001	Project Management			
ISE4024	Robotics and Automation Systems	Ref to syllabus in Part B		
ME41001	Automatic Control Systems	ME31002		
ME41006	Perceptual Robotics	ME31002		
ME42001	Artificial Intelligence in Products	ME31002 or ME41004		
ME42014	Networked Robots and IoT	ME35002		
ME46002	Numerical Methods for Engineers	AMA2111		
Training				
ME29004/ IC2117	Fundamentals of Engineering Design & Prototyping		2 (TRN)	
ME39002/ IC348	Appreciation of Manufacturing Processes	ME29004/IC2117	3 (TRN)	3 (TRN)
ME39003/ IC382	Multidisciplinary Manufacturing Project	ME39002/IC348	3 (TRN)	3 (TRN)
ME39004/ IC390	Applied Engineering Essentials: Python, Safety Practices, and Advanced CAD Modelling		2 (TRN)	
Total credits of Major Subjects:			86 + 10 (TRN)	53 + 6 (TRN)

Table 5.1.3: Major Subjects for BEngME

Subject Code	Subject Title	Pre-requisites (if any)	Credits	
			FY Entry	SY Entry
Compulsory Subjects				
AF3625	Engineering Economics		3	
AMA1110	Basic Mathematics I – Calculus and Probability & Statistics		3	
AMA1120 ⁽ⁱⁱ⁾	Basic Mathematics II– Calculus and Linear Algebra	AMA1110	3	
AMA2111	Mathematics I	AMA1120	3	
AMA2112	Mathematics II	AMA2111	3	3
AP10005 ⁽ⁱⁱ⁾⁽ⁱⁱⁱ⁾	Physics I		3	
AP10006 ⁽ⁱⁱⁱ⁾	Physics II		3	
CLC3421P	Professional Communication in Chinese	LCR-Chinese		2
EE2901S	Basic Electricity and Electronics		3	
ELC3531	Professional Communication in English for Engineering Students	LCR-English	2	2
ENG1004	Engineering Professionals in Society I		1	
ENG2001 ^(iv)	Fundamentals of Materials Science and Engineering		3	
ENG2002	Computer Programming		3	
ENG2003 ⁽ⁱⁱ⁾	Information Technology		3	
ENG3003	Engineering Management		3	3
ENG3004	Society and the Engineer			3
ENG3006	Engineering Professionals in Society II		2	
ME22003	Visualization and Communication in Design Engineering		3	
ME23001	Engineering Mechanics	AP10005	3	
ME31001	Dynamics and Vibrations	ME23001	3	3
ME31002	Linear Systems and Control	ME23001	3	3
ME32001	Manufacturing Fundamentals	ME23001 & ME32002	3	3
ME32002	Engineering Design Fundamentals	ME22003	3	
ME33001	Mechanics of Materials	ME23001 & ENG2001	3	3
ME34002	Engineering Thermodynamics	AP10005 & AMA2111	3	3
ME34004	Fluid Mechanics	AMA2112	3	3
ME34005	Fundamentals of Thermal Science & Engineering	AP10005 & AMA2111	3	
ME46002	Numerical Methods for Engineers	AMA2111		3
ME49001	Final Year Capstone Project	ME31001, ME31002, ME32001, ME33001, ME34002 & ME34004	6	6
Electives ^(v)				
Students are required to complete five subjects from a pool of electives as listed in Table 5.2(b))		Refer to syllabus in Part B	9	15
Training				
ME29004/IC2117	Fundamentals of Engineering Design & Prototyping		2 (TRN)	
ME39002/IC384	Appreciation of Manufacturing Processes	ME29004/IC2117	3 (TRN)	3 (TRN)
ME39003/IC382	Multidisciplinary Manufacturing Project	ME39002/IC348	3 (TRN)	3 (TRN)
ME39004/IC390	Applied Engineering Essentials: Python, Safety Practices, and Advanced CAD Modelling		2 (TRN)	
Total credits of Major Subjects:			86 + 10 (TRN)	55 + 6 (TRN)

Remarks:

AAE	Department of Aeronautical and Aviation Engineering
AMA	Department of Applied Mathematics
AP	Department of Applied Physics
CLC	Chinese Language Centre
COMP	Department of Computing
EE/EIE	Department of Electrical and Electronic Engineering
ELC	English Language Centre
ENG	Faculty of Engineering

IC	Industrial Centre
ISE	Department of Industrial and Systems Engineering
ME	Department of Mechanical Engineering
TRN	Training credits

* This is normally not required. Only students not meeting the equivalent standard of the Undergraduate Degree LCR (based on their previous studies in AD/HD programmes and their academic performance) will be required to take degree LCR subjects on top of the normal curriculum requirement.

- (i) Students of non-native Chinese language background, fulfilling any one of the following criteria with appropriate supporting certificate/document (if applicable), could be exempted from the CLCR at the time of admission. Students may use a CLCR subject or free elective to fulfil the credit requirement. They are also exempted from fulfilling the CR/CW of CAR:
- those with their first/native language as non-Chinese stated on the grade report of recognized tests (e.g. IELTS, TOEFL, etc.); OR*
 - those admitted with international qualifications without taking any Chinese subject(s) in their secondary/high school; OR*
 - those taken Chinese B or Chinese AB(SL) in IB Diploma; OR*
 - NCS status shown on the official proof provided by their secondary school.*

Otherwise, one subject will be pre-assigned to the students as Chinese LCR depending on their Chinese language proficiency and/or previous exam results. They might be given an assessment to ascertain that the pre-assigned subject is suitable for them

- (ii) Upon the completion of any two Faculty Electives among *AMA1120 Basic Mathematics II*, *AP10005 Physics I*, *AP10006 Physics II* and *ENG2003 Information Technology*, the remaining subjects will be counted as Major Subjects.

- (iii) Students who do **not** have Level 2 or above in HKDSE Physics (or Combined Science with a component in Physics), or the equivalent qualifications, *AP10001 Introduction to Physics* will be assigned before studying *AP10005 Physics I*. Upon the completion of *AP10001 Introduction to Physics*, students can fulfill a Free Elective (3 credits).

- (iv) Students must choose one subject from the list of subjects below:

Engineering Materials*:	(a) <i>ENG2001 Fundamentals of Materials Science and Engineering</i>
Biology:	(b) <i>ABCT1104 Fundamental Life Science</i>
	(c) <i>ABCT1303 Biotechnology and Human Health</i>
	(d) <i>BME11101 Bionic Human and the Future of Being Human</i>
Chemistry:	(e) <i>ABCT1301 Chemistry and Modern Living</i>
	(f) <i>ABCT1314 Chemistry and Sustainable Development</i>

* *ENG2001* will usually pre-assigned for students of *BEngIRE* & *BEngME*

- (v) Electives
Students are required to study three to five electives. They may choose any electives from Table 5.2(b). These elective subjects are classified into the following four clusters according to their specific disciplines:

- Aerospace Engineering (AE)*
- Robotics and Autonomous Systems (RAS)*
- Environmental and Energy Engineering (EE)*
- Manufacturing, Materials and Mechanics (MMM)*

5.2 Study Patterns of **BEngIRE** and **BEngME**

This section outlines the study patterns for the BEngIRE and BEngME programmes. These patterns are indicative and not mandatory. Students may choose alternative plans as long as they meet the credit requirements for their intended award within the maximum registration period. Each subject usually carries 3 credits, unless otherwise specified. *However, students whose study patterns deviate from this suggested plan may encounter class time conflicts and could extend the duration of their studies.*

Table 5.2.1: **BEngIRE (4-yr curriculum)** (for students with a Physics background⁽ⁱ⁾)

Table 5.2.2: **BEngIRE (4-yr curriculum)** (for students without a Physics background⁽ⁱ⁾)

Table 5.2.3: **BEngIRE (2-yr curriculum)**

Table 5.2.4: **BEngME (4-yr curriculum)** (for students with a Physics background⁽ⁱ⁾)

Table 5.2.5: **BEngME (4-yr curriculum)** (for students without a Physics background⁽ⁱ⁾)

Table 5.2.6: **BEngME (2-yr curriculum)**

Table 5.2.7: **BEngME (4-yr curriculum)** with a **Secondary Major** in Artificial Intelligent and Data Analytics (**AIDA**) (for students with a Physics background⁽ⁱ⁾⁽ⁱⁱ⁾)

Table 5.2.8: **BEngME (4-yr curriculum)** with a **Secondary Major** in Artificial Intelligent and Data Analytics (**AIDA**) (for students without a Physics background⁽ⁱ⁾⁽ⁱⁱ⁾)

Remarks:

- (i) Local students **without** a Level 2 or above in HKDSE Physics (or Combined Science with a Physics component) or equivalent, as well as non-local students from the Chinese Mainland who **do not have achieved** a pass (defined as 60% of the total marks) in Physics or Integrated Science in the Joint Entrance Examination for Universities, are required to study AP10001 Introduction to Physics. Their normal progression patterns differ slightly from those of students with prior Physics knowledge. These students can refer to the specific normal progression patterns for their programmes listed above.
- (ii) Please refer to Section 8.2 for the details of Secondary Major (only applicable to students who will opt BEng ME).

Table 5.2.1: BEngIRE Normal Progression Pattern (4-yr)

(120 academic credits + 10 training credits)
 (For students **with** a Physics background at admission)

Year 1 (34 + 2 training credits)	
Semester 1 (16 + 1 training credit)	Semester 2 (15 + 1 training credit)
AMA1110 Basic Mathematics I	AMA1120 Basic Mathematics II ⁺
AP10005 Physics I ⁺	APSS1L01 Tomorrow's Leaders
MM1031 Introduction to Innovation and Entrepreneurship	EIE1005 Fundamental AI and Data Analytics
CAR I ^{^^}	ENG1004 Professionals in Society I
Free Elective/CAR II ^{^^}	ENG2002 Computer Programming
LCR I English Language Subject	LCR II English Language Subject
Healthy Lifestyle ^{^^}	
ME29004/IC2117 Fundamentals of Engineering Design & Prototyping (training)	
Summer Semester (3 credits)	
Free Elective/CAR II ^{^^}	
Year 2 (29 + 5 training credits)	
Semester 1 (15 + 3 training credits)	Semester 2 (14 + 2 training credits)
AMA2111 Mathematics I	EE2901S Basic Electricity and Electronics
AP10006 Physics II ⁺	ENG2003 Information Technology ⁺
ENG2001 Fundamentals of Materials Science and Engineering	ENG3006 Professionals in Society II
ME22003 Visualization and Communication in Design Engineering	ME32002 Engineering Design Fundamentals
ME23001 Engineering Mechanics	ME33001 Mechanics of Materials
ME39002/IC348 Appreciation of Manufacturing Processes (training)	ME39004/IC390 Applied Engineering Essentials: Python, Safety Practices, and Advanced CAD modelling (training)
Year 3 (30 + 3 training credits)	
Semester 1 (15 + 1.5 training credits)	Semester 2 (15 + 1.5 training credits)
AMA3708 Differential Equations	EIE3320 Object-Oriented Design and Programming
COMP2011 Data Structure	ME31002 Linear Systems and Controls
ENG3003 Engineering Management	ME35002 Sensors and Actuators for Mechatronics
ME35001 Robotics and Computer Integrated Manufacturing	ME Elective A (Level 3)
CAR III ^{^^}	Service-learning ^{^^} (Recommendation: ME3S01 Engineering Design for Community)
ME39003/IC382 Multidisciplinary Manufacturing Project (training)	
Year 4 (26 credits)	
Semester 1 (14 credits)	Semester 2 (12 credits)
LCR III Chinese Language Subject	ME42011 Fundamentals of Robotics
ELC3531 Professional Communication in English for Engineering Students	ME42013 Robot Mechanisms and Motion
ME42012 Advanced Robotics Studio	Free Elective (Recommendation: ME Level 4 electives)
ME Elective B (Level 4)	
ME49007 Capstone Project	

+ Upon the completion of any two Faculty Electives among AMA1120 Basic Mathematics II/AP10005 Physics I/AP10006 Physics II/ENG2003 Information Technology, the remaining subjects will be counted as Major Subjects.

^{^^} The study pattern for these GUR subjects is indicative only.

Table 5.2.2: BEngIRE Normal Progression Pattern (4-yr)

(120 academic credits + 10 training credits)
 (For students **without** a Physics background at admission)

Year 1 (34 + 2 training credits)	
Semester 1 (14 + 1 training credit)	Semester 2 (17 + 1 training credit)
AMA1110 Basic Mathematics I	AMA1120 Basic Mathematics II ⁺
AP10001 Introduction to Physics I (<i>Free Elective</i>) [#]	AP10005 Physics I ⁺
ENG1004 Engineering Professionals in Society I	APSS1L01 Tomorrow's Leaders
MM1031 Introduction to Innovation and Entrepreneurship	EIE1005 Fundamental AI and Data Analytics
CAR I ^{^^}	ENG2002 Computer Programming
LCR I English Language Subject	LCR II English Language Subject
Healthy Lifestyle ^{^^}	
ME29004/IC2117 Fundamentals of Engineering Design & Prototyping (<i>training</i>)	
Summer Semester (3 credits)	
CAR II ^{^^}	
Year 2 (29 + 5 training credits)	
Semester 1 (15 + 3 training credits)	Semester 2 (14 + 2 training credits)
AMA2111 Mathematics I	EE2901S Basic Electricity and Electronics
AP10006 Physics II ⁺	ENG2003 Information Technology ⁺
ENG2001 Fundamentals of Materials Science and Engineering	ENG3006 Engineering Professionals in Society II
ME22003 Visualization and Communication in Design Engineering	ME32002 Engineering Design Fundamentals
ME23001 Engineering Mechanics	ME33001 Mechanics of Materials
ME39002/IC348 Appreciation of Manufacturing IC348 Processes (<i>training</i>)	ME39004/IC390 Applied Engineering Essentials: Python, Safety Practices, and Advanced CAD modelling (<i>training</i>)
Year 3 (30 + 3 training credits)	
Semester 1 (15 + 1.5 training credits)	Semester 2 (15 + 1.5 training credits)
AMA3708 Differential Equations	EIE3320 Object-Oriented Design and Programming
COMP2011 Data Structure	ME31002 Linear Systems and Controls
ENG3003 Engineering Management	ME35002 Sensors and Actuators for Mechatronics
ME35001 Robotics and Computer Integrated Manufacturing	ME Elective A (Level 3) [@]
CAR III ^{^^}	Service-learning ^{^^} (Recommendation: ME3S01 Engineering Design for Community)
ME39003/IC382 Multidisciplinary Manufacturing Project (<i>training</i>)	
Year 4 (26 credits)	
Semester 1 (14 credits)	Semester 2 (12 credits)
LCR III Chinese Language Subject	ME42011 Fundamentals of Robotics
ELC3531 Professional Communication in English for Engineering Students	ME42013 Robot Mechanisms and Motion
ME42012 Advanced Robotics Studio	Free Elective (Recommendation: ME Level 4 electives)
ME Elective B (Level 4) [@]	
ME49007 Capstone Project	

+ Upon the completion of any two Faculty Electives among AMA1120 Basic Mathematics II/AP10005 Physics I/AP10006 Physics II/ENG2003 Information Technology, the remaining subjects will be counted as Major Subjects.

^{^^} The study pattern for these GUR subjects is indicative only.

[#] Upon the completion of "AP10001 Introduction to Physics", students can fulfill a free elective (3 credits).

[@] Please refer to Table 5.2(a) for more details about the ME Electives.

Table 5.2.3: BEngIRE Normal Progression Pattern (SY)
(62 credits + 6 training credits)[#]

Senior Year 1 (33 + 6 training credits)	
Semester 1 (16 + 3 training credits)	Semester 2 (17 + 1.5 training credits)
AMA3708 Differential Equations	EIE3320 Object-Oriented Design and Programming
COMP2011 Data Structure	ME31002 Linear Systems and Controls
ENG3003 Engineering Management	ME35002 Sensors and Actuators for Mechatronics
ENG1004 Professionals in Society I	ENG3006 Professionals in Society II
ME35001 Robotics and Computer Integrated Manufacturing	ME Elective A (Level 3) [@]
CAR (A) ^{^^}	Service-learning ^{^^} (Recommendation: ME3S01 Engineering Design for Community)
ME39002/IC348 Appreciation of Manufacturing Processes (training)	ME39003/IC382 Multidisciplinary Manufacturing Project (training)
Summer Semester (1.5 training credits)	
(Con't) ME39003/IC382 Multidisciplinary Manufacturing Project (training)	
Senior Year 2 (29 credits)	
Semester 1 (14 credits)	Semester 2 (15 credits)
CAR (M) ^{^^}	ME42011 Fundamentals of Robotics
ELC3531 Professional Communication in English for Engineering Students (2 credits)	ME42013 Robot Mechanisms and Motion
ME42012 Advanced Robotics Studio	Elective B II (Level 4) [@]
Elective B I (Level 4) [@]	Elective B III (Level 4) [@]
ME49007 Capstone Project	

[#] The credits required and progression pattern presented above are for students who meet the equivalent standard of the Undergraduate Degree LCR based on their previous studies. Students who do not meet this standard are required to take the LCR subjects.

[^] The study pattern for these GUR subjects is indicative only. Students may take the subjects according to their own study plan.

[@] Please refer to Table 5.2(a) for more details about the ME Electives.

Table 5.2(a): Electives for BEngIRE⁽ⁱ⁾

Group A (Level 3)	Group B (Level 4)
COMP3011 Design Analysis of Algorithms	AAE4011 Artificial Intelligence in Unmanned Autonomous Systems
COMP3423 Human Computer Interaction	ENG4001 Project Management
EE3002 Electromechanical Energy Conversion	ISE4024 Robotics and Automation Systems
EIE3103 Digital Signals and Systems	ME41001 Automatic Control Systems
EIE3373 Microcontroller Systems and Interface	ME41006 Perceptual Robotics
ISE3018 Logistics Automation	ME42001 Artificial Intelligence in Products
ISE3021 Warehousing Technologies and Operations	ME42014 Networked Robots and IoT
	ME46002 Numerical Methods for Engineers

(i) The electives are updated from time to time to ensure the best development of the programme and to ensure the best career for our students. Since there is a minimum planned class size for each subject, the Department has the discretion to cease the offering of subjects that fail to enroll students up to the minimum class size.

Table 5.2.4: BEngME Normal Progression Pattern (4-yr)

(120 credits + 10 training credits)

(For students **with** a Physics background at admission)

Year 1 (34 + 2 training credits)	
Semester 1 (16 + 1 training credit)	Semester 2 (15 + 1 training credit)
AMA1110 Basic Mathematics I	AMA1120 Basic Mathematics II ⁺
AP10005 Physics I ⁺	APSS1L01 Tomorrow's Leaders
MM1031 Introduction to Innovation and Entrepreneurship	EIE1005 Fundamental AI and Data Analytics
CAR I ^{^^}	ENG1004 Engineering Professionals in Society I
Free Elective/CAR II ^{^^}	ENG2002 Computer Programming
LCR I English Language Subject	LCR II English Language Subject
Healthy Lifestyle ^{^^}	
ME29004/IC2117 Fundamentals of Engineering Design & Prototyping (training)	
Summer Semester (3 credits)	
Free Elective/CAR II ^{^^}	
Year 2 (29 + 5 training credits)	
Semester 1 (15 + 3 training credits)	Semester 2 (14 + 2 training credits)
AMA2111 Mathematics I	EE2901S Basic Electricity and Electronics
AP10006 Physics II ⁺	ENG2003 Information Technology ⁺
ENG2001 Fundamentals of Materials Science and Engineering	ENG3006 Engineering Professionals in Society II
ME22003 Visualization and Communication in Design Engineering	ME32002 Engineering Design Fundamentals
ME23001 Engineering Mechanics	ME33001 Mechanics of Materials
ME39002/IC348 Appreciation of Manufacturing IC348 Processes (training)	ME39004/IC390 Applied Engineering Essentials: Python, Safety Practices, and Advanced CAD modelling (training)
Year 3 (30 + 3 training credits)	
Semester 1 (15 + 1.5 training credits)	Semester 2 (15 + 1.5 training credits)
AF3625 Engineering Economics	ME31002 Linear Systems and Control
AMA2112 Mathematics II	ME32001 Manufacturing Fundamentals
ENG3003 Engineering Management	ME34002 Engineering Thermodynamics
ME31001 Dynamics and Vibrations	ME34004 Fluid Mechanics
ME34005 Fundamentals of Thermal Science and Engineering	Service-learning ^{^^} (Recommendation: ME3S01 Engineering Design for Community)
ME39003/IC382 Multidisciplinary Manufacturing Project (training)	
Year 4 (27 credits)	
Semester 1 (14 credits)	Semester 2 (13 credits)
ELC3531 Professional Communication in English for Engineering Students	ME Elective Subject II [@]
ME Elective Subject I [@]	ME Elective Subject III [@]
CAR III ^{^^}	Free Elective ^{^^} (Recommendation: ME electives)
LCR III Chinese Language Subject	
ME49001 Final Year Capstone Project	

+ Upon the completion of any two Faculty Electives among AMA1120 Basic Mathematics II/AP10005 Physics I/AP10006 Physics II/ENG2003 Information Technology, the remaining subjects will be counted as Major Subjects.

^{^^} The study pattern for these GUR subjects is indicative only.

[@] Please refer to Table 5.2(b) for more details about the ME Electives.

Table 5.2.5: BEngME Normal Progression Pattern (4-yr)

(120 credits + 10 training credits)

(For students without a Physics background at admission)

Year 1 (34 + 2 training credits)	
Semester 1 (14 + 1 training credit)	Semester 2 (17 + 1 training credit)
AMA1110 Basic Mathematics I	AMA1120 Basic Mathematics II ⁺
AP10001 Introduction to Physics I (<i>Free Elective</i>) [#]	AP10005 Physics I ⁺
ENG1004 Engineering Professionals in Society I	APSS1L01 Tomorrow's Leaders
MM1031 Introduction to Innovation and Entrepreneurship	EIE1005 Fundamental AI and Data Analytics
CAR I ^{^^}	ENG2002 Computer Programming
LCR I English Language Subject	LCR II English Language Subject
Healthy Lifestyle ^{^^}	
ME29004/IC2117 Fundamentals of Engineering Design & Prototyping (<i>training</i>)	
Summer Semester (3 credits)	
CAR II ^{^^}	
Year 2 (29 + 5 training credits)	
Semester 1 (15 + 3 training credits)	Semester 2 (14 + 2 training credits)
AMA2111 Mathematics I	EE2901S Basic Electricity and Electronics
AP10006 Physics II ⁺	ENG2003 Information Technology ⁺
ENG2001 Fundamentals of Materials Science and Engineering	ENG3006 Engineering Professionals in Society II
ME22003 Visualization and Communication in Design Engineering	ME32002 Engineering Design Fundamentals
ME23001 Engineering Mechanics	ME33001 Mechanics of Materials
ME39002/IC348 Appreciation of Manufacturing IC348 Processes (<i>training</i>)	ME39004/IC390 Applied Engineering Essentials: Python, Safety Practices, and Advanced CAD modelling (<i>training</i>)
Year 3 (30 + 3 training credits)	
Semester 1 (15 + 1.5 training credits)	Semester 2 (15 + 1.5 training credits)
AF3625 Engineering Economics	ME31002 Linear Systems and Control
AMA2112 Mathematics II	ME32001 Manufacturing Fundamentals
ENG3003 Engineering Management	ME34002 Engineering Thermodynamics
ME31001 Dynamics and Vibrations	ME34004 Fluid Mechanics
ME34005 Fundamentals of Thermal Science and Engineering	Service-learning ^{^^} (<i>Recommendation: ME3S01 Engineering Design for Community</i>)
ME39003/IC382 Multidisciplinary Manufacturing Project (<i>training</i>)	
Year 4 (27 credits)	
Semester 1 (14 credits)	Semester 2 (13 credits)
ELC3531 Professional Communication in English for Engineering Students	ME Elective Subject II [@]
ME Elective Subject I [@]	ME Elective Subject III [@]
CAR III ^{^^}	Free Elective ^{^^} (<i>Recommendation: ME electives</i>)
LCR III Chinese Language Subject	
ME49001 Final Year Capstone Project	

+ Upon the completion of any two Faculty Electives among AMA1120 Basic Mathematics II/AP10005 Physics I/AP10006 Physics II/ENG2003 Information Technology, the remaining subjects will be counted as Major Subjects.

^{^^} The study pattern for these GUR subjects is indicative only.

[#] Upon the completion of "AP10001 Introduction to Physics", students can fulfill a free elective (3 credits).

[@] Please refer to Table 5.2(b) for more details about the ME Electives.

Table 5.2.6: BEngME Normal Progression Pattern (SY)
(64 credits + 6 training credits)[#]

Senior Year 1 (33 + 6 training credits)			
Semester 1 (18 + 3 training credits)		Semester 2 (15 + 1.5 training credits)	
AMA2112	Mathematics II	ME31002	Linear Systems and Control
ME31001	Dynamics and Vibrations	ME32001	Manufacturing Fundamentals
ME33001	Mechanics of Materials	ME34004	Fluid Mechanics
ME34002	Engineering Thermodynamics	ME46002	Numerical Methods for Engineers
CAR (M) ^{^^}		CAR (A)	Specially-designed with Engl-Lang
Service-learning ^{^^} (Recommendation: ME3S01 Engineering Design for Community)		ME39003/IC382	Multidisciplinary Manufacturing Project (training)
ME39002/IC348	Appreciation of Manufacturing Processes (training)		
Essential Components of General Education			
Summer Semester (1.5 training credits)			
ME39003/IC382 Multidisciplinary Manufacturing Project (training)			
Senior Year 2 (31 academic credits)			
Semester 1 (16 credits)		Semester 2 (15 credits)	
CLC3241P	Professional Communication in Chinese*	ENG3004	Society and the Engineer
ELC3531	Professional Communication in English for Engineering Students	ME Elective Subject III [@]	
ENG3003	Engineering Management	ME Elective Subject IV [@]	
ME Elective Subject I [@]		ME Elective Subject V [@]	
ME Elective Subject II [@]			
ME49001 Final Year Capstone Project			

[#] Since students may be required to meet specific requirements at admission, the credits required for graduation will vary according to the academic background of students.

^{^^} The study pattern for this GUR subject is indicative only.

[@] Please refer to Table 5.2(b) for more details about the ME Electives.

Table 5.2(b) - Electives for BEngME⁽ⁱ⁾

Electives [^]	Pre-requisites (if any)	Clusters			
		AE	RAS	EE	MMM
ENG4001 Project Management		√	√	√	√
ISE404 Total Quality Management ⁽ⁱⁱⁱ⁾	Refer to syllabus in Part B				√
ME41001 Automatic Control Systems	ME31002		√		
ME41003 Principles of Sound and Vibration	ME34002			√	
ME41005 Noise Control Engineering ⁽ⁱⁱ⁾	ME34002 & ME34004			√	
ME41006 Perceptual Robotics	ME31002		√		
ME42001 Artificial Intelligence in Products	ME31002 or ME41004		√		
ME42004 Development of Green Products ⁽ⁱⁱ⁾	ME32001				√
ME42005 CAD/CAE Technologies for Product Development	AMA2111		√		√
ME42007 Design for Product Safety and Reliability ⁽ⁱⁱⁱ⁾	ME32002		√		
ME42011 Fundamentals of Robotics	ME31002 or ME41004		√		

Electives [^]	Pre-requisites (if any)	Clusters			
		AE	RAS	EE	MMM
ME43003 Product Testing Technology	ME33001				√
ME44002 Engine Technology	ME34002		√	√	
ME44004 Heat and Mass Transfer ⁽ⁱⁱ⁾	ME34002 & ME34004	√		√	
ME44008 Air Conditioning for Indoor Air Quality	ME34002			√	
ME45001 Aerodynamics	ME34004	√			
ME45002 Aircraft Systems	ME34004	√	√		
ME46002 Numerical Methods for Engineers	AMA2111	√	√	√	√
ME46003 Numerical Fluid Mechanics and Heat Transfer	ME34002 or ME34004			√	

- (i) The electives are updated from time to time to ensure the best development of the programme and to ensure the best career for our students. Since there is a minimum planned class size for each subject, the Department has the discretion to cease the offering of subjects that fail to enroll students up to the minimum class size.
- (ii) Students must have completed the following electives to be eligible for Scheme “A” in Environmental (ENV) discipline: a) *ME41005 Noise Control Engineering*, b) *ME42004 Development of Green Products* **AND** c) *ME44004 Heat and Mass Transfer*.
- (iii) Students must have completed the following electives to be eligible for Scheme “A” in Manufacturing, Industrial & Systems (MIS) Discipline: a) *ME42007 Design for Product Safety and Reliability* **OR** b) *ISE404 Total Quality Management*.

(More details about the Discipline Matching Results for Accredited Degrees (Scheme “A”) can be obtained from the HKIE’s website: <https://hkie.org.hk/en/quali/disciplineforaccreddeg/>.)

Table 5.2.7: BEngME + AIDA Normal Progression Pattern (4-yr)

(134 credits + 10 training credits)

(For students **with** a Physics background at admission)

Year 1 (34 + 2 training credits)			
Semester 1 (16 + 1 training credit)		Semester 2 (15 + 1 training credit)	
AMA1110	Basic Mathematics I [%]	AMA1120	Basic Mathematics II ⁺
AP10005	Physics I ⁺	APSS1L01	Tomorrow's Leaders
MM1031	Introduction to Innovation and Entrepreneurship	EIE1005	Fundamental AI and Data Analytics
CAR I ^{^^}		ENG1004	Engineering Professionals in Society I
Free Elective/CAR II ^{^^}		ENG2002	Computer Programming [%]
LCR I	English Language Subject	LCR II	English Language Subject
Healthy Lifestyle ^{^^}			
ME29004/IC2117 Fundamentals of Engineering Design & Prototyping (training)			
Summer Semester (3 credits)			
Free Elective/CAR II ^{^^}			
Year 2 (32 + 5 training credits)			
Semester 1 (15 + 3 training credits)		Semester 2 (17 + 2 training credits)	
AMA2111	Mathematics I [%]	EE2901S	Basic Electricity and Electronics
AP10006	Physics II ⁺	ENG2003	Information Technology ⁺
ENG2001	Fundamentals of Materials Science and Engineering	ENG3006	Engineering Professionals in Society II
ME22003	Visualization and Communication in Design Engineering	ME32002	Engineering Design Fundamentals
ME23001	Engineering Mechanics	ME33001	Mechanics of Materials
ME39002/IC348	Appreciation of Manufacturing IC348 Processes (training)	ME39004/IC390	Applied Engineering Essentials: Python, Safety Practices, and Advanced CAD modelling (training)
		DSAI2201	Structures and Algorithms
Year 3 (36 + 3 training credits)			
Semester 1 (18 + 1.5 training credits)		Semester 2 (18 + 1.5 training credits)	
AF3625	Engineering Economics	ME31002	Linear Systems and Control
AMA2112	Mathematics II	ME32001	Manufacturing Fundamentals
ENG3003	Engineering Management	ME34002	Engineering Thermodynamics
ME31001	Dynamics and Vibrations	ME34004	Fluid Mechanics
ME34005	Fundamentals of Thermal Science and Engineering	DSAI1102	Data Analytics Fundamentals /
		DSAI1201	Introduction to Data Analytics /
		EIE1003	Foundations of Data Science
COMP4431	Artificial Intelligence	DSAI4203	Machine Learning
ME39003/IC382 Multidisciplinary Manufacturing Project (training)			
Year 4 (32 credits)			
Semester 1 (17 credits)		Semester 2 (15 credits)	
ELC3531	Professional Communication in English for Engineering Students	ME Elective Subject III [@]	
ME46002	Numerical Methods for Engineering [%] (ME Elective Subject I)	AIDA Elective II [@]	
ME Elective Subject II [@]		Service-learning ^{^^}	
		(Recommendation: ME3S01 Engineering Design for Community)	
AIDA Elective I [@]		CAR III ^{^^}	
LCR III	Chinese Language Subject		
ME49006 Integrated Capstone Project*			

+ Upon the completion of any two Faculty Electives among AMA1120 Basic Mathematics II/AP10005 Physics I/AP10006 Physics II/ENG2003 Information Technology, the remaining subjects will be counted as Major Subjects.

^^ The study pattern for these GUR subjects is indicative only.

% Subject for double counting towards both Major in ME and Secondary Major in AIDA awards (not exceed 12 credits)

* Subject for double fulfillment: students completing ME49006 can also fulfill the Final Year Project of Major in ME.

@ Please refer to Table 5.2(b) for more details about the ME Electives. For AIDA Elective, please click [here](#).

Subjects for AIDA curriculum

Table 5.2.7: BEngME + AIDA Normal Progression Pattern (4-yr)

(134 credits + 10 training credits)

(For students **without** a Physics background at admission)

Year 1 (34 + 2 training credits)			
Semester 1 (16 + 1 training credit)		Semester 2 (15 + 1 training credit)	
AMA1110	Basic Mathematics I [%]	AMA1120	Basic Mathematics II ⁺
AP10001	Introduction to Physics I (<i>Free Elective</i>) [#]	AP10005	Physics I ⁺
MM1031	Introduction to Innovation and Entrepreneurship	APSS1L01	Tomorrow's Leaders
CAR I ^{^^}		EIE1005	Fundamental AI and Data Analytics
Free Elective/CAR II ^{^^}		ENG2002	Computer Programming [%]
LCR I	English Language Subject	LCR II	English Language Subject
Healthy Lifestyle ^{^^}			
ME29004/IC2117 Fundamentals of Engineering Design & Prototyping (<i>training</i>)			
Summer Semester (3 credits)			
CAR II ^{^^}			
Year 2 (32 + 5 training credits)			
Semester 1 (15 + 3 training credits)		Semester 2 (17 + 2 training credits)	
AMA2111	Mathematics I [%]	EE2901S	Basic Electricity and Electronics
AP10006	Physics II ⁺	ENG2003	Information Technology ⁺
ENG2001	Fundamentals of Materials Science and Engineering	ENG3006	Engineering Professionals in Society II
ME22003	Visualization and Communication in Design Engineering	ME32002	Engineering Design Fundamentals
ME23001	Engineering Mechanics	ME33001	Mechanics of Materials
ME39002/IC348	Appreciation of Manufacturing IC348 Processes (<i>training</i>)	ME39004/IC390	Applied Engineering Essentials: Python, Safety Practices, and Advanced CAD modelling (<i>training</i>)
		DSAI2201	Structures and Algorithms
Year 3 (36 + 3 training credits)			
Semester 1 (18 + 1.5 training credits)		Semester 2 (18 + 1.5 training credits)	
AF3625	Engineering Economics	ME31002	Linear Systems and Control
AMA2112	Mathematics II	ME32001	Manufacturing Fundamentals
ENG3003	Engineering Management	ME34002	Engineering Thermodynamics
ME31001	Dynamics and Vibrations	ME34004	Fluid Mechanics
ME34005	Fundamentals of Thermal Science and Engineering	DSAI1102	Data Analytics Fundamentals /
		DSAI1201	Introduction to Data Analytics /
		EIE1003	Foundations of Data Science
COMP4431	Artificial Intelligence	DSAI4203	Machine Learning
ME39003/IC382 Multidisciplinary Manufacturing Project (<i>training</i>)			
Year 4 (32 credits)			
Semester 1 (17 credits)		Semester 2 (15 credits)	
ELC3531	Professional Communication in English for Engineering Students	ME Elective Subject III [@]	
ME46002	Numerical Methods for Engineering [%] (<i>ME Elective Subject I</i>)	AIDA Elective II [@]	
ME Elective Subject II [@]		Service-learning ^{^^}	
		(<i>Recommendation: ME3S01 Engineering Design for Community</i>)	
AIDA Elective I [@]		CAR III ^{^^}	
LCR III	Chinese Language Subject		
ME49006 Integrated Capstone Project*			

+ Upon the completion of any two Faculty Electives among AMA1120 Basic Mathematics II/AP10005 Physics I/AP10006 Physics II/ENG2003 Information Technology, the remaining subjects will be counted as Major Subjects.

^^ The study pattern for these GUR subjects is indicative only.

% Subject for double counting towards both Major in ME and Secondary Major in AIDA awards (not exceed 12 credits)

* Subject for double fulfillment: students completing ME49006 can also fulfill the Final Year Project of Major in ME.

Upon the completion of AP10001 Introduction to Physics, students can fulfill a free elective (3 credits).

@ Please refer to Table 5.2(b) for more details about the ME Electives. For AIDA Elective, please click [here](#).

Subjects for AIDA curriculum

5.3 Work-Integrated Education (WIE)

In accordance with the University's policies, all full-time UGC-funded undergraduates should fulfill the mandatory requirement of Work-integrated Education (WIE). WIE is a work-based, structured and measurable learning experience in an industrial context which is relevant to the students' areas of studies. A student is required to spend at least 2 weeks (or 80 hours) on WIE before graduation. WIE is aimed at providing value-added education leading to the development of all-round students with professional competence.

Mandatory WIE activities are credit-bearing, but they are **not** included in the 120 academic credits required for graduation. The WIE components will **not** be counted towards GPA calculation except as stipulated below. For the completion of every two weeks of WIE activities, one credit will be earned. The WIE activities can be fulfilled by at least one of the following:

- (a) Integration into the *Final Year Capstone Project (ME49001)* / *Integrated Capstone Project (ME49007)* (the Project), which is industrially/commercially based. However, it is most important that the Project and WIE activities should be assessed separately. It is equally important that the WIE activities of students working in the same project team should be assessed **individually** as they can vary from student to student. In addition, the duration of the WIE activities is not necessarily the same as that of the Project. In these cases, the credit value of the Project incorporating the WIE component will be counted in full towards the GPA calculation.
- (b) Perform during a summer placement in industrial/commercial sector.
- (c) Conduct in firm proposed by students with the prior approval of the WIE coordinator.

Detailed guidelines for students on WIE are available on the ME website (<https://www.polyu.edu.hk/me/experience-and-opportunities/work-integrated-education/>).

5.4 Industrial Centre (IC) Training

Industrial Centre (IC) training is aimed at providing students with hands-on experience in dealing with different engineering projects under the supervision of academic and technical staff at IC of the University. They are graded in the same manner as other taught subjects from A+ to F and will be counted in the Grade Point Average (GPA). However, they will not be counted towards the credit requirement of the award or contribute to the Award/Weighted GPA. Students must pass the IC training stipulated in the curriculum to be considered for the award.

5.5 Student Exchange

Exchanges to universities overseas for a semester or an academic year are possible through various exchange schemes organized by the University, Faculty or Department. While the number of exchanges is limited, students are encouraged to participate to enhance their all-roundedness and broaden their experience.

Block credit transfer may be given to exchange-out students. However, in order to ensure attaining pre-requisite knowledge for smooth integration of study, students will be consulted on subject selections in the visiting universities before leaving for the exchange.

More information is available on ME's website: <https://www.polyu.edu.hk/me/experience-and-opportunities/student-exchange/>.

5.6 Curriculum Mapping with Programme Learning Outcomes

An analysis of the curriculum in terms of the coverage of the programme learning outcomes of BEngIRE and BEngME, respectively (*as shown in Sections 3.2 and 3.3, respectively*), is presented in **Tables 5.6.1** (a, b and c) and **5.6.2** (a, b and c). The tables indicate the Major subjects, electives and training natures that we **Teach (T)** students, give students **Practice (P)** and **Measure (M)** achievement of the Programme Learning Outcomes (PLOs).

In summary, the curriculum address two programme learning outcomes which students are expected to achieve (*A*) *professional/academic knowledge and skills (PAK)*, and (*B*) *professional outlook and workplace skills (POW)*. There are seven items for PAK and five items for POW. The electives are updated continually to meet the need of the ever-evolving industrial communities in Hong Kong and the South China region. Essentially, they cover most of the PLOs with variations of themes from subject to subject. The subject learning outcomes to be achieved by every subject of the programme are listed in the syllabi as shown in Part B of this document.

BEng in Intelligent Robotics Engineering (BEngIRE)

Table 5.6.1 (a): Curriculum Map for Major Subjects with PLOs

Subject Code	Programme Learning Outcomes (PLOs)											
	PAK							POW				
	a	b	c	d	e	f	g	a	b	c	d	e
AMA1110	TP	TP										
AMA1120	TP	TP										
AMA2111	TP	TP										
AMA3708	TPM	TP										T
AP10005		TP										
AP10006		TP										
COMP2011	TP				TP		TP					T
EE2901S	TP	TP	TP									
EIE3320	TP				TPM		T					T
ELC3531											TPM	
ENG1004								TPM	TPM	TPM		
ENG2001	TP	TP	TP									
ENG2002	TP				TP			TP				T
ENG2003	TP			TP	TP			TP				
ENG3003							TPM	TPM	TPM	TPM		
ENG3006								TPM	TPM	TPM		
ME22003		TP		TP	TP						TP	
ME23001	TP	TP	TP				TP				TP	
ME31002	TP	TP	TPM	TPM	TPM	TPM						
ME32002				TPM	TPM		TPM			TP	TP	
ME33001	TPM	TPM	TPM		TP	TP						
ME35001		TP			TPM		TPM	TPM			TP	
ME35002		TP	TPM	TP		TP		TP			TP	TPM
ME42011	TPM	TPM	TP	TP		TP		T	TP		TP	T
ME42012	TP		TP	TPM		TPM		TP	TPM	T	TP	
ME42013	TP	TP	TPM	TP		TPM		T	TP		TP	T
ME49007	TPM	TPM	TP	TPM	TP	TP	TP	TPM	TPM	TP	TPM	TPM

Table 5.6.1(b): Curriculum Map for Electives with PLOs

Subject Code	Programme Learning Outcomes (PLOs)											
	PAK							POW				
	a	b	c	d	e	f	g	a	b	c	d	e
Group A (Level 3) <i>(Students are required to study any ONE subject from this group.)</i>												
COMP3011	TP	TP			TP				TPM			T
COMP3423	TP			TP	TP	TP	TP		TPM			
EE3002	TP		TP						TPM			
EIE3103	TP	TP	TP		TP				TPM			
EIE3373	TP		TP	TP	TP				TPM			
ISE3018	TP			TP		TP	TP		TPM	TP		
ISE3021	TP			TP			TP	TP	TPM	TP		
Group B (Level 4) <i>(Students are required to study any ONE/TWO subjects from this group.)</i>												
AAE4011	TP			TP	TP		TP	TP	TP			
ENG4001		TP			TP				TP	TP	TP	
ISE4024	TP		TP		TP	TP		TP				T
ME41001	TP	TP			TP						P	
ME41006	TP	TP	TP	TP	TP			TP				
ME42001		TP		TP	TP	TP					P	TP
ME42014	TP	TP	TP		TP	TPM					TP	TPM
ME46002	TP	TP	TP	TP								

Table 5.6.1(c): Curriculum Map for Training subjects with PLOs

Subject Code	Programme Learning Outcomes (PLOs)											
	PAK							POW				
	a	b	c	d	e	f	g	a	b	c	d	e
ME29003/ IC2105		TP			TP	TP	TP					
ME29004/ IC2117				TP		TP	TP					
ME39002/ IC348				TP	TP	TP	TPM					
ME39004/ IC390		TP		TP	TP							
ME39003/ IC382				TPM	P	TPM	P		TPM	TPM	P	
WIE								P	P	P	P	P

BEng in Mechanical Engineering (BEngME)**Table 5.6.2(a): Curriculum Map for Major Subjects with PLOs**

Subject Code	Programme Learning Outcomes (PLOs)											
	PAK							POW				
	a	b	c	d	e	f	g	a	b	c	d	e
AF3625								TP	TP		TP	
AMA1110	TP	TP										
AMA1120	TP	TP										
AMA2111	TP	TP										
AMA2112	TP	TP										
AP10005		TP										
AP10006		TP										
CLC3241P											TPM	
EE2901S	TP	TP	TP									
ELC3531											TPM	
ENG1004								TPM	TPM	TPM		TPM
ENG2001	TP	TP	TP									
ENG2002	TP				TP			TP				T
ENG2003	TP			TP	TP			TP				
ENG3003							TPM	TPM	TPM	TPM		
ENG3004								TPM	TPM	TPM	TP	TPM
ENG3006								TPM	TPM	TPM		TPM
ME22003		TP		TP	TP						TP	
ME23001	TP	TP	TP				TP				TP	
ME31001	TPM	TP	TPM	TP		TP						
ME31002	TP	TP	TPM	TPM	TPM	TPM						
ME32001				TP	TP	TP	TPM		TP	TPM	TPM	
ME32002				TPM	TPM		TPM			TP	TP	
ME33001	TPM	TPM	TPM		TP	TP						
ME34002	TP	TPM	TPM			TPM					TP	
ME34004	TPM	TPM	TP		TPM	TPM						
ME34005	TPM	TPM			TPM			TP				TP
ME46002	TP	TPM			TPM			TPM			TP	TPM
ME49001	TPM	TPM	TP	TPM	TP	TP	TP	TPM	TPM	TP	TPM	TPM

Table 5.6.2(b): Curriculum Map for Electives with PLOs

Subject Code	Intended Learning Outcomes (ILOs) of BEngME											
	PAK							POW				
	a	b	c	d	e	f	g	a	b	c	d	e
ENG4001		TP			TP				TP	TP	TP	
ISE404		TP			TP				TP	TP	TP	
ME41001	TP	TP			TP						P	
ME41003	TP	TP		TP							P	
ME41005	TP	TP		TPM	TPM						P	
ME41006	TP	TP	TP	TP	TP			TP				
ME42001		TP		TP	TP	TP					P	TP
ME42004		TP		TPM				TP	TPM		P	TP
ME42005		TP		TP	TP	TP						
ME42007		TP			TP				TP	TP	TP	
ME42011	TP	TP	TP		TP			TP				T
ME43003			TP		TP	TP	TP			TP		
ME44002	TP	TP	TPM			TPM		TP				
ME44004	TP	TP	TP					TPM				TPM
ME44008		TP		TP		TPM			TP	TPM		
ME45001	TP	TP			TP			TP				
ME45002		TP	TP		TP					TP		T
ME46002	TP	TPM			TPM			TPM			TP	TPM
ME46003	TP	TP	TP		TP			TP				

Table 5.6.2(c): Curriculum Map for Training subjects with PLOs

Subject Code	Intended Learning Outcomes (ILOs) of BEngME											
	PAK							POW				
	a	b	c	d	e	f	g	a	b	c	d	e
ME29004/ IC2117				TP		TP	TP					
ME39002/ IC348				TP	TP	TP	TPM					
ME39003/ IC382				TPM	P	TPM	P		TPM	TPM	P	
ME39004/ IC390		TP		TP	TP							
WIE								P	P	P	P	P

6. ACADEMIC REGULATIONS AND ASSESSMENT

The academic regulations described below are based on the information known as of July 2022. They are subject to review and changes from time to time. Students will be informed of the changes as and when appropriate. Important information relating to students' study is also published in the Student Handbook (<https://www.polyu.edu.hk/ar/web/en/for-polyu-students/student-handbook/index.html>).

6.1 Subject Registration and Withdrawal

In addition to programme registration, students need to register for the subjects at specified periods prior to the commencement of the semester. An add/drop period will also be scheduled for each semester/term. Students may apply for withdrawal of their registration on a subject after the add/drop period if they have a genuine need to do so. The application should be made to the relevant programme offering department and will require the approval of both the subject teacher and the host department Programme Leader concerned (or an alternate academic staff authorised by the programme offering Department). Applications submitted **after week 12** will not be considered. For approved applications of subject withdrawal, the tuition fee paid for the subject will be forfeited and the withdrawal status of the subject will be shown in the assessment result notification and transcript of studies, but will not be counted in the calculation of the GPA.

The pre-requisite requirements of a subject must have been fulfilled before a student registers for that subject. However, the Subject Offering Department has the discretion to waive the pre-requisite requirements of a subject, if deemed appropriate. If the pre-requisite subject concerned forms part of the requirements for award, the subject has to be passed in order to satisfy the graduation requirements for the programme concerned, despite the waiving of the pre-requisite.

(For the information of the pre-requisite requirements, please refer to the respective syllabus listed in Part B.)

Subject to the maximum study load of 21 credits per semester and the availability of study places, students are allowed to take additional subjects on top of the prescribed credit requirement for award before they become eligible for graduation. Students will be allowed to take additional subjects for broadening purpose, after they fulfil the graduation requirements and for the following semester. However, they will still be subject to the maximum study load of 21 credits per semester and the availability of places in the subjects concerned, and their enrolment will be arranged as subject-based students only and be subject to the rules on 'Admission of Subject-based Students', except that graduates from UGC-funded programmes will not be restricted to taking only subjects from a self-financed programme.

6.2 Study Load

For students following the progression pattern specified for their programme, they have to take the number of credits and subjects, as specified in the Programme Requirement Document, for each semester. Students cannot drop those subjects assigned by the department unless prior approval has been given by the department.

The normal study load is 15 credits in a semester for full-time study. The maximum study load to be taken by a student in a semester is 21 credits, unless exceptional approval is given by the Head of the programme offering Department (or his/her delegates). For such cases, students should be reminded that the study load approved should not be taken as grounds for academic appeal.

To help improve the academic performance of students on **academic probation**, these students will be required to take a reduced study load in the following semester (Summer Term excluded). The maximum number of credits to be taken by the students varies according to the policies of individual Departments and will be subject to the approval of the authorities concerned.

Students are not allowed to take zero subject in any semester, including the mandatory summer term as required by some programmes, unless they have obtained prior approval from the programme

offering Department; otherwise they will be classified as having unofficially withdrawn from their programme. Students who have been approved for zero subject enrolment (i.e. taking zero subject in a semester) are allowed to retain their student status and continue using campus facilities and library facilities. Any semester in which the students are allowed to take zero subject will nevertheless be counted towards the total period of registration.

Students who have obtained approval to pace their studies and students on programmes without any specified progression pattern who wish to take more than the normal load of 15 credits in a semester should seek advice from the Department concerned before the selection of subjects.

6.3 Subject Exemption

Students may be exempted from taking any specified subjects, including mandatory General University Requirements (GUR) subjects, if they have successfully completed similar subjects previously in another programme or have demonstrated the level of proficiency/ability to the satisfaction of the Subject Offering Department. If students are exempted from taking a specified subject, the credits associated with the exempted subject will not be counted towards meeting the award requirements. It will therefore be necessary for the students to consult the programme offering department and take another subject in order to satisfy the credit requirement for the award.

6.4 Credit Transfer

Students may be given credits for recognised previous studies including mandatory General University Requirements (GUR) subjects; and the credits will be counted towards meeting the requirements for award/degree. Transferred credits may not normally be counted towards more than one degree. The granting of credit transfer is a matter of academic judgment.

Credit transfer may be done with or without the grade being carried over; the former should normally be used when the credits were gained from PolyU. Credit transfer with the grade being carried over may be granted for subjects taken from outside the University, if deemed appropriate, and with due consideration to the academic equivalence of the subjects concerned and the comparability of the grading systems adopted by the University and the other approved institutions. Subject credit transfer is normally decided by the Subject Offering Department. However, for applications which are submitted by students who have completed an approved student exchange programme, the decision will be made by the Programme Offering Department in consultation with the Subject Offering Departments.

The validity period of credits previously earned, is 8 years after the year of attainment. Normally, not more than 50% of the credit requirement for award may be transferable from approved institutions outside the University. For transfer of credits from programmes offered by PolyU, normally not more than 67% of the credit requirement for award can be transferred. In cases where both types of credits are being transferred (*i.e. from programmes offered by PolyU and from approved institutions outside the University*), not more than 50% of the credit requirement for award may be transferred.

All credit transfers approved will take effect only in the semester for which they are approved. A student who applies for transfer of credits during the re-enrolment or the add/drop period of a particular semester will only be eligible for graduation at the end of that semester, even if the granting of credit transfer will immediately enable the student to satisfy the credit requirement for the award.

Regarding credit transfer for GUR subjects, the Programme Host Department is the approval authority at the time of admission to determine the number of GUR credits which an Advanced Standing student will be required to complete for the award concerned. Programme Host Departments should make reference to the mapping lists of GUR subjects, compiled by the Committee on General University Requirements (CoGUR), on the eligibility of the subjects which can qualify as GUR subjects. Applications for credit transfer of GUR subjects after admission will be considered, on a case-by-case basis, by the Subject Offering Department or College of Undergraduate Studies (CUS)/Service-Learning and Leadership Office (SLLO), in consultation with the relevant Sub-committee(s) under CoGUR, as appropriate.

For credit transfer of retaken subjects, the grade attained in the last attempt should be taken in the case of credit transfer with grade being carried over. Students applying for credit transfer for a subject taken in other institutions are required to declare that the subject grade used for claiming credit transfer was attained in the last attempt of the subject in their previous studies. If a student fails in the last attempt of a retaken subject, no credit transfer should be granted, despite the fact that the student may have attained a pass grade for the subject in the earlier attempts.

Students should *not* be granted credit transfer for a subject which they have attempted and failed in their current study unless the subject was taken by the student as an exchange-out student in his current programme.

For students admitted to an Articulation Degree or Senior Year curriculum which is already a reduced curriculum, they should not be given credit transfer for any required GUR subjects, and are required to complete at least 60 credits in order to be eligible for a Bachelor's award.

6.5 Deferment of Study

Students may apply for deferment of study if they have a genuine need to do so such as illness or posting to work outside Hong Kong. Approval from the Department offering the programme is required. The deferment period will not be counted towards the total period of registration.

Application for deferment of study from students who have not yet completed the first year of a full-time programme will only be considered in exceptional circumstances.

Where the period of deferment of study begins during a stage for which fees have been paid, no refund of such fees will be made.

Students who have been approved for deferment are not entitled to enjoy any campus facilities during the deferment period.

6.6 General Assessment Regulations

Students progress by credit accumulation, i.e. credits earned by passing individual subjects can be accumulated and counted towards the final award.

A 'level' in a programme indicates the intellectual demand placed upon students and may characterise each subject with respect to its recommended sequencing within that programme. Upper level subjects should normally build on lower level subjects. Pre-requisite requirements, if any, must therefore be spelt out on a subject basis.

A 'subject' is defined as a discrete section of the programme which is assigned a separate assessment. A list of subjects, together with their level and weightings, shall be published in the Programme Requirement Document.

The language of assessment for all programmes/subjects shall be **English**, unless approval is given for it to be otherwise. Such approval shall normally be granted at the stage of validation.

6.7 Principles of Assessment

Assessment of learning and assessment for learning are both important for assuring the quality of student learning. Assessment of learning is to evaluate whether students have achieved the intended learning outcomes of the subjects that they have taken and have attained the overall learning outcomes of the academic programme at the end of their study at a standard appropriate to the award. Appropriate methods of assessment that align with the intended learning outcomes should be designed for this purpose. The assessment methods will also enable the teacher to differentiate students' different levels of performance within the subject. Assessment for learning is to engage students in productive learning activities through purposefully designed assessment tasks.

Assessment will also serve as feedback to students. The assessment criteria and standards should be made explicit to students before the start of the assessment to facilitate student learning, and feedback provided should link to the criteria and standards. Timely feedback should be provided to students so that they are aware of their progress and attainment for the purpose of improvement.

The ultimate authority in the University for the confirmation of academic decisions is the Senate, but for practical reasons, the Senate has delegated to the Faculty/School Boards the authority to confirm the decisions of Boards of Examiners provided these are made within the framework of the General Assessment Regulations. Recommendations from Board of Examiners which fall outside these Regulations shall be ratified by the Academic Planning and Regulations Committee (APRC) and reported to the Senate as necessary.

6.8 Assessment Methods

Students' performance in a subject can be assessed by continuous assessment and/or examinations, at the discretion of the individual Subject Offering Department. Where both continuous assessment and examinations are used, the weighting of each in the overall subject grade shall be clearly stated in the Programme Requirement Document. The Subject Offering Department can decide whether students are required to pass both the continuous assessment and examination components, or either component only, in order to obtain a subject pass, but this requirement (to pass both, or either, components) shall be specified in the Programme Requirement Document. Learning outcome should be assessed by continuous assessment and/or examination appropriately, in line with the outcome-based approach.

Continuous assessment may include tests, assignments, projects, laboratory work, field exercises, presentations and other forms of classroom participation. Continuous assessment assignments which involve group work should nevertheless include some individual components therein. The contribution made by each student in continuous assessment involving a group effort shall be determined and assessed separately, and this can result in different grades being awarded to students in the same group.

Assessment methods and parameters of subjects shall be determined by the Subject Offering Department.

At the beginning of each semester, the subject lecturer should inform students the details of the methods of assessments to be used, within the assessment framework as specified in the Programme Requirement Document.

6.9 Progression/Academic Probation/Deregistration

The Board of Examiners shall, at the end of each semester (except for the Summer Term unless there are students who are eligible to graduate after completion of Summer Term subjects or the Summer Term is mandatory for the programme), determine whether each student is:

- (i) eligible for progression towards an award; or
- (ii) eligible for an award; or
- (iii) required to be de-registered from the programme.

When a student has a Grade Point Average (GPA) lower than 1.70, he will be put on **academic probation** in the following semester. If a student is able to pull his GPA up to 1.70 or above at the end of the semester, the status of “academic probation” will be lifted. The status of “academic probation” will be reflected in the examination result notification but not in the transcript of studies.

A student will have “progressing” status unless he falls within any one of the following categories which shall be regarded as grounds for **deregistration** from the programme:

- (i) **the student has reached the final year of the normal period of registration for that programme, as specified in the Programme Requirement Document, unless approval has been given for extension; or**

- (ii) the student has reached the maximum number of retakes allowed for a failed compulsory subject; or
- (iii) the student's GPA is lower than 1.70 for *two* consecutive semesters and his Semester GPA in the second semester is also lower than 1.70; or
- (iv) the student's GPA is lower than 1.70 for *three* consecutive semesters.

When a student falls within any of the categories as stipulated above, except for category (i) with approval for extension, the Board of Examiners shall de-register the student from the programme without exception.

A student may be deregistered from the programme enrolled before the time frame specified in the above conditions (iii) or (iv) if his academic performance is poor to the extent that the Board of Examiners deems that his chance of attaining a GPA of 1.70 at the end of the programme is slim or impossible.

The progression of students to the following academic year will not be affected by the GPA obtained in the Summer Term, unless Summer Term study is mandatory for all students of the programme and constitutes a requirement for graduation, and is so specified in this Document.

If the student is not satisfied with the de-registration decision of the Board of Examiners, he can lodge an appeal. All such appeal cases will be referred directly to Academic Appeals Committee (AAC) for final decision. Views of Faculties/Schools/Departments will be sought and made available to AAC for reference.

6.10 Plagiarism and Bibliographic Referencing

The University views plagiarism and copying of copyright materials, without the licence of the copyright owner, as a serious disciplinary offence. The notes in the “Appendix 2. Plagiarism” section of the Student Handbook (<https://www.polyu.edu.hk/ar/students-in-taught-programmes/student-handbook/>) aim to help students comply with the University’s policy on plagiarism in continuous assessment, bibliographic referencing and photocopying of copyright materials. For details, please refer to: https://www.polyu.edu.hk/ous/docdrive/Academic_Integrity/Student_Guide.pdf

6.11 Recording of Disciplinary Actions in Students’ Records

With effect from Semester One of 2015/16, disciplinary actions against students’ misconducts will be entered in students’ records.

Students who are found guilty of academic dishonesty or non-compliance with examination regulations will be subject to the penalty of having the subject result concerned disqualified and be given a failure grade with a remark denoting ‘Disqualification of result due to academic dishonesty/non-compliance with examination regulations’. The remark will be shown in the students’ record as well as the assessment result notification and transcript of studies, until their leaving the University.

Students who have committed disciplinary offences (covering both academic and non-academic related matters) will be put on ‘disciplinary probation’. The status of ‘disciplinary probation’ will be shown in the students’ record as well as the assessment result notification, transcript of studies and testimonial during the probation period, until their leaving the University. The disciplinary probation is normally one year unless otherwise decided by the Student Discipline Committee.

The University reserves the right to withhold the issuance of any certificate of study to a student/graduand who has unsettled matters with the University, or is subject to disciplinary action.

6.12 Retaking of Subjects

Students may only retake a subject which they have failed (i.e. Grade F or S or U). Retaking of subjects is with the condition that the maximum study load of 21 credits per semester is not exceeded.

The number of retakes of a subject should be restricted to two, i.e. a maximum of three attempts for each subject is allowed.

In cases where a student takes another subject to replace a failed elective subject, the fail grade will be taken into account in the calculation of the GPA, despite the passing of the replacement subject. Likewise, students who fail in a Cluster Area Requirement (CAR) subject may need to take another subject from the same Cluster Area in order to fulfill this part of the GUR, since the original CAR subject may not be offered, in such cases, the fail grade of the first CAR subject will be taken into account in the calculation of the GPA, despite the passing of the second CAR subject. (*Note 1*)

Students need to submit a request to the Faculty/School Board for the second retake of a failed subject.

Students who have failed a compulsory subject after two retakes and have been de-registered can submit an appeal to the Academic Appeals Committee (AAC) for a third chance of retaking the subject. In case AAC does not approve further retakes of a failed compulsory subject or the taking of an equivalent subject with special approval from the Faculty, the student concerned would be de-registered and the decision of the AAC shall be final within the University.

Note 1: In these circumstances when students do not have a choice to retake a failed subject, such as when the failed subject has been phased out, a 'tie-subject' arrangement can be made with the approval of the Faculty/School Board. Under the arrangement, another appropriate subject can be taken as equivalent to the subject which is not offered. Upon passing the equivalent subject, the fail grade of the original subject will be replaced by the latest grade of the retake subject and the failure grade of the original subject will not be taken into account in the calculation of the GPA.

6.13 Exceptional Circumstances

Absence from an assessment component

If a student is unable to complete all the assessment components of a subject, due to illness or other circumstances which are beyond his control and considered by the Subject Offering Department as legitimate, the Department will determine whether the student will have to complete a late assessment and, if so, by what means. This late assessment shall take place at the earliest opportunity, and normally before the commencement of the following academic year (except that for Summer Term, which may take place within 3 weeks after the finalisation of Summer Term results). If the late assessment cannot be completed before the commencement of the following academic year, the Faculty/School Board Chairman shall decide on an appropriate time for completion of the late assessment.

The student concerned is required to submit his application for late assessment in writing to the Head of Department offering the subject, within five working days from the date of the examination, together with any **original** supporting documents. Approval of applications for late assessment and the means for such late assessments shall be given by the Head of Department offering the subject or the subject teacher concerned, in consultation with the Programme Leader. Verification of the supporting documents with the issuing authority may be conducted by the Subject Offering Department as part of the approval process.

Aegrotat award

If a student is unable to complete the requirements of the programme in question for the award due to very serious illness, or other very special circumstances which are beyond his control, and considered by the Board of Examiners as legitimate, the Faculty/School Board will determine whether the student will be granted an aegrotat award. Aegrotat award will be granted under very exceptional circumstances.

A student who has been offered an aegrotat award shall have the right to opt either to accept such an award, or request to be assessed on another occasion to be stipulated by the Board of Examiners; the student's exercise of this option shall be irrevocable.

The acceptance of an aegrotat award by a student shall disqualify him from any subsequent assessment for the same award.

An aegrotat award shall normally not be classified, and the award parchment shall not state that it is an aegrotat award. However, the Board of Examiners may determine whether the award should be classified, provided that they have adequate information on the students' academic performance.

6.14 Grading

Assessment grades shall be awarded on a criterion-referenced basis. A student's overall performance in a subject shall be graded as follows:

Subject Grade	Short Description	Elaboration on Subject Grading Description
A+ A A-	Excellent	Demonstrates excellent achievement of intended subject learning outcomes by being able to skillfully use concepts and solve complex problems. Shows evidence of innovative and critical thinking in unfamiliar situations, and is able to express the synthesis or application of ideas in a logical and comprehensive manner.
B+ B B-	Good	Demonstrates good achievement of intended subject learning outcomes by being able to use appropriate concepts and solve problems. Shows the ability to analyse issues critically and make well-grounded judgements in familiar or standard situations, and is able to express the synthesis or application of ideas in a logical and comprehensive manner.
C+ C C-	Satisfactory	Demonstrates satisfactory achievement of intended subject learning outcomes by being able to solve relatively simple problems. Shows some capacity for analysis and making judgements in a variety of familiar and standard situations, and is able to express the synthesis or application of ideas in a manner that is generally logical but fragmented.
D+ D	Pass	Demonstrates marginal achievement of intended subject learning outcomes by being able to solve relatively simple problems. Can make basic comparisons, connections and judgments and express the ideas learnt in the subject, though there are frequent breakdowns in logic and clarity.
F	Fail	Demonstrates inadequate achievement of intended subject learning outcomes through a lack of knowledge and/or understanding of the subject matter. Evidence of analysis is often irrelevant or incomplete.

“F” is a subject failure grade, whilst all others (“D” to “A+”) are subject passing grades. No credit will be earned if a subject is failed.

Indicative descriptors for modifier grades

Main Grade (solid)	The student generally performed at this level, indicating mastery of the subject intended learning outcomes at this level.
+ (exemplary)	The student consistently performed at this level and exceeded the expectations of this level in some regards, but not enough to claim mastery at the next level.
- (marginal)	The student basically performed at this level, but the performance was inconsistent or fell slightly short in some regards.

Note: The above indicative descriptors for modifier grades are not applicable to the pass grades D and D+

A numeral grade point is assigned to each subject grade. The grade points assigned to subject grades attained by students are as follows:

Grade	Grade Point for grades attained
A+	4.3
A	4.0
A-	3.7
B+	3.3
B	3.0
B-	2.7
C+	2.3
C	2.0
C-	1.7
D+	1.3
D	1.0
F	0.0

At the end of each semester/term, a Grade Point Average (GPA) will be computed, as follows, and based on the grade point of all the subjects:

$$\text{GPA} = \frac{\sum_{n=1}^N \text{Subject Grade Point}_n \times \text{Subject Credit Value}_n}{\sum_{n=1}^N \text{Subject Credit Value}_n}$$

where N = number of all subjects (inclusive of failed subjects) taken by the student up to and including the latest semester/term. For subjects which have been retaken, only the grade point obtained in the final attempt will be included in the GPA calculation.

In addition, the following subjects will be excluded from the GPA calculation:

- (i) Exempted subjects
- (ii) Ungraded subjects
- (iii) Incomplete subjects
- (iv) Subjects for which credit transfer has been approved, but without any grade assigned
- (v) Subjects from which a student has been allowed to withdraw (i.e. those with the grade “W”)

Subject which has been given an “S” code, i.e. absent from all assessment components, will be included in the GPA calculation and will be counted as “zero” grade point. GPA is thus the unweighted cumulative average calculated for a student, for all relevant subjects taken from the start of the programme to a particular point of time. GPA is an indicator of overall performance, and ranges from 0.00 to 4.30.

Different Types of GPA

GPA will be calculated for each Semester including the Summer Term. This **Semester GPA** will be used to determine students' eligibility to progress to the next Semester alongside with the ‘cumulative GPA’. However, the Semester GPA calculated for the Summer Term will not be used for this purpose, unless the Summer Term study is mandatory for all students of the programme concerned and constitutes part of the graduation requirements.

The GPA calculated after the second Semester of the students' study is therefore a “**cumulative**” GPA of all the subjects taken so far by students, and without applying any level weighting.

Along with the “cumulative” GPA, a **weighted GPA** will also be calculated, to give an indication to the Board of Examiners on the award classification which a student will likely get if he makes steady progress on his academic studies. Weighted GPA will be computed as follows:

$$\text{Weighted GPA} = \frac{\sum_{n=1}^N \text{Subject Grade Point}_n \times \text{Subject Credit Value}_n \times W_n}{\sum_{n=1}^N \text{Subject Credit Value}_n \times W_n}$$

where W_n = weighting to be assigned according to the level of the subject

N = number of all subjects counted in GPA calculation, except any subjects passed after the graduation requirement has been met or subjects taken on top of the prescribed credit requirements for award.

For calculating the weighted GPA (and award GPA) to determine the Honours classification of students who satisfy the graduation requirements of Bachelor's degree awards, a University-wide standard weighting will be applied to all subjects of the same level, with a weighting of 2 for Level 1 and 2 subjects and a weighting of 3 for Level 3 and above subjects. Although the Industrial Centre/Mechanical Engineering training credits are counted in the GPA calculation, they are excluded from the calculation of weighted GPA and award GPA. Same as for GPA, weighted GPA ranges from 0.00 to 4.30.

When a student has satisfied the requirements for award, an **award GPA** will be calculated to determine his award classification. GUR subjects will be included in the calculation of award GPA for all programmes.

For students taking the Major/Minor study route, a separate GPA will be calculated for their Major and Minor programmes. The **Major GPA** will be used to determine his award classification, which will be so reflected on the award parchment. The **Minor GPA** can be used as a reference for Board of Examiners to moderate the award classification for the Major, as explained further in Section 8.

For students taking the Major/Secondary Major study route, there is no separate “Secondary Major GPA”. The Major GPA is the weighted GPA of all subjects contributing to the Major and Secondary Major.

6.15 University Graduation Requirements

To be eligible for a Bachelor's Degree award under the 4-year full-time undergraduate curriculum, a student must:

1. Complete successfully the requisite number of credits as defined in Section 5.
2. Earn a cumulative GPA of 1.70 or above at graduation.
3. Complete successfully the mandatory Work-Integrated Education (WIE) component.
4. Satisfy the residential requirement for at least one-third of the normal credit requirement for the award unless the professional bodies concerned stipulate otherwise.
5. Satisfy any other requirements, including Free Electives, as specified in the Programme Requirement Document.
6. Satisfy the following GUR requirements:

Normal Year 1 Intake:

(a) Language and Communication Requirements (LCR)	9 credits
(b) Artificial Intelligence and Data Analytics Requirement (AIDA)	2 credits
(c) Innovation and Entrepreneurship Requirement (IE)	1 credit
(d) Leadership Education and Development (LEAD)	3 credits
(e) Service-Learning (SL)	3 credits
(f) Cluster Areas Requirement (CAR) <i>Student are required to fulfil the English and Chinese Reading & Writing Requirements (ER/EW & CR/CW).</i>	9 credits (3 credits from each of the three cluster areas)
(g) Healthy Lifestyle (HLS)	Non-credit bearing
	Total = 27 credits

Senior Year Intake:

(a) Language and Communication Requirements (LCR)*	---
(b) Artificial Intelligence and Data Analytics Requirement (AIDA)	3 credits
(c) Innovation and Entrepreneurship Requirement (IE)	---
(d) Leadership Education and Development (LEAD)	---
(e) Service-Learning (SL)	---
(f) Cluster Areas Requirement (CAR) 3 credits from CAR (M) “Chinese History and Culture” with Chinese Reading and Writing Requirements (CR/CW) 3 credits from a specially-designed CAR (A) “Human Nature, Relations and Development” – English language with English Reading and Writing Requirements (ER/EW)	6 credits
(g) Essential Components of General Education	Non-credit bearing
	Total = 9 credits

* This is normally not required. Only those students not meeting the equivalent standard of the Undergraduate Degree LCR (based on their previous studies in AD/HD programmes and their academic performance) will be required to take degree LCR subjects on top of the normal curriculum requirement.

(a) Language and Communication Requirements (LCR)**English**

All 4-year undergraduate students must successfully complete two[#] 3-credit English language subjects as stipulated by the University, according to their English language proficiency level (Table A).

Table A English LCR subjects (each 3 credits)

English language competence level/ Subject	ELC1011 Practical English for University Studies	ELC1012/ELC1013 English for University Studies	Any LCR Proficient level elective subject in English (Table B)
HKDSE Level 4 and above or equivalent	---	Subject 1	Subject 2
HKDSE Level 3 or equivalent	Subject 1	Subject 2	---

Table B Proficient level electives for DSE Level 4 students and above (or equivalent) (each 3 credits)

LCR Proficient level electives	ELC2011 Advanced English Reading and Writing Skills
	ELC2012 Persuasive Communication
	ELC2013 English in Literature and Film
	ELC2014 Advanced English for University Studies

Students entering the University with specified attainment grades in certain public examinations can be given credit transfer or exemption for one or both LCR English subjects. For the subject exempted, students must take another subject to make up the 3 credits. For the subject granted credit transfer, student do not need to take any other subject to make up the credits. Please refer to the CUS website for more information: <https://www.polyu.edu.hk/cus/>.

Chinese

All 4-year undergraduate students must successfully complete one* 3-credit Chinese language subject as stipulated by the University, according to their Chinese language proficiency level. All Chinese-speaking students will be required to take the same Chinese LCR subject.

Cantonese will be used as the Medium of Instruction (MOI) of a certain proportion of Chinese LCR subject (Table C). Students taking the Cantonese version of the subjects will be offered a 39-hour non-credit bearing e-Learning course in Putonghua (optional).

Table C Chinese LCR subjects (each 3 credits)

Subject Code	Subject Title	Mol
CLC1104C	University Chinese	Cantonese
CLC1104P	University Chinese	Putonghua

* Students entering the University with specified attainment grades in certain public examinations can be given credit transfer or exemption of the Chinese LCR subject. For the subject exempted, students must take any other subject to make up the 3 credits. For the subject granted credit transfer, students do not need to take any other subject to make up the credits. Please refer to the CUS website for more information: <https://www.polyu.edu.hk/cus/>.

For non-Chinese speaking students or students whose Chinese standards are at junior secondary level or below:

Depending on students' Chinese language background and/or previous exam results, students might be exempted from the CLCR at the time of admission. They may use a CLCR subject or free elective to fulfil the credit requirement. Students are also exempted from fulfilling the CR/CW of CAR.

Otherwise, one subject from Table D will be pre-assigned to students as Chinese LCR depending on their Chinese language proficiency and/or previous exam results. They might be given an assessment to ascertain that the pre-assigned subject is suitable for them.

Table D Chinese LCR Subjects for non-Chinese speakers or students whose Chinese standards are at junior secondary level or below

Subject Code	Subject Title
CLC1151	Chinese I (for non-Chinese speaking students)
CLC1152	Chinese II (for non-Chinese speaking students)
CLC2151	Chinese III (for non-Chinese speaking students)
CLC2154	Chinese IV (for non-Chinese speaking students)
CLC2152	Chinese Literature - Linguistics and Cultural Perspectives (for non-Chinese speaking students)

For the updated information, please refer to the website of the College of Undergraduate Studies (CUS): <https://www.polyu.edu.hk/cus/>.

Writing Requirement

In addition to the LCR in English and Chinese explained above, all students must also, among the Cluster Areas Requirement (CAR) subjects they take (see Section (f) below), pass one subject that includes the requirement for a substantial piece of writing in English and one subject with the requirement for a substantial piece of writing in Chinese.

Reading Requirement

All students must, among the CAR subjects they take, pass one subject that includes the requirement for reading an extensive text in English and one with the requirement for reading an extensive text in Chinese.

A list of approved CAR subjects for meeting the Writing Requirement (with a “W” designation) and for meeting the Reading Requirement (with an “R” designation) is shown at: <https://www.polyu.edu.hk/cus/>.

Non-Chinese speakers and those students whose Chinese standards are at junior secondary level or below will by default be exempted from the DSR – Chinese and CAR – Chinese Reading and Writing Requirements. However, students whose Chinese standards are at junior secondary level or below will still be required to take one Chinese LCR subject to fulfil their Chinese LCR. Students who fulfil the following criteria are exempted from the CLCR requirements:

- (i) those with their first/native language as non-Chinese stated on the grade report of recognized tests (e.g. IELTS, TOEFL, etc.); OR
- (ii) those admitted with international qualifications without taking any Chinese subject(s) in their secondary/ high school; OR
- (iii) those taken Chinese B or Chinese AB(SL) in IB Diploma; OR
- (iv) NCS status shown on the official proof provided by their secondary school.

(b) Artificial Intelligence and Data Analytics (GUR-AIDA) Requirement

GUR-AIDA subjects aim to help students demonstrate an understanding of the foundational concepts of AIDA, acquire basic skills in using AIDA technologies and applications, articulate examples of how the adoption AIDA could enhance students’ chosen disciplines and demonstrate an awareness of global contemporary ethical issues and impact from AIDA applications in daily life.

4-year degree students are required to complete a 2-credit for GUR-AIDA within the first year of study. Senior Year intakes students are required to take the mandatory subject “*Essential Components of General Education*” for the e-module on GUR-AIDA.

(c) Innovation and Entrepreneurship (GUR-IE) Requirement

GUR-IE subjects aim to help students to demonstrate an elementary understanding of innovation and entrepreneurship, appreciate the importance of innovation and entrepreneurship in local and global community, appreciate the applications and implications of the latest technologies on entrepreneurship and innovation in their chosen disciplines and identify ethical issues in entrepreneurship and innovation.

4-year degree students are required to complete a 1-credit for GUR-IE within the first year of study. Senior Year intakes students are required to take the mandatory subject “*Essential Components of General Education*” for the e-module on GUR-IE.

(d) Leadership Education and Development (LEAD)

Leadership Education and Development aims to introduce students to the concepts and theories of leadership and to prepare them to become ethical leaders. The “*Online Tutorial on Academic Integrity*”, National Education and the “learning-to-learn” component are embedded in the LEAD subjects.

“*APSSIL01 Tomorrow's Leaders*” will be pre-assigned for 4-year degree students. Senior Year intakes students are *not* required to complete this subject.

(e) Service-Learning (SL)

Service-Learning aims to help students develop into a competent professional with a heart to serve others. This subject contains a significant service component, which allows them to apply their skills and knowledge to deal with complex issues in social services, develop a stronger empathy for people in need and develop a stronger sense of civic responsibility.

All students must complete one 3-credit Service-Learning subject. Some Service-Learning subjects are embedded in the major study curriculum and need to be taken as Major subjects. Please check Section 5 for details.

Service-Learning subjects are administered by the Service-Learning and Leadership Office (SLLO). For subject offering and further information, please visit the SLLO website:
<https://www.polyu.edu.hk/sllo/>.

(f) Cluster Areas Requirement (CAR)

To provide students with a balanced and all-rounded educational experience. CAR subjects aim to nurture students’ intellectual capacity, global outlook, communication and critical thinking skills from a multidisciplinary perspective. There are three different Cluster Areas of CAR:

- *Human Nature, Relations and Development (CAR (A))*
- *Chinese History and Culture (CAR (M))*
- *Cultures, Organisation, Societies and Globalisation (CAR (N))*

4-year degree students must complete at least one 3-credit subject in each of these Cluster Areas, among which students have to fulfil the Reading and Writing Requirements in Chinese and English (CR/CW and ER/EW). Senior Year intakes students must complete one specially-designed CAR (A) – English Language subject (with embedded English Reading and Writing Requirements (ER/EW)) within the first year of study, and one CAR M subject, and fulfil the Chinese Reading and Writing Requirements (CR/CW).

A list of CAR subjects under each of the four Cluster Areas is available at:
<https://www.polyu.edu.hk/cus/>.

(g) Healthy Lifestyle (HLS)

4-year degree students are required to complete all the components of the Healthy Lifestyle Programme, including Introductory Lecture, Sports Training or Sports Participation, E-Learning and Wrap-up Lecture, before graduation, preferably, in your first two years of study. The HLS programme is a set of courses and activities covering different dimensions of health that aims to help you build up a healthy lifestyle. Through this programme, you will acquire knowledge on healthy living that contributes to one's well-being and be able to make responsible health decisions. The current HLS programme consists of four components that need to be completed in sequence. Details of the programme can be found at: <https://www.polyu.edu.hk/cus/>.

Senior Year intakes students are *not* required to take the Health Lifestyle Programme.

(h) Essential Components of General Education

To allow Senior Year intakes students to acquire the basic knowledge of the following:

- *Academic Integrity (OTAI)*
- *Artificial Intelligence and Data Analytics (AIDA)*
- *Innovation and Entrepreneurship (IE)*
- *National Education (NE)*

Senior Year intakes students are required to take “**Essential Components of General Education**”, and complete and pass the individual e-modules of the four components within the first year of study (Semesters 1 and 2). The “**Online Tutorial on Academic Integrity**” should be completed **by Week 5 of Semester 1**. 4-year degree students are *not* required to complete this component.

6.16 Award

A student is required to graduate as soon as he satisfies the graduation requirements. The student concerned is required to apply for graduation, in the semester in which he is able to fulfil all his graduation requirements, and after the add/drop period for that semester has ended.

Subject to the maximum study load of 21 credits per semester and the availability of study places, students are allowed to take additional subjects on top of the prescribed credit requirement for award before they become eligible for graduation.

The awards will be classified based upon the Award GPA. Any subjects passed after the graduation requirement has been met or subjects taken on top of the prescribed credit requirements for award shall not be taken into account in the grade point calculation for award classification. However, if a student attempts more electives (or optional subjects) than the requirement for graduation in or before the semester in which he becomes eligible for award, the electives (or optional subjects), except for subjects which are selected by students to fulfill the free electives requirement for graduation, with a higher grade/contribution shall be included in the grade point calculation (i.e. the excessive subjects with lower grade/contribution, including failed subjects, will be excluded).

The following are guidelines for Boards of Examiners' reference in determining award classifications:

Honours Degrees	Guidelines
1st	The student's performance/attainment is outstanding and identifies him as exceptionally able in the field covered by the programme in question.
2:i	The student has reached a standard of performance/ attainment which is more than satisfactory but less than outstanding.
2:ii	The student has reached a standard of performance/ attainment judged to be satisfactory, and clearly higher than the 'essential minimum' required for graduation.

Honours Degrees	Guidelines
3rd	The student has attained the 'essential minimum' required for graduation at a standard ranging from just adequate to just satisfactory.

Under exceptional circumstances, a student who has completed an Honours degree programme, but has not attained Honours standard, may be awarded a Pass-without-Honours degree. A Pass-without-Honours degree award will be recommended, when the student has demonstrated a level of final attainment which is below the 'essential minimum' required for graduation with Honours from the programme in question, but when he has nonetheless covered the prescribed work of the programme in an adequate fashion, while failing to show sufficient evidence of the intellectual calibre expected of Honours degree graduates. For example, if a student in an Honours degree programme has a Grade Point Average (GPA) of 1.70 or more, but his Weighted GPA is less than 1.70, he may be considered for a Pass-without-Honours classification. A Pass-without-Honours is an unclassified award, but the award parchment will not include this specification.

Students who have committed academic dishonesty or non-compliance with examination regulations will be subject to the penalty of the lowering of award classification by one level. For undergraduate students who should be awarded a Third class Honours degree, they will be downgraded to a Pass-without-Honours. The minimum of downgraded overall result will be kept at a Pass.

The following are the award GPA ranges for determining award classifications:

<i>Honours classification</i>	<i>Award GPA</i>
1st	3.60 – 4.30
2:i	3.00 – 3.59
2:ii	2.40 – 2.99
3rd	1.70 – 2.39

Decisions by the Boards of Examiners on award classifications to be granted to each student on completion of the programme shall be ratified by the Faculty/School Board (of Examiners). For cases the decisions of which do not conform to the above indicative GPA range, they should be referred, by the Faculty/School Board (of Examiners), to the Academic Planning and Regulations Committee (APRC) for ratification.

7. SCHEME/PROGRAMME OPERATION AND MANAGEMENT

7.1 Departmental Undergraduate Programme Committee (DUPC)

The Departmental Undergraduate Programme Committee will exercise the overall academic and operational responsibility for the Scheme/Programme.

7.2 Programme Executive Group

The day-to-day operation of the Scheme/Programme will be carried out by the Programme Executive Group, which consists of the Scheme/Programme Leader and Deputy Scheme/Programme Leader. The Group will report the operation back to the Departmental Undergraduate Programme Committee.

7.3 Student-Staff Consultative Group (SSCG)

The Student-Staff Consultative Group consists of Student Representatives together with the Scheme/Programme Leader. The Committee is normally chaired by the Scheme/Programme Leader and meets at least twice a year. Issues to be kept under consideration include: student workload, teaching methods, balance between subject areas, training matter and other areas of mutual concern.

7.4 Academic Advising

Academic advising at PolyU aims to help students to make informed and intelligent academic decisions/choices about their study at PolyU that suit their intellectual, professional and personal goals. It is instrumental to promoting student success, and plays a vital role in enhancing students' overall learning experience at PolyU. The specific aims are:

- (a) To build up an early connection between the students and the University, and to promote their sense of affiliation to the Faculties/Departments/Schools, and the University,
- (b) To provide students with accurate information about the academic regulations and requirements regarding their Major/programme, and the General University Requirements (GUR),
- (c) To assist students to explore their interests, abilities and values on academic pursuits, and formulate appropriate intellectual, professional and personal goals,
- (d) To provide advice and guidance to students that enable them to develop and pursue a study plan for their 4 years of study appropriate for meeting their professional and personal goals at PolyU,
- (e) To connect students to resources, opportunities and support within and outside the University to enhance their educational experiences and success.

Every student will be assigned an Academic Advisor from the ME Department. The main responsibilities of the academic advisor will include:

- Building rapport with the students, serving as a bridge that connects them to the department,
- Being accessible and available to students, and responding to their questions and concerns,
- Helping students to consider and clarify their intellectual, professional and personal goals,
- Helping students to develop an appropriate study plan and assisting in their selection of appropriate courses in their major study and GUR so as to achieve their identified goals and holistic development,
- Helping students to understand academic regulations and requirements,
- Identifying students with special learning needs or early signs of learning problems, and referring/encouraging them to seek help or support.

8. MINOR, SECONDARY MAJOR AND FAST-TRACK

Under the framework of the 4-year undergraduate degree programmes, students can work for either a single discipline Major, a Major plus a Minor or a Major plus a Secondary Major (i.e. X + Scheme).

8.1 Minor Study

Minor study will be a free choice by students and not mandatory. Each student can be applied to take not more than one Minor. This option will not be applicable to students who are admitted to the advanced stage of the programme. Students who opt for Minor study will be subject to the following regulations:-

- (i) A Minor programme is a collection of subjects totaling 18 credits with at least 50% (9 credits) of the subjects at Level 3 or above.
- (ii) Students must apply to and obtain approval from the Minor-offering Department, no later than the commencement of second year of study.
- (iii) Only students with a GPA of 2.50 or above can be considered for Minor study enrolment. The Minor-offering Department may set a quota and additional requirements for their Minor.
- (iv) Subject to approval by the Minor-offering Department, students may count up to 6 credits from their Major/GUR subjects [including Language Communication Requirement (LCR) subjects at proficiency level] towards their chosen Minor. Nevertheless, students must take at least 6 credits from their chosen Minor programme in order to satisfy the residential requirement of their chosen Minor. In addition, to be eligible for the Major and Minor awards, the total number of credits taken by the students for their Major-Minor studies must not be lower than the credit requirement of the single discipline Major programme.
- (v) Credit transfer can be given for not more than 9 credits of a Minor programme if the previous credits were earned from approved institutions outside of the university; and not more than 12 credits of a Minor programme if the previous credits were earned from programmes offered by PolyU.
- (vi) Students are expected to complete their approved Minor as part of their graduation requirements. Students who wish to withdraw from a Minor need to apply for approval officially from the Minor-offering Department, normally before the end of the add/drop period of the last Semester of study.
- (vii) Students are required to obtain a GPA of at least 1.70 to satisfy the requirement for graduation with a Major plus a Minor.
- (viii) There is no guarantee that a clash-free timetable can be provided for all students who pursue Minor study.

8.2 Secondary Major (only applicable to students who will opt BEng ME)

The University aims at nurturing and developing students with attributes that will prepare graduates to become “preferred graduates” for the professions. To benefit students by providing a broad-based curriculum and multidisciplinary study, apart from the Major in Mechanical Engineering, students are encouraged to take a Secondary Major in Artificial Intelligence and Data Analytics (AIDA). Students who opt Major in Product Analysis and Engineering Design are not eligible to select AIDA as their Secondary Major.

Studying on a Secondary Major is a free choice by students and not mandatory. Normally, this option to study for a Secondary Major will not be applicable to students admitted to the advanced stage of a programme, nor to students admitted to an articulation degree programme. Each student may take not more than one Secondary Major. Students who opt for a Secondary Major will be subject to the following regulations:-

- (i) Students are expected to complete the “X + Secondary Major” within the normal duration of the major programme.
- (ii) Students may count up to 12 credits of their Major/GUR subjects towards the Secondary Major. Nevertheless, students must take at least 12 credits from their chosen Secondary Major in order to satisfy the residential requirement of the chosen Secondary Major. Students who have

- completed more than 12 credits of subjects that are eligible for double counting will need to apply for graduation and indicate the subjects intended for double counting.
- (iii) Students must apply to and obtain approval from the programme offering Department, normally no later than the commencement of the second year of study, to be admitted to the Secondary Major.
 - (iv) Only students with a Cumulative GPA of 2.70 or above may be considered for Secondary Major enrolment. Each Secondary Major may stipulate additional selection criteria for admission.
 - (v) Students must complete the Secondary Major as part of their graduation requirements. Students who wish to withdraw from the Secondary Major must obtain approval from the programme offering Department normally before the end of the add/drop period of the last semester of study.
 - (vi) If deemed appropriate by the programme offering Department, students are allowed to take a Major with a Secondary Major and a Minor. Subjects already double-counted for the Major and Secondary Major cannot be used to fulfil the Minor requirement.
 - (vii) Curriculum is available at Secondary Major in AIDA website:
<https://www.polyu.edu.hk/comp/study/ug-programmes/aida/>.

Table 8.2.1: Credit Requirements and Subject List of Secondary Major in AIDA

Artificial Intelligence and Data Analytics (AIDA)	Credits
<u>Core</u>	30
Mathematics I for AIDA	(3)
• <i>AMAI110 Basic Mathematics I – Calculus and Probability & Statistics</i>	
Mathematics II for AIDA	(3)
• <i>AMA2111 Mathematics I</i>	
Programming I: Programming Fundamentals	(3)
• <i>ENG2002 Computer Programming</i>	
Programming II: Data Structures and Algorithms	(3)
• <i>DSAI2201 Data Structures and Algorithms</i>	
Fundamentals of Data Analytics	(3)
• <i>DSAI1102 Data Analytics Fundamentals; <u>OR</u></i>	
• <i>DSAI1201 Introduction to Data Analytics; <u>OR</u></i>	
• <i>EIE1003 Foundations of Data Science</i>	
Machine Learning	(3)
• <i>DSAI4203 Machine Learning</i>	
Artificial Intelligence	(3)
• <i>COMP4431 Artificial Intelligence</i>	
DSR-AIDA Bridging Subject	(3)
• <i>ME46002 Numerical Methods for Engineers</i>	
Integrated Capstone Project	(6)
• <i>ME49006 Integrated Capstone Project</i>	
<u>Electives</u>⁽ⁱ⁾	6
Total:	36

(i) Please refer to the curriculum of the Secondary Major in AIDA at <https://www.polyu.edu.hk/comp/study/ug-programmes/aida/>.

8.3 Fast-track Integrated Bachelor's and Master's Degree Programme

The fast-track programmes offer a pathway for high-achieving and ambitious students to complete both undergraduate and taught postgraduate studies at an accelerated pace. Students can enrol in the programme either in Year 1 or Year 3, depending on their academic standing.

- i) Eligible entrants will receive an offer of admission to the undergraduate programme, as well as a conditional offer to the taught postgraduate programme. Upon successful completion of both the undergraduate and taught postgraduate degrees, students will be awarded two separate degrees: a Bachelor's degree and a Master's degree.
- ii) As part of the Fast-track Programme, students will take taught postgraduate courses that comprise no more than 30% of the taught postgraduate curriculum (9 credit units), which can be taken as Majors or Free electives in the undergraduate curriculum.
- iii) To remain on the Fast-track Programme, students are required to maintain a yearly GPA of 3.30 or above.

More information about the Fast-track programme for ME students can be obtained from <https://www.polyu.edu.hk/me/study/fast-track-programme/>.

PART B SYLLABI

GUR Subject

Subject Description Form

Subject Code	APSS1L01																				
Subject Title	Tomorrow's Leaders																				
Credit Value	3																				
Level	1																				
GUR Requirements Intended to Fulfill	This subject intends to fulfill the following requirement(s): ☐ Healthy Lifestyle ☐ AI and Data Analytics (AIDA) ☐ Innovation and Entrepreneurship (IE) ☐ Languages and Communication Requirement (LCR) ☒ Leadership Education and Development (LEAD) ☐ Service-Learning ☐ Cluster-Area Requirement (CAR) ☐ Human Nature, Relations and Development [CAR A] ☐ Science, Technology and Environment [CAR D] ☐ Chinese History and Culture [CAR M] ☐ Cultures, Organizations, Societies and Globalization [CAR N] ☐ China-Study Requirement ☐ Yes or ☐ No ☐ Writing and Reading Requirements ☐ English or ☐ Chinese																				
Pre-requisite / Co-requisite/ Exclusion	Nil																				
Assessment Methods	<table border="1" style="width: 100%; border-collapse: collapse; text-align: center;"> <thead> <tr> <th style="width: 40%;">100% Continuous Assessment</th><th style="width: 30%;">Individual Assessment</th><th style="width: 30%;">Group Assessment</th></tr> </thead> <tbody> <tr> <td>1. Class Participation (including 5% “Learning to learn” self-reflection)</td><td>20%</td><td>--</td></tr> <tr> <td>2. Group Project</td><td>--</td><td>30%</td></tr> <tr> <td>3. Term Paper</td><td>50%</td><td>--</td></tr> <tr> <td>4. Online Academic Integrity Tutorial Test</td><td>Compulsory Pass Requirement</td><td></td></tr> <tr> <td>5. Law Abiding-Leadership Test</td><td>Compulsory Pass Requirement</td><td></td></tr> </tbody> </table>			100% Continuous Assessment	Individual Assessment	Group Assessment	1. Class Participation (including 5% “Learning to learn” self-reflection)	20%	--	2. Group Project	--	30%	3. Term Paper	50%	--	4. Online Academic Integrity Tutorial Test	Compulsory Pass Requirement		5. Law Abiding-Leadership Test	Compulsory Pass Requirement	
100% Continuous Assessment	Individual Assessment	Group Assessment																			
1. Class Participation (including 5% “Learning to learn” self-reflection)	20%	--																			
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3. Term Paper	50%	--																			
4. Online Academic Integrity Tutorial Test	Compulsory Pass Requirement																				
5. Law Abiding-Leadership Test	Compulsory Pass Requirement																				

	• The grade is calculated according to the percentage assigned; • The completion and submission of all component assignments are required for passing the subject; and • Student must pass all component(s) if he/she is to pass the subject.
Objectives	The course is designed to enable students to learn and integrate theories, research and concepts of the basic personal qualities (particularly intrapersonal and interpersonal qualities) of effective leaders. This subject also intends to help students develop and reflect on their intrapersonal qualities, interpersonal qualities and connection of learning to oneself. Finally, the subject cultivates students' appreciation of the importance of intrapersonal and interpersonal qualities in effective leadership.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. understand and integrate theories, research and concepts on the basic qualities (particularly intrapersonal and interpersonal qualities) of effective leaders; b. develop self-awareness and self-understanding; c. demonstrate self-leadership in pursuit of continual self-improvement; d. apply intrapersonal and interpersonal skills in daily lives; e. appreciate the importance of intrapersonal and interpersonal qualities in effective leadership, particularly the connection of learning in the subject to one's professional development and personal growth; f. recognize and accept their responsibility as professionals and citizens to the society and the world.
Subject Synopsis/ Indicative Syllabus	1. An overview of the personal attributes of effective leaders: roles of intrapersonal and interpersonal qualities in effective leadership and university graduates' employability in the service economy; compulsory requirements of the subject: "Learning to learn" assessment; Online Tutorial on Academic Integrity; law abiding-leadership assessment; group presentation; individual assignment; class participation. 2. Self-leadership in effective leaders: the importance of self-understanding and self-management; "Learning to learn" ability; life-long learning and leadership. 3. Cognitive competence (critical thinking): misinformation, disinformation, and propaganda; different types of thinking styles; critical thinking model; roles of cognitive competence, critical thinking and problem solving in effective leadership; learning to learn. 4. Social emotional competence: social awareness; relationship management; the application of social emotional competence in daily lives and in effective leadership. 5. Resilience and stress-coping: concepts and theories of resilience and stress-coping; relationship between resilience, stress and

	stress-coping; role of resilience in effective leadership; application of resilience and stress-coping on daily basis. 6. Morality and integrity: moral competence; role of morality in effective leadership; ethical leadership; importance of moral competence in different professions, academic integrity in university students (online tutorial on academic integrity). 7. Spirituality: connectedness to others, personal beliefs and values, meaning of life, spirituality and professional development, role of spirituality in effective leadership; spiritual practices in daily lives. 8. Law-abidance as a quality of leadership: basic concepts and theories related to law-abiding leadership and socially responsible leadership; importance of law-abiding leadership and socially responsible leadership to professionals and the general public; basic knowledge on national security and the Hong Kong; Hong Kong National Security Law and the Article 23 legislation and details of the Safeguarding National Security Ordinance; a brief overview of modern Chinese history, the Constitution, and the Basic Law. 9. Cultural competence and global citizenship: cultural competence in a globalized world; global citizenship and effective leadership; responsibilities of university students as both professionals and citizens of the society. 10. Effective communication: basic communication skills; importance of effective communication to daily life and leadership; care and compassion in effective leadership. 11. Team building: theories, concepts, skills and blocks of team building; role of team building in effective leadership; application of team building in different professions and daily lives. Note: For the topic on law abidance and the Hong Kong National Security Law, students are required to pass an online assessment with multiple-choice questions. Students can take the assessment with multiple attempts. The assessment does not carry any mark.
Teaching/Learning Methodology	Students taking this course are expected to be sensitive to their own behavior in intrapersonal and interpersonal contexts. Intellectual thinking, reflective learning, experiential learning and collaborative learning are emphasized in the course. Case studies on successful and fallen leaders will also be covered in the course. The teaching/learning methodology includes: 1. Lectures (including e-learning modules); 2. Experiential classroom activities; 3. Group project presentation; 4. Written assignment.

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					
			a	b	c	d	e	f
	1. Class Participation (including 5% “Learning to learn” self-reflection) ^	20%	✓	✓	✓	✓	✓	✓
	2. Group Project*	30%	✓	✓	✓	✓	✓	✓
	3. Term Paper^	50%	✓	✓	✓		✓	
	4. Academic integrity online module and assessment^	0%	✓			✓	✓	
	5. Quiz on law abidance and Hong Kong National Security Law^	0%	✓	✓	✓	✓	✓	✓
	Total	100 %						
*assessment is based on group effort ^assessment is based on individual effort								
Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:								
1. <u>Assessment of Class Participation (20%)</u> : It is expected that both online and classroom activities, and preparation for lectures can help students understand the subject matter and oneself, develop social skills, connect learning to oneself and promote an appreciation of the importance of intrapersonal and interpersonal leadership qualities. Hence, marks for class participation (including the participation in e-learning modules) and preparation for lectures will be given. Students will be assessed by: a) preparation for class (e.g., complete e-learning modules, online assignment, and dig up materials before class), b) participation in class and online learning activities (e.g., completion of worksheets and sharing in class, participation in online discussion forum) and c) volunteering to answer questions and join discussions. Also, students will be invited to rate the performance and learning of other group members in an honest and authentic manner. The marks will reflect the mastery of knowledge, self-reflection and quality of interpersonal skills (such as collaboration with other members and contribution to the group) of the group members								

	Peer assessment will contribute to marks in class participation. As the university strategic priority, “Learning to learn” has the aim to support the development of students’ ability to engage in the learning process, manage their own learning, and take their learning to a higher level. “Learning to learn” concept will be covered in the lectures and students are required to develop a personal development plan at the beginning of the course. To encourage students to reflect on their experience in achieving their learning goals set in the development plan, students are required to reflect on their learning to learn ability and related learning experience in a reflective journal (5%). 2. <u>Assessment of Group Project (30%)</u>: Group project presentation can give an indication of the students’ understanding and integration of theories and concepts on the personal qualities in effective leadership, personal and group reflections, interpersonal skills and degree of recognition of the importance of active pursuit of knowledge covered in the course. 3. <u>Assessment of Term Paper (50%)</u>: Individual paper can give an indication of the students’ understanding and integration of theories and concepts on the personal qualities in effective leadership, self-assessment, self-reflection, connection of the subject matter to oneself and degree of recognition of the importance of active pursuit of knowledge covered in the course. 4. <u>Quiz on Law Abidance and Hong Kong National Security Law (0%)</u>: As universities have the obligation to conduct education on the Constitution, Basic Law and Hong Kong National Security Law, students are required to take a 3-hour face-to-face lecture on law abiding leadership and 7 hours of self-study. Based on the related materials on modern Chinese history, the Constitution, the Basic Law, restoration of Hong Kong to mainland China, national security and the Hong Kong National Security Law, students have to take an assessment with 20 multiple choice questions. Students can pass the assessment if he/she has correct answers on at least 16 questions (multiple attempts allowed). A student will fail in this subject if he/she cannot pass this assessment component. 5. <u>Academic Integrity Online Module and Assessment (0%)</u>: As academic integrity is very important for university students, students are required to take an online Academic Integrity program lasting for two hours. First, students are required to take a multiple-choice test with 10 questions in the pre-test without a passing mark (multiple attempts allowed). After that, students need to study four online modules to understand the concepts of academic integrity and ways to avoid academic dishonesty. Finally, students are required to take another multiple-choice test with 20 questions in the post-test with a passing benchmark of 15 questions (multiple attempts allowed). A student will fail in this
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	subject if he/she cannot pass this assessment component. They must complete this component by the 5th week of the semester. Based on the implementation of this subject in the past ten academic years (2012-2022), evaluation findings consistently showed that this subject was able to achieve the intended learning outcomes in the students. The positive evaluation findings are documented as follows: Leung, H. (2016). Levels of reflection on teaching a leadership and positive youth development subject. <i>International Journal on Disability and Human Development</i>, 15(2), 211-220. Leung, H., Shek, D. T. L., & Mok, B. P. W. (2016). Post-lecture subjective outcome evaluation of a university subject on leadership and intrapersonal development. <i>International Journal of Child and Adolescence Health</i>, 9(2), 223-234. Li, X., & Shek, D. T. L. (2020). Objective outcome evaluation of a leadership course utilising the positive youth development approach in Hong Kong. <i>Assessment & Evaluation in Higher Education</i>, 45(5), 741-757. Ma, C. M. S., Shek, D. T. L., & Li, P. P. K. (2017). Evaluation of a leadership and intrapersonal development subject for university students: Experience in Hong Kong. <i>International Journal of Child and Adolescent Health</i>, 10(3), 337-346. Ma, C. M. S., Shek, D. T. L., Li, P. P. K., Mok, B. P. W. & Leung, E. Y. K. (2016). Qualitative evaluation of a leadership and intrapersonal development subject for university students in Hong Kong. <i>International Journal of Child and Adolescent Health</i>, 9(2), 217-224. Shek, D. T. L. (2012). Development of a positive youth development subject in a university context in Hong Kong. <i>International Journal on Disability and Human Development</i>, 11(3), 173-179. Shek, D. T. L. (2013). Promotion of holistic development in university students: A credit-bearing subject on leadership and intrapersonal development. <i>Best Practices in Mental Health</i>, 9(1), 47-61. Shek, D. T. L., Fok, H. K., Leung, C. T. L., & Li, P. P. K. (2016). Qualitative evaluation of a credit-bearing leadership subject in Hong Kong. <i>International Journal of Child and Adolescent Health</i>, 9(2), 173-183. Shek, D. T. L., & Leung, J. T. Y. (2014) Perceived benefits of a university subject on leadership and intrapersonal development. <i>International Journal on Disability and Human Development</i>. 13(4), 481-488. Shek, D. T. L., & Ma, C. M. S. (2014). Do university students change after taking a subject on leadership and intrapersonal
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	development? <i>International Journal on Disability and Human Development</i>, 13(4), 451-456. Shek, D. T. L., Sun, R. C. F., Tsien-Wong, T. B. K., Cheng, C. T., & Yim H. Y. (2013). Objective outcome evaluation of a leadership and intrapersonal development subject for university students. <i>International Journal on Disability and Human Development</i>, 12(2), 221-227. Shek, D. T. L., & Wu, F. K. Y. (2014). The role of teachers in youth development: Reflections of students. <i>International Journal on Disability and Human Development</i>, 13(4), 473-480. Shek, D. T. L., Wu, F. K. Y., Leung, C. T. L., Fok, H. K., & Li, P. P. K. (2016). Focus group evaluation of a subject on leadership and intrapersonal development in Hong Kong. <i>International Journal of Child and Adolescent Health</i>, 9(2), 185-194. Shek, D. T. L., & Yu, L. (2014). Post-course subjective outcome evaluation of a subject on leadership and intrapersonal development for university students in Hong Kong. <i>International Journal on Disability and Human Development</i>, 13(4), 457-464. Shek, D. T. L., & Yu, L. (2016). Student feedback on a subject on leadership and intrapersonal development for university students in Hong Kong. <i>International Journal on Disability and Human Development</i>, 15(3), 339-345 Shek, D. T. L., & Yu, L. (2017). An evaluation study on a university general education subject in Hong Kong. <i>International Journal of Adolescent Medicine and Health</i>, 29(1), 103-109. Shek, D. T. L., Yu, L., Lin, L., Li, X., Zhu, X., Dou, D., Chai, W., Chak, Y., Ho, W., Leung, E., Li, P., Mok, B., Shek, V., Shek, E., & Jin, T. (2021). Nurturing leadership qualities under COVID-19: Student perceptions of the qualities and effectiveness of online teaching and learning on leadership development. <i>International Journal of Child and Adolescent Health</i>, 14(1), 89-100. Shek, D. T. L., Zhu, X., Li, X., & Dou, D. (2022). Satisfaction with HyFlex teaching and law-abiding leadership education in Hong Kong university students under COVID-19. <i>Applied Research in Quality of Life</i>, 1-26. Yu. L., Shek, D. T. L., & Leung, E. Y. K. (2016). Post-lecture evaluation of a university subject on leadership and intrapersonal development. <i>International Journal of Child and Adolescent Health</i>, 9(2), 155-164.	
Student Study Effort Expected	Class contact:	
	Lectures and experiential/online learning activities	39 Hrs.

	Other student study effort:	
	▪ Group project preparation	20 Hrs.
	▪ Reading and writing term paper	61 Hrs.
	Total student study effort	120 Hrs.
Reading List and References	Basic References Catalano, R. F., Berglund, M. L., Ryan, J. A. M., Lonczak, H. S., & Hawkins, J. D. (2002). Positive youth development in the United States: Research findings on evaluations of positive youth development programs. <i>Prevention and Treatment</i>, 5(15), 1-106. Dalton, J., & Crosby, P. (2007). Being and having: Shouldn't excellence in higher education (and people) be a measure of what one does rather than what one has? <i>Journal of College and Character</i>, 9(1), 1-5. Davies, L. (2006). Global citizenship: abstraction or framework for action? <i>Educational Review</i>, 58(1), 5-25. Dugan, J. P. (2006). Involvement and leadership: A descriptive analysis of socially responsible leadership. <i>Journal of College Student Development</i>, 47(3), 335-343. Dugan, J. P. (2015). The measurement of socially responsible leadership: Considerations in establishing psychometric rigor. <i>Journal of Educational, Cultural and Psychological Studies</i>, 12, 23-42. Hong Kong Government. (2020, July 7). The Law of the People's Republic of China on Safeguarding National Security in the Hong Kong Special Administrative Region. Available at https://www.isd.gov.hk/nationalsecurity/eng/pdf/NSL_QnA_Book.pdf. Gilley, A., Gilley, J. W., McConnell, C. W., & Veliquette, A. (2010). The competencies used by effective managers to build teams: An empirical study. <i>Advances in Developing Human Resources</i>, 12(1), 29-45. Goleman, D. (1995). <i>Emotional Intelligence: Why it can matter more than IQ</i>. New York: Bantam Books. Houghton, J. D., & Yoho, S. K. (2005). Toward a contingency model of leadership and psychological empowerment: When should self-leadership be encouraged? <i>Journal of Leadership and Organizational Studies</i>, 11(4), 65-84. Kim, Y. H., Chiu, C. Y., & Zou, Z. M. (2010). Know thyself: Misperceptions of actual performance undermine achievement motivation, future performance, and subjective well-being. <i>Journal of Personality and Social Psychology</i>, 99(3), 395-409. Kohlberg, L. (1964). Development of moral character and moral ideology. In M. L. Hoffman, & L. W. Hoffman (Eds.), <i>Review of Child Development Research</i> (pp. 381-431). New York: Russell Sage Foundation. Lau, P. S. Y., & Wu, F. K. Y. (2012). Emotional competence as a positive youth development construct: A conceptual review. <i>The Scientific World Journal</i>, 2012, 8 pages. doi:10.1100/2012/975189	

	Marsh, H. W. (1990). A multidimensional, hierarchical self-concept: Theoretical and empirical justification. <i>Educational Psychological Review</i>, 2(2), 77-172. Masten, A. S., & Obradović, J. (2006). Competence and resilience in development. <i>Annals of the New York Academy of Sciences</i>, 1094(1), 13-27. Rockstuhl, T., Seiler, S., Ang, S., Van Dyne, L., & Annen, H. (2011). Beyond general intelligence (IQ) and emotional intelligence (EQ): The role of cultural intelligence (CQ) on cross-border leadership effectiveness in a globalized world. <i>Journal of Social Issues</i>, 67(4), 825-840. Rycek, R. F., Stuhr, S. L., McDermott, J., Benker, J., & Swartz, M. D. (1998). Adolescent egocentrism and cognitive functioning during late adolescence. <i>Adolescence</i>, 33(132), 745-749. Seligman, M. E. P., & Csikszentmihalyi, M. (2000). Positive psychology: An introduction. <i>American Psychologist</i>, 55(1), 5-14. Shek, D. T. L. (2010). Nurturing holistic development of university students in Hong Kong: Where are we and where should we go? <i>The Scientific World Journal</i>, 10, 563-575. Shek, D. T. L. (2012). Spirituality as a positive youth development construct: A conceptual review. <i>The Scientific World Journal</i>, 2012, 8 pages. doi:10.1100/2012/458953 Shek, D. T. L., & Leung, H. (2016a). Developing self-leadership and responsibility and moving away from egocentrism. <i>International Journal on Disability and Human Development</i>, 15(2), 157-164. Shek, D. T. L., & Leung, H. (2016b). Resilience as a focus of a subject on leadership and intrapersonal development. <i>International Journal on Disability and Human Development</i>, 15(2), 149-155. Shek, D. T. L., & Leung, J. T. Y. (2016). Developing social competence in a subject on leadership and intrapersonal development. <i>International Journal on Disability and Human Development</i>, 15(2), 165-173. Shek, D. T. L., & Ho, W. W. L. (2016). Nurturing moral competence in university students via a credit-bearing subject. <i>International Journal on Disability and Human Development</i>, 15(2), 181-186. Shek, D. T. L., & Ho, W. W. L. (2016). Spirituality as a key positive youth development construct for university students. <i>International Journal on Disability and Human Development</i>, 15(2), 175-180. Shek, D. T. L. & Ma, C. M. S. (2016). Emotional competence: A key leadership competence for university students. <i>International Journal on Disability and Human Development</i>, 15(2), 127-134. Shek, D. T. L., & Ng, E. C. W. (2017). Leadership training for university students in Hong Kong: A critical review. <i>International Journal of Child and Adolescent Health</i>, 10(1), 25-40. Shek, D. T. L., & Wu, F. K. Y. (2016). Clear and positive identity as an attribute of an effective leader. <i>International Journal on Disability and Human Development</i>, 15(2), 143-148. Shek, D. T. L., & Yu, L. (2016). Cognitive competence: A key positive youth development construct for university students. <i>International Journal on Disability and Human Development</i>,
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Kumru, A., & Thompson, R. A. (2003). Ego identity status and self-monitoring behavior in adolescents. *Journal of Adolescent Research*, 18(5), 481-495.

Leung, J. T. Y., & Shek, D. T. L. (2020). Theories of adolescent development: Overview. In D. T. L. Shek, & J. T. Y. Leung (Eds.), *The encyclopedia of child and adolescent development*. (Volume 7: History, theory, and culture in adolescence). (pp. 2887-2897). New York, NY: Wiley.

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Last Updated in July 2025

Subject Description Form

Subject Code	EIE1005
Subject Title	Fundamental AI and Data Analytics
Credit Value	2
Level	1
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	Data analytics is for extracting valuable knowledge from raw data. It is very critical to industry, business, and government. Artificial intelligence and machine learning are widely used to automate data analytics processes. This subject aims to provide students with the fundamental concepts of artificial intelligence and data analytics. In particular, it offers hands-on experiences and case studies in the applications of AI and data analytics, with a strong emphasis on their applications in the context of innovation and technological development. It also helps students appreciate how data analytics and artificial intelligence influence our daily life.
Intended Subject Learning Outcomes	Upon completion of the subject, students will be able to: 1. Demonstrate an understanding of the foundational concepts of Artificial Intelligence and Data Analytics (AIDA); 2. Acquire basic skills in using AIDA technologies and applications; 3. Articulate examples of how the adoption AIDA could enhance their chosen disciplines and contribute to technological progress; 4. Demonstrate an awareness of global contemporary ethical issues and impact from AIDA applications in daily life.
Subject Synopsis/ Indicative Syllabus	1. <u>Introduction</u> • Introduction to AI and data analytics • Introduction to machine learning • Ethical issues and impact from AI and data analytics applications 2. <u>Understanding and Presenting Data</u> • Data quality and preprocessing • Data analysis – Basic statistics, clustering, frequent pattern mining • Data visualization 3. <u>AI and Data Analytics in the Virtual World</u> • Development of game AI • Reward scheduling via data analytics • AI and data analytics in the Metaverse

	• Developing game AI with Unity ML-Agents 4. <u>AI and Data Analytics for Computer Vision</u> • Machine learning for computer vision • Predictive analytics for computer vision • Feature extraction • Pattern recognition • Convolutional neural network • Developing simple computer vision systems 5. <u>Conversational AI and Data Analytics</u> • Prescriptive analytics for human-computer interaction • Spoken language technologies • Natural language understanding • Understanding users' behaviours via data analytics • AI Chatbots
Teaching/Learning Methodology	e-Learning Module: The e-learning module is developed and delivered by the Department of Computing at PolyU, consisting of readings, exercises and assessments that are designed to introduce students to the basic concept and practice of AIDA. The e-learning module will provide basic foundation concepts about AIDA, as well as their potential global and societal context impacts. A brief understanding about the technology and applications will also be provided. Students are required to successfully complete the e-learning module (including video watching, an after-class exercise, and a lab with the AIDA interactive playground) within the first seven weeks of the semester in which they are taking the subject. Besides, online materials and video demonstrations are prepared for each of the software tools used in the workshops. Students can gain the background information about the tools and the skill to use them before they join the Tutorial Workshops. Lectures: The subject matters will be delivered through lectures. Students will be engaged in the lectures through Q&A, discussions, and specially designed classroom activities. Tutorial Workshops: Students will work on three AI and data analytics workshops using software tools. In each workshop, students may start from doing some small and easy tasks at the first stage. Students will solve a simple real-world problem commonly found in daily life in the second stage, using the knowledge and know-how that they have learnt from the first stage and the lectures. Tutorials are conducted in an interactive manner through discussions between students and tutors on the problems encountered during the workshop activities. The subject will offer the datasets used in the workshop. Assignment and Demonstration: Students will write a report for each workshop and demonstrate their work for one of the three workshops.

Assessment Methods in Alignment with Intended Subject Learning Outcomes	Specific Assessment Methods/Tasks	% Weighting	Intended Subject Learning Outcomes to be Assessed (Please tick as appropriate)			
			1	2	3	4
	1. Continuous Assessment (total: 100%)					
	• Workshop reports	40%	✓	✓	✓	
	• Workshop demonstration	15%	✓	✓		
	• e-learning module and seminars	10%	✓	✓		✓
	• Test and quizzes	35%	✓	✓		✓
	Total	100%				
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Tutorial Workshops will require students to apply what they have learnt to solve problems. There will be open-ended tasks that allow students to exercise their creativity in solution design. In the report, students need to write a reflection on how AIDA can be applied to their disciplines. Three workshop reports weight 40% of the whole assessment. In the workshop demonstration, students will think critically to discuss various aspects in the demonstration. The demonstration will be assessed individually and weights 15% of the whole assessment. Each group member will demonstrate the part he/she is responsible for in the workshop. Tests will assess students' achievement of the learning outcomes in a more formal manner. There will be a quiz after each workshop and a final test for the whole subject. Tests are assessed individually and weights 35% of the whole assessment. The workshops will use publicly available or proprietary software tools so that programming knowledge and mathematical skills are kept to a minimum, i.e., no programming and high-level mathematical background is assumed. Students will use available data sources, machine learning models, analytic algorithms, and evaluation methods from the tools to solve data analytics problems. Enthusiastic students could use the cloud-based API to perform more complex tasks. E-learning module aims to equip students with the basic concept and practice of AIDA.					

Student Study Effort Expected	Class contact (time-tabled):	
	• Lectures	11 Hours
	• Tutorial Workshops	14 Hours
	• Tests	1 Hour
	Other student study effort:	
	• Lecture: preview/review of notes; e-Learning; Self-studying	24 Hours
	• Workshop assignment and demonstration; preparation for tests	20 Hours
	Total student study effort:	70 Hours
Reading List and References	Reference Materials: 1. J. Moreira, T. Horvath, and A. Carvalho, A General Introduction to Data Analytics, Newark: John Wiley & Sons, Incorporated, 2018. 2. K. Moreland. The ParaView Tutorial, Version 5.6. Technical Report SAND 2018-11803 TR, Sandia National Laboratories, 2018. 3. A. Majumder, <i>Deep Reinforcement Learning in Unity: With Unity ML Toolkit</i>. Berkeley, CA: Apress, 2020. 4. D. Buckley, “Unity ML-Agents Tutorials – Complete Guide,” <i>GameDev Academy</i>, 2022 (online resource). 5. Deconstructing Chatbots: https://www.youtube.com/hashtag/deconstructingchatbots 6. A.R. Freed, <i>Conversational AI: Chatbots that work</i>. Manning, 2021 7. M. McTear, <i>Conversational Ai: Dialogue Systems, Conversational Agents, and Chatbots (Synthesis Lectures on Human Language Technologies)</i>. Morgan & Claypool, 2020 8. E. Ruane, A. Birhane, and A. Ventresque, “Conversational AI: Social and Ethical Considerations”, in <i>Proc. AICS</i>, 2019. 9. J. Lau,B. Zimmerman, and F. Schaub, “Alexa, are you listening?: Privacy perceptions, concerns and privacy-seeking behaviors with smart speakers,” in <i>Proc. Human-Computer Interaction</i>, 2018, pp. 1-31 10. A. Dertat, “Applied Deep Learning – Part 4: Convolutional Neural Networks,” https://towardsdatascience.com/applied-deep-learning-part-4-convolutional-neural-networks-584bc134c1e2 (online resources) 11. Vaibhav Verdhhan, <i>Computer Vision Using Deep Learning: Neural Network Architectures with Python and Keras</i>. Apress, 2021.	

July 2025

Subject Description Form

Subject Code	ME32101/ ME3S01
Subject Title	Engineering Design for the Community
Credit Value	3
Level	3
Pre-requisite / Co-requisite/ Exclusion	Fundamental knowledge in Design and Engineering Science
Objectives	The objectives of this subject are to: 1. Introduce to students the concept and practice of service learning. 2. Raise students' awareness of social issues in Hong Kong and educate them on the challenges and needs of underprivileged communities in Hong Kong. 3. Develop a systemic platform to facilitate engineering/design students to apply their knowledge/skills to serve the community. 4. Reinforce the students' problem solving skill through real-life design projects. 5. Enhance students' generic competencies of innovative problems solving, communication and teamwork. 6. Nurture students' sense of social awareness, responsibility and engagement.
Intended Learning Outcomes	Upon the completion of the subject, the students will be able to: 1. Concept and Practice of Service Learning a) Link their service learning activities and experiences with academic content of the subject. b) Demonstrate empathy for people in need and a strong sense of civic responsibility. c) Evaluate people's needs by considering the complex issues in the service setting. d) Understand the role and responsibility both as a professional in their chosen discipline and as a responsible citizen. e) Function effectively in a multi-disciplinary team. 2. Discipline-specific Concepts, Issues and Skills f) Identify and formulate a design problem by developing design specifications to achieve the planned goals.

	g) Apply knowledge of design, mathematics and engineering science to analyze and predict the life-cycle performance of a design. h) Assess the impacts of various factors including, materials, human, environment, safety and reliability of a design.
Subject Synopsis/ Indicative Syllabus	The topics in the course syllabus cover three major areas: 1. Concept and Practice of Service Learning (i) Understand the social responsibility (ii) Proper attitude and behaviours in service delivery (iii) Reflection as a tool for learning (iv) Ethical issues in service learning 2. Discipline-Specific Concepts, Issues and Skills Fundamental knowledge of engineering design for problem solving including: (i) Problem identification and analysis (ii) Develop a design project with the goal to solve the problem (iii) Develop design specifications (iv) Design for ergonomics (v) Application of materials (vi) Use of common engineering components (vii) Design for reliability, safety and environmental-friendly (viii) Proper use of engineering/computational tools to conduct the design project 3. Project-Specific Concepts, Issues and Skills Knowledge about and understanding of an identified target group of the community including: (i) Their human psychology and behavior - Human psychology and behavior focuses on the human factors that affect the elderly daily life operation. It involves their feeling in dealing with common tasks, their sensory and motion, their reaction to different materials. (ii) Social problems related to and social services provided for them (iii) Primary health and social care available (iv) Market situation for the possible preferable product
Teaching/Learning Methodology	The following teaching and learning methodology will be used for the first and second offerings of the subject and then reviewed before it is continued to offer: 1. E-learning of service learning (10 hours)

	2. Project-Specific Lectures, Tutorials, Seminars and/or Workshop • Study the psychology and behavior of the identified target group (1 lecture) • Understand public information in relation to the problems of the identified target group (1 lecture) 3. Service Learning Project • Identification of a target group at the beginning, during and at the end of the entire project to complete the following tasks (40 hours): (i) Identify the need for a design project (ii) Modify the design with input from the identified target group (iii) Train the identified target group to use the end product • Discuss with different professionals related in the field (2 lectures) • Develop and complete a design project to serve the identified target group in solving their problems which includes the following activities: (i) Visits and workshops (ii) Literature search and self-study (iii) Design team group discussions (iv) Consultations and discussions with professionals (v) Written report and oral presentation Examples of organizations that students may work with are: • The Hong Kong Senior Citizen Association (HKSCA) • The Hong Kong Government Elderly Commission • The Institution of Mechanical Engineers, Hong Kong Branch																																																
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods /tasks</th><th rowspan="2">% weighting</th><th colspan="8">Intended subject learning outcomes to be assessed</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th><th>f</th><th>g</th><th>h</th></tr><tr><td>1. E-learning Module and Project-specific seminars and workshops (individual)</td><td>20</td><td>√</td><td>√</td><td></td><td></td><td>√</td><td>√</td><td>√</td><td></td></tr><tr><td>2. Reflective Journal/report (individual)</td><td>30</td><td>√</td><td></td><td></td><td></td><td></td><td>√</td><td>√</td><td>√</td></tr><tr><td>3. Final Report (group) • Problem solving skill • Ability to assess </td><td>30</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td></tr></table>	Specific assessment methods /tasks	% weighting	Intended subject learning outcomes to be assessed								a	b	c	d	e	f	g	h	1. E-learning Module and Project-specific seminars and workshops (individual)	20	√	√			√	√	√		2. Reflective Journal/report (individual)	30	√					√	√	√	3. Final Report (group) • Problem solving skill • Ability to assess	30	√	√	√	√	√	√	√	√
Specific assessment methods /tasks	% weighting			Intended subject learning outcomes to be assessed																																													
		a	b	c	d	e	f	g	h																																								
1. E-learning Module and Project-specific seminars and workshops (individual)	20	√	√			√	√	√																																									
2. Reflective Journal/report (individual)	30	√					√	√	√																																								
3. Final Report (group) • Problem solving skill • Ability to assess	30	√	√	√	√	√	√	√	√																																								

• alternative design • Report writing skill • Helping prototype making • Providing service • Oral presentation								
4. Performance in Rendering Service (individual)	20	√	√	√				√
Total	100 %							

The students are expected to apply their knowledge and skills in performing an engineering design project to provide service to an identified target group, therefore their performance in both the service learning project and the engineering design project will be assessed.

Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes are shown below:

- In the preliminary report (group report), students will present their understanding of the community need and responsibility in performing the design task, as well as the appropriateness of the direction, goals and specifications of the design project.
- Making use of the reflective journal (individual writing), students are able to further elaborate the objectives of the design task in relation to provide service to an identified target group of the community and to demonstrate empathy for people in need and a strong sense of civic responsibility.
- In addition to those outcomes fulfilled by the preliminary report, students are expected to apply knowledge of design, mathematics and engineering science to analyze and predict the life-cycle performance of their design in the final report (group report). They are also required to consider the impacts of various factors including, materials, human, environment, safety and reliability of a design. In the oral presentation (group presentation), students' ability to explain precisely and concisely on their contribution will be assessed. Due to the time constraint, the technical details including problem formulation and analysis may not be emphasized.
- The students' attitude and performance in the rendering of service, their degree of engagement with the service recipients, their collaboration with other students, and interactions with the service recipients and/or collaborating NGOs are obviously indicators of their ability to communicate effectively with clients and stakeholders, their sense of responsibility, professional ethics and their empathy for people in need.
- Because of the nature of the subject, written examination seems not necessary.

Student Study Effort Expected	Class contact:	
	▪ Lectures and seminars	13 Hrs.
	▪ e-learning of service learning	10 Hrs.
	▪ Workshops	8 Hrs.
	▪ Discussions and consultations	8 Hrs.
	Other student study effort:	
	▪ Visits to the identified target group at different stages of the project	40 Hrs.
	▪ Literature review and marketing survey	9 Hrs.
	▪ Prepare preliminary report, final report, reflective journal and oral presentation, and Participate in prototype fabrication.	30 Hrs.
	▪ Self studying	10 Hrs.
	Total student study effort	128 Hrs.
Reading List and References	1. Mandell, B.R. and Schram, B., An introduction to human services: policy and practice, Pearson, latest edition. 2. Schriver, J.M., Human behavior and the social environment: shifting paradigms in essential knowledge for social work practice, Allyn and Bacon, latest edition. 3. Wayne, J.H., The social services: an introduction, F.E. Peacock Publishers, latest edition. 4. Ulrich, K.T., Product design and development, McGraw-Hill, latest edition. 5. Budynas, R.G. and Nisbett, J.K., Shigley's mechanical engineering design, McGraw-Hill, latest edition. 6. Boothroyd, G., Dewhurst, P. and Knight, W.A., Product design for manufacture and assembly, Boca Raton, CRC Press, latest edition. 7. Szalma, J.L. (2009). Individual differences in human-technology interaction: Incorporating variation in human characteristics into human factors and ergonomics research and design. Theoretical Issues in Ergonomics Science, 10(5), 381-397. doi: 10.1080/14639220902893613 8. Wickens, C.D., & Kramer, A. (1985). Engineering Psychology. Annual Review of Psychology, 36(1), 307-348. doi: 10.1146/annurev.ps.36.020185.001515	

Revised August 2019

Subject Description Form

Subject Code	MM1031
Subject Title	Introduction to Innovation and Entrepreneurship
Credit Value	1
Level	1
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	This subject introduces students to the essential aspects of innovation and entrepreneurship in a digital world. The objective is to prepare the first-year students with an entrepreneurial mindset and apply innovative strategies to find creative solutions that benefit both organizations and society in the age of digital transformation, while also bolstering Hong Kong's position as a global hub for innovation and technology.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - demonstrate an elementary understanding of innovation and entrepreneurship; - appreciate the importance of innovation and entrepreneurship in the local and global community; - appreciate the applications and implications of the latest technologies on entrepreneurship and innovation; and - identify ethical issues in entrepreneurship and innovation.
Subject Synopsis/ Indicative Syllabus	This subject is built upon three pillars – Nature and importance of innovation and entrepreneurship Defining innovation and entrepreneurship; differences between innovation and entrepreneurship; the importance of innovation and entrepreneurship in Hong Kong and beyond; entrepreneurship as a career path; ethical issues Innovation Technology and innovation; technology life cycle; diffusion of innovation; technology leadership and followership; assessing technology needs; making technology decisions; sourcing and acquiring new technologies; organizing for innovation Entrepreneurship Technology and entrepreneurship; design thinking; value proposition canvas; business model canvas; lean start-up

	Indicative Outline: <u>(A) Introduction</u> Videos (~10 minutes in total), plus discussion/activities/self-study in between the following topics • Defining innovation and entrepreneurship • Differences between innovation and entrepreneurship • The importance of innovation and entrepreneurship in Hong Kong and beyond • Entrepreneurship as a career path <u>(B) Innovation and entrepreneurship toolkit</u> Videos (~40 minutes in total), plus discussion/activities/self-study in between the following topics • Design Thinking • Value Proposition Canvas • Business Model Canvas • Lean Start-up (including MVP) <u>(C) Applications and implications of artificial intelligence on entrepreneurship and innovation</u> Videos (~40 minutes in total), plus discussion/activities/self-study in between the following topics • Hand-written digit recognition • Face detection • Stock price prediction • ROC Concept • Chatbot applications, e.g. customer service, enquiry handling in the customer journey • Latest A.I. development <u>(D) Applications and implications of blockchain technology on entrepreneurship and innovation</u> Videos (~40 minutes in total), plus discussion/activities/self-study in between the following topics • Defining blockchain technology • Background • Applications (e.g., verifying educational or employment credentials, intellectual property, smart contract, billing and revenue allocation, rights and royalties, history of ownership – critical minerals, diamond, fine art, garment, wine and spirits, supply chains, etc.) • Advantages and Disadvantages • Ethical implications (e.g., cryptojacking, co-ownership of illegal data, etc.) <u>(E) Applications and implications of Internet of Things technology on entrepreneurship and innovation</u> Videos (~40 minutes in total), plus discussion/activities/self-study in between the following topics • Defining Internet of Things technology • Background (from 1G to 5G) • Applications (e.g., daily life, manufacturing, retail, smart cities, etc.) • Advantages and Disadvantages
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	- Ethical implications (e.g., privacy, security, etc.) <u>(F) Other Applications:</u> Bio-technology and medical service Guest Lecture or video recording will be arranged (Both online and off-time students are eligible to attend) <u>(G) Managing technology for competitive advantage in a digital world</u> Videos (~10 minutes in total), plus discussion/activities/self-study in between the following topics - Technology life cycle - Diffusion of innovation - Technology leadership and followership - Assessing technology needs - Making technology decisions - Sourcing and acquiring new technologies - Organizing for innovation																																						
Teaching/Learning Methodology	This subject is designed to be interactive, with short videos, cases, in-class discussions and activities interspersed throughout an introductory session and thirteen 1-hour seminars. Students are encouraged to go beyond the understanding of concepts, and to reflect on their learning process. Learning from the responses and feedback from their peers is also critical.																																						
Assessment Methods in Alignment with Intended Learning Outcomes	<table border="1"> <thead> <tr> <th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="4">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr> <tr> <th>a</th><th>b</th><th>c</th><th>d</th></tr> </thead> <tbody> <tr> <td>1. Quizzes</td><td>10%</td><td>✓</td><td></td><td></td><td>✓</td></tr> <tr> <td>2. Participation in discussion forum / journal</td><td>30%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr> <tr> <td>3. Personal Reflection</td><td>60%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr> <tr> <td>Total</td><td>100 %</td><td colspan="4"></td></tr> </tbody> </table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Requiring students to answer multiple-choice questions at the end of each module is appropriate for helping the first-year students confirm their understanding of the concepts. The requirement of writing some textual responses is to assess the schema established by the students toward innovation and entrepreneurship.					Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)				a	b	c	d	1. Quizzes	10%	✓			✓	2. Participation in discussion forum / journal	30%	✓	✓	✓	✓	3. Personal Reflection	60%	✓	✓	✓	✓	Total	100 %				
Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)																																					
		a	b	c	d																																		
1. Quizzes	10%	✓			✓																																		
2. Participation in discussion forum / journal	30%	✓	✓	✓	✓																																		
3. Personal Reflection	60%	✓	✓	✓	✓																																		
Total	100 %																																						
Student Study Effort Expected	Class contact:																																						
	One online introductory session, plus four online video modules, combined with in-class discussions and activities, interspersed throughout			13 Hrs.																																			
	Other student study effort:																																						

	▪ Self-study and preparation	20 Hrs.
	▪ Assignment	10 Hrs.
	Total student study effort	43 Hrs.
Reading List and References	Bateman, T. S., & Konopaske, R. (2021). <i>Management: Leading & collaborating in a competitive world</i>. NY: McGraw-Hill. Bamford, C., & Bruton, G. (2022). <i>Entrepreneurship: The art, science, and process for success</i>. McGraw-Hill. Osterwalder, A., & Pigneur, Y. (2010). <i>Business model generation: A handbook for visionaries, game changers, and challengers</i>. Hoboken, NJ: John Wiley & Sons. Osterwalder, A., Pigneur, Y., Bernarda, G., & Smith, A. (2014). <i>Value proposition design: How to create products and services customers want</i>. Hoboken, NJ: John Wiley & Sons. Ries, E. (2011). <i>The lean start-up</i>. NY: Crown Business.	

July 2025

Core Subjects

Subject Description Form

Subject Code	AF3625
Subject Title	Engineering Economics
Credit Value	3
Level	3
Exclusion	AF2618
Objectives	This subject aims to equip students with 1. The fundamental concepts of micro- and macroeconomics related to the engineering industry; 2. The fundamental understanding of finance and costing for engineering operations, budgetary planning and control.
Intended Subject Learning Outcomes	Upon successful completion of this subject, students will be able to: 1. Understand how the relevant economic factors shape the environment within which an engineering company operates; 2. Evaluate the financial condition of a company; 3. Apply the basic cost accounting techniques in the planning and control of engineering and production activities.
Subject Synopsis/ Indicative Syllabus	<u>Economic Environment of a Firm</u> Microeconomic Factors Scarcity, choice and opportunity cost; Demand, supply and price; Profit-maximizing behavior of the firm; Organization of industry: perfect competition and monopoly Macroeconomic Factors International trade and globalization <u>Engineering Economics</u> Return on investment; Accounting profit versus economic profit <u>Fundamentals of Budgetary Planning and Control</u> Principle types of budgets for production and service operations; Approaches to budgeting and the budgeting process; Investment and source of finance; Cost of capital; Evaluation of investment alternatives
Teaching/ Learning Methodology	The two-hour lecture each week focuses on the introduction and explanation of key concepts of Engineering Economics. The one-hour tutorial provides students with directed studies to enhance their self-learning capacities. Individual and group activities including discussions and presentations are conducted to facilitate students' understanding and application of the concepts they have learned to tackling real-life problems in Engineering Economics.

Assessment Methods in Alignment with Intended Learning Outcomes	Specific Assessment Methods/Tasks	% Weighting	Intended Subject Learning Outcomes to be Assessed (Please tick as appropriate)		
			1	2	3
	Continuous Assessment	50%			
	1. In-class activities	15%	√	√	√
	2. Written assignments	15%	√	√	√
	3. Test	20%	√	√	√
	Final Examination	50%	√	√	√
	Total	100 %			
Student Study Effort Required	Class contact:				
	• Lecture				26 Hours
	• Tutorial				13 Hours
	Other student study effort:				
	• Study and self-learning				48 Hours
	• Presentation preparation and written assignments				18 Hours
	Total student study effort:				105 Hours
Reading List and References	Recommended Textbooks 1. Parkin and Bade, <i>Foundations of Microeconomics</i> , 8 th ed., Pearson, 2018. 2. Sullivan, Wicks and Koelling, <i>Engineering Economy</i> , 17 th ed., Pearson, 2019. References 1. Robert H. Frank, <i>The Economic Naturalist: Why Economics Explains Almost Everything?</i> , Basic Books, 2011.				
Last Updated	July 2023				
Prepared by	School of Accounting and Finance				

Subject Description Form

Subject Code	AMA1110
Subject Title	Basic Mathematics I – Calculus and Probability & Statistics
Credit Value	3
Level	1
Pre-requisite	Nil
Exclusion	Calculus and Linear Algebra (AMA1007) Calculus for Engineers (AMA1130) Calculus (AMA1131) Foundation Mathematics for Accounting and Finance (AMA1500) Calculus (AMA1702)
Objectives	This subject aims to introduce students to the basic concepts and applications of elementary calculus and statistics. Emphasis will be on the understanding of fundamental concepts and the use of mathematical techniques in handling practical problems in science and engineering.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) apply analytical reasoning to solve problems in science and engineering; (b) make use of the knowledge of mathematical/statistical techniques and adapt known solutions to various situations; (c) apply mathematical modeling in problem solving; (d) demonstrate abilities of logical and analytical thinking.
Subject Synopsis/ Indicative Syllabus	<u>Elementary calculus</u>: Limit and continuity, derivatives and their geometric meaning, rules of differentiation including chain rule, Leibniz's rule and L'Hopital's rule, exponential and logarithmic functions, trigonometric functions and their inverses, hyperbolic and inverse hyperbolic functions, applications of differential calculus. <u>Elementary Probability and Statistics</u>: Descriptive statistics, random variables, probability and probability distributions, binomial, Poisson and normal distributions, applications. Population and random samples. Sampling distributions related to sample mean, sample proportions, and sample variances. Concepts of a point estimator and a confidence interval. Point and interval estimates of a mean and the difference between two means.

Teaching/Learning Methodology	Basic concepts and elementary techniques of differential and integral calculus and elementary statistics will be taught in lectures. These will be further enhanced in tutorials through practical problem solving.					
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)			
			a	b	c	d
	1.Assignments and mid-term tests	40%	✓	✓	✓	✓
	2. Examination	60%	✓	✓	✓	✓
	Total	100 %				
	Continuous Assessment comprises of assignments, in-class quizzes, online quizzes and a mid-term test. An examination is held at the end of the semester.					
	Questions used in assignments, quizzes, tests and examinations are used to assess students’ level of understanding of the basic concepts and their ability to use mathematical techniques in solving problems in science and engineering.					
Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:						
<i>The subject focuses on understanding of basic concepts and application of techniques in differential/integral calculus, elementary statistics. As such, an assessment method based mainly on examinations/tests/quizzes is considered appropriate. Furthermore, students are required to submit homework assignments regularly in order to allow subject lecturers to keep track of students’ progress in the course.</i>						
Student Study Effort Expected	Class contact:					
	▪ Lecture			26 Hrs.		
	▪ Tutorial			13 Hrs.		
	Other student study effort:					
	▪ Homework and self-study			81 Hrs.		
	Total student study effort			120 Hrs.		
Reading List and References	Chung, K.C. <i>A Short Course in Calculus and Matrices</i> , McGraw Hill 2013					

	Hung, K.F., Kwan, Wilson, Pong, T.Y. <i>Foundation Mathematics & Statistics</i>, McGraw Hill 2013 Larson, R., Edwards, B. <i>Single Variable Calculus</i>, Brooks/Cole 2012 Walpole, R.E., Myers, R.H., Myers, S.L. Ye, K. <i>Probability and Statistics for Engineers and Scientists</i>, Prentice Hall, 2012
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Subject Description Form

Subject Code	AMA1120																																
Subject Title	Basic Mathematics II –Calculus and Linear algebra																																
Credit Value	3																																
Level	1																																
Pre-requisite	Basic Mathematics I – Calculus and Probability & Statistics (AMA1110)																																
Objectives	This subject aims to introduce students to the basic concepts and applications of elementary calculus and statistics. Emphasis will be on the understanding of fundamental concepts and the use of mathematical techniques in handling practical problems in science and engineering.																																
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) apply analytical reasoning to solve problems in science and engineering; (b) make use of the knowledge of mathematical/statistical techniques and adapt known solutions to various situations; (c) apply mathematical modeling in problem solving; (d) demonstrate abilities of logical and analytical thinking.																																
Subject Synopsis/ Indicative Syllabus	<u>Elementary calculus</u>: Mean Value Theorem with applications to optimization and curve sketching. Definite and indefinite integrals, fundamental theorem of calculus, methods of integration (integration by substitution, integration by parts, integration of rational functions using partial fractions and integration of trigonometric and hyperbolic functions), reduction formulas, applications to geometry and physics. Improper Integrals. <u>Linear algebra</u>: Basic properties of matrices and determinants, linear systems, Gaussian elimination, inverse of a square matrix, Cramer's rule, vectors in 2-space or in 3-space, applications to geometry.																																
Teaching/Learning Methodology	Basic concepts and elementary techniques of differential and integral calculus and linear algebra will be taught in lectures. These will be further enhanced in tutorials through practical problem solving.																																
Assessment Methods in Alignment with Intended Learning Outcomes	<table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="4">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr> <tr> <th>a</th><th>b</th><th>c</th><th>d</th></tr> </thead> <tbody> <tr> <td>1.Assignments and tests</td><td>40%</td><td style="text-align: center;">✓</td><td style="text-align: center;">✓</td><td style="text-align: center;">✓</td><td style="text-align: center;">✓</td></tr> <tr> <td>2. Examination</td><td>60%</td><td style="text-align: center;">✓</td><td style="text-align: center;">✓</td><td style="text-align: center;">✓</td><td style="text-align: center;">✓</td></tr> <tr> <td>Total</td><td>100 %</td><td colspan="4"></td></tr> </tbody> </table> Continuous Assessment comprises of assignments and tests. An examination is held at the end of the semester.					Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)				a	b	c	d	1.Assignments and tests	40%	✓	✓	✓	✓	2. Examination	60%	✓	✓	✓	✓	Total	100 %				
Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)																															
		a	b	c	d																												
1.Assignments and tests	40%	✓	✓	✓	✓																												
2. Examination	60%	✓	✓	✓	✓																												
Total	100 %																																

	Questions used in assignments, tests and examinations are used to assess students' level of understanding of the basic concepts and their ability to use mathematical techniques in solving problems in science and engineering. Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: <i>The subject focuses on understanding of basic concepts and application of techniques in differential/integral calculus, elementary statistics and elementary linear algebra. As such, an assessment method based mainly on examinations/tests is considered appropriate. Furthermore, students are required to submit homework assignments regularly in order to allow subject lecturers to keep track of students' progress in the course.</i>	
Student Study Effort Expected	Class contact:	
	▪ Lecture	26 Hrs.
	▪ Tutorial	13 Hrs.
	Other student study effort:	
	▪ Homework and self-study	81 Hrs.
	Total student study effort	120 Hrs.
Reading List and References	Chung, K.C. <i>A Short Course in Calculus and Matrices</i>, McGraw Hill 2013 Hung, K.F., Kwan, Wilson, Pong, T.Y. <i>Foundation Mathematics & Statistics</i>, McGraw Hill 2013 Larson, R., Edwards, B. <i>Single Variable Calculus</i>, Brooks/Cole 2012 Larson, R. <i>Elementary Linear Algebra</i>, Brooks/Cole 2013	

Subject Description Form

Subject Code	AMA2111
Subject Title	Mathematics I
Credit Value	3
Level	2
Pre-requisite	Calculus and Linear Algebra (AMA1007) or Basic Mathematics II – Calculus and Linear Algebra (AMA1120) or Calculus for Engineers (AMA1130) or Calculus (AMA1131) or Foundation Mathematics for Accounting and Finance (AMA1500)
Exclusion	Intermediate Calculus and Linear Algebra (AMA2007/AMA2707) Mathematics for Engineers (AMA2131/AMA2308) Engineering Mathematics (AMA2380) Applied Mathematics I (AMA2511) Mathematics for Scientists and Engineers (AMA2882) Engineering Mathematics (AMA290)
Objectives	This subject aims to introduce students to the basic principles and techniques of engineering mathematics. Emphasis will be on the understanding of fundamental concepts as well as applications of mathematical methods in solving practical problems in science and engineering.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: 1. apply mathematical reasoning to analyze essential features of different problems in science and engineering; 2. extend their knowledge of mathematical and numerical techniques and adapt known solutions in various situations; 3. develop and extrapolate the mathematical concepts in synthesizing and solving new problems 4. demonstrate abilities of logical and analytical thinking; 5. search for useful information in the process of problem solving.
Subject Synopsis/ Indicative Syllabus	1. <u>Algebra of complex numbers</u> Complex numbers, geometric representation, complex exponential functions, n-th roots of a complex number. 2. <u>Linear algebra</u>

	Systems of linear equations, vector spaces, inner product and orthogonality, eigenvalues and eigenvectors, applications. 3. <u>Ordinary differential equations</u> ODE of first and second order, linear systems, Laplace transforms, Convolution theorem, applications to mechanical vibrations and simple circuits. 4. <u>Differential calculus of functions of several variables</u> Partial derivatives, total differential, chain rule, Taylor’s expansion, maxima and minima, directional derivatives, Lagrange multipliers, implicit differentiation, applications.																																	
Teaching/Learning Methodology	The subject will be delivered mainly through lectures and tutorials. The lectures aim to provide the students with an integrated knowledge required for the understanding and application of mathematical concepts and techniques. Tutorials will mainly be used to develop students’ problem solving ability.																																	
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="5">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>1</th><th>2</th><th>3</th><th>4</th><th>5</th></tr><tr><td>1.Homework, quizzes and mid-term test</td><td>40%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>2. Examination</td><td>60%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Total</td><td>100%</td><td colspan="5"></td></tr></table> Continuous Assessment comprises of assignments, in-class quizzes, online quizzes and a mid-term test. An examination is held at the end of the semester. Questions used in assignments, quizzes, tests and examinations are used to assess students’ level of understanding of the basic concepts and their ability to use mathematical techniques in solving problems in science and engineering. Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: <i>The subject focuses on understanding of basic concepts and application of techniques in engineering mathematics. As such, an assessment method based mainly on examinations/tests/quizzes is considered appropriate. Furthermore, students are required to submit homework assignments regularly in order to allow subject lecturers to keep track of students’ progress in the course.</i>	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					1	2	3	4	5	1.Homework, quizzes and mid-term test	40%	✓	✓	✓	✓	✓	2. Examination	60%	✓	✓	✓	✓	✓	Total	100%					
Specific assessment methods/tasks	% weighting			Intended subject learning outcomes to be assessed (Please tick as appropriate)																														
		1	2	3	4	5																												
1.Homework, quizzes and mid-term test	40%	✓	✓	✓	✓	✓																												
2. Examination	60%	✓	✓	✓	✓	✓																												
Total	100%																																	
	<table><tr><td>Class contact:</td><td></td></tr><tr><td>• Lecture</td><td>26 Hours</td></tr></table>	Class contact:		• Lecture	26 Hours																													
Class contact:																																		
• Lecture	26 Hours																																	

Student Study Effort Expected	• Tutorial	13 Hours
	• Mid-term test and examination	
	Other student study effort	
	• Assignments and Self study	78 Hours
	Total student study effort:	117 Hours
Reading List and References	1. C.K. Chan, C.W. Chan and K.F. Hung, <i>Basic Engineering Mathematics</i>, McGraw-Hill, 2015. 2. Anton, H. <i>Elementary Linear Algebra</i> (11th edition). Wiley, 2014. 3. Kreyszig, E. (2011). <i>Advanced Engineering Mathematics</i>, 10th ed. Wiley. 4. James, G. (2015). <i>Modern Engineering Mathematics</i>, 5th ed. Pearson Education Limited 5. Thomas, G. B., Weir, M. D. & Hass, J. R. <i>Thomas' Calculus</i>, 14th ed. Pearson Education 2017	

Subject Description Form

Subject Code	AMA2112
Subject Title	Mathematics II
Credit Value	3
Level	2
Pre-requisite	Mathematics I (AMA2111)
Exclusion	Intermediate Calculus and Linear Algebra (AMA2007/AMA2707) Introduction to Differential Equations (AMA2008)
Objectives	This subject is a continuation of AMA2111. It aims to introduce students to the basic principles and techniques of engineering mathematics. Emphasis will be on the understanding of fundamental concepts as well as applications of mathematical methods in solving practical problems in science and engineering.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: 1. apply mathematical reasoning to analyze essential features of different problems in science and engineering; 2. extend their knowledge of mathematical and numerical techniques and adapt known solutions in various situations; 3. develop and extrapolate the mathematical concepts in synthesizing and solving new problems 4. demonstrate abilities of logical and analytical thinking; 5. search for useful information in the process of problem solving.
Subject Synopsis/ Indicative Syllabus	1. <u>Multiple integrals</u> Double and triple integrals, change of variables, applications to problems in geometry and mechanics. 2. <u>Vector calculus</u> Vector and scalar fields, the del operator, line and surface integrals, the theorems of Green, Gauss and Stokes, applications to electromagnetic theory and fluid mechanics. 3. <u>Series expansion</u> Infinite series, Taylor's expansion, Fourier series expansion of a periodic function.

	4. <u>Partial differential equations</u> Formulation of PDE of mathematical physics, separation of variables, initial-boundary value problems, introduction to Fourier transforms.						
Teaching/Learning Methodology	The subject will be delivered mainly through lectures and tutorials. The lectures aim to provide the students with an integrated knowledge required for the understanding and application of mathematical concepts and techniques. Tutorials will mainly be used to develop students' problem solving ability.						
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)				
			1	2	3	4	5
	1. Assignments, quizzes and mid-term test	40%	✓	✓	✓	✓	✓
	2. Examination	60%	✓	✓	✓	✓	✓
	Total	100%					
	Continuous Assessment comprises of assignments, in-class quizzes, online quizzes and a mid-term test. An examination is held at the end of the semester. Questions used in assignments, quizzes, tests and examinations are used to assess students' level of understanding of the basic concepts and their ability to use mathematical techniques in solving problems in science and engineering. Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: <i>The subject focuses on understanding of basic concepts and application of techniques in engineering mathematics. As such, an assessment method based mainly on examinations/tests/quizzes is considered appropriate. Furthermore, students are required to submit homework assignments regularly in order to allow subject lecturers to keep track of students' progress in the course.</i>						
Student Study Effort Expected	Class contact:						
	• Lecture				26 Hours		
	• Tutorial				13 Hours		
	• Mid-term test and examination						
	Other student study effort						
	• Assignments and Self study				78 Hours		
	Total student study effort:				117 Hours		

Reading List and References	1. C.K. Chan, C.W. Chan and K.F. Hung, <i>Basic Engineering Mathematics</i>, McGraw-Hill, 2015. 2. Anton, H. <i>Elementary Linear Algebra</i> (11th edition). Wiley, 2014. 3. Kreyszig, E. (2011). <i>Advanced Engineering Mathematics</i>, 10th ed. Wiley. 4. James, G. (2015). <i>Modern Engineering Mathematics</i>, 5th ed. Pearson Education Limited 5. Thomas, G. B., Weir, M. D. & Hass, J. R. <i>Thomas' Calculus</i>, 14th ed. Pearson Education 2017
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Subject Description Form

Subject Code	AMA3708
Subject Title	Differential Equations
Credit Value	3
Level	3
Pre-requisite	▪ Intermediate Calculus and Linear Algebra (AMA2007/AMA2707) or ▪ Mathematics I (AMA2111) or ▪ Mathematics for Engineers (AMA2131/AMA2308) or ▪ Engineering Mathematics (AMA2380) or ▪ Applied Mathematics II (AMA2512) or ▪ Multivariable Calculus (AMA2702) or ▪ Mathematics for Scientists and Engineers (AMA2882) or ▪ Engineering Mathematics (AMA290)
Exclusion	Mathematics II (AMA2112)
Objectives	The objective of this subject is to provide students with a comprehensive understanding of the ideas and techniques used in the topic of differential equations. Students will be able to apply their knowledge to real-world problems in disciplines such as physics, engineering, finance, and economics.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) Gain a fundamental understanding and knowledge of ordinary and partial differential equations, including their types, order and degree. (b) Apply various methods for solving ordinary and partial differential equations. (c) Analyze and interpret solutions to differential equations, including investigating their properties, stability, and the behavior of solutions over time. (d) Apply techniques for solving differential equations to real-world problems in fields such as physics, engineering, finance, and economics, and identify the most suitable method to solve the problem.
Subject Synopsis/ Indicative Syllabus	<u>Ordinary differential equations</u> Modeling using ordinary differential equations; First and second order linear differential equations; System of linear differential equations; Laplace transform and convolution theorem. Applications to real-world problems. <u>Partial differential equations</u> Definition and classification of PDEs; Introduction of several classical PDEs (heat, wave and Laplace equations); Initial and boundary value problem of PDEs; Methods of separation of variables; Green functions and fundamental solutions; Fourier transform. Applications to real-world problems.
Teaching/Learning Methodology	This course will use lectures, tutorials, and learning activities to teach students about differential equations. Lectures will introduce the concepts, while tutorials will provide support. Learning activities will reinforce the concepts and encourage critical thinking. Students will engage in self-directed learning, and assessment will be through homework, quizzes, and exams. This methodology balances theory and application, encourages participation.

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)			
			a	b	c	d
	1. Assignments/Quizzes	20%	✓	✓	✓	✓
	2. Midterm Test	20%	✓	✓	✓	
	3. Examination	60%	✓	✓	✓	
	Total	100%				
	The subject focuses on knowledge, skill and understanding of Differential Equations, thus, Exam-based assessment is the most appropriate assessment method, including 20% test and 60% examination. Moreover, 20% worth of assignments/in-class quizzes are included as a component of continuous assessment so as to keep the students in progress. Continuous Assessment comprises of assignments/quizzes and/or tests. A written examination is held at the end of the semester.					
Student Study Effort Required	Class contact:					
	▪ Lecture					26 Hrs.
	▪ Tutorial					13 Hrs.
	Other student study effort:					
	▪ Assignments/Quizzes					35 Hrs.
	▪ Self-study					35 Hrs.
	Total student study effort					109 Hrs.
Reading List and References	<u>References:</u>					
	Chan, C.K., Chan, C.W. & Hung, K.F.	Basic Engineering Mathematics.	McGraw Hill 2013			
	Kreyszig, E.	Advanced Engineering Mathematics 10th edition	Wiley 2011			
	James, G.	Advanced Modern Engineering Mathematics 3rd edition	Prentice Hall 2008			
	Walter A. Strauss	Partial Differential Equations. An introduction. Second edition	Wiley 2007			
	Grant B. Gustafson	Differential Equations and Linear Algebra: Undergraduate Mathematics, Science and Engineering, First edition (2022)	2022			

	Gilbert Strang Kenneth B. Howell	Differential Equations and Linear Algebra Ordinary Differential Equations: An Introduction to the Fundamentals, 2nd edition	Wellesley-Cambridge Press CRC Press, 2019
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AMA3708

August 2023

Subject Description Form

Subject Code	AP10001
Subject Title	Introduction to Physics
Credit Value	3
Level	1
Pre-requisite/ Co-requisite/ Exclusion	Exclusion: AP1D03 "How Things Work: the Physics of Everyday Life" & AP1D05 "Introduction to Physics"
Objectives	This is a subject designed for students with no background in physics studies. Fundamental concepts in major topics of physics (mechanics, heat, wave and electromagnetism) will be discussed. The aim of this subject is to equip students with some basic physics knowledge, and to appreciate its applications in various branches of science and technology.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) solve simple problems in kinematics Newton's law and Energy; (b) solve problems in heat capacity and latent heat; (c) explain phenomena related to the wave character of light; (d) apply the superposition of waves; (e) understand electrostatic field and potential; (f) solve problems on interaction between current and magnetic field; and (g) describe and demonstrate the phenomenon of electromagnetism.
Subject Synopsis/ Indicative Syllabus	Mechanics: scalars and vectors; kinematics and dynamics; Newton's laws; momentum, impulse, work and energy; conservation of momentum and conservation of energy. Thermal physics: heat and internal energy; heat capacity; conduction, convection and radiation; latent heat. Waves: nature of waves; wave motion; reflection and refraction; image formation by mirrors and lenses; superposition of waves; standing waves; diffraction and interference; electromagnetic spectrum; sound waves. Electromagnetism: charges; Coulomb's law; electric field and potential; current and resistance; Ohm's law; magnetic field; magnetic force on moving charges and current-carrying conductors; Faraday's law and Lenz's law.
Teaching/Learning Methodology	Lecture: Fundamentals in mechanics, waves and electromagnetism will be explained. Examples will be used to illustrate the concepts and ideas in the lecture. Students are free to request help. Homework problem sets will be given. Student-centered Tutorial: Students will work on a set of problems in tutorials. Students are encouraged to solve problems and to use their own knowledge to verify their solutions before seeking assistance. These problem sets provide them opportunities to apply their knowledge gained from the lecture. They also help the students to consolidate what they have learned. Furthermore, students can develop a deeper understanding of the subject in

	relation to daily life phenomena or experience. e-learning: In order to enhance the effectiveness of teaching and learning processes, electronic means and multimedia technologies would be adopted for presentations of lectures; communication between students and lecturer; delivery of handouts, homework and notices etc.								
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)						
			a	b	c	d	e	f	g
	(1) Continuous assessment	40	✓	✓	✓	✓	✓	✓	✓
	(2) Examination	60	✓	✓	✓	✓	✓	✓	✓
	Total	100							
	Continuous assessment: The continuous assessment includes assignments, quizzes and test(s) which aim at checking the progress of students’ study throughout the course, assisting them in fulfilling the learning outcomes. Assignments in general include end-of-chapter problems, which are used to reinforce and assess the concepts and skills acquired by the students; and to let them know the level of understanding that they are expected to reach. At least one test would be administered during the course of the subject as a means of timely checking of learning progress by referring to the intended outcomes, and as means of checking how effective the students digest and consolidate the materials taught in the class. Examination: This is a major assessment component of the subject. It would be a closed-book examination. Complicated formulas would be given to avoid rote memory, such that the emphasis of assessment would be put on testing the understanding, analysis and problem solving ability of the students.								
Student Study Effort Expected	Class contact:								
	• Lecture		33 h						
	• Tutorial		6 h						
	Other student study effort:								
	• Self-study		81 h						
	Total student study effort		120 h						
Reading List and References	John D. Cutnell & Kenneth W. Johnson, Introduction to Physics , 9th edition, 2013, John Wiley & Sons.								
	Hewitt, Conceptual Physics , 11th edition, 2010, Benjamin Cummings.								
	Radi, Hafez A., and John O. Rasmussen. Principles of Physics for Scientists and Engineers . Berlin; New York: Springer, 2013. Undergraduate Lecture Notes in Physics. Web.								

Subject Description Form

Subject Code	AP10005
Subject Title	Physics I
Credit Value	3
Level	1
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	This course provides a broad foundation in mechanics and thermal physics to those students who are going to study science, engineering, or related programmes.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) solve simple problems in single-particle mechanics using calculus and vectors; (b) solve problems in mechanics of many-particle systems using calculus and vectors; (c) understand simple harmonic motion and solve simple problems; (d) solve problems related to acoustic standing waves; (e) calculate changes in frequency received due to Doppler's effect; (f) apply ideal gas laws to solve problems; (g) apply the first law of thermodynamics to simple processes; and (h) solve simple problems related to the cyclic processes.
Subject Synopsis/ Indicative Syllabus	Mechanics: calculus-based kinematics, dynamics and Newton's laws; calculus-based Newtonian mechanics, involving the application of impulse, momentum, work and energy, etc.; conservation law; gravitational force; systems of particles; collisions; rigid body rotation; angular momentum; oscillations and simple harmonic motion; pendulum; statics; longitudinal and transverse waves; travelling wave and standing wave; Doppler effect; sound waves. Thermal physics: ideal gas and kinetic theory; work, heat and internal energy; first law of thermodynamics; entropy and the second law of thermodynamics; Carnot cycle; heat engine.
Teaching/Learning Methodology	Lecture: Fundamentals in mechanics, waves and electromagnetism will be explained. Examples will be used to illustrate the concepts and ideas in the lecture. Students are free to request help. Homework problem sets will be given. Student-centered Tutorial: Students will work on a set of problems in tutorials. Students are encouraged to solve problems and to use their own knowledge to verify their solutions before seeking assistance. These problem sets provide them opportunities to apply their knowledge gained from the lecture. They also help the students to consolidate what they have learned. Furthermore, students can develop a deeper understanding of the subject in relation to daily life phenomena or experience. e-learning: In order to enhance the effectiveness of teaching and learning processes, electronic means and multimedia technologies would be adopted for presentations of lectures; communication between students and lecturer; delivery of handouts, homework and notices etc.

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)								
			a	b	c	d	e	f	g	h	
	(1) Continuous assessment	40	✓	✓	✓	✓	✓	✓	✓	✓	
	(2) Examination	60	✓	✓	✓	✓	✓	✓	✓	✓	
	Total	100									
	Continuous assessment: The continuous assessment includes assignments, quizzes and test(s) which aim at checking the progress of students’ study throughout the course, assisting them in fulfilling the learning outcomes. Assignments in general include end-of-chapter problems, which are used to reinforce and assess the concepts and skills acquired by the students; and to let them know the level of understanding that they are expected to reach. At least one test would be administered during the course of the subject as a means of timely checking of learning progress by referring to the intended outcomes, and as means of checking how effective the students digest and consolidate the materials taught in the class. Examination: This is a major assessment component of the subject. It would be a closed-book examination. Complicated formulas would be given to avoid rote memory, such that the emphasis of assessment would be put on testing the understanding, analysis and problem solving ability of the students.										
Student Study Effort Expected	Class contact:										
	• Lecture		33 h								
	• Tutorial		6 h								
	Other student study effort:										
	• Self-study		81 h								
	Total student study effort:		120 h								
Reading List and References	John W. Jewett and Raymond A. Serway, “Physics for Scientists and Engineers”, 2014, 9th edition, Brooks/Cole Cengage Learning.										
	Hafez A. Radi, John O. Rasmussen, “Principles of physics: for scientists and engineers”, 2013, Springer.										
	W. Bauer and G.D. Westfall, “University Physics with Modern Physics”, 2011, McGraw-Hill.										

Subject Description Form

Subject Code	AP10006
Subject Title	Physics II
Credit Value	3
Level	1
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	To provide students with fundamental knowledge in physics focusing on the topics of waves and electromagnetism. This course prepares students to study science, engineering or related programmes.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) apply simple laws in optics to explain image formation; (b) understand phenomena related to the wave character of light; (c) solve problems in electrostatics; (d) solve problems on interaction between current and magnetic field; (e) apply electromagnetic induction to various phenomena; and (f) solve problems in simple circuits.
Subject Synopsis/ Indicative Syllabus	Waves and optics: nature of light, reflection and refraction; Snell's law; image formation by mirrors and lenses; compound lens; microscope and telescope; superposition of waves; Huygen's principle; interference and diffraction; diffraction grating; Rayleigh's criterion and optical resolution; polarization. Electromagnetism: charge and Field; Coulomb's law and Gauss' law; electrostatic field and potential difference; capacitors and dielectric; current and resistance; Ohm's law; electromotive force, potential difference; Lorentz force; magnetic force on moving charges and current; Hall effect; Biot-Savart law and Ampere's law; Faraday's law and Lenz's law; induction; transformers; AC circuits and applications.
Teaching/Learning Methodology	Lecture: The fundamentals in optics and electromagnetism will be explained. Examples will be used to illustrate the concepts and ideas in the lecture. Students are free to request help. Homework problem sets will be given. Student-centered Tutorial: Students will work on a set of problems in tutorials. Students are encouraged to solve problems and to use their own knowledge to verify their solutions before seeking assistance. These problem sets provide them opportunities to apply their knowledge gained from the lecture. They also help the students to consolidate what they have learned. Furthermore, students can develop a deeper understanding of the subject in relation to daily life phenomena or experience. e-learning: In order to enhance the effectiveness of teaching and learning processes, electronic means and multimedia technologies would be adopted for presentations of lectures; communication between students and lecturer; delivery of handouts, homework

	and notices etc.								
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)						
			a	b	c	d	e	f	
	(1) Continuous assessment	40	✓	✓	✓	✓	✓	✓	
	(2) Examination	60	✓	✓	✓	✓	✓	✓	
	Total	100							
	Continuous assessment: The continuous assessment includes assignments, quizzes and test(s) which aim at checking the progress of students’ study throughout the course, assisting them in fulfilling the learning outcomes. Assignments in general include end-of-chapter problems, which are used to reinforce and assess the concepts and skills acquired by the students; and to let them know the level of understanding that they are expected to reach. At least one test would be administered during the course of the subject as a means of timely checking of learning progress by referring to the intended outcomes, and as means of checking how effective the students digest and consolidate the materials taught in the class. Examination: This is a major assessment component of the subject. It would be a closed-book examination. Complicated formulas would be given to avoid rote memory, such that the emphasis of assessment would be put on testing the understanding, analysis and problem solving ability of the students.								
Student Study Effort Expected	Class contact:								
	• Lecture				33 h				
	• Tutorial				6 h				
	Other student study effort:								
	• Self-study				81 h				
	Total student study effort				120 h				
Reading List and References	John W. Jewett and Raymond A. Serway, “Physics for Scientists and Engineers”, 2014, 9th edition, Brooks/Cole Cengage Learning.								
	Hafez A. Radi, John O. Rasmussen, “Principles of physics: for scientists and engineers”, 2013, Springer.								
	W. Bauer and G.D. Westfall, “University Physics with Modern Physics”, 2011, McGraw-Hill.								

Subject Description Form

Subject Code	CLC3241P (2019-20 onward) CBS3241P (2018-19 and before)
Subject Title	Professional Communication in Chinese
Credit Value	2
Level	3
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite / Co-requisite: Chinese LCR subjects (in Semester 2 of Year 3 or Semester 1 of Year 4)
Objectives	This subject aims to develop the language competence for professional communication in Chinese required by students to communicate effectively with various parties and stakeholders in regard to engineering-related project proposals and reports.
Subject Intended Learning Outcomes	Upon completion of the subject, and in relation to effective communication with a variety of intended readers/audiences in Chinese, students will be able to - plan, organize and produce professionally acceptable project proposals and reports with appropriate text structures and language for different intended readers - plan, organize and deliver effective project-related oral presentations with appropriate interactive strategies and language for different intended audiences - adjust the style of expression and interactive strategies in writing and speaking in accordance with different intended readers/audiences
Subject Synopsis/ Indicative Syllabus	- Project proposals and reports in Chinese - Planning and organizing project proposals and reports - Explaining the background, rationale, objectives, scope and significance of a project - Referring to the literature to substantiate project proposals - Describing the methods of study - Describing and discussing project results, including anticipated results and results of pilot study - Presenting the budget, schedule and/or method of evaluation - Writing executive summaries/abstracts and/or professional reports - Oral presentations of projects - Selecting content for audience-focused presentations - Choosing language and style appropriate to the intended audience - Using appropriate transitions and maintaining coherence in team presentations - Using effective verbal and non-verbal interactive strategies

Teaching/Learning Methodology	<u>Learning and teaching approach</u> The subject is designed to develop the students’ Chinese language skills, both oral and written, that students need to communicate effectively and professionally with a variety of stakeholders of engineering-related projects. It builds upon the language and communication skills covered in GUR language training subjects. The study approach is primarily seminar-based. Seminar activities include instructor input as well as individual and group work, involving drafting and evaluating texts, mini-presentations, discussions and simulations. The learning and teaching activities in the subject will focus on a course-long project which will engage students in proposing and reporting on an engineering-related project to different intended readers/audiences. During the course, students will be involved in: - planning and researching the project - writing project-related documents such as project proposals and reports - giving oral presentations to intended stakeholders of the project																							
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="3">Intended subject learning outcomes to be assessed</th></tr><tr><th>a</th><th>b</th><th>c</th></tr><tr><td>1. Project proposal and report in Chinese</td><td>60%</td><td>✓</td><td></td><td>✓</td></tr><tr><td>2. Oral presentation of project proposal and report</td><td>40%</td><td></td><td>✓</td><td>✓</td></tr><tr><td>Total</td><td>100%</td><td colspan="3"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: The assessments will arise from the course-long engineering-related project. • Students will be assessed on written documents and oral presentations targeted at different intended readers/audiences. This facilitates assessment of students’ ability to select content and use language and style appropriate to the purposes and intended readers/audiences. • Students will collaborate in groups in planning, researching, discussing and giving oral presentations on the project. The written proposals will be individual work to ensure that students will be rigorously engaged in the application of language skills for the entire document. Students may use GenAI in assessment unless they have been told otherwise before the assessment. The work submitted for assessment must be students’ own work. The submitted work may not be accepted for the purpose of assessment if its authenticity is questionable. Third-party materials (included AI-generated materials) used in the work must be properly referenced. Submitting third-party materials as their own work is an act of academic dishonesty and will lead to disciplinary actions. Departments and subject teachers have the discretion to request students to make an honour declaration for the authenticity of the work they submit for assessment.	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed			a	b	c	1. Project proposal and report in Chinese	60%	✓		✓	2. Oral presentation of project proposal and report	40%		✓	✓	Total	100%			
Specific assessment methods/tasks	% weighting			Intended subject learning outcomes to be assessed																				
		a	b	c																				
1. Project proposal and report in Chinese	60%	✓		✓																				
2. Oral presentation of project proposal and report	40%		✓	✓																				
Total	100%																							

Student Study Effort Expected	Class contact:	
	▪ Seminars	26 Hrs.
	Other student study effort:	
	▪ Researching, planning, writing, and preparing the project	44 Hrs.
	Total student study effort	70 Hrs.
Reading List and References	1. 張永康(1994)：《工程應用文》，三民出版社。 2. 香港城市大學語文學部 (2001)：《中文傳意寫作篇》，香港城市大學出版社。 3. 朱志強主編 (2007)：《管理信息系統——原理、開發及應用》，復旦大學出版社。 4. 于成鯤、陳瑞端、秦扶一、金振邦主編 (2010)：《中國現代應用文寫作規範叢書：科教文與社交文書寫作規範》，復旦大學出版社。 5. 夏曉鳴(2011)：《應用文寫作（第三版）》，復旦大學出版社。 6. 孫永忠、曾紀剛、詹千慧、鍾宗憲 (2019)：《新編應用文（第3版）》，洪葉文化事業有限公司。 7. 郭東強主編(2013)：《現代管理信息系統》，清華大學出版社。 8. 戴國良(2017)：《企劃案撰寫：理論與案例》，大碩教育。 9. 郭莉、郝麗霞 (2017)：《經濟應用文寫作 (第3版)（高等學校應用型特色規劃教材）》，清華大學出版社。 10. 劉平青 (2018)：《面試溝通巧技能》，電子工業出版社。 11. 亞倫·皮斯、芭芭拉·皮斯(2018)：《身體語言密碼》，光明日報出版社。 12. 何建民(2018)：《電業應用文寫作 120 例》，中國電力出版社。 13. 馬丁·耶特，于巧峰譯 (2020)：《終極求職簡歷》，中信出版集團股份有限公司。 14. 吳燦銘(2023)：《AI 提示工程師的 16 堂關鍵必修課：精準提問 x 優化提示 x 有效查詢 x 文字生成 x AI 繪圖》，博碩出版社。 15. 唐納德·高斯、傑拉爾德·溫伯格，褚耐安譯(2017)：《從需求到設計：如何設計出客戶想要的產品》，經濟新潮社。 16. 薛貴輝(2023)：《公文寫作 32 講：從思維構思到筆法語言》，清華大學出版社。 17. 吳仁麟、李慶芳 (2023)：《AI 時代的思考與寫作》，布克文化。 18. 郭泰(2024)：《企劃學：打造企劃腦，讓你的方案更有說服力》，時報出版。 19. 謝東江(2024)：《麥肯錫文案寫作與溝通技巧：構建邏輯、傳達清晰、影響決策，掌握商業文案和有效溝通的策略》，財經錢線文化有限公司。	

Subject Description Form

Subject Code	COMP2011					
Subject Title	Data Structures					
Credit Value	3					
Level	2					
Pre-requisite / Co-requisite / Exclusion	Pre-requisite: COMP1011/COMP1012/ENG2002 Exclusion: COMP2013, DSAI2201					
Objectives	The objectives of this subject are to: 1. introduce students to basic concepts of data structures and algorithms; and 2. teach students to apply simple data structures and algorithms in developing computer programs.					
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) understand the properties of basic data structures; (b) identify the strengths and weaknesses of different data structures; (c) acquire specialised knowledge of various typical algorithms; (d) design and employ appropriate data structures and algorithms for developing computer applications; and (e) think critically for improvement in the solutions.					
Subject Synopsis/ Indicative Syllabus	<table><tr><td>Topic</td></tr><tr><td>1. Programming and Algorithms Computer algorithms; types of algorithms; data structures; and abstract data types.</td></tr><tr><td>2. Data Structures: Representation and Algorithms Linear structures: linked-lists, stacks, queues; tree structures: binary trees, balanced trees, tree traversals; and other common data structures: priority queues, heaps.</td></tr><tr><td>3. Sorting Basic sorting algorithms: bubble sort, insertion sort, selection sort; and advanced sorting algorithms: quicksort, mergesort, heapsort.</td></tr><tr><td>4. Searching Common searching algorithms: sequential search, binary search; and advanced searching algorithms: tree search, dictionary and hashing.</td></tr></table>	Topic	1. Programming and Algorithms Computer algorithms; types of algorithms; data structures; and abstract data types.	2. Data Structures: Representation and Algorithms Linear structures: linked-lists, stacks, queues; tree structures: binary trees, balanced trees, tree traversals; and other common data structures: priority queues, heaps.	3. Sorting Basic sorting algorithms: bubble sort, insertion sort, selection sort; and advanced sorting algorithms: quicksort, mergesort, heapsort.	4. Searching Common searching algorithms: sequential search, binary search; and advanced searching algorithms: tree search, dictionary and hashing.
Topic						
1. Programming and Algorithms Computer algorithms; types of algorithms; data structures; and abstract data types.						
2. Data Structures: Representation and Algorithms Linear structures: linked-lists, stacks, queues; tree structures: binary trees, balanced trees, tree traversals; and other common data structures: priority queues, heaps.						
3. Sorting Basic sorting algorithms: bubble sort, insertion sort, selection sort; and advanced sorting algorithms: quicksort, mergesort, heapsort.						
4. Searching Common searching algorithms: sequential search, binary search; and advanced searching algorithms: tree search, dictionary and hashing.						

	5. Applications Practical program development using combination of various data structures and algorithms, e.g., friends-book; and efficiency of the various approaches.						
Teaching/ Learning Methodology	The course material will be delivered as a combination of mass lectures and small group supervised tutorial and laboratory sessions. Lectures will provide the required knowledge while tutorials and laboratory sessions allow students to acquire hands-on experience on programming with different algorithms. Programming project provides students with a chance to integrate their knowledge on applying appropriate data structures and algorithms to solve practical problems.						
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed				
			a	b	c	d	e
	Continuous Assessment (assignments / project / quizzes)	30%	✓	✓	✓	✓	✓
	Final Examination	70%	✓	✓	✓	✓	✓
	Total	100%					
Student Study Effort Expected	Class contact:						
	▪ Lecture				39 Hrs.		
	▪ Tutorial/Lab				13 Hrs.		
	Other student study effort:						
	▪ Assignments, Quizzes, Projects, Self-study				55 Hrs.		
	Total student study effort				107 Hrs.		
Reading List and References	Reference Books: 1. Goodrich, Michael T., Tamassia, Roberto, and Goldwasser, Michael H., <i>Data Structures and Algorithms in Java</i> , 6 th Edition, Wiley, 2014. 2. Sedgewick, Robert and Wayne, Kevin, <i>Algorithms</i> , 4 th Edition, Addison-Wesley, 2011. 3. Cormen, Thomas H., Leiserson, Charles E., Rivest, Ronald L. and Stein, Clifford, <i>Introduction to Algorithms</i> , 3 rd Edition, MIT Press, 2009.						

Subject Description Form

Subject Code	COMP4431
Subject Title	Artificial Intelligence
Credit Value	3
Level	4
Pre-requisite / Co-requisite / Exclusion	Pre-requisite: COMP2011/COMP2013/DSAI2201
Objectives	The objectives of this subject are to: 1. introduce the fundamental concepts of artificial intelligence; 2. equip students with the knowledge and skills in the programming of artificial intelligence techniques; 3. master the problem formulation and the typical intelligent algorithms; 4. understand the technical development of artificial intelligence; and 5. make students become aware of the ethical issues related to artificial intelligence.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: <u>Professional/academic knowledge and skills</u> (a) understand the history, development, and technical trend of artificial intelligence; (b) understand the important concepts in artificial intelligence and the problems addressed by intelligent techniques; (c) master programming languages and tools to implement artificial intelligence techniques; (d) able to apply the skills and techniques in machine learning, such as decision tree induction, support vector machine, and artificial neural networks; (e) learn the design rationale and the typical algorithms of problem-solving agents, in particular, search algorithms and their applications in real world; (f) deal with the ethical issues related to artificial intelligence;

	<u>Attributes for all-roundedness</u> (g) explore the nature of human intelligence and its role in problem-solving; and (h) deepen thoughts and understanding of human abilities such as learning, reasoning and planning						
Subject Synopsis/ Indicative Syllabus	<table><tr><th>Topic</th></tr><tr><td>1. Introduction to Artificial Intelligence Definition of artificial intelligence; basic concepts of human intelligence; scope of classical artificial intelligence problem; the birth, golden time, and the winter of artificial intelligence.</td></tr><tr><td>2. Agent and Knowledge Base Definition of agents and rationality; design an intelligent agent; knowledge-based agents and knowledge representation language; inference using forward chain and backward chain; uncertainty and Bayesian networks.</td></tr><tr><td>3. Problem-solving Agents Problem-solving agents; uninformed search strategies and typical algorithms; informed search strategies and typical algorithms; heuristic functions; hill-climbing search, simulated annealing search, genetic algorithms; constraint satisfaction problem and game problem.</td></tr><tr><td>4. Machine Learning Supervised learning, unsupervised learning, and reinforcement learning; classification, clustering, and regression; decision tree; support vector machine, artificial neural networks.</td></tr><tr><td>5. Applications and Ethical Issues Latest development of artificial intelligence; typical applications; ethical issues of artificial intelligence techniques and applications; benefits and risk to human society.</td></tr></table>	Topic	1. Introduction to Artificial Intelligence Definition of artificial intelligence; basic concepts of human intelligence; scope of classical artificial intelligence problem; the birth, golden time, and the winter of artificial intelligence.	2. Agent and Knowledge Base Definition of agents and rationality; design an intelligent agent; knowledge-based agents and knowledge representation language; inference using forward chain and backward chain; uncertainty and Bayesian networks.	3. Problem-solving Agents Problem-solving agents; uninformed search strategies and typical algorithms; informed search strategies and typical algorithms; heuristic functions; hill-climbing search, simulated annealing search, genetic algorithms; constraint satisfaction problem and game problem.	4. Machine Learning Supervised learning, unsupervised learning, and reinforcement learning; classification, clustering, and regression; decision tree; support vector machine, artificial neural networks.	5. Applications and Ethical Issues Latest development of artificial intelligence; typical applications; ethical issues of artificial intelligence techniques and applications; benefits and risk to human society.
Topic							
1. Introduction to Artificial Intelligence Definition of artificial intelligence; basic concepts of human intelligence; scope of classical artificial intelligence problem; the birth, golden time, and the winter of artificial intelligence.							
2. Agent and Knowledge Base Definition of agents and rationality; design an intelligent agent; knowledge-based agents and knowledge representation language; inference using forward chain and backward chain; uncertainty and Bayesian networks.							
3. Problem-solving Agents Problem-solving agents; uninformed search strategies and typical algorithms; informed search strategies and typical algorithms; heuristic functions; hill-climbing search, simulated annealing search, genetic algorithms; constraint satisfaction problem and game problem.							
4. Machine Learning Supervised learning, unsupervised learning, and reinforcement learning; classification, clustering, and regression; decision tree; support vector machine, artificial neural networks.							
5. Applications and Ethical Issues Latest development of artificial intelligence; typical applications; ethical issues of artificial intelligence techniques and applications; benefits and risk to human society.							
Teaching/ Learning Methodology	During the lecture, students will come across the concepts, algorithms and applications in artificial intelligence, and will be supplemented by exercises, labs, and project.						

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed							
			a	b	c	d	e	f	g	h
	1. Continuous Assessment	30%	✓	✓	✓	✓	✓	✓	✓	✓
	2. Final Examination	70%	✓	✓	✓	✓	✓	✓		
	Total	100 %								
Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Continuous assessments consist of a project, lab exercises, and quizzes, which are designed to facilitate students to achieve intended learning outcomes. Lab exercise is designed to encourage students to acquire good understanding of the relevant knowledge, practice in order to enrich their hands-on experience with various software tools. The project is designed to enhance students’ ability to acquire the understanding of and using different knowledge, principles, techniques, tools to solve a real problem through teamwork. Quizzes are to ensure the students understand the concepts.										
Student Study Effort Expected	Class contact:									
	• Class activities including lecturers, tutorial, and labs							39 Hrs.		
	Other student study effort:									
	• Coursework and Project							80 Hrs.		
	Total student study effort							119 Hrs.		
Reading List and References	Textbooks: 1. Russell, Stuart and Norvig, Peter, <i>Artificial Intelligence: A Modern Approach</i> , 3 rd Edition, Pearson, 2009.									
	Reference Books: 1. Pal, Sankar K. and Shiu, Simon C. K., <i>Foundations of Soft Case-Based Reasoning</i> , John Wiley, 2004.									
	2. Negnevitsky, Michael, <i>Artificial Intelligence: A Guide to Intelligent Systems</i> , 2 nd Edition, Addison Wesley, 2005.									

Subject Description Form

Subject Code	DSAI1102
Subject Title	Data Analytics Fundamentals
Credit Value	3
Level	1
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	This subject is to introduce to students the fundamental concepts of data analytics and some basic skills and tools used in the field.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) apply statistical reasoning to describe and analyze different types of data sets (b) extend their knowledge of statistical techniques and adapt analytical procedures to different situations (c) develop and extrapolate statistical concepts in synthesizing and solving problems (d) search for useful information and use data analytics software in solving problems (e) undertake the formulation of data analytics problems through continuous self-learning (f) demonstrate the abilities of logical and analytical thinking
Subject Synopsis/ Indicative Syllabus	<i>General introduction to data science</i> Data collection, types of data, data structures, selected case studies <i>Data preparation and exploration</i> Data cleaning/processing, data summary, frequency table, density plot, data visualization, computational tools of statistics and data analytics (e.g., R/Python) <i>Basic probability</i> Experiment, event and probability, random variable, distributions, measures of central tendency and dispersion, measures of association <i>Statistics</i> Estimation of mean and variance, confidence interval, hypothesis testing, regression, prediction, classification

Teaching/Learning Methodology	The subject will be delivered mainly through lectures and tutorials. The lectures will be conducted to introduce the basic data analytics concepts of the topics in the syllabus which are then reinforced by learning activities involving demonstration and tutorial exercise.							
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					
			(a)	(b)	(c)	(d)	(e)	(f)
	1. Assignments/Test	40%	✓	✓	✓	✓	✓	✓
	2. Examination	60%	✓	✓	✓	✓	✓	✓
	Total	100%						
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Continuous Assessment comprises of assignments and/or quizzes, and test(s). This will allow the instructor(s) to observe and assess individual student’s achievement of a particular learning outcome based on the coverage of the assigned problem set questions. A written examination is held at the end of the semester.							
Student Study Effort Expected	Class contact:							
	▪ Lectures/Tutorials				39 Hrs.			
	Other student study effort:							
	▪ Self-study				66 Hrs.			
	Total student study effort				105 Hrs.			
Reading List and Reference	Adhikari, A. and DeNero, J. (2019). Computational and Inferential Thinking: The Foundations of Data Science. GitBook. Davy, C., Meysman, A. D. B., and Ali, M. (2016) Introducing Data Science: Big Data, Machine Learning, and More, Using Python Tools. Manning Publications. Utts, J.M. (2014). Seeing Through Statistics (4th edition). Cengage Learning. Utts, J. M. and Heckard, R. F. (2015). Mind on Statistics (5th edition). Cengage Learning.							

Subject Description Form

Subject Code	DSAI1201					
Subject Title	Introduction to Data Analytics					
Credit Value	3					
Level	1					
Pre-requisite/ Co-requisite/ Exclusion	Nil					
Objectives	The objectives of this subject are to: 1. understand data analytics concepts; 2. apply data analytics tools; and 3. strengthen students' mathematics background for computing					
Intended Learning Outcomes	Upon completion of the subject, students will be able to: <u>Professional/academic knowledge and skills</u> (a) understand basic data analytics concepts; (b) manipulate, analyse and visualise data; and (c) understand and apply related mathematics operations					
Subject Synopsis/ Indicative Syllabus	<table border="1" style="width: 100%;"> <tr> <td style="text-align: center;">Topic</td></tr> <tr> <td> 1. Data Analytics Basics Defining data requirements, collecting data, processing data, cleaning data and analysing data </td></tr> <tr> <td> 2. Data Processing Data manipulation, data analysis, data visualisation </td></tr> <tr> <td> 3. Statistical Analysis Basic statistical functions, linear regression, time series analysis </td></tr> <tr> <td> 4. Linear Algebra and Calculus Vector basics, matrix basics, differentiation, integration, finding maxima and minima </td></tr> </table> The aforementioned topics will be taught with the aid of a suitable programming language such as R.	Topic	1. Data Analytics Basics Defining data requirements, collecting data, processing data, cleaning data and analysing data	2. Data Processing Data manipulation, data analysis, data visualisation	3. Statistical Analysis Basic statistical functions, linear regression, time series analysis	4. Linear Algebra and Calculus Vector basics, matrix basics, differentiation, integration, finding maxima and minima
Topic						
1. Data Analytics Basics Defining data requirements, collecting data, processing data, cleaning data and analysing data						
2. Data Processing Data manipulation, data analysis, data visualisation						
3. Statistical Analysis Basic statistical functions, linear regression, time series analysis						
4. Linear Algebra and Calculus Vector basics, matrix basics, differentiation, integration, finding maxima and minima						

Teaching/Learning Methodology	Lectures on data analytics and mathematics concepts (e.g., using R) will be given through lectures. There will be in-class activities for active learning. Hands-on lab/exercises will be arranged for students to practice data analytics tools. Students will also be required to study e-learning materials.				
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed		
			a	b	c
	Continuous Assessment	55%			
	1. Assignments, Test, Quizzes		✓	✓	✓
	Examination	45%	✓	✓	✓
	Total	100%			
	The assignments/test/quizzes (individual assessment) are used to assess learning outcomes (a) – (c) (e.g., how to apply R). Finally, students are assessed by a formal examination, covering learning outcomes (a) – (c).				
Student study effort expected	Class Contact:				
	▪ Class/ Learning Activities			39 Hrs.	
	Other student study effort:				
	▪ Self-study, Assignments, e-Learning			66 Hrs.	
	Total student study effort			105 Hrs.	
Reading list and References	Reference Books: 1. Beecher, K., <i>Computational Thinking</i> , BCS, 2017. 2. Teetor, P., <i>R Cookbook</i> , O’Reilly Media, 2011. 3. Wickham, H. and Grolemund, G., <i>R for Data Science</i> , O’Reilly Media, 2017. 4. Boyd, S. and Vandenberghe, L., <i>Introduction to Applied Linear Algebra</i> , Cambridge University Press, 2018. 5. Stewart, J., <i>Calculus: Early Transcendentals</i> , 8 th Edition, Cengage Learning, 2015.				

Subject Description Form

Subject Code	DSAI2201
Subject Title	Data Structures and Algorithms
Credit Value	3
Level	2
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: COMP1011/COMP1012/ENG2002/LGT3109 & AMA1110/AMA1501/AMA2634 & AMA1751/AMA2111
Objectives	To introduce students to data structures and algorithms, and how to use them to solve computational problems.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) understand the properties of basic data structures, and their strengths and weaknesses; (b) understand how to analyse and compare the efficiency of algorithms; (c) possess the knowledge of common algorithms, and common techniques for designing algorithms; (d) understand the basic steps in designing and implementing efficient data structures and algorithms for solving computational problems efficiently in a high-level programming language; and (e) get familiar with critical thinking for computational efficiency.
Subject Synopsis/ Indicative Syllabus	1. Analysis of algorithms • Asymptotic notations. • Efficiency analysis. 2. Data Structures • Linear structures: linked-lists, stacks, queues. • Tree structures: binary trees, balanced trees, tree traversals. 3. Searching and Sorting • Greedy sorting algorithms: bubble sort, insertion sort, selection sort. • Sorting algorithms based on divide and conquer: quicksort and mergesort. • Heapsort and related data structures: priority queues, heaps. • Hashing: hash functions, collision resolution. 4. Graphs • Definition and representation. • Depth-first search and breadth-first search. 5. Algorithmic Design Techniques • Greedy and dynamic programming.

Subject Synopsis/ Indicative Syllabus (Cont'd)	• Divide-and-conquer. • Recursion. 6. Applications • Practical program development using combination of multiple data structures and algorithms.																																																						
Teaching/Learning Methodology	The course material will be delivered as a combination of mass lectures and small group supervised tutorial and laboratory sessions. Lectures will provide the required knowledge while tutorial and laboratory sessions allow students to acquire hands-on experience on programming with different algorithms. Both written and programming assignments will be utilised in the course. Written assignments help students develop analysis and design skills, whereas programming assignments emphasise on implementation skills.																																																						
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="5">Intended subject learning outcomes to be assessed</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr><tr><td>Continuous Assessment</td><td>60%</td><td colspan="5"></td></tr><tr><td>1. Written Assignments</td><td>15%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>2. Programming Assignments</td><td>20%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>3. Quizzes</td><td>25%</td><td>✓</td><td>✓</td><td>✓</td><td></td><td>✓</td></tr><tr><td>Examination</td><td>40%</td><td>✓</td><td>✓</td><td>✓</td><td></td><td>✓</td></tr><tr><td>Total</td><td>100%</td><td colspan="5"></td></tr></table>	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed					a	b	c	d	e	Continuous Assessment	60%						1. Written Assignments	15%	✓	✓	✓	✓	✓	2. Programming Assignments	20%	✓	✓	✓	✓	✓	3. Quizzes	25%	✓	✓	✓		✓	Examination	40%	✓	✓	✓		✓	Total	100%					
Specific assessment methods/tasks	% weighting			Intended subject learning outcomes to be assessed																																																			
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2. Programming Assignments	20%	✓	✓	✓	✓	✓																																																	
3. Quizzes	25%	✓	✓	✓		✓																																																	
Examination	40%	✓	✓	✓		✓																																																	
Total	100%																																																						
Student Study Effort Expected	<table><tr><td>Class contact:</td><td></td></tr><tr><td>▪ Lecture/Tutorial/Lab</td><td>39 Hrs.</td></tr><tr><td>Other student study effort:</td><td></td></tr><tr><td>▪ Written Assignments, Programming Assignments, Quizzes, Self-study</td><td>101 Hrs.</td></tr><tr><td>Total student study effort</td><td>140 Hrs.</td></tr></table>	Class contact:		▪ Lecture/Tutorial/Lab	39 Hrs.	Other student study effort:		▪ Written Assignments, Programming Assignments, Quizzes, Self-study	101 Hrs.	Total student study effort	140 Hrs.																																												
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▪ Written Assignments, Programming Assignments, Quizzes, Self-study	101 Hrs.																																																						
Total student study effort	140 Hrs.																																																						

Reading List and References	1. Goodrich, Michael T., Tamassia, Roberto, and Goldwasser, Michael H., <i>Data Structures and Algorithms in Java</i>, 6th Edition, Wiley, 2014. 2. Goodrich, Michael T., Tamassia, Roberto, and Goldwasser, Michael H., <i>Data structures and algorithms in Python</i>, Wiley, 2013. 3. Sedgewick, Robert and Wayne, Kevin, <i>Algorithms</i>, 4th Edition, Addison-Wesley, 2011. 4. Cormen, Thomas H., Leiserson, Charles E., Rivest, Ronald L. and Stein, Clifford, <i>Introduction to Algorithms</i>, 4th Edition, MIT Press, 2022. 5. Frank M. Carrano, <i>Data Abstraction & Problem Solving with C++: Walls & Mirrors</i>, 7th Edition, Pearson, 2017
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Subject Description Form

Subject Code	DSAI4203
Subject Title	Machine Learning
Credit Value	3
Level	4
Pre-requisite / Co-requisite / Exclusion	Students are preferred to have some previous exposure to introductory Artificial Intelligence/ Data Analytics concepts and be familiar with basic notions in linear algebra and probability.
Objectives	The objectives of this subject are to: 1. present the basic principles, concepts and models of modern machine learning; and 2. introduce recent advances of machine learning technology with impactful applications in pattern recognition, computer vision and other areas.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) understand the major concepts of machine learning models and algorithms; (b) critically evaluate the effectiveness of machine learning techniques; (c) gain knowledge and abilities to apply machine learning techniques to various cutting-edge applications; and (d) design machine learning solutions to solve new challenging problems in practice.
Subject Synopsis/ Indicative Syllabus	Topic
	1. Part I: Machine Learning Fundamentals • Linear algebra and probability • Numerical computation and optimisation • Learning tasks: Regression, classification, etc. • Performance issues: Cross-validation, Overfitting and curse of dimensionality, Bias-variance dilemma, etc.

Subject Synopsis/ Indicative Syllabus (Cont'd)	2. Part II: Models and Techniques - Supervised learning: - Parametric vs non-parametric methods - Decision tree based methods - Bayesian models - Neural networks and support vector machines - Advanced models: CNN - Unsupervised learning: - k-means and hierarchical clustering - Spectral clustering and density-based clustering - Advanced models: autoencoder, embedding techniques, etc. - Regression and boosting - Feature selection and dimensionality reduction					
	3. Part III: Applications - Handwriting recognition challenge, e.g. MNIST - Object detection, recognition and tracking - Object feature descriptions: Engineering approach vs feature learning approach - Object detection examples, e.g. pixel clustering for face detection, etc. - Object recognition examples, e.g. face recognition via eigenface features - Object tracking examples, e.g. human motion tracking					
Teaching/ Learning Methodology	39 hours of class activities including lectures on the main concepts and models, together with applicational case studies, tutorials and class/group discussions, laboratory works and student presentations. Additional reading of research papers will be assigned, whenever appropriate.					
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed			
			a	b	c	d
	Continuous Assessment	55%				
	1. Assignments		✓	✓	✓	
	2. Tests/Quizzes		✓	✓	✓	
	3. Project				✓	✓
	Examination	45%	✓	✓	✓	✓
Total	100%					

Assessment Methods in Alignment with Intended Learning Outcomes (Cont'd)	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Assignment: After-class assessment of the continuous understanding of the concepts, issues, models and applications of machine learning techniques by providing answers to given questions. Test/Quiz: In-class assessment of the understanding of the concept, issues, models and applications of machine learning techniques by providing answers to given questions. Project: Assessment of problem solving ability in dealing with practical application problems by written reports and oral presentations. End-of-term Assessment: Assessment of the overall performance by a written examination.	
Student Study Effort Expected	Class contact:	
	▪ Lecture/Tutorial/Lab	39 Hrs.
	Other student study effort:	
	▪ Self-study	83 Hrs.
	Total student study effort	122 Hrs.
Reading List and References	Reference Books: - Hastie, T., Tibshirani, R. and Friedman, J., <i>The Elements of Statistical Learning</i>, 2nd Edition, Springer, 2009. - Alpaydin, E., <i>Introduction to Machine Learning</i>, 2nd Edition, MIT Press, 2010. - Shalev-Shwartz, S. and Ben-David, S., <i>Understanding Machine Learning: From Theory to Algorithms</i>, 2014. - Bousquet, O., Boucheron, S. and Lugosi, G., <i>Introduction to Statistical Learning Theory</i>, Advanced Lectures on Machine Learning. - Mohri, M., Rostamizadeh, A. and Talwalkar, A., <i>Foundations of Machine Learning</i>. USA, Massachusetts: MIT Press, 2012. - Vapnik, V. N., <i>The Nature of Statistical Learning Theory</i>. Springer, 2000. - Bishop, Christopher, <i>Pattern Recognition and Machine Learning</i>, Springer, 2006. - To be amended and updated at the beginning of the semester.	

Subject Description Form

Subject Code	EE2901S
Subject Title	Basic Electricity and Electronics
Credit Value	3
Level	2
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	1. To introduce students to the fundamental principles and analysis techniques for studying and operating basic electrical and electronic devices and circuits. 2. To introduce students to the appropriate skills and tools for experimenting with basic electrical and electronic devices and circuits.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Describe the fundamental principles and analysis techniques for studying and operating basic electrical and electronic devices and circuits. b. Use the appropriate skills and tools for experimenting with basic electrical and electronic devices and circuits.
Subject Synopsis/ Indicative Syllabus	DC Electrical Circuits Electromechanical analogies. Common Prefixes. How electricity works. Atoms and atomic structure. Basic electrical quantities: charge, potential, current, voltage, power, and energy. Resistance, Ohm's law, and resistors. Resistors in series and in parallel. Sign convention. Practical, ideal, independent, and dependent voltage and current sources. Voltage and current dividers. Use of basic test meters: voltmeters, ammeters, ohmmeters, and multimeters. Lumped circuit elements. Network description: branch, node, loop, and mesh. Kirchhoff's voltage and current laws. Tellegen's theorem. Mesh-current and node-voltage methods. Thévenin's and Norton's theorems. Loading effect and maximum power transfer. AC Electrical Circuits Tesla vs. Edison: the war of the currents. Differences between AC and DC. AC signal basics: time-dependent, periodic, and sinusoidal signals and sources. Worldwide mains electricity. Peak, average, and root-mean-square values. Inductors and capacitors. Sinusoidal steady-state analysis by time-domain method. Phasor and impedance basics: complex numbers, Euler's identity, phasor representation, impedance and admittance, phasor diagrams. Sinusoidal steady-state analysis by phasor-impedance method. Power, energy, and electricity bill. Instantaneous and average powers. Power in resistive, inductive, capacitive, and complex loads. Complex power and power factor. Power generation, transmission, and distribution. Three-phase power basics. Diodes and Diode Circuits Semiconductor basics: basic electrical and electronic materials, energy band diagrams, intrinsic and extrinsic semiconductors, electrons and holes, doping, donors and acceptors, n-type and p-type semiconductors. P-N junction diodes: basic structure, symbol, depletion region, barrier potential, forward bias, reverse bias, ideal current-voltage characteristics, ON and OFF states, ideal diode equation, breakdown characteristics, Zener diodes. Diode circuits: ideal and practical diode assumptions, basic and specific diode circuits.

	Bipolar Junction Transistors (BJTs) and BJT Amplifiers BJTs: basic structures, symbols, BJT semiconductor basics, BJTs as electric switches and amplifiers. DC biasing and circuits: modes of operation, DC biasing, input and output characteristics, DC equivalent circuits, DC analysis, load line, Q-point, DC biasing schemes. AC small signal and amplifiers: linear analog amplifiers, notation, basic characteristics of BJT amplifiers, DC and AC analyses with equivalent circuits, small-signal hybrid-π equivalent circuits with and without early effect, small-signal amplifier parameters, common-emitter amplifiers, DC and AC load lines. Laboratory Experiments: - E01: DC Circuits. - E02: Diode Circuits.																				
Teaching/Learning Methodology	<table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="2">Outcome</th></tr><tr><th>a</th><th>b</th></tr><tr><td>Lecture</td><td>✓</td><td></td></tr><tr><td>In-class Practice (Tutorial)</td><td>✓</td><td></td></tr><tr><td>Assignment</td><td>✓</td><td></td></tr><tr><td>Test</td><td>✓</td><td></td></tr><tr><td>Laboratory</td><td></td><td>✓</td></tr></table> Lecture: Students are introduced to the knowledge of the subject, and their comprehension is strengthened with interactive Q&A (outcome a). In-class Practice (Tutorial): Students apply what they have learned to solve the problems in the class (outcome a). Assignment: Students further test and develop their understanding and comprehension of the knowledge through after-class exercises (outcome a). Test: Students develop their actual understanding and comprehension of the knowledge (outcome a). Laboratory: Students acquire hands-on/simulated experience using the appropriate electrical and electronic devices, test equipment, and design software in circuits, apply what they have learned in the class to experimentally validate the theoretical investigations, and develop the experimental log and report writing skills (outcome b).	Teaching/Learning Methodology	Outcome		a	b	Lecture	✓		In-class Practice (Tutorial)	✓		Assignment	✓		Test	✓		Laboratory		✓
Teaching/Learning Methodology	Outcome																				
	a	b																			
Lecture	✓																				
In-class Practice (Tutorial)	✓																				
Assignment	✓																				
Test	✓																				
Laboratory		✓																			
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="2">Intended learning outcomes to be assessed</th></tr><tr><th>a</th><th>b</th></tr><tr><td>Continuous Assessment</td><td>50%</td><td>✓</td><td>✓</td></tr><tr><td>Examination</td><td>50%</td><td>✓</td><td></td></tr><tr><td>Total</td><td>100%</td><td colspan="2"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Overall Assessment: 0.5 × Continuous Assessment + 0.5 × End of Subject Examination	Specific assessment methods/tasks	% weighting	Intended learning outcomes to be assessed		a	b	Continuous Assessment	50%	✓	✓	Examination	50%	✓		Total	100%				
Specific assessment methods/tasks	% weighting			Intended learning outcomes to be assessed																	
		a	b																		
Continuous Assessment	50%	✓	✓																		
Examination	50%	✓																			
Total	100%																				

	Continuous Assessment covers all intended learning outcomes a and b, while Examination involves intended learning outcome a. Continuous Assessment (50%) contains Assignment (18%), Test (18%), and Laboratory (14%), aiming to provide timely feedback to students on various topics of the syllabus, including assignment works, laboratory skills, use of the appropriate electrical and electronic devices, test equipment, and design software in circuits, analysis and presentation of the experimental results, etc. Examination (50%) is a three-hour, closed-book, end-of-subject written examination, aiming to assess students' overall understanding and ability to apply the fundamental principles and analysis techniques.	
Student Study Effort Expected	Class contact:	
	▪ Lecture	28 Hrs.
	▪ In-class Practice (Tutorial)	3 Hrs.
	▪ Test	2 Hrs.
	▪ Laboratory	6 Hrs.
	Other student study efforts:	
	▪ Self-study	44 Hrs.
	▪ Assignment	12 Hrs.
	▪ Laboratory Log and Report	10 Hrs.
	Total student study effort	105 Hrs.
Reading List and References	Textbooks: 1. Giorgio Rizzoni and James Kearns, <i>Principles and Applications of Electrical Engineering</i>, 7th Edition, New York: McGraw Hill (2022). 2. Donald A. Neamen, <i>Microelectronics: Circuit Analysis and Design</i>, 6th Edition, New York: McGraw Hill (2016). Reference books: 1. William H. Hayt, Jack E. Kemmerly, Jamie D. Phillips, and Steven M. Durbin, <i>Engineering Circuit Analysis</i>, 10th Edition, New York: McGraw Hill (2024). 2. Alan H. Robbins and Wilhelm C. Miller, <i>Circuit Analysis: Theory and Practice</i>, 5th Edition, New York: Cengage Learning (2013). 3. Raymond A. DeCarlo and Pen-Min Lin, <i>Linear Circuit Analysis: Time Domain, Phasor, and Laplace Transform Approaches</i>, 2nd Edition, New York: Oxford University Press (2001).	

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Subject Description Form

Subject Code	EIE1003
Subject Title	Foundations of Data Science
Credit Value	3
Level	1
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	Being able to discover useful knowledge and information from a large amount of data is very critical to industry, business and government. This subject aims to provide students the fundamental concepts of data science and the basic technologies for data analytics. It provides hands-on experiences in data analytics and case studies in applications of data science in engineering, social science, healthcare, business and government. It also prepares students with the right mentality towards data and the ability to leverage data for decision-making.
Intended Subject Learning Outcomes	Upon completion of the subject, students will be able to: <u>Category A: Professional/academic knowledge and skills</u> 1. Understand the basic concepts and technologies of data science. 2. Acquire the basic technical know-how on data analytics. <u>Category B: Attributes for all-roundedness</u> 3. Understand the data-driven process for problem solving. 4. Demonstrate how to harness and process data for decision-making.
Subject Synopsis/ Indicative Syllabus	1. <u>Introduction to Data Science</u> • Data science vs. big data vs. data analytics • Benefits of data science • Skill sets required • Privacy, security and ethics • Example applications and case studies 2. <u>Technologies for Data Science</u> • Basic concepts in summary statistics • Graphs and plots for data analytics, e.g., box plots, scatter plots, histograms, run charts, etc. • Example case studies of exploratory data analytics for data science • Fundamental of machine learning for data science • Cloud technologies

	3. <u>Tools for Data Science</u> • Data cleaning, e.g., OpenRefine • Machine learning tools, e.g., Microsoft ML Studio, Weka • Data visualization tools, e.g., Google Chart, Tableau 4. <u>Applications with Case Studies</u> • Recommendation systems • Spam filtering • Stock prediction • Social networks • Sentiment analysis																																														
Teaching/Learning Methodology	Lectures: The subject matters will be delivered through lectures (both in-person and online ones). Students will be engaged in the lectures through Q&A, discussions and specially designed classroom activities. Practitioners and software vendors will be invited to give guest lectures. Tutorials and Workshops: Students will work on data analytics projects using software tools. Students will start from small and easy projects in the first half of the subject. In the second half, students will work on a more realistic project that solves real-world problems, using the knowledge and know-hows that they have learnt from the small projects. Assignment: Students will need to do a group-based mini-project on data science.																																														
Assessment Methods in Alignment with Intended Subject Learning Outcomes	<table><tr><th rowspan="2">Specific Assessment Methods/Tasks</th><th rowspan="2">% Weighting</th><th colspan="4">Intended Subject Learning Outcomes to be Assessed (Please tick as appropriate)</th></tr><tr><th>1</th><th>2</th><th>3</th><th>4</th></tr><tr><td colspan="6">1. Continuous Assessment (total: 100%)</td></tr><tr><td>• Mini-project (proposal, report and presentation)</td><td>30%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>• Exercises</td><td>16%</td><td>✓</td><td>✓</td><td></td><td></td></tr><tr><td>• Tests</td><td>30%</td><td>✓</td><td>✓</td><td></td><td></td></tr><tr><td>• Laboratory activities</td><td>24%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Total</td><td>100%</td><td colspan="4"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Laboratory exercises and mini-project will require students to apply what they have learnt to solve problems. There will be open-ended questions that allow students to exercise their creativity in solution	Specific Assessment Methods/Tasks	% Weighting	Intended Subject Learning Outcomes to be Assessed (Please tick as appropriate)				1	2	3	4	1. Continuous Assessment (total: 100%)						• Mini-project (proposal, report and presentation)	30%	✓	✓	✓	✓	• Exercises	16%	✓	✓			• Tests	30%	✓	✓			• Laboratory activities	24%	✓	✓	✓	✓	Total	100%				
Specific Assessment Methods/Tasks	% Weighting			Intended Subject Learning Outcomes to be Assessed (Please tick as appropriate)																																											
		1	2	3	4																																										
1. Continuous Assessment (total: 100%)																																															
• Mini-project (proposal, report and presentation)	30%	✓	✓	✓	✓																																										
• Exercises	16%	✓	✓																																												
• Tests	30%	✓	✓																																												
• Laboratory activities	24%	✓	✓	✓	✓																																										
Total	100%																																														

	design. Tests and Exercises assess students' achievement of the learning outcomes in a more formal manner. Mini-project is group-based and weights 30% of the whole assessment. Among the 30% weight, 7% is for proposal, 13% is for final report, and 10% is for presentation (in the form of a 10-minutes video). Proposal and report (20% in total) are evaluated based on group, while presentation (10%) is evaluated individually. Each group member will present the part he/she is responsible for in the mini-project. The mini-project will make use of publicly available tools such as Microsoft Azure Machine Learning Studio so that requirements on programming knowledge is kept to a minimum, i.e., no programming background is assumed. Students will perform drag and drop of data sources, machine learning models, analytic methods, and evaluation methods from the tool to solve data science problems. Enthusiastic students could use the cloud-based API to perform more complex tasks. Tests and Exercises weight 46% and they are individual assessments. Laboratory activities weight 24% and they are individual assessments. Overall, 80% of the assessment is individual assessment and 20% is group-based assessment.	
Student Study Effort Expected	Class contact (time-tabled):	
	• Lectures (In-person and online)	22 Hours
	• Tutorial/Laboratory/Practice Classes	17 Hours
	Other student study effort:	
	• Lecture: preview/review of notes; homework/assignment; preparation for test/quizzes	30 Hours
	• Tutorial/Laboratory/Practice Classes: preview of materials, revision and/or reports writing	36 Hours
	Total student study effort:	105 Hours
Reading List and References	Reference Materials: 1. L. Cao, <i>Data Science Thinking: The Next Scientific, Technological and Economic Revolution</i>. Cham: Springer International Publishing, 2018. 2. L. Igual and S. Sequi, <i>Introduction to Data Science: a Python Approach to Concepts, Techniques and Applications</i>. Cham, Switzerland: Springer, 2017. 3. S. Alan and V. Torra, <i>Data Science in Practice</i>. Cham, Switzerland: Springer 2019.	

	4. G. Rebala, A. Ravi, and S. Churiwala, <i>An Introduction to Machine Learning</i>. Cham, Switzerland: Springer 2019. 5. P. Kromer and R. Journey, <i>Big Data for Chimps</i>. O'Reilly, 2016. 6. T. Ojeda et al., <i>Practical Data Science Cookbook</i>. Packt Publishing Ltd, 2014. 7. A. Adhikari and J. DeNero, <i>Computational and Inferential Thinking: The Foundations of Data Science</i>. https://www.inferentialthinking.com/
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July 2025

Subject Description Form

Subject Code	EIE3320
Subject Title	Object-Oriented Design and Programming
Credit Value	3
Level	3
Pre-requisite/ Co-requisite/ Exclusion	• COMP2011 Data Structure or COMP2013 Data Structures and Algorithms • AND ENG2002 Computer Programming
Objectives	This subject will provide students with the principles of object-oriented software design and programming from the perspective of Java implementation and UML. Students are expected to learn the concepts of and practical approaches to object-oriented analysis, design and programming using UML and Java.
Intended Subject Learning Outcomes	Upon completion of the subject, students will be able to: <u>Category A: Professional/academic knowledge and skills</u> 1. Understand the principles of object-oriented design. 2. Apply Java in object-oriented software development. 3. Apply UML in object-oriented software modeling. 4. Apply object-oriented approach to developing computer software. <u>Category B: Attributes for all-roundedness</u> 5. Learn independently and be able to search for the information required in solving problems. 6. Present ideas and findings effectively. 7. Think critically. 8. Work in a team and collaborate effectively with others.
Subject Synopsis/ Indicative Syllabus	Syllabus: 1. <u>Introduction to Software Engineering</u> Software products; software processes; software process models. 2. <u>Java Programming Basic</u> Java technologies; Java platform; Java language basic: variables, operators, expressions, statements, blocks, control flow, methods, arrays. 3. <u>Object-Oriented Programming with Java</u> Objects and classes; class definition; fields, constructors and methods; object interaction; grouping objects; array and collections; designing classes; inheritance and polymorphism; managing

	inheritance: creating subclasses and super-classes, hiding member variables, overriding methods. Interfaces and packages. 4. <u>Web Programming with Java</u> JavaScript: Client-side Web programming; JavaScript and HTML; Object, events, and event handlers in JavaScript. Java Servlets: architecture of servlets, client interaction, life cycle of servlets, saving client states; servlet communications, session tracking, and using server resources. JavaServer Pages (JSP): basics of JSP, JSP lifecycle, scripting elements, directives, implicit objects, session management, and integration of JSP with servlets for dynamic web content generation 5. <u>Unified Modelling Language (UML)</u> Purposes of modelling. Structural Modelling: classes, relationships, class Diagrams, interfaces, packages, and object diagrams. Behavioural modelling interactions and use case diagrams. Architectural modelling: components, deployment, and collaborations. Mapping UML diagrams to Java Code. Laboratory Experiment: Students will be requested to use integrated development environment (IDE) to write and debug Java programs during tutorial and lab sessions.															
Teaching/ Learning Methodology	<table><tr><th>Teaching and Learning Method</th><th>Intended Subject Learning Outcome</th><th>Remarks</th></tr><tr><td>Lectures</td><td>1, 2, 3</td><td>Fundamental principles and key concepts of the subject are delivered to students</td></tr><tr><td>Quizzes/Tests</td><td>1, 2, 4</td><td>Students’ knowledge on understanding of certain topics can be easily estimated, and the corresponding teaching time will be adjusted accordingly</td></tr><tr><td>Assignments</td><td>2,4,5,7</td><td>Programming exercises are used to reinforce the knowledge taught in lectures.</td></tr><tr><td>Laboratory sessions</td><td>2,3,4,5,6,7,8</td><td>Students will need to design, develop, test, and document Java programs.</td></tr></table>	Teaching and Learning Method	Intended Subject Learning Outcome	Remarks	Lectures	1, 2, 3	Fundamental principles and key concepts of the subject are delivered to students	Quizzes/Tests	1, 2, 4	Students’ knowledge on understanding of certain topics can be easily estimated, and the corresponding teaching time will be adjusted accordingly	Assignments	2,4,5,7	Programming exercises are used to reinforce the knowledge taught in lectures.	Laboratory sessions	2,3,4,5,6,7,8	Students will need to design, develop, test, and document Java programs.
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Assignments	2,4,5,7	Programming exercises are used to reinforce the knowledge taught in lectures.														
Laboratory sessions	2,3,4,5,6,7,8	Students will need to design, develop, test, and document Java programs.														

Assessment Methods in Alignment with Intended Subject Learning Outcomes	Specific Assessment Methods/ Task	% Weighting	Intended Subject Learning Outcomes to be Assessed (Please tick as appropriate)								
			1	2	3	4	5	6	7	8	
	1. Continuous Assessment (Total: 100%)										
	• Assignments	8%		✓		✓	✓		✓		
	• Lab reports	20%		✓	✓	✓	✓	✓	✓	✓	
	• Knowledge Check Quizzes/ Practical Skill Test	15%	✓	✓		✓					
	• Quizzes/ Practical Test(s)	42%	✓	✓		✓					
	• Project	15%		✓	✓	✓	✓	✓	✓	✓	
	Total	100%									
	The continuous assessment consists of programming assignments, laboratory reports, knowledge tests/quizzes and practical tests.										
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:										
	Specific Assessment Methods/Tasks	Remark									
	Knowledge Check Quizzes, Written Tests	Short questions will be used to test and enhance students’ understanding about the topics covered in lectures. End-of-chapter problems will be used to evaluate students’ ability in applying concepts and skills learnt in the classroom.									
	Assignments	Students will be asked to write Java programs and test the programs. Students will need to think critically and creatively in order to come up with a good solution for an existing problem.									
	Lab reports, Project	Each group of students are required to produce a written report for the Laboratory sessions. Students will be assessed based on the quality of their programs and the clarity of their reports. Students will be asked to work as a team to develop a Java application. Each of them will									

		be responsible for part of the software. They will also need to use UML diagram to illustrate the structure of their programs. Students will need to think critically and creatively in order to come up with a good solution for an existing problem.
	Practical Skill Test	Students will be given basic programming problems and asked to write Java programs to solve them in the early stages of the subject. This will evaluate their ability to apply concepts and skills practically.
	Practical Tests	Students will be given programming problems and asked to write Java programs to solve the problems.
Student Study Effort Expected	Class contact (time-tabled):	
	• Lecture	30 Hours
	• Tutorial/Laboratory/Practice Classes	9 hours
	Other student study effort:	
	• Lecture: preview/review of notes; homework/assignment; preparation for test/quizzes	36 Hours
	• Tutorial/Laboratory/Practice Classes: preview of materials, revision and/or reports writing	30 Hours
	Total student study effort:	105 Hours
Reading List and References	Reference Books: 1. G. Booch, I. Jacobson and J. Rumbaugh, <i>The Unified Modeling Language User Guide</i>, 2nd ed., Addison-Wesley, 2005. 2. D.J. Barnes and M. Kolling, <i>Objects First with Java: A Practical Introduction using BlueJ</i>, 6th ed., Pearson, 2017. 3. Nell Dale, Daniel T. Joyce, and Chip Weems. <i>Object-Oriented Data Structures Using Java (4th. ed.)</i>. Jones and Bartlett Publishers, Inc., USA. 2018. 4. H.M. Deitel and P.J. Deitel, <i>Java: How To Program (Early Objects)</i>, 11th ed., Pearson, 2018. 5. J. Lewis and W. Loftus, <i>Java Software Solutions</i>, 9th Edition, Pearson, 2018. 6. J. Rumbaugh, I. Jacobson and G. Booch, <i>The Unified Modeling Language Reference Manual</i>, 2nd ed., Addison-Wesley, 2005.	

July 2025

Subject Description Form

Subject Code	ELC3531
Subject Title	Professional Communication in English for Engineering Students
Credit Value	2
Level	3
Pre-requisite / Co-requisite	English LCR subjects
Objectives	This subject aims to develop the language competence for professional communication in English required by students to communicate effectively with various parties and stakeholders in regard to engineering-related project proposals.
Intended Learning Outcomes	Upon completion of the subject, and in relation to effective communication with a variety of intended readers/audiences in English, students will be able to: a. plan, organise and produce professionally acceptable project proposals with appropriate text structures and language for different intended readers b. plan, organise and deliver effective project-related oral presentations with appropriate interactive strategies and language for different intended audiences c. adjust the style of expression and interactive strategies in writing and speaking in accordance with different intended readers/audiences
Subject Synopsis / Indicative Syllabus	<i>Synopsis</i> This subject enables students to develop the transferrable thinking, language, and communication skills that they will employ as aspiring professionals in the engineering field. Topics include analysis, clarity, appropriacy and persuasion in language and communication. Through a course-long engineering-related project, students will produce a professional project proposal on a creative solution which addresses problems and needs in the society, and deliver an effective pitch justifying the need for the project and the feasibility of the idea. In both tasks, students are required to demonstrate critical research and thinking when planning, organising and producing written and spoken discourses. They are also required to employ advanced language and communication strategies to convey meaning clearly, accurately, appropriately, and persuasively to different audiences.

	1. Project proposal in English • understanding and analysing problems, needs and requirements • analysing the structure and language of project proposals • extracting and evaluating information • discussing project ideas with the teacher and peers • developing and writing goals, objectives, and informed solutions based on critical analysis • integrating well-researched evidence and discipline specific knowledge clearly and convincingly • organising content logically and coherently • employing advanced language and communication strategies to convey meaning clearly, accurately, appropriately, and persuasively • producing a professional and reader-friendly document • peer-reviewing other proposals and reflecting on their project proposal 2. Project pitch in English • having a clear presentation purpose • selecting appropriate content and evidence • adapting language and style appropriate to the purpose, context and intended audience • employing advanced communication strategies and language features to convey meaning clearly, accurately, appropriately, and persuasively • speaking with clarity (including clear pronunciation) • speaking with fluency and confidence • using effective verbal and non-verbal interactive strategies • using visuals and text to support the spoken message • handling questions professionally • establishing rapport and connection with the audience
Teaching/Learning Methodology	The subject is designed to develop the English language skills, both oral and written, that students need to use to communicate effectively and professionally with a variety of stakeholders of engineering-related projects. It builds upon the language and communication skills covered in GUR language training subjects. Classes are seminar / workshop based. The lessons and materials help students to articulate and pitch their ideas in professionally acceptable language structures, text formats and registers. Activities include discussions, sample analysis, student-led

	investigations, process writing, peer reviews and mini-presentations. Online resources are integrated into the course for in-class and out-of-class learning. Students' personal integrity in the fast-changing technological landscape is also nurtured.																																															
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="6">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th></th><th></th><th></th></tr><tr><td>1. Project proposal in English</td><td>40%</td><td>√</td><td></td><td>√</td><td></td><td></td><td></td></tr><tr><td>2. Project pitch in English</td><td>60%</td><td></td><td>√</td><td>√</td><td></td><td></td><td></td></tr><tr><td>Total</td><td>100%</td><td colspan="6"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Project proposal in English The project proposal is used to assess a student’s essential writing skills relevant to their field. These skills include using discipline specific concepts and knowledge to justify their rationale and approach, writing with clarity and purpose by adopting a style, structure and design which meets the funder’s requirements, and using persuasive language, communication and writing strategies to win support. Embedded into this task is a consultation in which students explain the feasibility of their idea and the overall structure of their project proposal, and followed by a peer-review task in which students review and give actionable feedback to their peers. Project pitch in English The project pitch is applied to assess a student’s ability to deliver professional and persuasive presentations to an audience relevant to the engineering field. The assignment requires students to justify their project idea, and persuade the audience to take action. Students will need to speak with fluency, clarity and purpose, pitch ideas in a style and structure appropriate to the specific audience, engage the audience, and use persuasive language and communication strategies. <table><tr><th>Assessment type</th><th>Intended readers/audience</th><th>Timing</th></tr><tr><td>1. Project proposal in English Each team writes a proposal of 2000-2500 words</td><td>ELC Fund Assessment Panel (including engineering experts)</td><td>Week 8</td></tr><tr><td>2. Project pitch in English Each individual delivers a 5-6 minute project pitch followed by a question-and-answer session.</td><td>ELC Fund Assessment Panel and competitors</td><td>Weeks 12-13</td></tr></table>	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)						a	b	c				1. Project proposal in English	40%	√		√				2. Project pitch in English	60%		√	√				Total	100%							Assessment type	Intended readers/audience	Timing	1. Project proposal in English Each team writes a proposal of 2000-2500 words	ELC Fund Assessment Panel (including engineering experts)	Week 8	2. Project pitch in English Each individual delivers a 5-6 minute project pitch followed by a question-and-answer session.	ELC Fund Assessment Panel and competitors	Weeks 12-13
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Student Study Effort Expected	Class contact:	
	Seminars	26 Hrs.
	Other student study effort:	
	Researching, planning and writing the project proposal Rehearsing the presentation	52 Hrs.
	Total student study effort:	78 Hrs.
Reading List and References	1. D. F. Beer, Ed., <i>Writing and Speaking in the Technology Professions: A practical guide</i>, 2nd ed. Hoboken, NJ: Wiley, 2003. 2. R. Johnson-Sheehan, <i>Writing Proposals</i>, 2nd ed. New York: Pearson/Longman, 2008. 3. S. Kuiper and D. Clippinger, <i>Contemporary Business Reports</i>, 5th ed. Mason, OH: South-Western, 2013. 4. M. H. Markel, <i>Practical Strategies for Technical Communication</i>, 2nd ed. New York: Bedford/St. Martin's, 2016. 5. D. C. Reep, <i>Technical Writing: Principles, strategies, and readings</i>, 8th ed. Boston: Pearson/Longman, 2011. 6. E. D. Zanders and L. Macleod, <i>Presentation Skills for Scientists: A practical guide</i>, 2nd ed. Cambridge: Cambridge University Press, 2018.	

Subject Description Form

Subject Code	ENG1004
Subject Title	Engineering Professionals in Society I
Credit Value	1
Level	1
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	Through a series of seminars, this subject enables students to understand: 1. the features and career development opportunities of various engineering disciplines 2. the responsibilities and accountability of engineering professionals 3. the organizational activities of professional engineering institutions Furthermore, it will also: 4. Enthuse students about their major study in engineering as well as engage them in active, autonomous learning and deep understanding
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Describe the role and impact of engineers in a variety of professional fields in addressing contemporary societal needs at local, national and global levels b. Identify the qualities and competences required to become a successful professional engineer and explain why they are important for professional work c. Reflect on their professional aspirations and develop a personal development plan for pursuing their career goals in Engineering
Subject Synopsis/ Indicative Syllabus	1. <u>Features and Career Development Opportunities of Various Disciplines in Engineering</u> Features and career development opportunities of the disciplines under the Engineering Faculty, namely: Aeronautical and Aviation Engineering, Biomedical Engineering, Electrical and Electronic Engineering, Industrial and Systems Engineering, and Mechanical Engineering 2. <u>Responsibilities and Accountability of Engineering Professionals</u> Responsibilities and accountability of the engineering professionals in social, political, legal, economic, safety and health, and environmental domains; professional ethics, the work of the Independent Commission Against Corruption (ICAC), the work of the Environmental Protection Department 3. <u>Professional Institutions</u> International engineering agreements and the organizational activities of professional engineering institutions, professional accreditation, qualifications and criteria of engineering professionals

Teaching/Learning Methodology	The subject will be offered in terms of a series of seminars conducted by representatives from academic departments in PolyU with invited guest speakers from professional institutions; HKSAR Government organisations such as the Independent Commission Against Corruption (ICAC) and Environmental Protection Department; experts/practitioners/alumni from the industry, etc. In each seminar, the speaker will introduce to students the various disciplines and the different topics listed in the section “Subject Synopsis/Indicative Syllabus” while addressing the three learning outcomes (a), (b) and (c). “Flipped Classroom” approach will be adopted to nurture active learning for students with emphasis in student engagement. An active learning classroom will compose of various activities, namely: 1. Pre-seminar reading, research, and quizzes, 2. In-seminar case study, discussions, Question and Answer, 3. Post-seminar reflection, reviewing, and quizzes																																
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% Weighting</th><th colspan="3">Intended subject learning outcomes to be assessed</th></tr><tr><th>a</th><th>b</th><th>c</th></tr><tr><td>1. Pre-seminar quizzes</td><td>20%</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>2. Post-seminar quizzes</td><td>50%</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>3. Case Study</td><td>30%</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Total</td><td>100%</td><td>✓</td><td>✓</td><td>✓</td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: 1. Pre-seminar quizzes can encourage students to engage in the “flipped classroom” mode of learning, whereby they prepare for the oncoming seminar by advance reading, and research for background information. 2. Post-seminar quizzes can encourage students to participate actively in the seminars and do extended reading and information search afterwards. They can also assess students’ understanding about the various topics introduced in the seminars and hence their achievement of the intended learning outcomes. 3. One case study to be conducted by students can assess students’ in-depth understanding about the various topics introduced in the seminars. Pre-seminar and post-seminar quizzes assess foundational knowledge, measuring learning outcomes (a) and (b). To assess learning outcome (c), a case study evaluates students’ ability to define their career aspirations and create a personal development plan, guiding their journey as professional engineers, researchers, engineering entrepreneurs, etc.					Specific assessment methods/tasks	% Weighting	Intended subject learning outcomes to be assessed			a	b	c	1. Pre-seminar quizzes	20%	✓	✓	✓	2. Post-seminar quizzes	50%	✓	✓	✓	3. Case Study	30%	✓	✓	✓	Total	100%	✓	✓	✓
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3. Case Study	30%	✓	✓	✓																													
Total	100%	✓	✓	✓																													
Student Study Effort Expected	Class contact:																																
	• Seminars (7 sessions, 5 x 2 + 2 x 1.5 hours)				13 Hrs.																												
	Other student study efforts:																																
	• Pre-seminar reading and research for information				7 Hrs.																												
	• Post-seminar extended reading, information search, case report writing, and other learning and teaching activities				19 Hrs.																												
	Total student study efforts:				39 Hrs.																												

Reading List and References	Reference Books & Articles 1. Holbrook, J. Britt, (ed), <i>Ethics, science, technology, and engineering: a global resource</i>, Farmington Hills, Mich.: Gale, Cengage Learning, 2015 2. Alam, Firoz, <i>Engineering education : accreditation & graduate global mobility</i>, Leiden: CRC Press/Balkema, 2021 3. Video (PolyU Library), <i>Engineering – solving problems through science</i>, San Francisco, California, USA: Kanopy Streaming; 2014 Online Materials: 1. Faculty of Engineering website: https://www.polyu.edu.hk/en/feng/ 2. Departmental websites: a. https://www.polyu.edu.hk/en/aae/ b. https://www.polyu.edu.hk/en/bme/ c. https://www.polyu.edu.hk/en/eee/ d. https://www.polyu.edu.hk/en/ise/ e. https://www.polyu.edu.hk/en/me/ 3. HKIE Website: https://www.hkie.org.hk/en/ 4. Websites of professional societies and institutions, such as: a. The Chartered Institute of Logistics and Transport https://www.cilt.org.hk/en-US/Default.aspx b. CMI-HK Website: https://www.managers.org.uk/community/regional-networks/hong-kong/ 5. ICAC Website: https://www.icac.org.hk/en/home/index.html 6. Environmental Protection Department Website: https://www.epd.gov.hk/epd/english/top.html 7. TEDx Talks: Engineering for social impact: Randy Marsden at TEDx Edmonton, https://www.youtube.com/watch?v=UCDwNWSXFHk
Last Updated	May 2025
Prepared by	C.K. Choi

May 2025

Subject Description Form

Subject Code	ENG2001
Subject Title	Fundamentals of Materials Science and Engineering
Credit Value	3
Level	2
Pre-requisite / Co-requisite/ Exclusion	Nil
Objectives	1. To realize the impact of the development of engineering materials on human civilization; 2. To enable students to establish a broad knowledge base on the structure and properties of materials for solving engineering problems. 3. To enable students to understand the applications and selection of engineering materials based on the consideration of properties, cost, ease of manufacture, environmental issues and their in service performance.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. comprehend the importance of materials in engineering and society; b. explain the properties and behaviour of materials using fundamental knowledge of materials science. c. apply the knowledge of materials science to analyze and solve basic engineering problems related to stress, strain and fracture of materials; d. select appropriate materials for various engineering applications taking into consideration of issues in cost, quality and environmental concerns.
Subject Synopsis/ Indicative Syllabus	1. <u>Introduction</u> Historical perspective; Evolution of engineering materials; Materials science and engineering; Classification of materials 2. <u>Atomic Structure and Structures of Materials</u> Atomic structure; Bonding forces and energies; Primary interatomic bonds and secondary bonding; Crystalline and non-crystalline materials; Phase diagram and microstructure of alloys 3. <u>Electrical and Optical Properties of Materials</u> Conductors and insulators; Semi-conductor materials; N-type and P-type semiconductors; P/N junction; Light interactions with materials; Light emitting diode (LED) and photovoltaics; Light propagation in optical fibers; Liquid crystal; Photoelasticity

	4. <u>Mechanical Properties of Materials</u> Concept of stress and strain; Stress-strain behaviour; Elastic and plastic properties of materials; Concepts of dislocations and strengthening mechanisms; Tensile properties; Elastic recovery after plastic deformation; Hardness; Stress concentration; Impact energy, Fracture toughness; Design and safety factors 5. <u>Introduction to Failure Analysis and Prevention</u> Fundamentals of fracture: ductile, brittle, fatigue and creep; Corrosion; Nondestructive testing; Techniques for failure analysis and prevention 6. <u>Selection of Engineering Materials</u> Characteristics of metallic, polymeric, ceramic, electronic and composite materials; Economic, environmental and recycling issues																																																						
Teaching/Learning Methodology	The subject will be delivered mainly through lectures but tutorials, case studies and laboratory work will substantially supplement which. Practical problems and case studies of material applications will be raised as a focal point for discussion in tutorial classes, also laboratory sessions will be used to illustrate and assimilate some fundamental principles of materials science. The subject emphasizes on developing students’ problem solving skills.																																																						
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="6">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th></th><th></th></tr><tr><td>1. Assignments</td><td>15%</td><td>√</td><td>√</td><td>√</td><td>√</td><td></td><td></td></tr><tr><td>2. Test</td><td>20%</td><td></td><td>√</td><td>√</td><td>√</td><td></td><td></td></tr><tr><td>3. Laboratory report</td><td>5%</td><td></td><td>√</td><td>√</td><td></td><td></td><td></td></tr><tr><td>3. Examination</td><td>60%</td><td></td><td>√</td><td>√</td><td>√</td><td></td><td></td></tr><tr><td>Total</td><td>100 %</td><td colspan="6"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: The assignments are designed to reflect students’ understanding of the subject and to assist them in self-monitoring of their progress. The laboratory report is designed to assess the capability of students in analyzing and reporting experimental data relates to learning outcome (b). The test and examination are for determining students’ understanding of key concepts as well as for assessing their achievement of the learning outcomes.	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)						a	b	c	d			1. Assignments	15%	√	√	√	√			2. Test	20%		√	√	√			3. Laboratory report	5%		√	√				3. Examination	60%		√	√	√			Total	100 %						
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3. Examination	60%		√	√	√																																																		
Total	100 %																																																						
	Class contact:																																																						

Student Study Effort Expected	▪ Lectures, tutorials, practical	39Hrs.
	Other student study effort:	
	▪ Guided reading, assignments and reports	37Hrs.
	▪ Self-study and preparation for test and examination	47Hrs.
	Total student study effort	123Hrs.
Reading List and References	1. William D. Callister, Jr., David G. Rethwisch, <i>Fundamentals of materials science and engineering</i>, 4th edition, <i>E-Text</i> John Wiley & Sons; ISBN: 978-1-118-53126-6 2. William D. Callister, Jr., David G. Rethwisch, <i>Materials Science and Engineering</i>, 8th edition, <i>E-Text</i> John Wiley & Sons; ISBN: 978-1-118-37325-5 3. Materials World (Magazine of the Institute of Materials, Minerals and Mining)	

Revised (April 2014)

Subject Description Form

Subject Code	ENG2002
Subject Title	Computer Programming
Credit Value	3
Level	2
Pre-requisite/Co-requisite/Exclusion	Nil
Objectives	(i) To introduce the fundamental concepts of computer programming. (ii) To equip students with solid skills in Python programming. (iii) To equip students with techniques for developing structured and object-oriented computer programs. (iv) To demonstrate the techniques for implementing engineering applications using computer programs.
Intended Subject Learning Outcomes	Upon completion of the subject, students will be able to: 1. Familiarize themselves with at least one Python programming environment. 2. Be proficient in using the basic constructs of Python to develop a computer program. 3. Develop a structured and well-documented computer program. 4. Understand the basics of object-oriented programming and use them to write computer programs. 5. Apply computer programming to solve practical engineering problems.
Subject Synopsis/ Indicative Syllabus	Syllabus: 1. Introduction to Programming Components of a computer; Data representation in computers; Programming environment; Python IDE; Editing, saving, and running a script; Process of application development. 2. Bolts and Nuts of Python Data types; Variables and constants; Operators, expressions, and statements; Basic syntax; Functions and modules; Scope of variables; Python modules; Absolute and relative import. 3. Program Flow Control and Functions Branching and looping; Iterators; Unicode; Python functions; static functions; Lambda function; Position arguments and default arguments; args and kwargs; Interface with command line; argparse. 4. Program Design and Debugging Structured program design; Testing and debugging a program; Exception and assertion. 5. Strings and File I/O String encoding format; F-string; String operations; String and number conversion; File and directory manipulations; The “os”, “sys”, and “shutil” modules; Reading/writing text and numbers from/to a file. 6. Tuples, Lists, Dictionaries, and Sets Basic tuple and list operations; Dictionary literals; Basic dictionary and sets operations; Built-in tuple/list/dictionary/set methods and functions; Use of enumerate and zip. 7. Basic Object-Oriented Programming Objects and classes; Attributes and methods; Inheritance and polymorphism; Special methods and operator overloading.

	8. Data Analytics with Python Libraries Introduction to NumPy, Pandas, and Matplotlib; NumPy arrays, built-in methods, and mathematical operations; Reading/writing data files using Pandas; Pandas operations and functions; Data visualization with Matplotlib.							
Teaching/Learning Methodology	Teaching and Learning Method		Intended Subject Learning Outcome		Remarks			
	Lectures, supplemented with short quizzes		2,3,4		Students are introduced to the knowledge of computer programming through explanation and illustrative examples. Comprehension of the knowledge is strengthened with short quizzes. Students will be able to monitor the skills of using Python and apply the techniques of developing structured object-oriented applications.			
	Practical sessions/ Tutorials where problems are given to students for them to solve		1,2,3,4		Students apply what they have learnt in lectures and solve problems in exercises. The purpose is to ensure students have captured the important points. Tutors will aid the lecturer in helping the students finish the exercises, and interactive Q&A will take place.			
	Assignment, tests and final examination		1,2,3,4,5		By doing the assignment, students will develop a firm understanding and comprehension of the knowledge taught. They will analyse given Python applications and apply knowledge to solve problems. They will have to design solutions by evaluating different alternatives. To enhance the students' problem-solving skills in a given programming environment, open-book programming tests are arranged regularly. To assure students' understanding of fundamental concepts, a closed-book final examination is arranged.			
Assessment Methods in Alignment with Intended Learning Outcomes	Specific Assessment Methods/Tasks		% Weighting	Intended subject learning outcomes to be assessed				
				1	2	3	4	5
	1. In-class exercises and homework		10%	✓	✓	✓	✓	
	2. Quizzes		10%		✓	✓	✓	
	3. Programming tests		30%	✓	✓	✓	✓	✓
	4. Assignment		20%	✓	✓	✓	✓	✓

	5. Final examination	30%	✓	✓	✓	✓	✓
	Total	100%					
Student Study Effort Expected	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:						
	The quizzes are for assessing the understanding of fundamental concepts. The in-class exercises and homework are conducted to help students familiarize themselves with the programming language and skills. The programming tests are for assessing the ability of students to solve computer problems through programming within a specified period. Through doing assignments, students will be able to experience how to solve engineering problems and design solutions by using a systematic approach. The final examination is for assessing the students' ability to use the programming language and analyse computer programs.						
	Class contact:						
	• Lectures, Tests and Quizzes		26 Hours				
	• Practical sessions/Tutorial		13 Hours				
	Other student study effort:						
	• Self-studying and assignments		57 Hours				
• Homework		12 Hours					
Total student study effort:		108 Hours					
Reading List and References	Reference Books:						
	1. G. van Rossum and the Python development team, <i>Python Tutorial Release 3.10.0</i> , Nov. 2021. 2. C. Hill, <i>Learning Scientific Programming with Python</i> , (2nd ed.) Cambridge: Cambridge University Press, 2020. 3. C.P. Millike, <i>Python Projects for Beginners: a ten-week bootcamp approach to Python programming</i> . Berkeley, CA: Apress, 2020.						
Online resources:							
1. DataCamp online course https://app.datacamp.com/							
2. https://www.w3schools.com/python/							
3. https://www.online-python.com/							

(revised) July 2025

Subject Description Form

Subject Code	ENG2003
Subject Title	Information Technology
Credit Value	3
Level	2
Pre-requisite / Co-requisite/ Exclusion	Nil
Objectives	To provide the foundation knowledge in internet applications, computer networks, and database management that is essential to modern information system design
Intended Subject Learning Outcomes	Upon completion of the subject, students will be able to: <u>Category A: Professional/academic knowledge and skills</u> 1. Understand the functions and features of modern computing systems. 2. Understand the client-server architecture and be able to set up multiple internet applications. 3. Understand the principles of computer networks and be able to set up simple computer networks. 4. Understand the basic structure of a database system and be able to set up a simple database system. <u>Category B: Attributes for all-roundedness</u> 1. Solve problems using systematic approaches.
Subject Synopsis/ Indicative Syllabus	Syllabus: 1. <u>Introduction to computers</u> Introduction to information technology using Internet of Things and Cyber-physical systems as real life examples. Introduction to modern computing systems. 2. <u>Computer Networks</u> Introduction to computer networks (Client-Server Architecture). Study different internet applications (HTTP/FTP/DNS). Explain basic concepts on packet routing (Data Encapsulation/IP Addressing/Functions of Routers). Introduction to basic network security measures. 3. <u>Introduction to data processing and information systems</u> Database systems – architecture, relational database concept, structural query language (SQL), database management systems, Web and database linking, database application development. Introduction to Information systems. Workflow management. Case study: Database design, implementation and management.
Teaching/Learning Methodology	There will be a mix of lectures, tutorials, and laboratory sessions/workshops to facilitate effective learning. Students will be given case studies to understand and practice the usage of modern information systems.

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)				
			A1	A2	A3	A4	B1
	1. Quizzes (in lectures & tutorials)	20%	√	√	√	√	√
	2. Workshops	15%	√	√	√	√	√
	3. Mid-term Test	15%	√	√	√		√
	4. Examination	50%	√	√	√	√	√
	Total	100 %					
Student Study Effort Expected	Class contact:						
	• Lectures (16), tutorials (8), and workshops (15)						39 Hours
	Other student study effort:						
	• Workshops preparation (6/workshop)						30 Hours
	• Self study (3/week)						39 Hours
	Total student study effort						108 Hours
Reading List and References	1. B. Williams and S. Sawyer, <i>Using Information Technology: A Practical Introduction to Computers and Communications</i> , 11 th ed., McGraw-Hill, 2014. 2. J. F. Kurose and K. W. Ross, <i>Computer Networking: A Top-Down Approach</i> , 9 th ed., Pearson, 2025. 3. D. E. Comer, <i>Computer Networks and Internets</i> , 6 th ed., Pearson, 2015. 4. B. A. Forouzan, <i>Data Communications and Networking with TCP/IP Protocol Suite</i> , 6 th ed., McGraw-Hill, 2022. 5. W. Stalling, <i>Data and Computer Communications</i> , 10 th ed., Pearson, 2013. 6. C. Coronel and S. Morris, <i>Database Systems: Design, Implementation, and Management</i> , 13 th ed., Cengage Learning, 2018. 7. M. Mannino, <i>Database Design, Query, Formulation, and Administration: Using Oracle and PostgreSQL</i> , 8 th ed., SAGE Publications , 2022. 8. A. Silberschatz, H. F. Korth, S. Sudarshan, <i>Database System Concepts</i> , 7 th ed., McGraw-Hill, 2019. 9. Rawat, Danda B., Joel JPC Rodrigues, and Ivan Stojmenovic, eds. <i>Cyber-physical systems: from theory to practice</i> . CRC Press, 2015.						

(revised) July 2025

Subject Description Form

Subject Code	ENG3003
Subject Title	Engineering Management
Credit Value	3
Level	3
Pre-requisite/Co-requisite/Exclusion	Nil
Objectives	This subject provides students with: 1. A practical introduction to management and a comprehensive guide to the tools and techniques used in managing people and other resources. 2. Opportunities to trace the historical development and describe the functions of management, from planning, and decision making to organizing, staffing, leading, motivating, and controlling. It also includes a discussion on engineering ethics. 3. Opportunities to explore the core business strategy, technology, and innovation, and examine how these functions intertwine to play a central role in structural design, as well as supporting an organization's overall success.
Intended Learning Outcomes	Upon completion of the subject, students will be able to a. perform tasks in an organization related to organizing, planning, leading and controlling project and process activities; b. select appropriate management techniques for improving organizational structures, work procedures, and quality performance of operational tasks; c. analyze the factors that affect changes in the work environment, and be aware of the approaches in implementing change in an organization; d. be aware of the imperatives of ethical and business behaviors in engineering organizations in a fast-changing business environment.
Subject Synopsis/Indicative Syllabus	1. <u>Introduction</u> General management concepts in organizations; Functions and types of industrial organizations; Organizational structures; Corporate objectives, strategy, and policy 2. <u>Industrial Management</u> Roles of managers: Process of management, leadership, planning, organizing, motivating, and control of social and engineering activities; Quality management: Related tools and techniques 3. <u>Project Management</u>

	Project scope and objectives; Network analysis; Tools that support engineering operations and task scheduling 4. <u>Management of Change</u> Change leadership; Organizational change; Phases of planned change; Stress management; Factors that affect the execution of change 5. <u>Effects of Environmental Factors</u> The effects of extraneous factors on the operations of engineering organizations, such as ethics and corporate social responsibilities issues																																												
Teaching/Learning Methodology	A mixture of lectures, tutorial exercises, and case studies are used to deliver various topics in this subject. Some topics are covered by problem-based format whenever applicable in enhancing the learning objectives. Other topics are covered by directed study so as to develop students' "life-long learning" ability. The case studies, largely based on real experience, are designed to integrate the topics covered in the subject and to illustrate the ways various techniques are inter-related and applied in real life situations.																																												
Assessment Methods in Alignment with Intended Learning Outcomes	<table border="1"> <thead> <tr> <th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="6">Intended subject learning outcomes to be assessed</th></tr> <tr> <th>a</th><th>b</th><th>c</th><th>d</th><th></th><th></th></tr> </thead> <tbody> <tr> <td>1. Coursework • Group learning activities (10%) • Presentation (individual) (30%)</td><td>40%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td></tr> <tr> <td>2. Final examination</td><td>60%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td></tr> <tr> <td>Total</td><td>100%</td><td></td><td></td><td></td><td></td><td></td><td></td></tr> </tbody> </table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: The coursework of this subject involves students working in groups to study cases that reflect the realities of management situations in an engineering setting. Through such exercises, students' ability to apply and synthesize acquired knowledge can be assessed on the basis of their performance in group discussion, oral presentations, and the quality of their written reports on these case studies. A written final examination is also designed to assess the intended learning outcomes.							Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed						a	b	c	d			1. Coursework • Group learning activities (10%) • Presentation (individual) (30%)	40%	✓	✓	✓	✓			2. Final examination	60%	✓	✓	✓	✓			Total	100%						
Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed																																											
		a	b	c	d																																								
1. Coursework • Group learning activities (10%) • Presentation (individual) (30%)	40%	✓	✓	✓	✓																																								
2. Final examination	60%	✓	✓	✓	✓																																								
Total	100%																																												
Student Study Effort Expected	Class contact:																																												
	▪ Lectures and review						27 Hrs.																																						
	▪ Tutorials and presentations						12 Hrs.																																						
	Other student study effort:																																												

	▪ Research and preparation	30 Hrs.
	▪ Report writing	10 Hrs.
	▪ Preparation for oral presentation and examination	37 Hrs.
	Total student study effort	116 Hrs.
Reading List and References	1. John R. Schermerhorn, Jr., 2013, Introduction to Management, 12th Ed., John Wiley 2. Robbins, S P, DeCenzo, D A, and Coulter, M, 2013, Fundamentals of Management Essential Concepts and Applications, 8th Ed., Pearson 3. Morse, L C and Babcock, D L, 2010, Managing Engineering and Technology: an Introduction to Management for Engineers, 5th Ed., Prentice Hall 4. White, M A and Bruton, G D, 2011, The Management of Technology and Innovation: A Strategic Approach, 2nd Ed., South-Western Cengage Learning	

(revised) July 2015

Subject Description Form

Subject Code	ENG3004
Subject Title	Society and the Engineer
Credit Value	3
Level	3
Pre-requisite/Co-requisite/Exclusion	Nil
Objectives	This subject is designed for engineering students as a complementary subject on the role of the professional engineer in practice and their responsibilities toward the profession, colleagues, employers, clients, and the public. The objectives of the subject are to enable students to 1. appreciate the historical context of modern technology and the nature of the process whereby technology develops and the relationship between technology and the environment, as well as the implied social costs and benefits; 2. understand the social, political, legal, and economic responsibilities and accountability of the engineering profession and the organizational activities of professional engineering institutions; 3. be aware of the short-term and long-term effects related to safety and health, and the environmental impacts of technology; 4. observe professional conduct, as well as the legal and other applicable constraints, related to various engineering issues; and 5. develop a strong vision to optimize their contribution to sustainable development.
Intended Learning Outcomes	Upon completion of the subject, students will be able to a. identify and evaluate the effects of technology as it applies to the social, cultural, economic, legal, health, safety, and environmental dimensions of society; b. explain the importance of local and international professional training, professional conduct and ethics, and responsibilities in various engineering disciplines, particularly the Washington Accord; c. evaluate and estimate, in a team setting, the impact of contemporary issues, planned projects, and unforeseen technological advances related to engineers; effectively communicate and present the findings to laymen and peers.
Subject Synopsis/ Indicative Syllabus	1. <u>Impact of Technology on Society</u> Historical cases and trends of technological innovation explored through their impact on social and cultural developments of civilization and their commonalities.

	2. <u>Environmental Protection and Related Issues</u> Roles of the engineer in energy conservation, ecological balance, and sustainable development. 3. <u>Global Outlook for Hong Kong's Economy and Industries</u> Support organizations, policies and their impacts on industrial and economic development in Greater China, the Pacific Rim, and the world. 4. <u>Regulatory Organizations and Compliance</u> Discussion of engineer's responsibilities within different regulatory frameworks and environments; Examples from various entities such as the Labor Department and the Occupational Health and Safety Council; Legal dimensions to engineering such as liability, contract law, and industrial legislation. 5. <u>Professional Institutions</u> Local and overseas professional institutions; Washington Accord and the qualifications and criteria of professional engineers. 6. <u>Professional Ethics</u> Prevention of bribery and corruption; The work of the Independent Commission Against Corruption (ICAC); Social responsibilities of engineers.
Teaching/Learning Methodology	Class comprises short lectures to provide essential knowledge and information on the relationships between society and the engineer under a range of dimensions. Other methods include in-class discussions, case studies, and seminars to develop students' in-depth analysis of the relationships. Each student will submit two assignments based on their weekly learning activities, which will be part of the subject's evaluation. The assignments will deal with important issues of social, cultural, economic, legal, health, safety, and environmental dimensions of society. Students are assembled into groups; throughout the course, they will work on engineering cases by completing the following learning activities: 1. Case analysis where students explore the relationships between society and the engineering issues of a project under specific dimensions; 2. Construction and assembly of a case portfolio which includes i. Presentation slides ii. Feedback critiques iii. Individual Reflections 3. Final oral presentation

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks		% weighting	Intended subject learning outcomes to be assessed		
				a	b	c
	1. Continuous assessment		70%			
	• Group weekly learning activities		(20%)	✓	✓	✓
	• Individual Assignments (2)		(20%)	✓	✓	
	• Individual final presentation		(15%)	✓	✓	
	• Individual reflection statement		(5%)	✓	✓	
	• Group project		(10%)	✓	✓	✓
	2. Take-home Assignment		30%	✓	✓	
	Total		100%			
Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: The coursework requires students to work in groups to study cases from the perspectives of the eight dimensions in an engineering setting. Based on these exercises, students’ ability to apply and synthesize acquired knowledge can be assessed through their performance during groups’ discussion, oral presentations, and the quality of their portfolio reports on the case studies. The take-home assignment is used to assess students’ critical thinking and problem-solving skills when working on their own and give students more time and flexibility to complete an assignment. It provides students the opportunity to review and extend what they have learnt in class and to check their understanding and progress.						
Student Study Effort Expected	Class contact:					
	▪ Lectures and review			27 Hrs.		
	▪ Presentation			12 Hrs.		
	Other student study efforts:					
	▪ Research and preparation			55 Hrs.		
	▪ Report and Assignments writing			25 Hrs.		
	Total student study effort			119 Hrs.		

Reading List and References	Reference Books & Articles: 1. Education for Sustainable Development - An Expert Review of Processes and Learning, UNESCO, 2011 2. Poel, Ibo van de, and Lambèr M. M. Royakkers. Ethics, Technology, and Engineering : an Introduction. Wiley-Blackwell, 2011 3. Engineering-Issues, Challenges and Opportunities for Development, USECO, 2010 4. Engineering for Sustainable Development: Guiding Principles, Royal Academy of Engineering, 2005 5. Securing the future: delivering UK sustainable development strategy, 2005 6. Johnston, F S, Gostelow, J P, and King, W J, 2000, <i>Engineering and Society Challenges of Professional Practice</i>, Upper Saddle River, N.J.: Prentice Hall 7. Hjorth, L, Eichler, B, and Khan, A, 2003, <i>Technology and Society A Bridge to the 21st Century</i>, Upper Saddle River, N.J.:Prentice Hall 8. The Council for Sustainable Development in Hong Kong, http://www.enb.gov.hk/en/susdev/council/ 9. Poverty alleviation: the role of the engineer, http://publications.arup.com/publications/p/poverty_alleviation_the_role_of_the_engineer Reading materials: Engineering journals: - Engineers by The Hong Kong Institution of Engineers - Engineering and Technology by The Institution of Engineers and Technology Magazines: Time, Far East Economic Review Current newspapers: South China Morning Post, China Daily, Ming Pao Daily
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(revised) June 2021

Subject Description Form

Subject Code	ENG3006
Subject Title	Engineering Professionals in Society II
Credit Value	2
Level	3
Pre-requisite/Co-requisite/Exclusion	ENG1004 Engineering Professionals in Society I
Objectives	This subject is designed for engineering students as a complementary subject on the role of the professional engineer in practice and their responsibilities toward the profession, colleagues, employers, clients, and the public. The objectives of the subject are to enable students to: 1. appreciate the historical context of modern technology and the nature of the process whereby technology develops and the relationship between technology and the environment, as well as the implied social costs and benefits; 2. understand the roles of engineers in environmental protection and related issues; 3. be aware of the short-term and long-term effects related to safety and health, and the legal aspects related to engineering professions; 4. understand professional conduct and ethics as well as the legal and other applicable constraints, related to various engineering issues; and 5. develop a strong vision to optimize their contribution to sustainable development.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. identify and evaluate the effects of technology as it applies to the social, cultural, economic, legal, health, safety, and environmental dimensions of society; b. explain the importance of local and international professional training, professional conduct and ethics, and responsibilities in various engineering disciplines; c. <i>evaluate and estimate, in a team setting, the impact of contemporary issues, planned projects, and unforeseen technological advances related to engineers and effectively communicate and present the findings to laymen and peers;</i> d. <i>discuss how professionalism and professional ethics are manifested in the professional settings and practices of engineering; and</i> e. <i>engage in dialogues with community stakeholders to explore engineering-related issues.</i>
Subject Synopsis/ Indicative Syllabus	1. <u>Impact of Technology on Society</u> Historical cases and trends of technological innovation explored through their impact on social and cultural developments of civilization and their commonalities. 2. <u>Environmental Protection and Related Issues</u> Roles of the engineer in energy conservation, ecological balance, and sustainable development. 3. <u>Regulatory Organizations and Compliance</u>

	Discussion of engineer’s responsibilities within different regulatory frameworks and environments; Examples from various entities such as the Labour Department and the Occupational Health and Safety Council; Legal dimensions to engineering such as liability, contract law, and industrial legislation. 4. <u>Professional Conduct and Ethics</u> Foundations of ethics; Professional codes of conduct; Confidentiality and privacy; Conflict of interest; Corporate social responsibility; Professional integrity, and social responsibilities of engineers.																																																													
Teaching/Learning Methodology	Class comprises short lectures to provide essential knowledge and information on the relationships between society and the engineer under a range of dimensions. The “Flipped Classroom” approach will be adopted for students to take up a more proactive role in learning. Activities include pre-lecture and post-lecture quizzes will be arranged to enhance the learning effectiveness. Each student will submit a number of individual assignments, dealing with the various topics covered by the syllabus. Students are assembled into small groups; throughout the course, they will work on engineering cases by completing the following learning activities: 1. Case analysis where students explore the relationships between society and the engineering issues of a project under specific dimensions; 2. Construction and assembly of a case portfolio which includes: i. Presentation slides ii. Individual reflection report iii. Final oral presentation																																																													
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% Weighting</th><th colspan="5">Intended subject learning outcomes to be assessed</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr><tr><td>Continuous assessment</td><td>100%</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>• Quizzes</td><td>(30%)</td><td>✓</td><td>✓</td><td></td><td></td><td></td></tr><tr><td>• Individual assignments</td><td>(30%)</td><td>✓</td><td>✓</td><td></td><td>✓</td><td>✓</td></tr><tr><td>• Individual final oral presentation</td><td>(15%)</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td></tr><tr><td>• Individual reflection report</td><td>(10%)</td><td>✓</td><td>✓</td><td></td><td></td><td>✓</td></tr><tr><td>• Group project report</td><td>(15%)</td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td></tr><tr><td>Total</td><td>100%</td><td></td><td></td><td></td><td></td><td></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: The individual assignments are used to assess students’ critical thinking and problem-solving skills when working on their own and give students more time and flexibility to	Specific assessment methods/tasks	% Weighting	Intended subject learning outcomes to be assessed					a	b	c	d	e	Continuous assessment	100%						• Quizzes	(30%)	✓	✓				• Individual assignments	(30%)	✓	✓		✓	✓	• Individual final oral presentation	(15%)	✓	✓	✓	✓		• Individual reflection report	(10%)	✓	✓			✓	• Group project report	(15%)	✓	✓	✓			Total	100%					
Specific assessment methods/tasks	% Weighting			Intended subject learning outcomes to be assessed																																																										
		a	b	c	d	e																																																								
Continuous assessment	100%																																																													
• Quizzes	(30%)	✓	✓																																																											
• Individual assignments	(30%)	✓	✓		✓	✓																																																								
• Individual final oral presentation	(15%)	✓	✓	✓	✓																																																									
• Individual reflection report	(10%)	✓	✓			✓																																																								
• Group project report	(15%)	✓	✓	✓																																																										
Total	100%																																																													

	complete at home. Students will also work in groups to study contemporary issues and cases in an engineering and technology setting. Activities will include in-class discussions, oral presentation and preparation of the group report and an individual reflection report. By completing the projects, students will be able to evaluate the effects of technology relating to social, cultural, economic, legal, health, safety, and environmental dimensions of society, and understand the importance of engineers' responsibilities.	
Student Study Effort Expected	Class contact:	
	• Lectures and review	18 Hrs.
	• Presentation	8 Hrs.
	Other student study efforts:	
	• Research and preparation	34 Hrs.
	• Report and Assignments writing	18 Hrs.
	Total student study efforts	78 Hrs.

Reading List and References	Reference Books & Articles: 1. Education for Sustainable Development - An Expert Review of Processes and Learning, UNESCO, 2011 2. Poel, Ibo van de, and Lambèr M. M. Royakkers. Ethics, Technology, and Engineering: An Introduction. Wiley-Blackwell, 2011 3. Engineering-Issues, Challenges and Opportunities for Development, USECO, 2010 4. Securing the future: delivering UK sustainable development strategy, 2005 5. Johnston, F S, Gostelow, J P, and King, W J, 2000, <i>Engineering and Society Challenges of Professional Practice</i>, Upper Saddle River, N.J.: Prentice Hall 6. Hjorth, L, Eichler, B, and Khan, A, 2003, <i>Technology and Society A Bridge to the 21st Century</i>, Upper Saddle River, N.J.: Prentice Hall Reading materials: Engineering journals: - Engineers by The Hong Kong Institution of Engineers - Engineering and Technology by The Institution of Engineers and Technology Current newspapers: South China Morning Post, China Daily, Ming Pao Daily
Last Updated	April 2025
Prepared by	C.K. Choi

April 2025

Subject Description Form

Subject Code	ME22003
Subject Title	Visualization and Communication in Design Engineering
Credit Value	3
Level	2
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	1. To enhance the students' three-dimensional visualization skills and equip them with range of visual communication tools; and 2. To provide students with fundamental understanding of the engineering design process.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Appreciate the qualities of a well-designed product and have awareness of the need of specific disciplinary knowledge to develop functional products that meet customer's and manufacturer's need; b. Communicate engineering design details with confidence using range of visualization and communication tools including hand-sketching, CAD geometric models, mechanism simulations, engineering drawings and physical prototypes; c. Work in a team to modify or improve an existing consumer product; and d. Apply project management techniques in meeting planned schedule.
Subject Synopsis/ Indicative Syllabus	<i>Appreciation of Design Engineering</i> – Characteristics of good design, customer's perspective, manufacturer's perspective, role of materials, investigation of manufacturing methods, skills of a design engineer, examples of successful design. <i>Creativity and Design process</i> – What is design? Basic introduction to engineering design process, creativity in design, Visual thinking (brainstorming, concept mapping) and its relationship with design innovation, applications of computer in design. <i>Visualization and Communication techniques for Design Engineering</i> – hand-sketching, 3-D solid modelling, assembly modeling and simulation, engineering drawings, prototyping (virtual prototypes, 3-D printing, physical prototypes). <i>Working in a Team Environment</i> – Multi-disciplinary project team: its importance in modern industry/community. Functions of design project team: team roles, design logbook, project scheduling, review and assessment of design stages, project outcome communication including interim report, oral presentation, final report and prototype presentation.

Teaching/ Learning Methodology	This introductory course aims at arousing students’ interest in design engineering and related skills important for engineering design. It also aims at developing interest and curiosity in all relevant subsequent subjects. Students are learnt to appreciate the qualities of well-designed products through direct interaction with the products. The importance of design visualization and communication will be experienced with in-class group/individual activities. Design communication tools including hand- sketching, CAD geometric modeling, mechanism simulation, engineering drawings, and 3D printing will be introduced with the use of real product examples. The focus of software introduction is not mere training but education in fundamentals to equip the students with sound skills for lifelong career. The intended learning outcomes of the subject are mainly achieved through a group design improvement project. Students will work in groups of 4-5 members. All the learning activities, including CAD skill development are centered around a product given to each student group. Students begin the learning by studying the function, design features, materials, manufacturing methods, technology, etc. of the product. They then develop visualization techniques by sketching the product. The product is then dissected to learn about the internal mechanisms, take part measurements and develop CAD geometric models. The geometric model is used for assembly and mechanism visualization, basic design analysis and use creativity techniques to make a simple design improvement to improve the product function/cost reduction/improve its aesthetic qualities, etc. The details of the final design are then presented using different visualization and communication techniques such as photo rendering, 3-D printed physical models, hand crafted models, and 2-D engineering drawings. Final outcome will be presented orally and through a written technical report. Following is the suggested study plan: <table border="1" data-bbox="411 1294 1457 2016"> <tr> <td>Week 1</td><td>Introduction of the subject and the mini team-project</td></tr> <tr> <td>Week 2</td><td>Appreciation of design engineering; Introduction to solid modelling: Part modelling</td></tr> <tr> <td>Week 3</td><td>Creativity and design process; Introduction to solid modelling: Assembly modelling and 2-D Engineering documentation</td></tr> <tr> <td>Week 4</td><td>Practice and consultation: (for weeks 1, 2 & 3 including planning of the mini team- project)</td></tr> <tr> <td>Week 5</td><td>Solid Modelling with mechanism</td></tr> <tr> <td>Week 6</td><td>Surface Modelling and sheet metal modelling</td></tr> <tr> <td>Week 7</td><td>Practice and consultation: (for weeks 5 & 6 including checking milestones of the mini team- project)</td></tr> <tr> <td>Week 8</td><td>Basic simulation of mechanism motion</td></tr> <tr> <td>Week 9</td><td>Basic Structural/static analysis</td></tr> <tr> <td>Week 10</td><td>3-D printing and Solid Modelling with photo rendering</td></tr> <tr> <td>Week 11-13</td><td>Practice and consultation: (to complete and present the mini team-project)</td></tr> </table>	Week 1	Introduction of the subject and the mini team-project	Week 2	Appreciation of design engineering; Introduction to solid modelling: Part modelling	Week 3	Creativity and design process; Introduction to solid modelling: Assembly modelling and 2-D Engineering documentation	Week 4	Practice and consultation: (for weeks 1, 2 & 3 including planning of the mini team- project)	Week 5	Solid Modelling with mechanism	Week 6	Surface Modelling and sheet metal modelling	Week 7	Practice and consultation: (for weeks 5 & 6 including checking milestones of the mini team- project)	Week 8	Basic simulation of mechanism motion	Week 9	Basic Structural/static analysis	Week 10	3-D printing and Solid Modelling with photo rendering	Week 11-13	Practice and consultation: (to complete and present the mini team-project)
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Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed			
			a	b	c	d
	1. Individual assignments	30%	√	√		
	2. Group assignments	20%	√	√	√	
	3. Oral communication: Group oral presentation	20%	√	√	√	
	4. Written communication: Group technical report (with peer evaluation)	30%	√	√	√	√
	Total	100 %				
Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Overall Assessment: 1.0 x Continuous Assessment Written examination is not suitable for this course since a specific set of knowledge is not the intended learning outcome. Focus is more on the skill development, critical thinking and creativity improvement. Individual assessment of students can be achieved through individual assignments, oral presentation and peer evaluation. Project assessment consists of an oral presentation and a written report. The assessment is performed by a panel of assessors that includes guest assessors from the industry. Project assessment is mainly based on the performance of the whole project group to encourage team spirit. Intra-group peer evaluation is performed to elicit the individual contribution for project activities.						
Student Study Effort Expected	Class contact:					Time
	▪ Lectures					15 Hrs.
	▪ Small group activities/Workshops/Consultations					24 Hrs.
	Other student study effort:					Time
	▪ CAD and sketching skills development					40 Hrs.
	▪ Independent study and homework					20 Hrs.
	▪ Project					20 Hrs.
	Total student study effort					119 Hrs.

Reading List and References	1. D.K. Lieu and S. Sorby, Visualization, Modeling, and Graphics for engineering Design, CENGAGE Learning, Latest Edition. 2. M.N. Horenstein, Design Concepts for Engineers, Prentice Hall, latest edition. 3. T. Taura, Creative design engineering : introduction to an interdisciplinary approach, Academic Press, Latest Edition.
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Modified July 2017

Subject Description Form

Subject Code	ME23001
Subject Title	Engineering Mechanics
Credit Value	3
Level	2
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: AP10005 Physics I
Objectives	To provide students the fundamental mechanics concepts of equilibrium and motion for rigid structural systems.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Apply the fundamental knowledge of mechanics to solve for forces and moments in simple systems. - Distinguish the basic differences between diverse engineering systems and select the suitable design in achieving the engineering purposes. - Employ engineering mechanics to solve the problems encountered in assignments and projects. - Collaborate with peers in experiments and projects and present effectively the results of experiment or project.
Subject Synopsis/ Indicative Syllabus	Fundamentals of Mechanics - Basic concepts of mechanics; Scalar and vectors; Vector algebra and vector components; Position, unit of force vectors; Two and three-dimensional force systems; Moment of a force about a point; Moment of a force about a line. Statics - Equilibrium of a particle and the associated free-body diagrams; Equilibrium of a rigid body and the associated free-body diagram; Two and three force members equilibrium in three dimensions; Simple trusses: the method of joints; the method of sections; zero-force members; Internal forces developed in structural members; Shear and moment equations and diagrams in structural members; Relations between distributed load, shear and moment; Theory of dry friction; Systems with friction; Wedges; Belt friction; Rolling resistance. Equivalent Systems - Determination of the resultant concurrent forces; Equivalent force/couple systems; Centre of gravity and centroid: by composite parts, by integration; Resultant of a general distributed force system; Moment of inertia of areas; Parallel-axis theorem for an area; Radius of gyration of an area; Calculation of moments of areas: by composite areas, by integration; Product of inertia for an area; Principles of virtual work. Dynamics - Kinematics and kinetics of particles; rectilinear motion; plane curvilinear motion; relative motion; equation of motion.

Teaching/Learning Methodology	Lectures are used to deliver the fundamental knowledge in relation to the topics as described in the section subject synopsis (Outcomes a, b and c). Tutorials are used to illustrate the application of fundamental knowledge to practical situations (Outcomes a, b and c). Experiments are used to relate the concepts to practical applications and students are exposed to hand-on experience, proper use of equipment and application of analytical skills on interpreting experimental results (Outcomes c and d). <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="4">Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>Lecture</td><td>√</td><td>√</td><td>√</td><td></td></tr><tr><td>Tutorial</td><td>√</td><td>√</td><td>√</td><td></td></tr><tr><td>Experiment/Projects</td><td></td><td></td><td>√</td><td>√</td></tr></table>	Teaching/Learning Methodology	Outcomes				a	b	c	d	Lecture	√	√	√		Tutorial	√	√	√		Experiment/Projects			√	√										
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**Reading List and
References**

1. R.C. Hibbeler, Engineering Mechanics – Statics, Prentice Hall, latest edition.
2. A. Pytel, J. Kiusalaas, Engineering Mechanics – Statics, Stamford, CT: Cengage Learning, latest edition.

Revised June 2020

Subject Description Form

Subject Code	ME31001
Subject Title	Dynamics and Vibrations
Credit Value	3
Level	3
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: ME23001 Engineering Mechanics
Objectives	To teach students basic concepts of rigid body planar motion and mechanical vibration.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Formulate and solve planar motion problems in rigid body dynamics by applying knowledge of dynamic analyses and mathematics. - Formulate and solve vibration problems in single DOF mechanical systems by applying knowledge of vibration theory and mathematics. - Analyse and interpret data obtained from experiments in dynamics and vibrations. - Present effectively in completing written reports of laboratory work.
Subject Synopsis/ Indicative Syllabus	Dynamics - Kinematics of rigid bodies, translational and rotational motion, relative velocity, instantaneous centre of zero velocity, relative acceleration. <i>Plane kinetics of rigid bodies</i>, force, mass and acceleration, general equation of motion, applications, e.g., four-bar linkage and slider-crank mechanisms, gear trains. <i>Work and energy</i> methods. Vibration of Single-degree-of-freedom System - Free vibration of particles, equation of motion, damping effects, forced vibration of particles, vibration of rigid bodies, energy methods in vibration analysis, computer simulations of the free and forced vibration response of a single-degree-of-freedom system, transmissibility, vibration isolation. Vibration of Continuous System - Vibration of string, equation of motion, resonance, vibration modes. Laboratory Experiment The course includes one 2-hour laboratory session where students will have the opportunity to apply the theoretical concepts learned in the classroom. Typical Experiments: - Natural frequency of a vibrating system - Forced vibration - Vibration modes of a beam

Teaching/Learning Methodology	Lectures aim at providing students with an integrated knowledge required for understanding dynamics and single-degree-freedom vibration systems. Theories and examples will be presented to cover the syllabus on kinematics and kinetics of rigid bodies; equation of motions, work and energy, impulse and momentum, and one DOF vibrations. (Outcomes a and b) Tutorials aim at enhancing the analytical skills of the students. Examples will be provided to teach students the skills of solving different engineering problems using the knowledge of dynamics and single-degree-freedom vibration systems. Students will be able to solve real-world problems using the knowledge they acquired in the class. (Outcomes a and b) Experiments will provide students with experience on gear train systems, forced vibration systems and whirling of shafts. These experiments are designed to train students how to apply theories to practical applications, how to analyze and present experimental data. (Outcomes c and d) <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="4">Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>Lecture</td><td>√</td><td>√</td><td></td><td></td></tr><tr><td>Tutorial</td><td>√</td><td>√</td><td></td><td></td></tr><tr><td>Laboratory</td><td></td><td></td><td>√</td><td>√</td></tr></table>	Teaching/Learning Methodology	Outcomes				a	b	c	d	Lecture	√	√			Tutorial	√	√			Laboratory			√	√																
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Student Study Effort Expected	Class contact:	
	▪ Lecture	30 Hrs.
	▪ Laboratory/Tutorial	9 Hrs.
	Other student study effort:	
	▪ Reading and review	40 Hrs.
	▪ Homework assignment	20 Hrs.
	▪ Laboratory report	6 Hrs.
	Total student study effort	105 Hrs.
Reading List and References	1. F.P. Beer and E.R. Johnson, Vector Mechanics for Engineers: Dynamics, McGraw-Hill, latest edition. 2. J.L. Meriam and L.G. Kraige, Engineering Mechanics, John Wiley, latest edition. 3. S. Graham Kelly, Fundamentals of Mechanical Vibrations, McGraw Hill, latest edition. 4. W.T. Thomson, Theory of Vibration with Applications, Prentice Hall, latest edition.	
Last Update	July 2024	

Subject Description Form

Subject Code	ME31002
Subject Title	Linear Systems and Control
Credit Value	3
Level	3
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: ME23001 Engineering Mechanics
Objectives	To teach students time and frequency responses of dynamic systems to different inputs and the feedback control of such systems using PID controllers
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Model physical elements in dynamic systems and find the transfer function of a system comprising mechanical and other physical components. - Predict the output response of a first- or second-order system both in time and frequency domains subject to typical input signals. - Complete a given task in linear system control, such as an assignment or a project, by applying concepts in dynamics and control systems. - Analyze and interpret the data obtained from a control experiment. - Design a first-order and second-order system with suitable parameters and/or PID controller that will be stable and has the required system performance.
Subject Synopsis/ Indicative Syllabus	<i>Dynamic Responses of First-Order and Second-Order Systems</i> - Mathematical modelling of system elements, interconnection of elements in systems by differential equations, parameters of first-order and second-order systems, system response analysis due to step, ramp and impulse inputs using Laplace transform, simulation of dynamic systems using Matlab. <i>Frequency Response of First-Order and Second-Order Systems</i> - Harmonic response, Bode diagrams, frequency domain specifications, frequency response applications. <i>Introduction to Feedback Control</i> - Analysis of open-loop and closed-loop systems, transfer functions and block diagrams, time-domain specifications, system stability analysis, time-domain analysis of control systems. <i>Feedback Control Systems</i> - Automatic controllers, basic P, PD, PID controllers, Routh-Hurwitz stability criterion, numerical computations for the frequency-domain analysis of dynamical systems. Laboratory Experiment There are two 2-hour laboratory sessions. Typical Experiments: - Digital simulation of feedback control systems - DC servomechanism - Water level control

Teaching/Learning Methodology	Lectures aim at providing students with an integrated knowledge required for understanding and analyzing feedback control systems. (Outcomes a, b, c and e) Tutorials aim at enhancing analytical skills of students. Examples on system modeling, transient and frequency response of dynamic systems, and performance and stability of control systems will be involved. Students will be able to solve real-world problems using the knowledge they acquired in the class. (Outcomes a, b, c and e) Experiments will provide students with hands-on experience on the instrumentation and measurement of physical variables such as motor speed and water level, and their control. It also trains students in the analysis and presentation of experimental data. (Outcome d) <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="5">Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr><tr><td>Lecture</td><td>√</td><td>√</td><td>√</td><td></td><td>√</td></tr><tr><td>Tutorial</td><td>√</td><td>√</td><td>√</td><td></td><td>√</td></tr><tr><td>Experiment</td><td></td><td></td><td></td><td>√</td><td></td></tr></table>	Teaching/Learning Methodology	Outcomes					a	b	c	d	e	Lecture	√	√	√		√	Tutorial	√	√	√		√	Experiment				√																			
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Student Study Effort Expected	Class contact:	
	▪ Lecture	31 Hrs.
	▪ Laboratory/Tutorial	8 Hrs.
	Other student study effort:	
	▪ Self-study	45 Hrs.
	▪ Homework assignment	20 Hrs.
	▪ Laboratory report	8 Hrs.
	Total student study effort	112 Hrs.
Reading List and References	1. K. Ogata, Modern Control Engineering, Prentice Hall, latest edition. 2. N.S. Nise, Control Systems Engineering, John Wiley, latest edition. 3. C.L. Phillips and R.D. Harbor, Feedback Control Systems, Prentice-Hall, latest edition. 4. M.R. Driels, Linear Control Systems Engineering, McGraw-Hill, latest edition.	
Last Update	July 2025	

Subject Description Form

Subject Code	ME32001
Subject Title	Manufacturing Fundamentals
Credit Value	3
Level	3
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: ME22001 Engineering Design Fundamentals, or ME32002 Engineering Design Fundamentals, and ME23001 Engineering Mechanics
Objectives	To provide students with the fundamental knowledge of manufacturing processes and to teach students on how to apply manufacturing processes in product design and development.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Understand the basic working principles and rationales of common manufacturing processes and the related tooling for product development. - Select appropriate manufacturing processes for product fabrication at up-front design stage. - Present the completed mini-project related to manufacturing.
Subject Synopsis/ Indicative Syllabus	Machining - The principle, operation, mechanisms and the related machines of boring, drilling, facing, grinding, milling, planning, turning, sawing, ECM and EDM. Finishing - The principles and realization of anodizing, honing, painting, plating and polishing and their related facilities. Bulk Plastic Deformation - The principles, rationales and realization related to facilities of extrusion, forging, rolling, bar drawing, wire drawing processes. Sheet Metal Forming - The principles, design rationales and the process realization of drawing, blanking, bending, punching, shearing and spinning processes. Casting - The operation, realization and principles of die casting, investment casting, permanent mold casting, sand casting, and centrifugal casting. Polymer Processing - The process, principles and the realization of blow molding, casting, compression molding, extrusion, injection molding, and thermoforming. Assembly - Introduction to the process principle of welding (fusion, brazing & soldering, solid state), adhesive bonding and mechanical fastening. Process determination, die and tooling design, plastic deformed components design and product quality for bulk metal forming, sheet metal forming, casting and polymer processing.

Teaching/Learning Methodology	Lectures are used to deliver the fundamental knowledge related to conventional and advanced manufacturing processes. (Outcomes a – b). Tutorials and case studies are used to illustrate the application of fundamental knowledge to practical situations (Outcomes a – c). Mini-project/study report is used to enhance the understanding and use of the learned knowledge (Outcomes a – c). <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="3">Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th></tr><tr><td>Lecture</td><td>√</td><td>√</td><td></td></tr><tr><td>Tutorials</td><td>√</td><td>√</td><td>√</td></tr><tr><td>Mini-project</td><td>√</td><td>√</td><td>√</td></tr><tr><td>Study report</td><td>√</td><td>√</td><td>√</td></tr></table>	Teaching/Learning Methodology	Outcomes			a	b	c	Lecture	√	√		Tutorials	√	√	√	Mini-project	√	√	√	Study report	√	√	√										
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Mini-project	√	√	√																															
Study report	√	√	√																															
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	▪ Course work	23 Hrs.
	▪ Literature search and private study	22 Hrs.
	Total student study effort	104 Hrs.
Reading List and References	1. S. Kalpakjian, S. Schmid, manufacturing engineering and technology, Prentice Hall, latest edition. 2. B. Benhabib, Manufacturing: Design, Production, Automation and Integration, Marcel Dekker, latest edition. 3. J.Y.H. Fuh, Y.F. Zhang, A.Y.C. Nee, M.W. Fu, Computer-aided injection mold design and manufacture, Marcel Dekker, Inc, latest edition. 4. Jiri Tlusty, Manufacturing processes and equipment, Prentice Hall, latest edition. 5. Robert H. Wagoner, Jean-Loup Chenot, Fundamental of metal forming, New York: Wiley, latest edition. 6. MW Fu, Design and development of metal-forming processes and products aided by finite element simulation, Springer, 2017	

Revised August 2017

Subject Description Form

Subject Code	ME32002
Subject Title	Engineering Design Fundamentals
Credit Value	3
Level	2
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: ME22003 Visualization and Communication in Design Engineering
Objectives	To provide students with an extensive knowledge in product design and development process, and professional obligations of an engineer with the use of real world open-ended engineering problems.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Identify, formulate and solve engineering design problems and search for design related/needed data, information and knowledge for decision-making and design solution generation in product design and development. - Explore up-to-date information on product design, materials and manufacturing processes. - Recognize the design for X, human factors/ergonomics, product safety and reliability, and the professional and ethical responsibilities in product design and development. - Use different modes of communications effectively to present outcomes of design activities.
Subject Synopsis/ Indicative Syllabus	Overview of Design Process - Design problem formulation, identifying product function and establishing engineering specifications, generating and evaluating design alternatives, detail design, product testing and prototyping, communicating the design outcome: virtual/physical prototypes, engineering drawings, oral and written reports Types of Design Problems – Selection design, configuration design, variant design, adaptive design, original design, redesign Design of Common Mechanical Components – Load and stress analysis, material selection, strategies for failure prevention, safety factors, overview and design/selection design of common engineering components (gears, bearings, couplings, belt and train drives, shafts, keys, springs, etc.) Design for X – Design for manufacture, assembly, tolerance, affordability, safety & reliability, quality, environment, human factors / ergonomics Cost Estimation in Design Engineering – Time value of money, design to cost, cost and price estimation, break-even economics Ethics in Design – Professional obligations, codes of ethics

Teaching/Learning Methodology	This subject provides students with the opportunity to develop essential skills required for a professional design engineer and understanding of key concepts through activity-, project-, and problem-based (APPB) learning approach. APPB-learning challenges students to continually hone their interpersonal skills, creative abilities and understanding of the design process. It also allows students to develop strategies to enable and direct their own learning, which is the ultimate goal of education. Lectures are used to deliver the fundamental knowledge related to product engineering design and development (Outcomes a – c). Group/individual design activities and case studies are used to illustrate the application of fundamental knowledge to practical situations (Outcomes a – c). Group project is used to apply concepts learned to develop design solution/s for real-world open-ended engineering problem and enhance team-working skills, communication skills, project management skills, etc. (Outcomes a – d).																																																															
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="4">Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>Lecture</td><td>√</td><td>√</td><td>√</td><td></td></tr><tr><td>Small group activities</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>Project</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>Homework/assignment</td><td>√</td><td>√</td><td>√</td><td>√</td></tr></table> <table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="4">Intended subject learning outcomes to be assessed</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>1. Group Project a. Individual concept solutions b. Final group report c. Group oral presentation d. Peer and facilitator evaluation</td><td>60 %</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>2. Homework/Assignments</td><td>15 %</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>3. Test</td><td>25 %</td><td>√</td><td>√</td><td>√</td><td></td></tr><tr><td>Total</td><td>100 %</td><td colspan="4"></td></tr></table>	Teaching/Learning Methodology	Outcomes				a	b	c	d	Lecture	√	√	√		Small group activities	√	√	√	√	Project	√	√	√	√	Homework/assignment	√	√	√	√	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed				a	b	c	d	1. Group Project a. Individual concept solutions b. Final group report c. Group oral presentation d. Peer and facilitator evaluation	60 %	√	√	√	√	2. Homework/Assignments	15 %	√	√	√	√	3. Test	25 %	√	√	√		Total	100 %				
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	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Overall Assessment: 1.0 x Continuous Assessment The group project is intended to be carried out by student groups of 3-4 members. The outcome of the group design project is presented orally in group oral presentation and in a detailed design project report. Other than the contribution for group activities each student in the group need to develop individual conceptual designs which will be assessed individually. Peer and facilitator assessments will be used to decide the individual contribution by group members for project activities. Individual homework assignments are used to enhance students' comprehension and assimilation of selected design concepts. Test is adopted for assessment of individual student's overall understanding and the ability of applying engineering design concepts.	
Student Study Effort Required	Class contact:	
	▪ Lectures	24 Hrs.
	▪ Small group activities/ Workshops/ Consultations	15 Hrs.
	Other student study effort:	
	▪ Reading and review	20 Hrs.
	▪ Homework assignment	20 Hrs.
	▪ Project	40 Hrs.
	Total student study effort	119 Hrs.
Reading List and References	1. C.L. Dym and P. Little, Engineering Design – A Project-Based Introduction, Jon Wiley & Sons, latest edition 2. D.G. Ullman, The Mechanical Design Process, McGraw Hill Education, latest edition 3. R.J. Eggert, Engineering design, Prentice Hall, latest edition. 4. B. Hyman, Fundamentals of Engineering Design, Prentice Hall, latest edition. 5. George E. Dieter, Engineering Design, McGraw-Hill International Editions, latest edition. 6. G. Pahl and W. Beitz, Engineering Design-A systematic approach, Springer, latest edition. 7. R.C. Juvinall, Fundamentals of machine component design, John Wiley & sons, latest edition 8. J.A. Collins, Mechanical design of machine elements and machines: a failure prevention perspective, John Wiley & sons, latest edition	

Revised March 2017

Subject Description Form

Subject Code	ME32101/ ME3S01
Subject Title	Engineering Design for the Community
Credit Value	3
Level	3
Pre-requisite / Co-requisite/ Exclusion	Fundamental knowledge in Design and Engineering Science
Objectives	The objectives of this subject are to: 1. Introduce to students the concept and practice of service learning. 2. Raise students' awareness of social issues in Hong Kong and educate them on the challenges and needs of underprivileged communities in Hong Kong. 3. Develop a systemic platform to facilitate engineering/design students to apply their knowledge/skills to serve the community. 4. Reinforce the students' problem solving skill through real-life design projects. 5. Enhance students' generic competencies of innovative problems solving, communication and teamwork. 6. Nurture students' sense of social awareness, responsibility and engagement.
Intended Learning Outcomes	Upon the completion of the subject, the students will be able to: 1. Concept and Practice of Service Learning a) Link their service learning activities and experiences with academic content of the subject. b) Demonstrate empathy for people in need and a strong sense of civic responsibility. c) Evaluate people's needs by considering the complex issues in the service setting. d) Understand the role and responsibility both as a professional in their chosen discipline and as a responsible citizen. e) Function effectively in a multi-disciplinary team. 2. Discipline-specific Concepts, Issues and Skills f) Identify and formulate a design problem by developing design specifications to achieve the planned goals.

	g) Apply knowledge of design, mathematics and engineering science to analyze and predict the life-cycle performance of a design. h) Assess the impacts of various factors including, materials, human, environment, safety and reliability of a design.
Subject Synopsis/ Indicative Syllabus	The topics in the course syllabus cover three major areas: 1. Concept and Practice of Service Learning (i) Understand the social responsibility (ii) Proper attitude and behaviours in service delivery (iii) Reflection as a tool for learning (iv) Ethical issues in service learning 2. Discipline-Specific Concepts, Issues and Skills Fundamental knowledge of engineering design for problem solving including: (i) Problem identification and analysis (ii) Develop a design project with the goal to solve the problem (iii) Develop design specifications (iv) Design for ergonomics (v) Application of materials (vi) Use of common engineering components (vii) Design for reliability, safety and environmental-friendly (viii) Proper use of engineering/computational tools to conduct the design project 3. Project-Specific Concepts, Issues and Skills Knowledge about and understanding of an identified target group of the community including: (i) Their human psychology and behavior - Human psychology and behavior focuses on the human factors that affect the elderly daily life operation. It involves their feeling in dealing with common tasks, their sensory and motion, their reaction to different materials. (ii) Social problems related to and social services provided for them (iii) Primary health and social care available (iv) Market situation for the possible preferable product
Teaching/Learning Methodology	The following teaching and learning methodology will be used for the first and second offerings of the subject and then reviewed before it is continued to offer: 1. E-learning of service learning (10 hours)

	2. Project-Specific Lectures, Tutorials, Seminars and/or Workshop • Study the psychology and behavior of the identified target group (1 lecture) • Understand public information in relation to the problems of the identified target group (1 lecture) 3. Service Learning Project • Identification of a target group at the beginning, during and at the end of the entire project to complete the following tasks (40 hours): (i) Identify the need for a design project (ii) Modify the design with input from the identified target group (iii) Train the identified target group to use the end product • Discuss with different professionals related in the field (2 lectures) • Develop and complete a design project to serve the identified target group in solving their problems which includes the following activities: (i) Visits and workshops (ii) Literature search and self-study (iii) Design team group discussions (iv) Consultations and discussions with professionals (v) Written report and oral presentation Examples of organizations that students may work with are: • The Hong Kong Senior Citizen Association (HKSCA) • The Hong Kong Government Elderly Commission • The Institution of Mechanical Engineers, Hong Kong Branch																																																
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods /tasks</th><th rowspan="2">% weighting</th><th colspan="8">Intended subject learning outcomes to be assessed</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th><th>f</th><th>g</th><th>h</th></tr><tr><td>1. E-learning Module and Project-specific seminars and workshops (individual)</td><td>20</td><td>√</td><td>√</td><td></td><td></td><td>√</td><td>√</td><td>√</td><td></td></tr><tr><td>2. Reflective Journal/report (individual)</td><td>30</td><td>√</td><td></td><td></td><td></td><td></td><td>√</td><td>√</td><td>√</td></tr><tr><td>3. Final Report (group) • Problem solving skill • Ability to assess </td><td>30</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td></tr></table>	Specific assessment methods /tasks	% weighting	Intended subject learning outcomes to be assessed								a	b	c	d	e	f	g	h	1. E-learning Module and Project-specific seminars and workshops (individual)	20	√	√			√	√	√		2. Reflective Journal/report (individual)	30	√					√	√	√	3. Final Report (group) • Problem solving skill • Ability to assess	30	√	√	√	√	√	√	√	√
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alternative design									
• Report writing skill • Helping prototype making • Providing service • Oral presentation									
4. Performance in Rendering Service (individual)	20	√	√	√				√	
Total	100 %								

The students are expected to apply their knowledge and skills in performing an engineering design project to provide service to an identified target group, therefore their performance in both the service learning project and the engineering design project will be assessed.

Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes are shown below:

- In the preliminary report (group report), students will present their understanding of the community need and responsibility in performing the design task, as well as the appropriateness of the direction, goals and specifications of the design project.
- Making use of the reflective journal (individual writing), students are able to further elaborate the objectives of the design task in relation to provide service to an identified target group of the community and to demonstrate empathy for people in need and a strong sense of civic responsibility.
- In addition to those outcomes fulfilled by the preliminary report, students are expected to apply knowledge of design, mathematics and engineering science to analyze and predict the life-cycle performance of their design in the final report (group report). They are also required to consider the impacts of various factors including, materials, human, environment, safety and reliability of a design. In the oral presentation (group presentation), students' ability to explain precisely and concisely on their contribution will be assessed. Due to the time constraint, the technical details including problem formulation and analysis may not be emphasized.
- The students' attitude and performance in the rendering of service, their degree of engagement with the service recipients, their collaboration with other students, and interactions with the service recipients and/or collaborating NGOs are obviously indicators of their ability to communicate effectively with clients and stakeholders, their sense of responsibility, professional ethics and their empathy for people in need.
- Because of the nature of the subject, written examination seems not necessary.

Student Study Effort Expected	Class contact:	
	▪ Lectures and seminars	13 Hrs.
	▪ e-learning of service learning	10 Hrs.
	▪ Workshops	8 Hrs.
	▪ Discussions and consultations	8 Hrs.
	Other student study effort:	
	▪ Visits to the identified target group at different stages of the project	40 Hrs.
	▪ Literature review and marketing survey	9 Hrs.
	▪ Prepare preliminary report, final report, reflective journal and oral presentation, and Participate in prototype fabrication.	30 Hrs.
	▪ Self studying	10 Hrs.
	Total student study effort	128 Hrs.
Reading List and References	1. Mandell, B.R. and Schram, B., An introduction to human services: policy and practice, Pearson, latest edition. 2. Schriver, J.M., Human behavior and the social environment: shifting paradigms in essential knowledge for social work practice, Allyn and Bacon, latest edition. 3. Wayne, J.H., The social services: an introduction, F.E. Peacock Publishers, latest edition. 4. Ulrich, K.T., Product design and development, McGraw-Hill, latest edition. 5. Budynas, R.G. and Nisbett, J.K., Shigley's mechanical engineering design, McGraw-Hill, latest edition. 6. Boothroyd, G., Dewhurst, P. and Knight, W.A., Product design for manufacture and assembly, Boca Raton, CRC Press, latest edition. 7. Szalma, J.L. (2009). Individual differences in human-technology interaction: Incorporating variation in human characteristics into human factors and ergonomics research and design. Theoretical Issues in Ergonomics Science, 10(5), 381-397. doi: 10.1080/14639220902893613 8. Wickens, C.D., & Kramer, A. (1985). Engineering Psychology. Annual Review of Psychology, 36(1), 307-348. doi: 10.1146/annurev.ps.36.020185.001515	

Revised August 2019

Subject Description Form

Subject Code	ME33001
Subject Title	Mechanics of Materials
Credit Value	3
Level	3
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: ME23001 Engineering Mechanics; and ENG2001 Fundamentals of Materials Science and Engineering
Objectives	To introduce the fundamental mechanics knowledge of solid materials under basic loading conditions. And to introduce practical approaches to solve for the stress and strain/deformation of solid materials under external mechanical loadings.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Solve for external forces and moments applied on a structure and determine the distribution of internal forces and moments in the structure by using free body diagrams and the laws of equilibrium. - Recognize the crucial material and geometrical properties for a structural component under different types of loading, and solve for stress and deformation in a structural component due to axial loading, torsion, and bending acting individually or in combination. - Evaluate the principal stresses in structural components subjected to a combined state of loading. - Formulate and solve problems involving tension, compression, torsion or bending for statically indeterminate structural components.
Subject Synopsis/ Indicative Syllabus	Fundamentals - Free Body Diagram; Equilibrium of a deformable body; General state of stress; Strain; Mechanical properties of materials. Axial Load - Saint-Venant's Principle; Axial elastic deformation; Principle of superposition; Statically indeterminate axially loaded member; Thermal stress. Torsion - Torsional deformation; Torsional Stress; Angle of twist; Statically indeterminate torque-loaded members. Bending - Equilibrium of beams; Shear force and bending moments; Flexural stresses; Beam deflection; Slope and deflection by method of superposition; Statically indeterminate systems. Combined Loading - Transformation of stresses; Principle stresses and maximum shear stress; Mohr's circle. Thin walled pressure vessels; Cylinders and spheres under internal and external pressures; Compounded cylinder; Stress distribution in beams; Stresses due to combined loads.

	Laboratory Experiment There are two 2-hour laboratory sessions. Typical Experiments: 1. Torsion test 2. Deflection of beam																																								
Teaching/Learning Methodology	Lectures are used to deliver the fundamental knowledge in relation to the topics as described in the section subject synopsis (Outcomes a to d). Tutorials are used to illustrate the application of fundamental knowledge to practical situations (Outcomes a to d). Experiments are used to relate the concepts to practical applications and students are exposed to hand-on experience, proper use of equipment and application of analytical skills on interpreting experimental results (Outcomes a and d). <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="4">Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>Lecture</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>Tutorial</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>Experiment</td><td>√</td><td></td><td></td><td>√</td></tr></table>	Teaching/Learning Methodology	Outcomes				a	b	c	d	Lecture	√	√	√	√	Tutorial	√	√	√	√	Experiment	√			√																
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4. Examination	60%	√	√	√	√																																				
Total	100%																																								

Student Study Effort Expected	Class contact:	
	▪ Lecture	33 Hrs.
	▪ Tutorial/Laboratory	6 Hrs.
	Other student study effort:	
	▪ Course work	23 Hrs.
	▪ Self-study	42 Hrs.
	Total student study effort	104 Hrs.
Reading List and References	1. R.C. Hibbeler, Mechanics of Materials, Pearson Prentice Hall, latest edition. 2. F.P. Beer, E.R. Johnston and Jr. J.T. DeWolf, Mechanics of Materials, McGraw-Hill, latest edition. 3. A.C. Ugural, A.C. and S.K. Fenster, Advanced Strength and Applied Elasticity, Prentice Hall, latest edition.	

Revised August 2014

Subject Description Form

Subject Code	ME34002
Subject Title	Engineering Thermodynamics
Credit Value	3
Level	3
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: AP10005 Physics I, AMA2111 Mathematics I, and ME34005 Fundamentals of Thermal Science and Engineering Exclusion: ME34001 Engineering Thermodynamics
Objectives	To extend students' knowledge of fundamental of thermodynamics to various energy conversion systems, including combustion processes, steam and gas power cycles, refrigeration cycle, air-conditioning and fuel cells.
Intended Learning Outcomes	a. Understand the fundamental principles of thermodynamic systems. b. Formulate and solve thermodynamic problems relating to combustion, steam power, gas power and refrigeration cycles, and air-conditioning. c. Analyze and interpret data obtained from experiments in engineering thermodynamics. d. Present effectively in completing written reports of laboratory work and the given task.
Subject Synopsis/ Indicative Syllabus	<i>Review of Basic Concepts of Thermodynamics</i> - First and second law of thermodynamics. Non-flow and steady-flow processes. Carnot cycle. Entropy. <i>Power Cycles</i> - Vapour cycles: Rankine cycle. Superheat and reheat. Air standard engine cycles: Otto cycle and Diesel cycle. Gas turbine cycles. <i>Refrigeration Cycles</i> - Refrigerator and heat pump. Vapor compression cycle. Coefficient of Performance. <i>Thermodynamic Property Relations</i> – The Maxwell Relations. The Clapeyron Equation. General Relation for change of internal energy, enthalpy, entropy and specific heat. The Joule-Thomson Coefficient. <i>Psychrometry and Air Conditioning</i> – Gas-vapor mixture. Psychrometry. Psychrometric chart. Air conditioning processes. <i>Combustion</i> - Hydrocarbon fuels. Combustion equations. Air fuel ratio. Lean and rich mixture. Theoretical and actual combustion processes. Enthalpy of combustion. Adiabatic flame temperature. <i>Fuel Cells</i> - Basic principle. Thermodynamics and reaction kinetics. Polarization. Water and thermal management. Laboratory Experiment There are two 2-hour laboratory sessions with the typical experiments: 1. Refrigeration system 2. Combustion

Teaching/Learning Methodology	Lectures are used to deliver the fundamental knowledge in relation to thermodynamics and heat transfer (outcomes a and b). Tutorials are used to illustrate the application of fundamental knowledge to practical situations (outcomes a, b and d). Experiments are used to relate the concepts to practical applications and students are exposed to hand-on experience, proper use of equipment and application of analytical skills on interpreting experimental results (outcomes c and d). <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="4">Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>Lecture</td><td>√</td><td>√</td><td></td><td></td></tr><tr><td>Tutorial</td><td>√</td><td>√</td><td></td><td></td></tr><tr><td>Experiment</td><td>√</td><td></td><td>√</td><td>√</td></tr></table>	Teaching/Learning Methodology	Outcomes				a	b	c	d	Lecture	√	√			Tutorial	√	√			Experiment	√		√	√																						
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Experiment	√		√	√																																											
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="4">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>1. Examination</td><td>50%</td><td>√</td><td>√</td><td></td><td></td></tr><tr><td>2. Tests</td><td>20%</td><td>√</td><td>√</td><td></td><td></td></tr><tr><td>3. In-class activity/quiz</td><td>10%</td><td>√</td><td>√</td><td></td><td></td></tr><tr><td>4. Thermal Design Project</td><td>10%</td><td>√</td><td>√</td><td></td><td>√</td></tr><tr><td>5. Laboratory Work/Reports</td><td>10%</td><td></td><td></td><td>√</td><td>√</td></tr><tr><td>Total</td><td>100%</td><td colspan="4"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Overall Assessment: 0.5 × End of Subject Examination + 0.5 × Continuous Assessment Examination is adopted to assess students on the overall understanding and the ability of applying the concepts and knowledge. It is supplemented by the tests (usually two), design project (and report) and laboratory work (and reports) which provide timely feedbacks to both lecturers and students on various topics of the syllabus.	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)				a	b	c	d	1. Examination	50%	√	√			2. Tests	20%	√	√			3. In-class activity/quiz	10%	√	√			4. Thermal Design Project	10%	√	√		√	5. Laboratory Work/Reports	10%			√	√	Total	100%				
Specific assessment methods/tasks	% weighting			Intended subject learning outcomes to be assessed (Please tick as appropriate)																																											
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2. Tests	20%	√	√																																												
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4. Thermal Design Project	10%	√	√		√																																										
5. Laboratory Work/Reports	10%			√	√																																										
Total	100%																																														

Student Study Effort Expected	Class contact:	
	▪ Lecture	33 Hrs.
	▪ Tutorial / Experiment	6 Hrs.
	Other student study effort:	
	▪ Course work	40 Hrs.
	▪ Self-study and Literature Search	38 Hrs.
	Total student study effort	117 Hrs.
Reading List and References	1. Y.A. Cengel, M. A. Boles, M. Kanoglu. Thermodynamics: An Engineering Approach, McGraw Hill, latest edition, in SI units. 2. R.E. Sonntag, C. Borgnakke and G.J.V. Wylen, Fundamentals of Thermodynamics, John Wiley and Son, latest edition. 3. T.D. Eastop and A. McConkey, Applied Thermodynamics for Engineering Technologists, Pearson, latest edition. 4. K. Wark, and D. Richards, Thermodynamics, McGraw-Hill, latest edition. 5. K.D. Hagen, Heat Transfer with Applications, Prentice Hall, latest edition. 6. F.D. Incropera, and D.P. Dewitt, Introduction to Heat Transfer, Wiley, latest edition.	
Last Update	March 2025	

Subject Description Form

Subject Code	ME34004
Subject Title	Fluid Mechanics
Credit Value	3
Level	3
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: AMA2112 Mathematics II
Objectives	1. To teach fundamental concepts and knowledge of fluid mechanics. 2. To provide fundamental concepts and knowledge of inviscid and viscous flows, low-Reynolds number and high-Reynolds number flows, incompressible and compressible flows, and their applications in mechanical engineering.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Formulate and solve flow problems by applying knowledge of fluid mechanics and mathematics. b. Analyze and interpret data obtained from experiments in fluid mechanics. c. Search for updated technology in fluid mechanics in completing assignments related to fluid systems. d. Communicate effectively in completing written reports of laboratory work and assigned tasks.
Subject Synopsis/ Indicative Syllabus	Basic Concepts – Define fluid and its important properties; viscosity and shear stress; Newton’s Law of viscosity; Newtonian and non-Newtonian fluids; compressibility; incompressible and compressible fluids; introduction to shock waves. Fluid Statics - Fluid pressure; Pascal’s law and pressure-height relation; forces on submerged surfaces and buoyancy; manometers; stability of unconstrained body in fluid. General Description & Equations of Motion of Fluid Flow - Flow: steady and unsteady, uniform and non-uniform, incompressible and compressible, laminar and turbulent; subsonic and supersonic; shock waves; streamline; Continuity, Euler’s, Bernoulli’s and Momentum Equations; Pitot and Pitot-static tubes, venture-meter and orifice; force on stationary/moving object caused by a fluid jet. Dimensional Analysis - Principle of dimensional analysis; fundamental dimensions; Buckingham’s π theorem; dimensionless groups and their physical significance; similarity and model testing. Conservation Equations - Navier-Stokes equations and Energy equation; exact solutions on solving N-S equations applied to fluid systems: Couette flow and Poiseuille flow; introduction on solving N-S equations by CFD software and numerical simulation models. Internal Flow - Fully developed flow and entrance length in a pipe; Darcy’s law; Moody chart; primary (frictional) and minor losses in pipe system; design for pipes in parallel and in series.

	External Flow - Viscosity and viscous stress; laminar and turbulent boundary layers over a flat plate; effects of adverse pressure gradient and flow separation; velocity profiles and characteristics of flow over bluff body and streamline body; lift, friction and profile drag; boundary layer theory; boundary layer thicknesses; laminar boundary layer profiles; skin friction coefficient; turbulent boundary layer profiles, power law and laws of walls. Laboratory Experiment There are two 2-hour laboratory sessions and the typical Experiments are: 1. Compressible flow nozzle 2. Potential Flow Visualization (Hele-Shaw Experiment) 3. Wind Tunnel Testing of Cylinder and aerofoil 4. Universal velocity Profile 5. Boundary Layer Experiment																													
Teaching/Learning Methodology	Lectures aim to deliver the fundamental knowledge in relation to fluid mechanics (Outcome a). Tutorials are deployed to illustrate the application of fundamental knowledge to practical situations (Outcomes a and c). Experiments are arranged to relate the concepts to practical applications and students are exposed to hand-on experience, proper use of equipment and application of analytical skills on interpreting experimental results (Outcomes b, c and d). Assignments are arranged to provide opportunity for students to search information, to analyze fluid systems with knowledge obtained, and to present the completed tasks (Outcomes a, c and d). <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="4">Learning Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>Lectures</td><td>√</td><td></td><td></td><td></td></tr><tr><td>Tutorials</td><td>√</td><td></td><td>√</td><td></td></tr><tr><td>Laboratory Work</td><td></td><td>√</td><td>√</td><td>√</td></tr><tr><td>Assignments</td><td>√</td><td></td><td>√</td><td>√</td></tr></table>	Teaching/Learning Methodology	Learning Outcomes				a	b	c	d	Lectures	√				Tutorials	√		√		Laboratory Work		√	√	√	Assignments	√		√	√
Teaching/Learning Methodology	Learning Outcomes																													
	a	b	c	d																										
Lectures	√																													
Tutorials	√		√																											
Laboratory Work		√	√	√																										
Assignments	√		√	√																										

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/ tasks	% weighting	Intended subject learning outcomes to be assessed			
			a	b	c	d
	1. Examination	50%	√	√		
	2. Assignments/ Laboratory Reports	30%	√	√	√	√
	3. Test	20%	√	√		
	Total	100%				
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Overall Assessment: 0.50 × End of Subject Examination + 0.50 × Continuous Assessment Examination is adopted to assess students on the overall understanding and the ability of applying the concepts. It is supplemented by the tests, assignments (including analyses of fluid systems and problem-solving) and laboratory reports which provide timely feedbacks to both lecturers and students on various topics of the syllabus.					
Student Study Effort Expected	Class contact:					
	• Lecture		33 Hrs.			
	• Tutorial / Laboratory		6 Hrs.			
	Other student study effort:					
	• Assignments		38 Hrs.			
	• Laboratory Reports		10 Hrs.			
	• Self-study and information search		30 Hrs.			
	Total student study effort		117 Hrs.			
Reading List and References	1. Y.A. Cengel J.M. Cimbala, Fluid Mechanics (Fundamentals and Applications), McGraw-Hill, latest edition.					
	2. F.M. White, Fluid Mechanics, McGraw-Hill, latest edition.					
	3. J.F. Douglas, J.M. Gasiorek and J.A. Swaffield, Fluid Mechanics, Pearson, latest edition					
	4. M.C. Potter, and D.C. Wiggert, Mechanics of Fluids, Prentice-Hall, latest edition.					

September 2017

Subject Description Form

Subject Code	ME34005
Subject Title	Fundamentals of Thermal Science and Engineering
Credit Value	3
Level	3
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: AP10005 Physics I AMA2111 Mathematics I
Objectives	Thermal Science and Engineering (TSE) is a branch of science and engineering, covering topics in thermodynamics and heat transfer. These topics form the basic foundations that govern processes in engineering applications. This course is an introduction to TSE and how the principles can be applied to understand/design thermal processes. The specific course objectives are: (1) understand and apply principles of thermodynamics and heat transfer to engineering applications, and (2) form a foundation for subsequent studies in engineering thermodynamics, heat transfer, building services, material science, marine engineering, environmental engineering, power engineering, energy conversion, energy system and other areas.
Intended Learning Outcomes	a. Understand the fundamental principles of engineering thermodynamics and heat transfer. b. Formulate and solve thermal problems. c. Analyze and interpret data in thermal science and engineering. d. Present effectively in completing written reports.
Subject Synopsis/ Indicative Syllabus	Basic Concepts: Thermodynamics and applications; Systems; Properties; Extensive and intensive properties; Measured properties; Energy; Equilibrium; Zeroth law of thermodynamics; State and state principle; Process and cycle; Quasiequilibrium process. Properties of pure substances: Pure substances; Phase-change process; Property diagrams; Critical point and critical properties; Enthalpy; Property tables; Reference state; State equations for vapor phase; Ideal gases. Heat and work: Forms of energy; Energy transfer by heat; Adiabatic processes; Energy transfer by work; Mechanical forms of work; Moving boundary work; Polytropic processes. Energy analysis of closed systems: Specific heats; Internal energy; Enthalpy; First law of thermodynamics; Energy balance. Mass and energy analysis of open systems: Control volumes; Conservation of mass; Conservation of energy; Flow work and total energy of a flowing fluid; Steady-state and unsteady-flow processes; Applications in nozzles, diffusers, turbines, compressors, valves, mixing chambers and heat exchangers. Second law of thermodynamics: Thermal reservoirs; Power cycles; Efficiency; Refrigeration and heat pump cycles; Coefficients of performance; The Kelvin-Planck statement; The Clausius statement; Reversible and irreversible processes;

	Carnot principle; Carnot cycle; Thermodynamic temperature scale; Maximum performance measures for cycles; Quality of energy. Entropy: Clausius inequality; Entropy; Thermal entropy flux; Entropy production; Entropy of substances; Entropy changes; Entropy balance for closed systems; Principle of entropy increase; Entropy rate balance for open systems. Heat conduction and convection: Heat; Conduction; Forced convection; Natural convection; Fourier law of heat conduction; Thermal conductivity; Heat conduction equation; Newton law of cooling; Convective heat transfer coefficient; Nondimensional numbers; Empirical relations. Heat exchangers: Double-pipe heat exchangers; Shell-and-tube heat exchangers; Cross-flow heat exchangers; Overall heat-transfer coefficient; Log mean temperature difference; Effectiveness-NTU method; Design and applications.																																		
Teaching/Learning Methodology	Lectures are used to deliver the fundamental knowledge and application in relation to thermodynamics and heat transfer (outcomes a-d). Tutorials are used to illustrate the application of fundamental knowledge to practical situations (outcomes a-d). <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="4">Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>Lecture</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>Tutorial</td><td>√</td><td>√</td><td>√</td><td>√</td></tr></table>	Teaching/Learning Methodology	Outcomes				a	b	c	d	Lecture	√	√	√	√	Tutorial	√	√	√	√															
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2. Tests	20%	√	√	√	√																														
3. Homework	20%	√	√	√	√																														
Total	100%																																		

Student Study Effort Expected	Class contact:	
	▪ Lecture	33 Hrs.
	▪ Tutorial	6 Hrs.
	Other student study effort:	
	▪ Course work	41 Hrs.
	▪ Self-study	40 Hrs.
	Total student study effort	120 Hrs.
Reading List and References	1. Y.A. Cengel, M. A. Boles, M. Kanoglu. Thermodynamics: An Engineering Approach, McGraw Hill, latest edition, in SI units. 2. R.E. Sonntag, C. Borgnakke and G.J.V. Wylen, Fundamentals of Thermodynamics, John Wiley and Son, latest edition. 3. T.D. Eastop and A. McConkey, Applied Thermodynamics for Engineering Technologists, Pearson, latest edition. 4. K. Wark, and D. Richards, Thermodynamics, McGraw-Hill, latest edition. 5. J. P. Holman, Heat Transfer, McGraw Hill, latest edition. 6. Y. A. Cengel, Heat transfer: A Practical Approach, McGraw Hill, latest edition. 7. K.D. Hagen, Heat Transfer with Applications, Prentice Hall, latest edition. 8. F.D. Incropera, and D.P. Dewitt, Introduction to Heat Transfer, Wiley, latest edition.	
Develop Date	August 2024	

Subject Description Form

Subject Code	ME35001																								
Subject Title	Robotics and Computer Integrated Manufacturing																								
Credit Value	3																								
Level	3																								
Pre-requisite/ Co-requisite/ Exclusion	Prerequisite: Students should have basic knowledge of mechanical systems																								
Objectives	To provide students with the fundamental knowledge of advanced manufacturing process and to teach students how intelligence can be applied																								
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Understand the basic working principles of common manufacturing process b. Explain the fundamental principles of Boolean logic, fuzzy logic (FL) systems, artificial neural networks (ANN). c. Apply simple machine learning algorithms to intelligent control, planning and scheduling d. Present effectively in completing written reports of laboratory work and the given task.																								
Subject Synopsis/ Indicative Syllabus	Manufacturing – Working Principle and Applications of Articulated, SCARA, Delta, Cartesian, and Polar robots, Traditional and modern manufacturing process Machine (Artificial) Intelligence – Boolean Logic; Fuzzy Logic; Regression; Classification; Supervised Learning; Unsupervised Learning, Deep Learning Applied AI for Manufacturing – Enhance Precision through Control, Planning and Scheduling for multi-agent systems																								
Teaching/Learning Methodology	Lectures aim at providing students with an integrated knowledge required for understanding and applying intelligence to enhance the efficiency and performance of robots for manufacturing (Outcomes a to d) Tutorials aim at enhancing students’ analytical and problem-solving skills. Students will be able to solve real-world problems using the knowledge they acquired in the class. (Outcomes b to c) The project/experiment aims to have knowledge of computer simulations and hand-on experience on robots for manufacturing. (Outcomes a to d) <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="4">Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>1. Lectures</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>2. Tutorials</td><td></td><td>√</td><td>√</td><td></td></tr><tr><td>3. Project or experiments</td><td>√</td><td>√</td><td>√</td><td>√</td></tr></table>	Teaching/Learning Methodology	Outcomes				a	b	c	d	1. Lectures	√	√	√	√	2. Tutorials		√	√		3. Project or experiments	√	√	√	√
Teaching/Learning Methodology	Outcomes																								
	a	b	c	d																					
1. Lectures	√	√	√	√																					
2. Tutorials		√	√																						
3. Project or experiments	√	√	√	√																					

Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="4">Intended subject learning outcomes to be assessed</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>1. Assignments</td><td>10%</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>2. Tests</td><td>20%</td><td>√</td><td>√</td><td>√</td><td></td></tr><tr><td>3. Projects</td><td>20%</td><td></td><td>√</td><td>√</td><td>√</td></tr><tr><td>4. Examinations</td><td>50%</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>Total</td><td>100 %</td><td colspan="4"></td></tr></table>		Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed				a	b	c	d	1. Assignments	10%	√	√	√	√	2. Tests	20%	√	√	√		3. Projects	20%		√	√	√	4. Examinations	50%	√	√	√	√	Total	100 %								
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	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:																																													
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The examination is adopted to assess students on their overall understanding and ability to apply the concepts. It is supplemented by continuous assessment including assignments, closed-book tests, and projects. The continuous assessment is aimed at enhancing the students’ comprehension and assimilation of various topics of the syllabus.																																														
Student Study Effort Expected	Class contact:																																													
	▪ Lecture				33Hrs.																																									
	▪ Tutorial				6Hrs.																																									
	Other student study effort:																																													
	▪ Reading and review				45Hrs.																																									
	▪ Coursework (assignments, project)				36Hrs.																																									
	Total student study effort				120Hrs.																																									
Reading List and References	1. Ray Asfahl, Robots and Manufacturing Automation, applications, Wiley, latest edition. 2. Wilson M, Implementation of Robot Systems, An Introduction to Robotics, Automation, and Successful Systems Integration in Manufacturing, latest edition M. W. Spong S. Hutchinson, and M. Vidyasagar, Robot Modeling and Control, Wiley, latest edition. 5. Campesato O., Artificial Intelligence, Machine Learning and Deep Learning, Mercury Learning & Information, latest edition																																													
Develop Date:	June 2024																																													

Subject Description Form

Subject Code	ME35002
Subject Title	Sensors and Actuators for Mechatronics
Credit Value	3
Level	3
Pre-requisite/ Co-requisite/ Exclusion	Prerequisite: Students should have basic knowledge of mechanical systems
Objectives	To provide students with the knowledge in designing mechatronic and robotic systems
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Formulate and solve problems related to sensors and actuators in mechatronic systems. - Understand the importance of signal conditioning and data acquisition in mechatronic systems. - Design and analyze a given task or project in a mechatronics system by applying knowledge acquired in the subject and information obtained through a literature search. - Present effectively in completing written reports of laboratory work and the given task.
Subject Synopsis/ Indicative Syllabus	Sensors - Instrumentation and measurement principles; principles of different sensors for linear and angular measurements on position, velocity, and acceleration; force and pressure sensors, principle of different cameras, 3D measurements, Time-of-flight sensors, etc. Actuators – Different types of motors: DC, AC, servo, stepper motors; motor control shields; electric linear actuators; pneumatic systems; hydraulic systems, etc. Signal Conditioning - Concepts and principles; complex number in electronics; operational amplifier; conversion between analog and digital signals, multiplexing; data acquisition principles, signal filtering, active and passive filters; modulation and demodulation, etc.

Teaching/Learning Methodology	Lectures aim at providing students with an integrated knowledge required for understanding and analyzing different components (sensors and actuators) in a mechatronics system (Outcomes a to d) Tutorials aim at enhancing students’ analytical and problem-solving skills related to sensing or actuation principles. Students will be able to solve real-world problems using the knowledge they acquired in the class. (Outcomes b to c) The project/experiment aims to have knowledge of computer simulations and hands-on experience on mechatronics systems (Outcomes a to d) <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="4">Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>1. Lectures</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>2. Tutorials</td><td></td><td>√</td><td>√</td><td></td></tr><tr><td>3. Project or experiments</td><td>√</td><td>√</td><td>√</td><td>√</td></tr></table>	Teaching/Learning Methodology	Outcomes				a	b	c	d	1. Lectures	√	√	√	√	2. Tutorials		√	√		3. Project or experiments	√	√	√	√																
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2. Tutorials		√	√																																						
3. Project or experiments	√	√	√	√																																					
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Reading List and References	1. Bolton, W., Mechatronics: Electronic Control Systems in Mechanical Engineering, Prentice Hall, latest edition. 2. Shetty, D. and Kolk, R. A., Mechatronic System Design, PWS Publishing Company, latest edition. 3. Alciatore, D. G. and Hstand, M. B., Introduction to Mechatronics and Measurement Systems, McGraw Hill, latest edition.
Develop Date	June 2024

Subject Description Form

Subject Code	ME42011
Subject Title	Fundamentals of Robotics
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: ME31002 Linear systems and control; or ME41004 Mechatronics and Control
Objectives	1. To provide students with the concepts and techniques for the design, modeling, analysis of robots. 2. To provide students with the fundamental knowledge of machine vision for robot guidance and automation.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Identify different types of robots and their applications in the industry. b. Construct the kinematics and dynamics equations of robots for motion analysis. c. Apply motion planning algorithms to generate the path for robots. d. Apply different machine vision and image processing algorithms to automate robots.
Subject Synopsis/ Indicative Syllabus	Robot Modelling. Degrees of freedom, coordinate frame and homogeneous transformation, Denavit-Hartenberg (DH) convention, forward and inverse kinematics, Jacobian matrix, singularity, robot dynamic models, state estimation, motion planning and obstacle avoidance. Computer Vision. Image formation, acquisition, histogram, edge and line detections, image enhancement, filtering, object recognition, stereo vision, camera modeling and calibration, deep neural networks in computer vision. Laboratory Work There is at least 6-hour laboratory sessions or an equivalent project. Typical Experiments are: 1. Programming and control of robots. 2. Path planning of mobile robots for collision avoidance. 3. Programming of image processing and computer vision.

Teaching/Learning Methodology	Lectures aim at providing students with an integrated knowledge required for understanding and analyzing different robots, including system modeling, trajectory planning and image processing (Outcomes a to d) Tutorials aim at enhancing students’ analytical and problem solving skills on robotics. Students will be able to solve real-world problems using the knowledge they acquired in the class. (Outcomes a to d) The project/experiments aims to have hand-on experience to automation of a robot system with vision or other functions. (Outcomes a to d) <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="4">Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>1. Lectures</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>2. Tutorials</td><td></td><td>√</td><td>√</td><td>√</td></tr><tr><td>3. Homework assignments</td><td></td><td>√</td><td>√</td><td>√</td></tr><tr><td>4. Project or experiments</td><td>√</td><td>√</td><td>√</td><td>√</td></tr></table>	Teaching/Learning Methodology	Outcomes				a	b	c	d	1. Lectures	√	√	√	√	2. Tutorials		√	√	√	3. Homework assignments		√	√	√	4. Project or experiments	√	√	√	√					
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Student Study Effort Expected	Class contact:	
	▪ Lecture	30 Hrs.
	▪ Tutorial/Laboratory	9 Hrs.
	Other student study effort:	
	▪ Reading and review	36 Hrs.
	▪ Coursework (assignments, project)	40 Hrs.
	Total student study effort	115 Hrs.
Reading List and References	1. Introduction to Robotics: Analysis, Control, Applications, Saeed B. Niku, John Wiley and Sons, latest edition. 2. Modern Robotics: Mechanics, Planning, and Control, Kevin M. Lynch, Frank C. Park, latest edition. 3. Introduction to Autonomous Mobile Robots, Roland Siegwart, Illah Reza Nourbakhsh and Davide Scaramuzza, latest edition. 4. Introduction to Autonomous Robots: Mechanisms, Sensors, Actuators, and Algorithms, Nikolaus Correll, Bradley Hayes, Christoffer Heckman and Alessandro Roncone, latest edition. 5. R. C. Gonzalez and R. E. Woods, Digital Image Processing, Prentice Hall, latest edition. 6. Deep Learning, Ian Goodfellow, Yoshua Bengio and Aaron Courville, latest edition.	

Revised July 2021

Subject Description Form

Subject Code	ME42012
Subject Title	Advanced Robotics Studio
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: EIE3320 Object-Oriented Design and Programming
Objectives	To acquire knowledge and hands-on experience of the latest technologies used in robotics. To understand the design process of robots and the systematic approach to solve engineering problems.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - systematically analyse a given engineering problem and devise an effective robotic solution to address it. - understand and utilise the latest computational technologies to command the motion of robots. - develop computer-aided designs and simulations of robotic systems, mechanisms, and environments. - develop a working robotic prototype by integrating sensors, actuators, mechanisms, and computer systems. - effectively work in teams, design systems, conduct experiments, and deliver effective solutions.
Subject Synopsis/ Indicative Syllabus	Robot Design Process. Robotics as a scientific discipline. Discuss different cases of study and the need for robots in society. Introduce the theory and principles for a methodological robot design. Problem formulation, requirements, and constraints. Computational Tools. Mathematical and numerical preliminaries. Introduce different computer libraries used in robotics. Computer-aided simulation of robotic systems and mechanisms. Digital twins and physics engines. Implementation of common algorithms for perception and control. Hardware Systems. Study the fundamental differences between a computational models and real robotic mechanisms. Study common hardware components (materials, mechanisms, sensors, electronics) used to develop real robotic platforms. Implement Sim2Real strategies to command the motion of real robotic platforms. Course Projects. Conduct hands-on experimental work where students can apply different concepts studied during the course. Form teams and design innovative mechanisms and automated systems that address a particular challenge provided in class. Organise a final course teams' competition where each team presents their unique solution. Field Trip. Group visit to a company, research centre, or institution relevant to the development of advanced robotic systems.

Teaching/Learning Methodology	1. Lectures aim at providing students with fundamental knowledge required to design and develop efficient robotics solutions to real-world problems. (Outcomes a to d) 2. Tutorials aim at enhancing students’ technical skills in robotic systems. Students will be able to use the latest robotics technologies with the knowledge acquired in class. (Outcomes b to d) 3. The experiments/projects aim to provide hands-on experience for developing robotic systems and reinforcing the acquired knowledge. (Outcomes b to e) <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="5">Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr><tr><td>1. Lectures</td><td>√</td><td>√</td><td>√</td><td>√</td><td></td></tr><tr><td>2. Tutorials</td><td></td><td>√</td><td>√</td><td>√</td><td></td></tr><tr><td>3. Project or experiments</td><td></td><td>√</td><td>√</td><td>√</td><td>√</td></tr></table>	Teaching/Learning Methodology	Outcomes					a	b	c	d	e	1. Lectures	√	√	√	√		2. Tutorials		√	√	√		3. Project or experiments		√	√	√	√											
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Reading List and References	• Lecture notes, articles, and handouts. • Practical and Experimental Robotics. F. Sahin and P. Kachroo, CRC Press, Latest Edition. • Engineering Design, by Rudolph J. Eggert. High Peak Press, Latest Edition. • Practical Field Robotics: A Systems Approach, by Robert H. Sturges. Wiley Press, Latest Edition. • Programming Robots with ROS: A Practical Introduction to the Robot Operating System, by Morgan Quigley. O'Reilly Media, Latest Edition. • Arduino Programming Notebook. B. Evans, Latest Edition.
Develop Date	June 2024

Subject Description Form

Subject Code	ME42013
Subject Title	Robot Mechanisms and Motion
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: ME23001 Engineering Mechanics AND Students should have basic knowledge of Dynamics.
Objectives	This course covers topics including classifications of mechanisms; velocity, acceleration and force analysis (e.g. for linkages, cranks, sliders, and cams); balancing of rotating and reciprocating machinery; gears and gear-trains; graphical and computer-oriented methods of analysis for mechanisms; applications of different mechanisms in mechanical systems (e.g., engines, manufacturing systems).
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Recognize concepts and applications of mechanisms (e.g., cam, flywheels, gear and gear trains, etc.) in mechanical systems. - Conduct graphical and analytical analyses and/or use computer programs to obtain position, velocity, and acceleration of a given mechanism. - Apply graphical and analytical methods to obtain static, inertia, and dynamic forces on planar mechanisms. - Analyze motions and solve for forces in simple, compound, and planetary gear trains. - Design four-bar and/or slider crank mechanisms for function generation or rigid body guidance.
Subject Synopsis/ Indicative Syllabus	This is a course that develops the knowledge and skills mechanical engineers need to design and analyze mechanisms in mechanical systems. The gained knowledge and experiences in this course will be essential for any future project activities. Introduction to Mechanisms and Kinematics - Types of mechanisms, Definitions and terminologies, Kinematic Chains and Pairs, Mechanism Mobility and Inversion. Mechanics of Rigid Bodies - Relative velocity and acceleration, Limit positions and time ratio of mechanisms, Transmission angle, Instantaneous centre of velocity, Kennedy's theorem, Kinetics, Equations of equilibrium. Graphical Kinematics Analysis - Velocity analysis using instantaneous centres, Velocity polygon analysis, Velocity image, Acceleration polygon analysis, Acceleration image. Analytical Kinematics - Loop closure equation, Complex vector analysis with one and multiple loops.

	Graphical Force Analysis - Two and three force member, Force transmission, Force polygons, Static force analysis, principle of superposition, Dynamic force analysis. Analytical Force and Balancing - Force analysis (e.g., four-bar mechanism, slider crank mechanism), Unbalance and balancing, Force balancing of mechanisms. Flywheels and gyroscopic forces - Mathematical formulation for flywheels, Branched systems, Gyroscopic effects. Cam design - Nomenclature, Pressure angle, Displacement diagram, Follower motions, Disc cam profile, Positive motion cams, Program cam design Gears and gear train - Fundamental law of gearing, Involute and Cycloidal tooth gearing, Nonstandard spur, bevel, helical, and work gearing, Production of gears, Gear trains: ordinary gear trains, planetary gear trains, harmonic drives Synthesis of mechanism - Design of four-bar and slider crank mechanisms: function generation and rigid body guidance.																													
Teaching/Learning Methodology	- Lectures aim at providing students with fundamental knowledge required for understanding and analysing different networked systems, including its mathematical models, controller design, and algorithms. (Outcomes a to e) - Tutorials aim at enhancing students’ analytical and problem-solving skills in networked robots and IoT. Students will be able to solve real-world problems using the knowledge they acquired in the class. (Outcomes b to e) - The experiments/project aim to provide hands-on experience for developing networked robots and IoT and reinforcing the acquired knowledge. (Outcomes a to e) <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="5">Intended subject learning outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr><tr><td>1. Lectures</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>2. Tutorials</td><td></td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>3. Project report and presentation</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td></tr></table>	Teaching/Learning Methodology	Intended subject learning outcomes					a	b	c	d	e	1. Lectures	√	√	√	√	√	2. Tutorials		√	√	√	√	3. Project report and presentation	√	√	√	√	√
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1. The continuous assessment will comprise three components: team project (20%), quizzes (20%) and homework (15%). The team project, quizzes and homework are aimed at evaluating their understandings on robotic mechanisms and motion systems and enhancing the integration of their knowledge learnt.																																																						
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Reading List and References	<u>Required Course Materials</u> Recommended Textbook: Mechanics of Machines, 2nd Edition, William Cleghorn & Nikolai Dechev, Oxford University Press, 2014, ISBN: 978-0195384086																																																					
	<u>Optional Course Materials</u> 1. Theory Of Machines and Mechanisms, 4th Edition, Gordon R. Pennock & Joseph E. Shigley John J. Uicker, Oxford University Press, 2010, ISBN: 978-0195371239 2. Machines and Mechanisms: Applied Kinematic Analysis, 4th Edition, David H. Myszka, Prentice Hall, 2012, ISBN: 978-0132157803 3. Design Simulation Technologies ® Working Model-2D Guide https://www.design-simulation.com/documents/WM/wmdemotoolkitpro.pdf 4. MathWorks® Matlab User Guide http://www.mathworks.com/help/pdf_doc/matlab/getstart.pdf																																																					
Develop Date	June 2024																																																					

Subject Description Form

Subject Code	ME46002
Subject Title	Numerical Methods for Engineers
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: AMA2111 Mathematics I
Objectives	To teach students numerical methods of solving typical engineering problems.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Formulate simple engineering problems with knowledge in data analysis and engineering mathematics. - Solve non-linear equations, simultaneous linear algebraic equations, eigenvalue problems, using numerical methods. - Perform numerical differentiation and integration and analyze the errors. - Know the contemporary issues, and understand the impact of artificial intelligence in a global and societal context. - Use MATLAB or other data analysis tools to compute the solutions of engineering problems using the appropriate numerical methods.
Subject Synopsis/ Indicative Syllabus	Introduction to Mathematical Modelling and Computational Methods – Importance of computational modelling in engineering. Data representation and errors. Applications of commercial software packages such as MATLAB. Functions and plotting using MATLAB. Computer Solution of Non-linear Equations - Bracketing Methods. Bisection Method. Open Methods. Newton-Raphson Method. Secant Method. Convergence of methods. Determination of multiple roots. Engineering applications. Simultaneous Linear Equations - Solving simultaneous linear equations by Matrix Inversion. Cramer's Rule. Gauss Elimination. Gauss-Jordan Elimination. LU decomposition method. Engineering applications and choice of methods. Eigenvalues Problems. Optimization - Unconstrained optimization. Multi-dimensional optimization. Unconstrained optimization. Curve Fitting and Data Analysis - Interpolation using splines. Linear Least-Squares Regression. Nonlinear Regression. Introduction to Machine Learning Algorithms. Numerical Differentiation and Integration - Taylor's series expansion. Finite differences for the first derivative and the second derivative. High-accuracy differentiation formulas. Trapezoidal rule. Simpson's rule. High-order Newton-Cotes formulas. Applications of numerical differentiation and integration in heat transfer, solid mechanics and fluid flow problems.

Teaching/Learning Methodology	Lectures are used to deliver the fundamental knowledge in relation to numerical methods. (Outcomes a - d) Tutorials will be conducted in small groups to facilitate discussions. (Outcomes a - d) Computational workshops provide hands-on experience in using software to solve numerical problems. (Outcomes b - e) <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="5">Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr><tr><td>Lecture</td><td>√</td><td>√</td><td>√</td><td>√</td><td></td></tr><tr><td>Tutorial</td><td>√</td><td>√</td><td>√</td><td>√</td><td></td></tr><tr><td>Computational workshop</td><td></td><td>√</td><td>√</td><td>√</td><td>√</td></tr></table>	Teaching/Learning Methodology	Outcomes					a	b	c	d	e	Lecture	√	√	√	√		Tutorial	√	√	√	√		Computational workshop		√	√	√	√											
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	▪ Private study	25 Hrs.
	Total student study effort	116 Hrs.
Reading List and References	1. S.C. Chapra and R.R. Canale, Numerical Methods for Engineers, McGraw-Hill, latest edition. 2. S.S. Rao, Applied Numerical Methods for Engineers and Scientists, Prentice-Hall, latest edition. 3. S.C. Chapra, Applied Numerical Methods with MATLAB for Engineers and Scientists, McGraw-Hill, latest edition. 4. D.M. Etter, Engineering Problem Solving with Matlab, Prentice-Hall, latest edition.	

Revised December 2021

Subject Description Form

Subject Code	ME49001
Subject Title	Final Year Capstone Project
Credit Value	6
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: ME31001 Dynamics and Vibrations; ME31002 Linear Systems and Control; ME32001 Manufacturing Fundamentals; ME33001 Mechanics of Materials; ME34002 Engineering Thermodynamics; and ME34004 Fluid Mechanics Exclusion: ME49006 Integrated Capstone Project
Objectives	To provide students with an opportunity of integrating and applying knowledge from different disciplines of mechanical engineering to conduct an engineering project that is open-ended and requires team collaboration for its completion.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Formulate the problem and suggest a practical solution to solve an open-ended real-world engineering problem. - Utilize knowledge from different disciplines of engineering to solve problems encountered in conducting the team project. - Design, plan and carry out scientific and engineering experiments (physical tests and/or computer numerical simulations) to prove the feasibility of their designed solutions. - Design the test apparatus, rigs, assemblies or systems as required by the project. - Apply appropriate engineering tool (analytical, experimental, and/or computational) for carrying out tasks in the development and implementation of a designed solution. - Work in a professional manner and comply with all applicable standards and regulations in conducting the project. - Select and employ the appropriate manufacturing methods in the production and fabrication of components and assemblies required by the project. - Evaluate the potential impact of their designed solution on performance, safety, cost and environment. - Participate and lead in a multi-functional team. - Take into account of safety, legal, environmental protection considerations in an engineering project. - Communicate their project work to sponsors (if any), supervisors, other peer teams, and even non-technical audience and articulate the results and findings with scientific and logical arguments. - Conduct literature search including patents, books, archived publications and product catalogues, and to perform the state-of-the-art and benchmark studies.

Subject Synopsis/ Indicative Syllabus	A project group consisting normally of three students will be expected to complete a substantial project of a major mechanical engineering task. The task can be an analytical study, an experimental investigation, a design project or a numerical simulation aimed at solving an engineering problem. The students are expected to go through the following stages of work: • Problem identification • Literature review • Methodology of study • Project execution • Report writing • Project presentation																																																																																																												
Teaching/Learning Methodology	The subject is taught through guided studies. The students are given the project title, objectives and description. The students are guided by the project supervisor to go through the different stages of the project as shown in the Subject Synopsis/Indicative Syllabus. (Outcomes a – l) <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="12">Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th><th>f</th><th>g</th><th>h</th><th>i</th><th>j</th><th>k</th><th>l</th></tr><tr><td>Guided study</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td></tr></table>													Teaching/Learning Methodology	Outcomes												a	b	c	d	e	f	g	h	i	j	k	l	Guided study	√	√	√	√	√	√	√	√	√	√	√	√																																																										
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Total	100%																																																																																																												

	report is assessed by the independent assessor. The final report is assessed by both the supervisor and the independent assessor. As part of the assessment process, each group member is required to specify his/her own contribution in completing the project when compared to his/her team mates (peer assessment). In case of an industrial-based project, comments will be invited from the industrial supervisor but he/she will not be required to perform the formal assessment. 3. The supervisor monitors and assesses the overall and individual progresses through regular meetings. The interim report should be submitted to the independent assessor at around week 8 of the first semester. The final report submitted before the end-of-year examination is assessed by both the supervisor and the independent assessor. Due consideration of each student’s individual contribution and performance will be taken into account. 4. During the oral examination, every group member is required to present the project especially on his/her significant contribution to the whole project, and respond to the questions addressed to him/her by the examination panel. Marks for oral examination are awarded to individual student by taking into account the group’s overall performance. 5. The assessment system is summarized as shown in the following table: <table><tr><th rowspan="2">Assessor</th><th colspan="5">Assessment Component (% of the total)</th></tr><tr><th>Continuous Monitoring (15)</th><th>Interim Report (10)</th><th>Final Report (25)</th><th>Final Report (25)</th><th>Oral Examination (25)</th></tr><tr><td>Supervisor</td><td>√</td><td></td><td>√</td><td></td><td></td></tr><tr><td>Independent Assessor</td><td></td><td>√</td><td></td><td>√</td><td></td></tr><tr><td>Examination Panel</td><td></td><td></td><td></td><td></td><td>√</td></tr></table>	Assessor	Assessment Component (% of the total)					Continuous Monitoring (15)	Interim Report (10)	Final Report (25)	Final Report (25)	Oral Examination (25)	Supervisor	√		√			Independent Assessor		√		√		Examination Panel					√
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Independent Assessor		√		√																										
Examination Panel					√																									
Student Study Effort Expected	<table><tr><td>Class contact:</td><td></td></tr><tr><td>▪ Guided study</td><td>26 Hrs.</td></tr><tr><td>Other student study effort:</td><td></td></tr><tr><td>▪ Conducting project</td><td>154 Hrs.</td></tr><tr><td>▪ Literature search and private study</td><td>72 Hrs.</td></tr><tr><td>Total student study effort</td><td>252 Hrs.</td></tr></table>	Class contact:		▪ Guided study	26 Hrs.	Other student study effort:		▪ Conducting project	154 Hrs.	▪ Literature search and private study	72 Hrs.	Total student study effort	252 Hrs.																	
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Reading List and References	To be advised by supervisor																													

Revised March 2022

Subject Description Form

Subject Code	ME49006
Subject Title	Integrated Capstone Project
Credit Value	6
Level	4
Pre-requisite / Co-requisite/ Exclusion	Pre-requisite: ME31001 Dynamics and Vibrations; ME31002 Linear Systems and Control; ME32001 Manufacturing Fundamentals; ME33001 Mechanics of Materials; ME34002 Engineering Thermodynamics; ME34004 Fluid Mechanics; ENG2002 Computer Programming; and ME46002 Numerical Methods for Engineers. Exclusion: ME49001 Final Year Capstone Project
Objectives	The objectives of this subject are to: (a) Provide students with an opportunity of integrating and applying knowledge from different disciplines of mechanical engineering (ME) with artificial intelligence and data analytics (AIDA) to conduct an engineering project that is open-ended and requires team collaboration for its completion; (b) Develop the capabilities of a student in analyzing and solving complex and possibly real-life problems using ME and AIDA knowledge; (c) Train students with skills on systematic development and documentation of a significant piece of work.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: <u>Professional/academic knowledge and skills</u> (a) Conduct literature search including patents, books, archived publications and product catalogues, and to perform the state-of-the-art and benchmark studies; (b) Formulate the problem precisely, and utilize knowledge and skills from ME and AIDA to generate good solutions to the open-ended real-world problems encountered in the project; (c) Think critically the formulation of alternative models and solutions to the problem, in the analysis of approaches to the solution and their implementation; (d) Design, plan and carry out scientific and engineering experiments (physical tests and/or computer numerical simulations) to prove the feasibility of their designed solutions; (e) Design the test apparatus, rigs, assemblies or systems as required by the project; (f) Apply appropriate engineering tool (analytical, experimental, and/or computational) for carrying out tasks in the development and implementation of a designed solution; (g) Select and employ the appropriate manufacturing methods in the production and fabrication of components and assemblies required by the project; <u>Attributes for all-roundedness</u> (h) Work in a professional manner and comply with all applicable standards and regulations in conducting the project; (i) Take into account of safety, legal, environmental protection considerations in an engineering project

	(j) Participate and lead in a multi-functional team, and communicate their project work to sponsors (if any), supervisors, other peer teams, and even non-technical audience and articulate the results and findings with scientific and logical arguments; (k) Evaluate the potential impact of their designed solution on performance, safety, cost and environment, and evaluate the final outcome in an objective manner; (l) Improve presentation and communication skills via oral presentation, and enhance technical report writing skills with proper organization of materials.																																																																																																
Subject Synopsis/ Indicative Syllabus	1. In-depth Study of a Topic Typically Proposed by the Supervisor 2. Project Meeting and Planning 3. Proposal Writing 4. Regular Progress Checking and Reporting 5. Project Documentation 6. Presentation and Demonstration Students are expected to identify a project topic with a supervisor with artificial intelligence and data analytics expertise in ME Department. Students need to demonstrate their knowledge in both ME and AIDA in the project, receiving advice from supervisor. The project should represent requisite effort in analysing and interpreting the data/information obtained, using the principles and techniques learnt from various related subjects. Students are also expected to demonstrate significant analytical and, preferably, research ability in the chosen application domain.																																																																																																
Teaching/Learning Methodology	The Integrated Capstone Project spans across the academic year for two consecutive semesters. The teaching/learning activities include regular project meetings with the supervisor, guided study of project materials, independent project development work and other project management tasks. <table><tr><td>Teaching/Learning Methodology</td><td colspan="12">Outcomes</td></tr><tr><td></td><td>a</td><td>b</td><td>c</td><td>d</td><td>e</td><td>f</td><td>g</td><td>h</td><td>i</td><td>j</td><td>k</td><td>l</td></tr><tr><td>Regular project meetings and guided study</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr></table>	Teaching/Learning Methodology	Outcomes													a	b	c	d	e	f	g	h	i	j	k	l	Regular project meetings and guided study	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																																																									
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Examination Panel					✓																													
Student Study Effort Expected	Class contact:																																	
	• Guided study				26 Hrs.																													
	Other student study effort:																																	
	• Conducting project				154 Hrs.																													
	• Literature search and private study				72 Hrs.																													
	Total student study effort				252 Hrs																													
Reading List and References	To be advised by supervisor.																																	

August 2023

Subject Description Form

Subject Code	ME49007
Subject Title	Capstone Project
Credit Value	6
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: ME31002 Linear Systems and Control; ME35001 Robotics and Computer Integrated Manufacturing; ME35002 Sensors and Actuators for Mechatronics; EIE3320 Object-Oriented Design and Programming; ME33001 Mechanics of Materials; AND ME39003/IC382 Multidisciplinary Manufacturing Project Exclusion: ME49006 Integrated Capstone Project
Objectives	To provide students with an opportunity of integrating and applying knowledge from different areas of intelligent robotics engineering to conduct a project that is open-ended and requires team collaboration for its completion.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Formulate the problem and suggest a practical solution to solve an open-ended real-world robotics automation problem. - Utilize knowledge from different disciplines of engineering to solve problems encountered in conducting the team project. - Design, plan and carry out scientific and engineering experiments (physical tests and/or computer numerical simulations) to prove the feasibility of their designed solutions. - Design the test apparatus, rigs, assemblies or systems as required by the project. - Apply appropriate engineering tool (analytical, experimental, and/or computational) for carrying out tasks in the development and implementation of a designed solution. - Work in a professional manner and comply with all applicable standards and regulations in conducting the project. - Select and employ the appropriate manufacturing methods in the production and fabrication of components and assemblies required by the project. - Evaluate the potential impact of their designed solution on performance, safety, cost and environment. - Participate and lead in a multi-functional team. - Take into account of safety, legal, environmental protection considerations in an engineering project. - Communicate their project work to sponsors (if any), supervisors, other peer teams, and even non-technical audience and articulate the results and findings with scientific and logical arguments. - Conduct literature search including patents, books, archived publications and product catalogues, and to perform the state-of-the-art and benchmark studies.

Subject Synopsis/ Indicative Syllabus	A project group consisting normally of three students will be expected to complete a substantial project by using intelligent robotics technologies. The project can be a detailed simulation study, the fabrication of a robotic prototype, the development of an intelligent system to automate a task, etc. The students are expected to go through the following stages of work: • Problem identification • Literature review • Methodology of study • Project execution • Report writing • Project presentation																																																																																																
Teaching/Learning Methodology	The subject is taught through guided studies. The students are given the project title, objectives and description. The students are guided by the project supervisor to go through the different stages of the project as shown in the Subject Synopsis/Indicative Syllabus. (Outcomes a – l) <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="12">Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th><th>f</th><th>g</th><th>h</th><th>i</th><th>j</th><th>k</th><th>l</th></tr><tr><td>Guided study</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td></tr></table>	Teaching/Learning Methodology	Outcomes												a	b	c	d	e	f	g	h	i	j	k	l	Guided study	√	√	√	√	√	√	√	√	√	√	√	√																																																										
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Student Study Effort Expected	Class contact: ▪ Guided study Other student study effort: ▪ Conducting project ▪ Literature search and private study Total student study effort 26 Hrs. 154 Hrs. 72 Hrs. 252 Hrs.																													
Reading List and References	To be advised by supervisor																													
Develop Date	June 2024																													

Intended Blank

Elective Subjects

Subject Description Form

Subject Code	AAE4011
Subject Title	Artificial Intelligence in Unmanned Autonomous Systems
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: AAE2003 Introduction to Aircraft Systems
Objectives	This subject will provide students with 1. The main concepts, ideas, and techniques of advanced artificial intelligence (AI) in unmanned autonomous systems, e.g. unmanned aerial vehicles (UAV), unmanned ground vehicles (UGV); 2. The major components of typical unmanned autonomous systems fulfilling a certain function, such as environment inspection using UAVs; and 3. Expansive view into the technological trend of AI and its application in unmanned autonomous systems.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Apply AI algorithms or adopt AI tools in solving engineering problems in unmanned autonomous systems; b. Understand the relationship between multiple functions of unmanned autonomous systems, including perception, path planning, decision making, and control; c. Design partial of the functions of typical unmanned autonomous systems, such as positioning, and path planning; and d. Improve the existing AI algorithms to specific unmanned autonomous systems applications.
Subject Synopsis/ Indicative Syllabus	<u>Introduction to Artificial Intelligence:</u> The topic mainly includes the basic knowledge of machine learning such as conventional classification and regression together with high-level AI, such as convolutional neural network (CNN) for image segmentation. <u>Introduction to Unmanned Autonomous Systems:</u> The topic mainly includes the major existing applications of unmanned autonomous systems, such as UAV and UGV. Meanwhile, the topic will include the basic knowledge of typical unmanned autonomous systems. <u>Optimisation Algorithm to Unmanned Autonomous Systems:</u> The topic mainly includes the optimisation algorithms such as Gauss-Newton used to solve the engineering problems related to unmanned autonomous systems. <u>Sensors for Unmanned Autonomous Systems:</u> The topic mainly introduces the typical sensors applicable to unmanned autonomous systems. The sensors include the light detection and ranging (LiDAR), inertial measurement unit (IMU), and camera. Basic algorithms for sensors-based positioning will be introduced.

	Navigation for Unmanned Autonomous Systems: The topic mainly include positioning and navigation for the unmanned autonomous system using simultaneous localisation and mapping (SLAM) using LiDAR sensors together with point cloud processing, registration, AI-aided Navigation for Unmanned Autonomous Systems: The topic mainly includes the application of AI in LiDAR SLAM using object detection in unmanned autonomous systems. Case Study (mini-group projects): A design project will be carried out for students to learn the deployment of AI in unmanned autonomous systems through practice.					
Teaching/Learning Methodology	Teaching is conducted through lectures and case studies (mini-group projects). Lectures are used to deliver advanced knowledge concerning various aspects of AI, data analysis, and its applications in unmanned autonomous systems The basic knowledge, research methodology, and theoretical models will be introduced. Case study will provide the understanding of how to address and formulate problems by using mathematical programming, artificial intelligence algorithms, and optimisation techniques in unmanned autonomous systems.					
	Research methodology, data analytics skills, and algorithm design skills are taught in class as well as the related real-life scenarios using data to enhance their research abilities.					
	Teaching/Learning Methodology	Intended subject learning outcomes to be covered				
		a	b	c	d	
		1. Lecture	✓	✓	✓	✓
2. Case Study	✓	✓	✓			
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed			
			a	b	c	d
	1. Homework assignment	20%	✓	✓	✓	✓
	2. Mini-group project	15%			✓	✓
	3. Test	15%	✓	✓		
	4. Examination	50%	✓	✓	✓	✓
	Total	100 %				

	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Overall Assessment: $0.50 \times \text{End of Subject Examination} + 0.50 \times \text{Continuous Assessment}$ The examination is adopted to assess students on the overall understanding and the ability to apply the concepts. It is supplemented by continuous assessment including assignments, closed-book tests, and mini-group projects. The continuous assessment is aimed at enhancing the students' comprehension and assimilation of various topics of the syllabus. In particular, a group mini-project is used to assess the students' capacities of self-learning and problem-solving, and effective communication skills in English to fulfill the requirements of being aircraft design engineers.	
Student Study Effort Expected	Class contact:	
	▪ Lecture	33 Hrs.
	▪ Case Study	6 Hrs.
	Other student study effort:	
	▪ Literature review / case study / reading	36 Hrs.
	▪ Self-study / preparation	36 Hrs.
	The total student study effort	111 Hrs.
Reading List and References	1. Guido, S., & Müller, A. (2016). Introduction to machine learning with python (Vol. 282). O'Reilly Media. 2. Marsland, S. (2015). Machine learning: an algorithmic perspective. CRC press. 3. Zhang, Tao, Qing Li, Chang-shui Zhang, Hua-wei Liang, Ping Li, Tian-miao Wang, Shuo Li, Yun-long Zhu, and Cheng Wu. "Current trends in the development of intelligent unmanned autonomous systems." Frontiers of information technology & electronic engineering 18, no. 1 (2017): 68-85. 4. Barfoot, Timothy D. State estimation for robotics. Cambridge University Press, 2017. 5. Thrun, S. (2002). Probabilistic robotics. Communications of the ACM, 45(3), 52-57.	

Mar 2023

Subject Description Form

Subject Code	COMP3011				
Subject Title	Design and Analysis of Algorithms				
Credit Value	3				
Level	3				
Pre-requisite / Co-requisite / Exclusion	Pre-requisite: COMP2011/COMP2013/DSAI2201/EIE3320 or equivalent				
Objectives	The objectives of this subject are to: 1. provide students with in-depth knowledge on algorithm design techniques; and 2. introduce and practice advanced algorithms for various data types.				
Intended Learning Outcomes	Upon completion of the subject, students will be able to: <u>Professional/academic knowledge and skills</u> (a) understand common techniques for designing algorithms; (b) acquire the skills to design efficient algorithms for solving computational problems; (c) analyse and compare the efficiency of algorithms; (d) design and implement efficient algorithms for solving computing problems in a high-level programming language (e.g., C++ or Java); <u>Attributes for all-roundedness</u> (e) solve problems independently; and (f) think critically for improvement in solutions.				
Subject Synopsis/ Indicative Syllabus	<table><tr><td>Topic</td></tr><tr><td>1. Analysis of algorithms Mathematical techniques; big-O notation; efficiency analysis; recurring relations.</td></tr><tr><td>2. Advanced Algorithmic Design Techniques Dynamic programming, divide-and-conquer, branch-and-bound, greedy algorithm.</td></tr><tr><td>3. Advanced Analysis Techniques Introduction to randomised algorithms, probabilistic analysis, amortised analysis.</td></tr></table>	Topic	1. Analysis of algorithms Mathematical techniques; big-O notation; efficiency analysis; recurring relations.	2. Advanced Algorithmic Design Techniques Dynamic programming, divide-and-conquer, branch-and-bound, greedy algorithm.	3. Advanced Analysis Techniques Introduction to randomised algorithms, probabilistic analysis, amortised analysis.
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3. Advanced Analysis Techniques Introduction to randomised algorithms, probabilistic analysis, amortised analysis.					

	4. Computational Geometry Algorithms Spatial range searching, indexing of spatial objects, convex hull, closest pairs																																																										
	5. NP-Complete Problems Complexity classes, NP-completeness, reduction, approximation algorithms.																																																										
Teaching/ Learning Methodology	Lectures provide students the main concepts of the topic, together with comprehensive examples for easy understanding. Tutorials and lab sessions offer an opportunity to students for practicing their algorithmic analysis, design, and implementation techniques. Both written and programming assignments will be utilised in the course. Written assignments help students develop analysis and design skills, whereas programming assignments emphasise on implementation skills.																																																										
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighti ng</th><th colspan="6">Intended subject learning outcomes to be assessed</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th><th>f</th></tr><tr><td>Continuous Assessment</td><td rowspan="3">30%</td><td colspan="6"></td></tr><tr><td>1. Assignments</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td></tr><tr><td>2. Mid-Term / Tests</td><td>✓</td><td>✓</td><td>✓</td><td></td><td>✓</td><td>✓</td></tr><tr><td>Examination</td><td>70%</td><td>✓</td><td>✓</td><td>✓</td><td></td><td>✓</td><td>✓</td></tr><tr><td>Total</td><td>100%</td><td colspan="6"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: All three items are relevant to the assessment of the use of algorithms for problem solving, as well as their efficiency analysis (for items a, b, c). In addition, programming exercises in assignments are used to assess implementation skills (for item d); whereas the mid-term / tests and the examination are used to assess independent problem solving and critical thinking skills (for items e, f).							Specific assessment methods/tasks	% weighti ng	Intended subject learning outcomes to be assessed						a	b	c	d	e	f	Continuous Assessment	30%							1. Assignments	✓	✓	✓	✓	✓		2. Mid-Term / Tests	✓	✓	✓		✓	✓	Examination	70%	✓	✓	✓		✓	✓	Total	100%						
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Student Study Effort Expected	Class contact: <table><tr><td>▪ Lecture</td><td>26 Hrs.</td></tr><tr><td>▪ Tutorial/Lab</td><td>13 Hrs.</td></tr></table> Other student study effort: <table><tr><td>▪ Assignments (Written and Programming)</td><td>65 Hrs.</td></tr><tr><td>Total student study effort</td><td>104 Hrs.</td></tr></table>							▪ Lecture	26 Hrs.	▪ Tutorial/Lab	13 Hrs.	▪ Assignments (Written and Programming)	65 Hrs.	Total student study effort	104 Hrs.																																												
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Reading List and References	Textbook: 1. Cormen, Thomas H., Leiserson, Charles E., Rivest, Ronald L. and Stein, Clifford, <i>Introduction to Algorithms</i>, 3rd Edition, MIT Press, 2009. Reference Books: 1. Goodrich, M.T., and Tamassia, R., <i>Data Structures and Algorithms in Java</i>, 3rd Edition, John Wiley, 2005. 2. Carrano, Frank M., <i>Data Abstraction & Problem Solving with C++: Walls & Mirrors</i>, Addison Wesley, 2007. 3. Jon M. Kleinberg, Éva Tardos: Algorithm design. Addison-Wesley 2006, ISBN 978-0-321-37291-8.
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Subject Description Form

Subject Code	COMP3423
Subject Title	Human Computer Interaction
Credit Value	3
Level	3
Pre-requisite / Co-requisite / Exclusion	Pre-requisite: COMP1011/COMP1012/ENG2002 Exclusion: COMP2222
Objectives	The objectives of this subject are to: 1. provide students with a broad view of both theoretical and practical issues in human factors for design of human-computer interfaces; 2. equip students with knowledge and understanding of the nature of human computer interactions, human characteristics, computer system and interface architecture; 3. equip students with sound skills in design and evaluation of user interfaces; 4. equip students with computing techniques and paradigms in interface and interaction development; and 5. provide students with a broad view of the state of interactive software development in today's industry.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: <u>Professional/academic knowledge and skills</u> (a) identify the human factors and the theoretical issues involved in human-computer interaction design; (b) apply the theoretical design principles to the design and evaluation of user interfaces; (c) collect user requirements, design an innovative human-computer interface according to these requirements, and evaluate the design; (d) design and develop computer systems for different kinds of human interaction; <u>Attributes for all-roundedness</u> (e) solve problems by using systematic approaches; (f) solve problems in groups; and (g) write technical reports and present the findings.

Subject Synopsis/ Indicative Syllabus	Topic
	1. Introduction to Human Computer Interaction (HCI) Definitions and importance of HCI; historical context of HCI; roles various disciplines play within HCI.
	2. Human Information Processing Perception and representation; models and limits of human memory; mental models; use of metaphors; user aspects of language, social and organisational aspects; input and output devices: response time, performance characteristics (human and system).
	3. Visual Perception, Computer Graphics and Multimodal Interface Characteristics of human visual system; multimedia and multimodal interface; psychological aspects of multimodal interface; advancements in computer graphics and their impact on multimodal interfaces.
	4. Language, Communication and Dialogue Dialogue interaction: types and techniques; non-graphical dialogues; prototyping and user-centered design; structural HCI design; design rationale; speech dialogue and basic natural language processing.
	5. Ergonomics and I/O Devices Definitions and importance of ergonomics; physical ergonomics and cognitive ergonomics; input devices; output devices; advancements in I/O devices.
	6. Evaluation and Testing Importance of evaluation; standards and performance metrics; evaluation techniques; variable types; experiments design; basic statistics related to evaluation.
Teaching/ Learning Methodology	<u>Lectures, Tutorials and Labs</u> The subject material will be delivered through lectures, tutorials and labs. Lectures will provide the main body of the subject materials. Where possible, guest lectures and/or case studies will be used to give the subject material more relevancy to real-world scenarios. Tutorials and labs will provide students with more in-depth opportunities to explore the lecture materials and practice the lecture concepts. Where possible, a hands-on, interactive approach will be used. <u>Projects and Assignments</u> Projects and assignments will provide students with in-depth opportunities to practice the lecture concepts, as well as to assess their ability to apply these concepts in practical scenarios. <u>Examinations</u> Examinations will assess students on their grasp of subject material.

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed						
			a	b	c	d	e	f	g
	Continuous Assessment	30%							
	1. Assignments and/or Project		✓	✓	✓	✓	✓	✓	✓
	Final Examination	70%	✓	✓		✓	✓		
	Total	100%							
	The course will be accessed by assignments, projects, and tests.								
	Assignments are designed to reinforce the concepts and methods learned in the class. Projects are used to develop students’ analytic and problem-solving skills. The written part of the assignments and projects helps student develop their organisation and documentation skills. The oral part of the coursework allows students to present their ideas and communicate effectively to the audience. Examinations are used to assess independent problem solving and critical thinking skills.								
Student Study Effort Expected	Class contact:								
	▪ Lecture, Tutorials, Workshops and Labs						39 Hrs.		
	Other student study effort:								
	▪ Assignments, Coursework, Reading, Exam						66 Hrs.		
	Total student study effort						105 Hrs.		
Reading List and References	Textbook:								
	1. Shneiderman, B., Plaisant, C., Cohen, M., Jacobs, S., Elmqvist, N., and Diakopoulos, N. Designing the User Interface: Strategies for Effective Human-Computer Interaction, 6th Edition, Pearson, 2016.								
	Reference Books:								
	1. Dix, J. Finlay, Abowd, G., and Beale, R., <i>Human-Computer Interaction</i> , 3 rd Edition, Prentice Hall, 2004.								
	Reading List:								
	1. Azmandian, M., Hancock, M., Benko, H., Ofek, E., & Wilson, A. D. (2016). Haptic retargeting: Dynamic repurposing of passive haptics for enhanced virtual reality experiences. In Proceedings of the 2016 chi conference on human factors in computing systems (pp. 1968-1979).								
	2. Hoorn, J. F. (2014). Stakeholder logistics of an interactive system. Journal of Systems and Software, 95, 52-69. doi:10.1016/j.jss.2013.11.1095								

	3. Slater, M. (2009). Place illusion and plausibility can lead to realistic behaviour in immersive virtual environments. Philosophical Transactions of the Royal Society B: Biological Sciences, 364(1535), 3549-3557.
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Subject Description Form

Subject Code	EE3002 / EE3002A / EE3002B
Subject Title	Electromechanical Energy Conversion
Credit Value	3
Level	3
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite for EE3002: EE2002 Pre-requisite for EE3002A: EE2002A Pre-requisite for EE3002B: EE2002B
Objectives	1. To provide students a general knowledge on common types of electric machines. 2. To provide students the basic techniques of steady-state electric machine analysis.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Explain the construction, operating principles, performance characteristics, control and applications of major types of rotating electric machines. b. Analyse the steady-state performance of electric machines using appropriate equivalent circuit models. c. Operate practical electric machines and to conduct relevant tests and experiments. d. Present results of electric machine studies in the form of tables, graphs, and written reports.
Subject Synopsis/ Indicative Syllabus	1. Introduction: Principles of motors and generators. Materials for electric machines. Types of electric machines and applications. Losses and efficiency. 2. Machine rating: Temperature rise and cooling methods. Heating and cooling curves. Thermal ratings. Machine nameplate. 3. Windings: Phase and commutator windings. Winding factors. E.M.F. equation. Harmonics. Production of rotating magnetic field. 4. D.C. machines: Construction. E.M.F equation. Armature reaction and commutation. Characteristics of shunt, series and compound machines. Testing. Speed control. Universal motor. Brushless d.c. motor. 5. Synchronous machines: Construction. Synchronous impedance. Voltage regulation. Synchronising. Performance on infinite busbars. Power/load angle relationship. Stability. Synchronous motor. 6. Induction machines: Squirrel cage and wound-rotor types. Equivalent circuit. Torque-slip relationship. Starting, braking and generating. Testing. Speed control. Single-phase induction motors. Laboratory Experiments: Load test, efficiency and speed control of a d.c. motor. Performance evaluation of a three-phase cage induction motor. Synchronous generator synchronization.

Teaching/Learning Methodology	Delivery of the subject is mainly through formal lectures and complemented by tutorials. Excel programmes are used to clarify concepts of electric machines learnt and for conducting ‘what-if’ analysis. Laboratory work provides students hands-on experience in operation and control of practical machines, while report-writing enables students to practise written and graphic presentation skills.					
	Teaching/Learning Methodology		Outcomes			
			a	b	c	d
	Lectures	✓	✓	✓		
	Tutorials	✓	✓			
	Laboratory work		✓	✓	✓	
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed			
			a	b	c	d
	1. Examination	60%	✓	✓	✓	✓
	2. Mid-term Test	24%	✓	✓	✓	
	3. Laboratory work and reports	16%		✓	✓	✓
	Total	100%				
	It is a fundamental subject on electric machines and transformers. The outcomes on concepts, operating principles and applications are assessed by the usual means of assignment, tests, and examination. The outcomes on practical operation of electric machines and technical communication are evaluated by laboratory work and reports.					
Student Study Effort Expected	Class contact:					
	▪ Lecture/Tutorial			33 Hrs.		
	▪ Laboratory			6 Hrs.		
	Other student study effort:					
	▪ Revision, self-study, and assignment			48 Hrs.		
	▪ Write-up of laboratory reports			18 Hrs.		
	Total student study effort			105 Hrs.		
Reading List and References	Reference books: 1. M.S. Sarma And M.K.Pathak, “Electric Machines”, Cengage Learning, 2012. 2. S.A. Nasar, Schaum’s Outline of Theory and Problems of Electric Machines and Electromechanics, 2 nd Edition, McGraw-Hill, 1998					

July 2024

Subject Description Form

Subject Code	EIE3103
Subject Title	Digital Signals and Systems
Credit Value	3
Level	3
Pre-requisite	EIE2108 Fundamentals of Internet and Multimedia Technologies or EIE2112 Foundation Techniques In Artificial Intelligence
Co-requisite/ Exclusion	Nil
Objectives	1. To provide students with basic concepts and techniques for the modelling and analysis of discrete-time signals and systems. 2. To provide students with an analytical foundation for further studies in Communication Engineering and Digital Signal Processing.
Intended Subject Learning Outcomes	Upon completion of the subject, students will be able to: <u>Category A: Professional/academic knowledge and skills</u> 1. Understand the representations and classifications of digital signals and systems. 2. Understand the modelling of linear discrete-time systems. 3. Use different techniques to analyze and design discrete-time systems. 4. Apply software tools to laboratory exercises for experimenting with theories, and to the analysis and design of discrete-time systems. 5. Appreciate the advantages and disadvantages of using the different representations and modelling approaches. <u>Category B: Attributes for all-roundedness</u> 6. Present ideas and findings effectively.
Subject Synopsis/ Indicative Syllabus	Syllabus: 1. <u>Fourier Representations for Discrete-time Signals</u> Mathematical Description of Discrete-Time Signals. Discrete Fourier Series. Discrete-Time Fourier Transform. Discrete Fourier Transform. Relationship Among Various Fourier Transforms. 2. <u>Discrete-Time Systems</u> Time-Domain Analysis of Discrete-Time Systems. Unit pulse response. Difference Equation Representation. Convolution. 3. <u>System Analysis</u> Frequency Response of LTI Discrete-Time Systems. Concept of Filtering: Lowpass, Bandpass and Highpass Filters. FIR Filters and IIR Filters. Linear and Circular Convolution. FIR Filter Analysis. Filtering Examples to Different Signals. 4. <u>z-Transform</u> Definition and Properties of z-Transform. Inverse z-Transform: Power Series Expansion, Partial-Fraction Expansion. z-Transfer Analysis of LTI Systems. 5. <u>Filter design</u> FIR filter design using windows, FIR design by frequency sampling, etc. Laboratory Experiments:

	1. Linear Time-Invariant Discrete-time Systems 2. Fourier Analysis of Discrete-time Signals 3. Convolution and Correlation 4. Application of Digital Filters							
Teaching/ Learning Methodology	Teaching and Learning Method	Intended Subject Learning Outcome	Remarks					
	Lectures	1, 2, 3, 5	Fundamental principles and key concepts of the subject are delivered to students.					
	Tutorials	1, 2, 3, 5	These are supplementary to lectures; Students will be able to clarify concepts and to gain a deeper understanding of the lecture material; Problems and application examples are given and discussed.					
	Laboratory sessions	4, 6	Students will make use of the software MATLAB and/or LabView to simulate various theories and visualize the results.					
Assessment Methods in Alignment with Intended Subject Learning Outcomes	Specific Assessment Methods/ Task	% Weighting	Intended Subject Learning Outcomes to be Assessed (Please tick as appropriate)					
			1	2	3	4	5	6
	1. Continuous Assessment	50%						
	• Laboratory sessions	14%				✓		✓
	• Short quizzes	18%	✓	✓	✓		✓	
	• Tests	18%	✓	✓	✓		✓	✓
	2. Examination	50%	✓	✓	✓		✓	✓
	Total	100%						
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:							
	Specific Assessment Methods/Tasks	Remark						
	Short quizzes	These can measure the students' understanding of the theories and concepts as well as their comprehension of subject materials.						
	Tests and examination	End-of-chapter-type problems are used to evaluate the students' ability in applying concepts and skills learnt in the classroom; Students need to think critically and to learn independently in order to come up with an alternative solution to an existing problem.						

	Laboratory sessions	Oral examination based on the laboratory exercises will be conducted to evaluate student's technical knowledge and communication skills.
Student Study Effort Expected	Class contact (time-tabled):	
	• Lecture	24 Hours
	• Tutorial/Laboratory/Practice Classes	15 Hours
	Other student study effort:	
	• Lecture: preview/review of notes; homework/assignment; preparation for test/quizzes/examination	36 Hours
	• Tutorial/Laboratory/Practice Classes: preview of materials, revision and/or reports writing	30 Hours
	Total student study effort:	105 Hours
Reading List and References	References: 1. M.J. Roberts, <i>Fundamentals of Signals & Systems</i> , McGraw-Hill, 2008. 2. James H. McClellan, Ronald W. Schafer and Mark A. Yoder, <i>DSP First: A Multimedia Approach</i> , Prentice-Hall, 1999.	
Last Updated	July 2024	
Prepared by	Dr Chris Chan	

Subject Description Form

Subject Code	EIE3373
Subject Title	Microcontroller Systems and Interface
Credit Value	3
Level	3
Pre-requisite/ Co-requisite/ Exclusion	EIE2261 Logic Design
Objectives	To provide students with the concepts and techniques required in designing computer hardware interfaces and embedded software for microcontrollers.
Intended Subject Learning Outcomes	Upon completion of the subject, students will be able to: <u>Category A: Professional/academic knowledge and skills</u> 1. Understand the architecture of 8-bit and 32-bit microcontrollers. 2. Use the C programming language in developing programs for the use of microcontrollers. 3. Apply basic skills for interfacing common devices to microcontrollers. <u>Category B: Attributes for All-roundedness</u> 4. Present ideas and findings effectively. 5. Think critically and creatively.
Subject Synopsis/ Indicative Syllabus	Syllabus: 1. Overview of Typical Microcontrollers: Features and architectures of 8-bit and 32-bit microcontrollers; hardware connections, hex file and flash loaders; overview of different built-in devices in a microcontroller; 2. Software Development Environment: Understand C compilers, microcontroller programming in C. 3. Microcontroller Programming: I/O programming, timer/counter programming, interrupt programming, serial port programming, programming for other (built-in) devices connected to microcontrollers. 4. Laboratory Exercises: I/O programming, timer/counter programming, interrupt programming, serial port programming, programming for other (built-in) devices connected to microcontrollers.

Teaching/ Learning Methodology	Teaching and Learning Method	Intended Subject Learning Outcome	Remarks				
	Lectures	1,2,3	Fundamental principles and key concepts of the subject are delivered to students				
	Laboratory sessions	1,2,3,4,5	Students will make use of software and hardware tools to carry out laboratory assignments				

Assessment Methods in Alignment with Intended Subject Learning Outcomes	Specific Assessment Methods/ Task	% Weightin g	Intended Subject Learning Outcomes to be Assessed (Please tick as appropriate)				
			1	2	3	4	5
	1. Continuous Assessment (Total: 50%)						
	• Laboratory Exercises	20%	✓	✓	✓	✓	✓
	• Tests	30%	✓	✓	✓		
	2. Examination	50%	✓	✓	✓		
	Total	100%					
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:						
	Specific Assessment Methods/Tasks	Remark					
	Assignments	Enhance the understanding of the taught materials in the lectures					
Tests and examination	End-of chapter type problems are used frequently to evaluate students’ ability in applying concepts and skills learned in class The students are also needed to think critically and creatively in the process of solving problems						
Laboratory sessions	Each student is required to illustrated their achievement and produce a detailed work record when presenting his/her demonstrations						

		Students are also needed to think critically and creatively to accomplish certain laboratory assignments
Student Study Effort Expected	Class contact (time-tabled):	
	• Lecture	24 Hours
	• Tutorial/Laboratory/Practice Classes	33 Hours
	Other student study effort:	
	• Lecture: preview/review of notes; homework/assignment; preparation for test/quizzes/examination	24 Hours
	• Tutorial/Laboratory/Practice Classes: preview of materials, revision and/or reports writing	24 Hours
	Total student study effort:	105 Hours
Reading List and References	Reference Books: 1. The AVR Microcontroller and Embedded Systems: Using Assembly and C, M. A. Mazidi, S. Naimi, and S. Naimi, Pearson, 2014. 2. The Definitive Guide To The ARM Cortex-M3, Joseph Yiu, 2nd edition, Newnes, 2010.	

July 2025

Subject Description Form

Subject Code	ENG4001
Subject Title	Project Management
Credit Value	3
Level	4
Pre-requisite/Co-requisite/Exclusion	Nil
Objectives	This subject provides students with knowledge in: 1. project management tools in business organizations, taking into account the time-cost relationships, resources, processes, risks, the project life cycle, organization, and management principles; 2. project management methodologies and their application; 3. choosing project variables for effective project management; and 4. various developments of project management.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. demonstrate good understanding of definition of a project, the characteristics and project life cycle; b. identify appropriate project variables and practices that are applicable to engineering projects; c. perform project planning, cost/resources estimation, evaluate and monitor of project progress; and d. propose project management solutions, taking into consideration the project objectives and constraints.
Subject Synopsis/ Indicative Syllabus	1. <u>Project Overview, Management Principles, and the Systems Approach</u> Characteristics of projects and project management. Management principles. Project organization. Team development. Systems concepts and principles. Conflict management. 2. <u>Project Methodologies and Planning Techniques</u> Constraints: time, cost, and technical performance. Work breakdown structure. Management of scope. Scheduling tools: Gantt charts, network analysis techniques, time-phased networks, CPA, PERT, and resource smoothing. 3. <u>Cost Estimation and Cost Control for Projects</u> Types of estimates. Budgeting project costs. Experience curve. Cost schedules and forecasts. Cost control systems. 4. <u>Evaluation and Control of Projects</u> Earned value measurement system. Managing project risks. Status reporting. Project closeout and termination.

Teaching/Learning Methodology	A mixture of lectures, tutorial exercises, case studies, and laboratory work are used to deliver the various topics in this subject. Some material is covered using a problem-based format where this advances the learning objectives. Other material is covered through directed study to enhance the students’ “learning to learn” ability. Some case studies are from best practices of projects, based on a literature review. They are used to integrate the topics and demonstrate to students how the various techniques are interrelated and applied in real-life situations.																																													
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="4">Intended subject learning outcomes to be assessed</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>1. Tutorial exercises/ written report</td><td>10%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>2. Oral presentation</td><td>10%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>3. End Term Test</td><td>20%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>4. Written examination</td><td>60%</td><td></td><td>✓</td><td>✓</td><td></td></tr><tr><td>Total</td><td>100%</td><td colspan="4"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Continuous assessment (1), (2), and (3): Test, written reports, oral presentation, and tutorial exercises are used to assess students’ understanding and application of the knowledge that they have learnt relative to learning outcomes (a), (b), (c) and (d). Written examination: questions are designed to assess learning outcomes (b) and (c).						Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed				a	b	c	d	1. Tutorial exercises/ written report	10%	✓	✓	✓	✓	2. Oral presentation	10%	✓	✓	✓	✓	3. End Term Test	20%	✓	✓	✓	✓	4. Written examination	60%		✓	✓		Total	100%				
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Total	100%																																													
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	4. Project Management Institute, 2013, <i>A Guide to the Project Management Body of Knowledge (PMBOK® Guide)</i>, Fifth Edition. 5. Smith, NJ (ed.) 2008. <i>Engineering Project Management</i>, Blackwell, Oxford
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(Revised) June 2024

Subject Description Form

Subject Code	ISE3018
Subject Title	Logistics Automation
Credit Value	3
Level	3
Pre-requisite/Co-requisite/Exclusion	Nil
Objectives	This subject provides students with 1. the understanding of artificial intelligence and automation concept; 2. the understanding of artificial intelligence and automation applications in facilitating logistics operations; and 3. the ability to identify potential applications of artificial intelligence and automation in logistics operations in practice.
Intended Learning Outcomes	Upon completion of the subject, students will be able to a. evaluate and improve the performance of terminal operations based on logistics engineering approach; b. analyze and design a solution to enhance warehouse operation efficiency and accuracy by using robots and automation technology; and c. analyze and develop a solution to support last-mile delivery operations by using drones and unmanned guided vehicles.
Subject Synopsis/ Indicative Syllabus	1. <u>Artificial Intelligence and Automation</u> Introduction to artificial intelligence and automation. 2. <u>Container Terminal Automation</u> Introduction to container terminal industries and container terminal operations. Applications of automation technologies, (e.g., automated cranes, automated vehicles, information systems) in yard, quay, and gate operations. 3. <u>Warehouse Automation</u> Warehouse and distribution center operations. Applications of autonomous robots, e.g., order picking robots and autonomous mobile robots, to enhance order picking, and goods storage and retrieval operations. 4. <u>Last-mile Automation</u> Last-mile delivery operations. Applications of drones and unmanned

	guided vehicles in last-mile delivery.				
Teaching/Learning Methodology	The teaching pedagogy of this subject is a combination of classroom teaching and experiential learning. It consists of a series of lectures and well-organized laboratory activities. Emphasizing is put on the acquisition of required skills and knowledge in artificial intelligence and automation applied in logistics industries. The lectures provide the basics and theories while the laboratory activities cover the skills following an interest-based approach.				
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed		
			a	b	c
	Quizzes and Midterm Test	40%	✓	✓	✓
	Laboratory Exercises	20%		✓	
	Final Examination	40%	✓	✓	✓
	Total	100%			
	Quizzes and midterm test are used to assess students’ understanding on the concepts of applying logistics automations in logistics operations. The laboratories are used to assess their ability on using logistics automation technology. At the end of the subject, an examination is given to students to assess their learning outcomes.				
Student Study Effort Expected	Class contact:				
	▪ Lecture	3 hours/week for 7 weeks		21 Hrs.	
	▪ Tutorial/Laboratory	3 hours/week for 6 weeks		18 Hrs.	
	Other student study effort:				
	Self-study				
	▪ Laboratory Reports			39 Hrs.	
	▪ Preparation for Tests and Examination			39 Hrs.	
	▪ Total student study effort			117 Hrs.	
Reading List and References	1. Jerry Kaplan 2016, <i>Artificial Intelligence</i> , New York, NY: Oxford University Press. 2. Eric Su 2016, <i>Operational Risk Management in Container Terminals</i> , Abingdon, Oxon, New York, NY: Routledge. 3. Ning Zhao, Yuan Liu, Weijian Mi 2020, <i>Digital Management of Container Terminal Operations</i> , Singapore: Springer. 4. Gwynne Richards 2018, <i>Warehouse Management: A Complete Guide to Improving Efficiency and Minimizing Costs in the Modern Warehouse</i> , London, United Kingdom: Kogan Page Limited.				

	5. Christian Wurst, Luca Graf 2021, <i>Disrupting Logistics: Startups, Technologies, and Investor Building Future Supply Chains</i>, Springer International Publishing. 6. Albert Causo 2020, <i>Advances on Robotic Item Picking: Applications in Warehousing & e-commerce Fulfillment</i>, Springer.
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27.7.2022

Subject Description Form

Subject Code	ISE3021
Subject Title	Warehousing Technologies and Operations
Credit Value	2
Level	3
Pre-requisite/Co-requisite/Exclusion	Nil
Objectives	This subject provides students with 1. the basic knowledge of warehouse technologies and operations; 2. various techniques to improve warehouse operations including hardware and software; 3. skills to establish performance measures in relation to warehouse operations.
Intended Learning Outcomes	Upon completion of the subject, students will be able to a. understand warehouse operations and applications of advance hardware; b. apply various techniques to improve warehouse operations; c. formulate performance measures for warehouse to monitoring.
Subject Synopsis/ Indicative Syllabus	1. <u>Introduction to warehouse operations and management</u> Purpose and functions of warehouse. Role of warehouse in supply chain. Basic facilities in warehouse. Warehouse Management Systems (WMS). Simulation of warehouse operations/inventory control. 2. <u>Techniques for warehouse operations and management</u> Inventory management and optimization (EOQ, Safety Stock, EPQ, and Quantity Discount model). Stock management policies (random storage, delegated storage, stacking, and ABC analysis). Applications of dynamic programming in inventory control. 3. <u>Warehouse performance measures</u> Formulation of good Key Performance Indicators [KPIs]. KPIs for warehouse activity efficiency, inventory management, service quality, order and cost control. Analysis of warehouse performance indicators. 4. <u>Advance hardware applications and programming</u> Use of robots, image processing and applications, utilizations of hardwares in Automated Storage and Retrieval Systems (ASRS)

	operations.							
Teaching/Learning Methodology	The learning method of this subject consists of class lectures and laboratory exercises. The lectures are employed to help students to understand the fundamental principles and techniques in related to field of warehouse technologies and operations. Midterm test is given to testify issues may arise from the lectures and to assist the understanding of analytical skills. Laboratory exercises are incorporated to give students more practical familiarity on relevant areas and to exercise the applications of simulation in warehouse operations.							
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weightin g	Intended subject learning outcomes to be assessed					
			a	b	c			
	Laboratory Reports	40%	✓	✓				
	Midterm Test	20%		✓	✓			
	Final Examination	40%	✓	✓	✓			
	Total	100%						
	Laboratory reports will be given at each laboratory class, students need to submit them later, which measure the learning outcome a and b. Midterm test will be given to check the students’ performance of learning outcomes b, and c, the testing results will be analyzed and some improvements will be made during the rest of lectures. At the end of this subject, an examination will be given to obtain the students’ all learning outcomes.							
Student Study Effort Expected	Class contact:							
	▪ Lecture/Seminar /Tutorial	2 hours/week for 10 weeks				20 Hrs.		
	▪ Laboratory	2 hours/week for 3 weeks				6 Hrs.		
	Other student study effort:							
	▪ Self Study					30 Hrs.		
	▪ Laboratory Report/Tutorial					30 Hrs.		
	Total student study effort					86 Hrs.		
Reading List and References	1. Gwynne Richards 2011, <i>Warehouse Management: A Complete Guide to Improving Efficiency and Minimizing Costs in the Modern Warehouse</i> , Kogan Page [ISBN-10: 0749460741] 2. David J. Piasecki 2021, <i>Inventory Accuracy: People, Processes, & Technology</i> , OPS Publishing, 2 nd Edition [ISBN- 978-0972763134]							

	3. John J. Bartholdi, III, Steven T. Hackman 2017, <i>Warehouse & Distribution Science</i>, www.warehouse-science.com. 4. Harold R. Kerzner 2013, <i>Project Management Metrics, KPIs, and Dashboards: A Guide to Measuring and Monitoring Project Performance</i>, Wiley [ISBN-10: 1118524667]
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27.3.2023

Subject Description Form

Subject Code	ISE4024
Subject Title	Robotics and Automation Systems
Credit Value	3
Level	4
Pre-requisite/Co-requisite/Exclusion	ISE3019 Industrial Control Systems and Their Applications
Objectives	This subject provides students with 1. the knowledge and skills to develop low cost automation systems; 2. the techniques to evaluate the feasibility and effectiveness of alternative automation strategies; and 3. the principle and application of industrial robots.
Intended Learning Outcomes	Upon completion of the subject, students will be able to a. develop low cost automation systems using computational, mechanical, and electrical means; b. apply industrial robots, programmable logic controllers, and microprocessors to develop solutions for robotics and automation systems; and c. apply computing vision control technology for industrial process automation.
Subject Synopsis/ Indicative Syllabus	1. <u>Automation Process</u> Association of smart sensors and system elements to support automated process in systems. Facile methods for machine vision. 2. <u>Motor Control Mechanism for Industrial Applications</u> DC motor control using Pulse Width Modulation (PWM) technique, Stepper motor control, Servo motor control, Linear actuator control in Industrial applications. 3. <u>Programmable Logic Controllers and Automatic Assembly Systems</u> Introduction to programmable logic applications in automation systems. Computer applications in industrial systems. Assembly process automation: classification of assembly systems, transfer and feeding mechanisms, and robotic assembly.

Teaching/Learning Methodology	The learning method of this subject consists of a series of class lectures and laboratory exercises. Emphasis is put on development of practical hand-on skills and context in automation with theoretical backup. The lectures facilitate students in understanding the fundamental concepts, working principles and techniques in related to automation processes while laboratory exercises are incorporated to give students practical familiarity on relevant areas.				
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed		
			a	b	c
	Assignment and Quizzes	25%	✓	✓	
	Midterm Exam	15%	✓	✓	
	Laboratory Exercise	20%	✓	✓	✓
	Final Exam	40%	✓	✓	✓
	Total	100%			
	Each laboratory exercise would be divided into teams of size of 2 to 3; the individual report can be hand-in afterwards. Tutorials would be provided to students to improve their understanding of the subject materials and final examination will be given to assist the assessment of students' learning outcomes.				
Student Study Effort Expected	Class contact:				
	▪ Lecture/Seminar	3 hours/week for 7 weeks	21 Hrs.		
	▪ Tutorial/Laboratory	3 hours/week for 6 weeks	18 Hrs.		
	Other student study effort:				
	Self-study				
	▪ Laboratory Reports/Tutorial		39 Hrs.		
	▪ Preparation for Mid-term Test and Examination		39 Hrs.		
	Total student study effort				117 Hrs.

Reading List and References	1. Boothroyd, G, Knight, W, Dewhurst, P 2010, <i>Product Design for Manufacture and Assembly</i>, Third Edition (Manufacturing Engineering and Materials Processing), CRC Press 2. Kilian C T 2004, <i>Modern Control Technology</i>, 3rd Edition, Thomson Delmar Learning 3. Clarence, W. de Silva 2007, <i>Sensors and Actuators: Control System Instrumentation</i>, CRC Press 4. Ebel, F, Idler, S, Prede, G, Scholz, D 2008, <i>Fundamentals of Automation Technology</i>, Festo Didactic GmbH & Co. KG, 73770 Denkendorf, Germany
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27.7.2022

Subject Description Form

Subject Code	ISE404
Subject Title	Total Quality Management
Credit Value	3
Level	4
Pre-requisite/Co-requisite/Exclusion	Students who do not have background knowledge in quality control and quality engineering should be prepared to do additional reading.
Objectives	This subject provides students with the knowledge to 1. understand the philosophy and core values of Total Quality Management (TQM); 2. determine the voice of the customer and the impact of quality on economic performance and long-term business success of an organization; 3. apply and evaluate best practices for the attainment of total quality.
Intended Learning Outcomes	Upon completion of the subject, students will be able to a. select and apply appropriate techniques in identifying customer needs, as well as the quality impact that will be used as inputs in TQM methodologies; b. measure the cost of poor quality and process effectiveness and efficiency to track performance quality and to identify areas for improvement; c. understand proven methodologies to enhance management processes, such as benchmarking and business process reengineering; d. choose a framework to evaluate the performance excellence of an organization, and determine the set of performance indicators that will align people with the objectives of the organization.
Subject Synopsis/ Indicative Syllabus	1. <u>Principles of Total Quality</u> Concepts of quality; Core values and paradigms for TQM, including corporate citizenship and protection of the environment; Models for performance excellence: Deming Prize, Baldrige Quality Award, European Quality Award 2. <u>Customer Needs</u> Internal and external customers; Voice of the customer; Customer satisfaction; Customer loyalty; Service recovery; Crisis management 3. <u>Economics of Quality</u> Classification and analysis of quality costs; Implementing quality costing

	systems; Economic value of customer loyalty and employee loyalty																																																					
	4. <u>TQM Methodologies</u> Quality Function Deployment (QFD); Benchmarking; Business process reengineering; Process improvement																																																					
	5. <u>Learning and Growth</u> Organizational learning; Organizational renewal; Change management; Employee empowerment																																																					
	6. <u>Strategic Quality Management</u> Vision, strategy, goals, and action plans; Measurement of organizational performance																																																					
Teaching/Learning Methodology	A mixture of lectures, group discussions (tutorials), and mini-case studies are used to achieve the objectives of this subject. Some topics are taught in the classroom environment; students have to learn these topics by themselves in the process of writing problem-based assignments. Directed study is also used to develop the self-learning ability of students.																																																					
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="6">Intended subject learning outcomes to be assessed</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th></th><th></th></tr><tr><td>1. Assignments/Case Studies</td><td>40%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td></tr><tr><td>2. Tests</td><td>15%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td></tr><tr><td>3.Examination</td><td>45%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td></tr><tr><td>Total</td><td>100%</td><td colspan="6"></td></tr></table> The assignments, reflective journals, essays, and case studies facilitate the application of concepts and skills learned in analyzing and attaining total quality while emphasizing factors that may affect decisions. Examination/tests allow students to demonstrate the extent of their understanding of concepts, as well as their abilities to analyze and solve problems related to the subject.								Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed						a	b	c	d			1. Assignments/Case Studies	40%	✓	✓	✓	✓			2. Tests	15%	✓	✓	✓	✓			3.Examination	45%	✓	✓	✓	✓			Total	100%						
Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed																																																				
		a	b	c	d																																																	
1. Assignments/Case Studies	40%	✓	✓	✓	✓																																																	
2. Tests	15%	✓	✓	✓	✓																																																	
3.Examination	45%	✓	✓	✓	✓																																																	
Total	100%																																																					
Student Study Effort Expected	Class contact:																																																					
	▪ Lecture/Tutorial		2 hours/week for 13 weeks				26 Hrs.																																															
	▪ Tutorial/Case Study		1 hour/week for 13 weeks				13 Hrs.																																															
	Other student study effort:																																																					

	▪ Studying and self learning	50 Hrs.
	▪ Assignment and report writing	28 Hrs.
	Total student study effort	117 Hrs.
Reading List and References	1. Oakland, John S, 2014, <i>Total Quality Management and Operational Excellence: Text with Cases</i>, Taylor and Francis 2. Besterfield, DH, et.al. 2003, <i>Total Quality Management</i>, 3rd edn, Prentice Hall 3. Goetsch, DL & Davis, B 2006, <i>Quality Management: Introduction to Total Quality Management for Production, Processing and Services</i>, 5th edn, Pearson 4. Gryna FM 2001, <i>Quality Planning & Analysis</i>, 4th edn, Jr., McGraw-Hill 5. Selected articles in Quality Progress and the web site of American Society for Quality	

10.1.2022

Subject Description Form

Subject Code	ME41001
Subject Title	Automatic Control Systems
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: ME31002 Linear Systems and Control
Objectives	To provide students with the fundamental knowledge of controller design for automatic control systems.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Formulate and solve problems relating to modeling of linear mechanical systems, analysis of system relative stabilities; determining specifications for open- or closed-loop control systems and designing controllers or compensators for mechanical systems. - Complete a given task such as a project in system modeling or controller design by applying knowledge acquired in the subject and information obtained through literature search. - Analyze and interpret data obtained from experiments in system modeling, stability analysis or frequency-domain analysis of mechanical systems. - Present effectively in completing written reports of laboratory work and the given task.
Subject Synopsis/ Indicative Syllabus	<i>Time Domain Controller Design</i> - Multi-mode controllers; Optimum controller settings; Ratio, cascade and feedforward control. <i>Frequency Domain Compensator Design</i> - Nyquist criterion; Phase and gain margins; Multiple design constraints; Characteristics of lead, lag and lag-lead elements; Compensator design via Bode plots. <i>State-Space Representation of Dynamic Systems</i> - State variables of a dynamic system; State differential equations; State-space form equations from transfer functions; Canonical forms and decoupled systems; Relationship between eigenvalues and system poles. <i>Control System Analysis Using State Variable Method</i> - Direct numerical solution of state equation; Solution using state transition matrix; System stability; Controllability and observability. <i>Control System Design Using State Variable Method</i> - State variable feedback; Direct calculation of gains by comparison with characteristic equation; Pole placement via control canonical form of state equations; Pole placement via Ackermann's formula. Laboratory Experiment

	There are two 2-hour laboratory sessions. Typical Experiments: 1. Twin-rotor control 2. Inverted pendulum control 3. DC servo control																																								
Teaching/Learning Methodology	Lectures aim at providing students with an integrated knowledge required for understanding controller or compensator design, analyzing and designing state-space control systems (Outcomes a and b). Tutorials aim at enhancing the analytical skills of the students. Examples on time-domain controller design, frequency domain compensator design, state-space system representation, analysis and controller design are used to illustrate the application of integrated knowledge to solve real-world problems (Outcomes a and b). Experiments will provide the students with experience on the use of simulation tools for the computer-aided analysis and controller design of typical state-space dynamic systems. It also trains students in the measurement and instrumentation, the analysis and presentation of experimental data (Outcomes c and d). <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="4">Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>Lecture</td><td>√</td><td>√</td><td></td><td></td></tr><tr><td>Tutorial</td><td>√</td><td>√</td><td></td><td></td></tr><tr><td>Experiment</td><td></td><td></td><td>√</td><td>√</td></tr></table>	Teaching/Learning Methodology	Outcomes				a	b	c	d	Lecture	√	√			Tutorial	√	√			Experiment			√	√																
Teaching/Learning Methodology	Outcomes																																								
	a	b	c	d																																					
Lecture	√	√																																							
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Experiment			√	√																																					
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="4">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>1. Examination</td><td>50%</td><td>√</td><td>√</td><td></td><td></td></tr><tr><td>2. Test</td><td>25%</td><td>√</td><td>√</td><td></td><td></td></tr><tr><td>3. Assignment</td><td>15%</td><td>√</td><td>√</td><td></td><td></td></tr><tr><td>4. Laboratory report</td><td>10%</td><td></td><td></td><td>√</td><td>√</td></tr><tr><td>Total</td><td>100%</td><td colspan="4"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Overall Assessment: 0.50 × End of Subject Examination + 0.50 × Continuous Assessment Examination is adopted to assess students on the overall understanding and the ability of applying the concepts. It is supplemented by the tests, assignments and laboratory reports which provide timely feedbacks to both lecturers and students on various topics of the syllabus.	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)				a	b	c	d	1. Examination	50%	√	√			2. Test	25%	√	√			3. Assignment	15%	√	√			4. Laboratory report	10%			√	√	Total	100%				
Specific assessment methods/tasks	% weighting			Intended subject learning outcomes to be assessed (Please tick as appropriate)																																					
		a	b	c	d																																				
1. Examination	50%	√	√																																						
2. Test	25%	√	√																																						
3. Assignment	15%	√	√																																						
4. Laboratory report	10%			√	√																																				
Total	100%																																								

Student Study Effort Expected	Class contact:	
	▪ Lecture	31 Hrs.
	▪ Tutorial/Laboratory	8 Hrs.
	Other student study effort:	
	▪ Course work	26 Hrs.
	▪ Self-study	45 Hrs.
	Total student study effort	110 Hrs.
Reading List and References	1. M. Gopal, Control Systems, Principles and Design, McGraw-Hill, latest edition. 2. N.S. Nise, Control Systems Engineering, Wiley, latest edition. 3. K. Ogata, Modern Control Engineering, Prentice Hall, latest edition.	

Revised July 2014

Subject Description Form

Subject Code	ME41003
Subject Title	Principles of Sound and Vibration
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: ME34002 Engineering Thermodynamics
Objectives	To provide students with the fundamental knowledge of generation and measurement of sound and vibration and the sound propagation.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Understand the physics of the vibration of simple structure and sound propagation in the acoustic medium, in duct and in room. - Formulate and solve the sound and vibration problem relating to vibration of string, beam and plate, sound radiation from the source, sound reflection and transmission through a junction and a flat interface of acoustic media by applying knowledge in noise mitigation method. - Understand the mechanisms of basic measurement devices for sound and vibration, analyze and interpret the measured data from the experiments of noise and vibration.
Subject Synopsis/ Indicative Syllabus	Fundamentals of Sound - Fluid compressibility, wave equation, sound pressure level and sound power, addition of sounds of different frequencies, octave bands and one-third octave bands, conservation of acoustic energy flux at the absence of a mean flow. Vibration of Continuous Systems - Vibration of string, rod, beams and plates; energy transmission through structures, natural modes, free and forced vibrations. Sources of Sound - Radiation of sound by pistons (1D, 2D), impedance, radiation efficiency, monopole and dipole, critical frequency, sound radiation by 2D structures. Sound Propagation - Single travelling wave and properties of standing wave, reflection of sound at pipe junctions and at interface of two media. Sound and Vibration Measurement - Measuring systems, microphones, sound level meters, background noise, measurement of sound intensity, reverberation time and absorption coefficient; accelerometers, calibration and mounting of accelerometers; shakers, hammers, force transducers and amplifiers; damping measurement, experimental modal analysis. Laboratory Measurement - Sound propagation in anechoic chamber - Impedance tube measurement - Experimental modal analysis of a vibrating beam - Traffic noise measurement

Teaching/Learning Methodology	Lectures are aimed at providing students with the knowledge of acoustics and vibration. (Outcomes a to c). Tutorials are aimed at enhancing students’ skills necessary for analyzing the physics of sound and vibration system (Outcomes a and b). Laboratory experiments are conducted to improve students’ ability to apply their knowledge to implement real engineering systems (Outcomes b and c). <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="3">Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th></tr><tr><td>Lecture</td><td>√</td><td>√</td><td>√</td></tr><tr><td>Tutorial</td><td>√</td><td>√</td><td></td></tr><tr><td>Experiment</td><td></td><td>√</td><td>√</td></tr></table>				Teaching/Learning Methodology	Outcomes			a	b	c	Lecture	√	√	√	Tutorial	√	√		Experiment		√	√														
Teaching/Learning Methodology	Outcomes																																				
	a	b	c																																		
Lecture	√	√	√																																		
Tutorial	√	√																																			
Experiment		√	√																																		
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="3">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th></tr><tr><td>1. Class test</td><td>20%</td><td>√</td><td>√</td><td>√</td></tr><tr><td>2. Homework</td><td>20%</td><td>√</td><td>√</td><td>√</td></tr><tr><td>3. Laboratory report</td><td>10%</td><td></td><td>√</td><td>√</td></tr><tr><td>4. Examination</td><td>50%</td><td>√</td><td>√</td><td>√</td></tr><tr><td>Total</td><td>100%</td><td colspan="3"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Overall Assessment: 0.50 × End of Subject Examination + 0.50 × Continuous Assessment Examination is used to assess students on the overall understanding and the ability of applying the knowledge. It is supplemented by tests, assignments and laboratory reports which provide timely feedbacks to both lecturers and students.				Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)			a	b	c	1. Class test	20%	√	√	√	2. Homework	20%	√	√	√	3. Laboratory report	10%		√	√	4. Examination	50%	√	√	√	Total	100%			
Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)																																			
		a	b	c																																	
1. Class test	20%	√	√	√																																	
2. Homework	20%	√	√	√																																	
3. Laboratory report	10%		√	√																																	
4. Examination	50%	√	√	√																																	
Total	100%																																				
Student Study Effort Expected	Class contact:																																				
	▪ Lecture		32 Hrs.																																		
	▪ Tutorial/Laboratory		7 Hrs.																																		
	Other student study effort:																																				
	▪ Reading and review		43 Hrs.																																		
	▪ Homework assignment		10 Hrs.																																		
	▪ Laboratory report		10 Hrs.																																		
	Total student study effort		102 Hrs.																																		

Reading List and References	1. L.E. Kinsler, et al., Fundamentals of Acoustics, Wiley, latest edition. 2. M.P. Norton, Fundamentals of Noise and Vibration Analysis for Engineers, Cambridge University Press, latest edition. 3. H. Benaroya, Mechanical Vibration: Analysis, Uncertainties and Control, Prentice-Hall, latest edition. 4. A.P. Dowling and J.E. Ffowcs Williams, Sound and Sources of Sound, Chichester: E. Horwood, latest edition. 5. L.L. Beranek, Noise and Vibration Control Engineering: Principles and Applications, Wiley, latest edition.
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Revised July 2014

Subject Description Form

Subject Code	ME41005
Subject Title	Noise Control Engineering
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: ME34002 Engineering Thermodynamics; and ME34004 Fluid Mechanics
Objectives	To provide students with fundamental concepts and knowledge of acoustic noise and control, including sound generation mechanism, noise abatement technology and applications
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Understand the sound generation mechanisms and the method to analyze the type of noise source. - Understand the simple sound fields and identify the noise sources and their respective mitigation measures. - Understand the importance and usage of the noise assessment criteria for typical problems such as duct and room noise applications. - Apply the state-of-the-art noise abatement technology and design elementary reactive and absorptive noise control device, analyze and interpret its performance from measurement. Understand basic principles in structural noise and aero-acoustic noise.
Subject Synopsis/ Indicative Syllabus	Noise Characteristics and its modeling – Sound and noise characterization, sound measure in time frequency domain, elementary noise source, modelling of acoustic waves, and various types of sound source models. Overview of control strategy for different frequency ranges. Sound Reflection and Absorption – Sound propagation in different acoustic media, typical sound propagation phenomena and characterization, duct acoustics, sound reflection by expansion chamber, Helmholtz resonator, sound absorbing materials and absorbers, design of reactive silencers, acoustic enclosures etc. Flow-induced Noise and Control – Von Karman vortices, turbulence noise, cavitations, jet noise, fan noise etc. Structure-induced Noise and Control – Basic sound radiation phenomena, vibration isolation and absorption, sound transmission and mass law. Environmental Noise and Control – Basic concepts of sound propagation outdoors, absorption of sound in air; attenuation of sound over ground, temperature gradient etc. Noise reduction by sound barriers, Maekawa formula. Train noise, etc. Room Acoustic Control – Basic concepts of room acoustics, direct and diffuse sound field, reverberation time, Sabin formula, prediction of internal sound field and noise mitigation measures.

	Laboratory Experiment There is one 1-hour laboratory session. Typical experiment: 1. Helmholtz resonator 2. Expansion chamber						
Teaching/Learning Methodology	Lectures are aimed at providing students with the knowledge of acoustics and noise control for achieving the subject outcomes. (Outcomes a to d)						
	Tutorials are aimed at enhancing students’ skills necessary for analyzing and designing the noise control method. (Outcomes a, b and d)						
	Laboratory experiments are conducted to improve students’ ability to apply their knowledge to implement real engineering systems, to develop the students’ interest and curiosity in the design of noise control method. (Outcomes b to d)						
	Teaching/Learning Methodology		Outcomes				
			a	b	c	d	
	Lecture		√	√	√	√	
	Tutorial		√	√		√	
Experiment			√	√	√		
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks		% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)			
				a	b	c	d
	1. Homework		20%	√	√		√
	2. Class test		20%	√	√	√	√
	3. Lab report		10%		√	√	√
	4. Examination		50%	√	√	√	√
	Total		100%				
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:						
Overall Assessment: 0.50 × End of Subject Examination + 0.50 × Continuous Assessment							
Examination is applied to assess students on understanding and the ability to apply the concepts. It is supplemented by the homework and laboratory reports which provide timely feedbacks to both lecturers and students on various topics of the syllabus.							

Student Study Effort Expected	Class contact:	
	▪ Lecture	33 Hrs.
	▪ Tutorial/Laboratory	6 Hrs.
	Other student study effort:	
	▪ Reading and review	44 Hrs.
	▪ Homework assignment	12 Hrs.
	▪ Laboratory report	10 Hrs.
	Total student study effort	105 Hrs.
Reading List and References	1. A.D. Pierce, Acoustics: an Introduction to its Physical Principles and Applications, Acoustical Society of America, Woodbury, N.Y., latest edition. 2. A.P. Dowling and J.E. Ffowcs Williams, Sound and Sources of Sound, Chichester: E. Horwood, latest edition. 3. L.L. Beranek, Noise and Vibration Control Engineering: Principles and Applications, Wiley, latest edition. 4. D.A. Bies and C.H. Hansen, Engineering Noise Control: Theory and Practice, E & FN Spon, latest edition.	
Last Update	January 2024	

Subject Description Form

Subject Code	ME41006
Subject Title	Perceptual Robotics
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: ME31002 Linear Systems and Control
Objectives	The subject aims to equip students with knowledge of: 1. Artificial robot perception 2. Perception-guided control 3. Adaptive robot behaviour 4. Perception-aided algorithms
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a) Able to mathematically model the different perceptual modalities used for robotic systems; b) Able to design perception-guided motion controls for mechanical robots c) Able to use perceptual feedback to implement adaptive robot behaviours d) Able to design perception-aided methods for learning properties about the environment e) Able to conduct experiments with perceptual and robotic systems
Subject Synopsis/ Indicative Syllabus	Artificial robot perception. Vision sensors (monocular perception and RGB-D sensors), thermal imaging (models and principles), touch (force and tactile imaging), proximity (different ranging methodologies), audio sensing. Perception-guided control. Sensor-motion coordination problem, derivation of sensorimotor models (analytical and computational), formulation of sensor servoing controls (vision-based, thermal-based, touch-based, proximity-based). Adaptive robot behaviour. Braitenberg machines, reactive motion paradigms (potential fields, subsumption architecture, etc.), hybrid paradigms, multi-agent systems, robot babbling, bug algorithms, sensor-based navigation. Perception-aided algorithms. Iterative closest point (ICP), simultaneous localisation and mapping (SLAM), sensor-based model learning, and image registration. Practical work. Various laboratory sessions will be delivered to provide students with hands-on experience.

Teaching/Learning Methodology	1. Lectures aim at providing students with fundamental knowledge required for understanding and analysing different perceptual robotic systems, including its mathematical models, controller design, and algorithms. (Outcomes a to d)						
	2. Tutorials aim at enhancing students’ analytical and problem solving skills on robotics. Students will be able to solve real-world problems using the knowledge they acquired in the class. (Outcomes a to d)						
	3. The experiments/project aim to provide hands-on experience for developing perceptual robots, and reinforcing the acquired knowledge. (Outcomes a to e)						
	Teaching/Learning Methodology	Intended Subject Learning Outcomes to be assessed					
		a	b	c	d	e	
1. Lecture	√	√	√	√			
2. Tutorial	√	√	√	√			
3. Experiments/Project	√	√	√	√	√		
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)				
			a	b	c	d	e
	1. Class test	10%	√	√	√	√	√
	2. Coursework: May include assignments, project, experiments, lab report	50%	√	√	√	√	√
	3. Final Examination	40%	√	√	√	√	
	Total	100 %					
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:						
	Overall Assessment:						
	0.40 x End of Subject Examination + 0.60 x Continuous Assessment.						
	1. The continuous assessment aims at evaluating the progress of the students’ study, assisting them in self-monitoring the respective learning outcomes, and applying the knowledge learnt in practical situations.						
2. The examination is used to assess the knowledge acquired by the students for understanding and analysing the problems critically and independently; as well as to determine the degree of achieving the subject learning outcomes.							
Student Study Effort Expected	Class contact:						
	▪ Lectures					33 Hrs.	
	▪ Tutorials / Laboratory					6 Hrs.	
	Other student study effort:						

	▪ Course work	40 Hrs.
	▪ Self-learning	36 Hrs.
	Total student study effort	115 Hrs.
Reading List and References	1. Lecture Notes, Articles, and Handouts 2. Computer Vision: A Modern Approach, David A. Forsyth and Jean Ponce, latest edition. 3. Introduction to AI Robotics, Robin Murphy, MIT Press Cambridge, MA, USA, latest edition. 4. Principles of Robot Motion: Theory, Algorithms, and Implementations, Howie Choset et al, MIT, latest edition. 5. Vehicles: Experiments in Synthetic Psychology, Valentino Braitenberg, MIT Press Ltd, latest edition. 6. Robotics Modelling, Planning and Control, Bruno Siciliano et al, latest edition.	
Last Update	July 2025	

Subject Description Form

Subject Code	ME42001
Subject Title	Artificial Intelligence in Products
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: ME31002 Linear Systems and Control ; or ME41004 Mechatronics and Control
Objectives	To provide students with basic knowledge on Artificial Intelligence (AI) for product design and development.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Apply knowledge of mathematics, expert systems, fuzzy systems and learning models to aid the design and development of intelligent products and systems. - Understand applications of AI in different fields. - Work effectively as a member to tackle a multi-disciplinary design project involving the application of AI. - Appreciate the state-of-the-art applications of AI in products and present a design project via computer programming and written report.
Subject Synopsis/ Indicative Syllabus	Expert Systems - Principles of expert systems; Knowledge representations; Inference mechanisms; Graph search algorithms; Rule-based expert systems; Application of expert systems to product design and product data management using Prolog or available software packages. (Delete some topics) Fuzzy Inference Systems - Fuzzy sets and crisp sets; Membership functions; Properties of fuzzy sets; Operations on fuzzy sets; Operations on fuzzy relations; Fuzzy if-then statements; Inference rules; Developing fuzzy inference systems using Matlab or available software packages. Learning Models – Introduction to different learning algorithms and models; Regression; Classification; Supervised learning; Unsupervised learning; Reinforcement learning; Neural Network; Deep learning; Developing learning models using Python or available software packages

Teaching/Learning Methodology	1. The lectures are aimed at providing fundamental knowledge on artificial intelligence for product design and development. (Outcomes a and b) 2. The tutorials/computer labs are aimed at enhancing applicable skills of the students. Examples of machine intelligence and other forms of AI in commercial products will be involved. (Outcomes a and b) 3. The project is aimed at integrating the knowledge that will be applied through a team project on product design and development with intelligence systems. (Outcomes a - d) <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="4">Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>Lecture</td><td>√</td><td>√</td><td></td><td></td></tr><tr><td>Tutorial</td><td>√</td><td>√</td><td></td><td></td></tr><tr><td>Project</td><td>√</td><td>√</td><td>√</td><td>√</td></tr></table>	Teaching/Learning Methodology	Outcomes				a	b	c	d	Lecture	√	√			Tutorial	√	√			Project	√	√	√	√																
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Student Study Effort Expected	Class contact:	
	▪ Lecture	30 Hrs.
	▪ Tutorial / Computer Labs	9 Hrs.
	Other student study effort:	
	▪ Reading and review	20 Hrs.
	▪ Homework assignment	10 Hrs.
	▪ Project Report	36 Hrs.
	Total student study effort	105 Hrs.
Reading List and References	1. Luger, G.F., and Stubblefield, W.A., Artificial Intelligence and the Design of Expert Systems, The Benjamin/Cummings Publishing Co., latest edition. 2. Clocksin, W. F., Programming in Prolog, Berlin; New York: Springer-Verlag, latest edition. 3. Ross, Timothy J., Fuzzy logic with engineering applications, Chichester; Hoboken, NJ: Wiley, latest edition. 4. Campesato, O., Artificial Intelligence, Machine Learning, and Deep Learning, Mercury Learning & Information, latest edition.	

Revised Jun 2020

Subject Description Form

Subject Code	ME42004
Subject Title	Development of Green Products
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: CEE370 Environmental Science I; or ME22002 Integrated Product Development Fundamentals; or ME32001 Manufacturing Fundamentals; or ME32003 Design and Manufacturing; or ISE386 Integrated Design for Manufacture
Objectives	To enhance students' awareness of environmental issues and provide them with necessary knowledge in green product development.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Appreciate the environmental impact of product manufacturing, distribution, use and disposal. b. Critically evaluate the environmental impacts of products during their life cycle and suggest appropriate actions to minimize/mitigate the impacts. c. Apply green design concepts in designing/re-designing products to fulfill the needs of green product market. d. Evaluate existing products/processes/technologies in terms of their environmental performance, and present the findings via oral presentation and written report.
Subject Synopsis/ Indicative Syllabus	<i>Environmental Issues of Concern</i> - Depletion and degradation of natural resources, environmental pollution and history of responses to pollution, waste and waste disposal issues, global warming, ozone layer depletion, acid rains, desertification, climate change, consumerism and its effect on global environment , individual and social preference for green living. <i>Environmental Impact of Products</i> - Life-cycle of a product, environmental impact of products over its life-cycle, environmental impact of packaging, strategies for minimizing environmental impact, drivers for green product design <i>Green and Sustainable Product Development Process</i> - Concept of green and sustainable product development: product design, planning and innovation for environment, concept of eco-design, eco-labelling and energy-labelling, international environmental management standards. <i>Material Selection and Procurement for Green Product Development</i> – Material selection for green design: Material selection process steps for green design, material selection methods, and material assessments. Green Procurement: Benefits of green procurement, green procurement process steps, evaluation of suppliers, green procurement programmes.

	Environmental Assessment of Green Products - Criteria on the global warming, stratospheric ozone depletion, photochemical ozone formation, acidification, nutrient enrichment, ecotoxicity, human toxicity, resource consumption and working environment. Normalisation and weighting in the environmental assessment of products, life-cycle impact assessment (LCA) of products. The Green Future - Green consumerism, opportunities from green technologies, green taxes and their effect on product development and marketing.																									
Teaching/Learning Methodology	1. The lectures are aimed at providing students with an integrated knowledge required for understanding the need for a green design approach, developing green products, assessing environmental impact of products and highlighting the opportunities arising from green consumerism. They provide a necessary framework for subsequent self-learning and group-learning activities. (Outcomes a to c) 2. The tutorials are aimed at enhancing the students’ skills necessary for analyzing the environmental impact of existing products and packaging solutions using various tools and develop solution strategies to minimize impact. Therefore, students will be able to solve real-world problems using the knowledge they acquired in the class. (Outcomes a to c) 3. The mini-project is aimed at enhancing the written and oral communication skills and teamwork spirit of the students. The students are expected to utilize the knowledge acquired in class to analyze the environmental impact of a selected existing product and systematically redesign it to enhance its green attributes in order to strategically place the product in rapidly developing green market. (Outcomes c and d) 4. The assignments and case studies are aimed at providing students with learning opportunities to study the practical implementations of green product and process assessments and developments. (Outcomes a, b and d) <table><tr><th>Teaching/Learning Methodology</th><th colspan="4">Outcomes</th></tr><tr><th></th><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>Lecture/Tutorial</td><td>√</td><td>√</td><td>√</td><td></td></tr><tr><td>Mini-project report & presentation</td><td></td><td></td><td>√</td><td>√</td></tr><tr><td>Homework assignments/Case studies</td><td>√</td><td>√</td><td></td><td>√</td></tr></table>	Teaching/Learning Methodology	Outcomes					a	b	c	d	Lecture/Tutorial	√	√	√		Mini-project report & presentation			√	√	Homework assignments/Case studies	√	√		√
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Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed			
			a	b	c	d
	1. Homework assignments/ Case studies	10%	√	√		√
	2. Test	20%	√	√	√	
	3. Mini-project report & presentation	20%			√	√
	4. Examination	50%	√	√	√	
	Total	100%				
Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Overall Assessment: 0.50 × End of Subject Examination + 0.50 × Continuous Assessment. 1. The continuous assessment will comprise three components: homework assignments & case studies (10%), test (20%) and mini-project report & presentation (20%). The homework assignments and test are aimed at evaluating the progress of students study and assisting them in fulfilling the respective subject learning outcomes. The mini-project and case studies are to assess students learning outcomes while providing them with opportunities to apply their learnt knowledge, enhance written & oral communication skills and team-work spirit. 2. The examination (50%) will be used to assess the knowledge acquired by students independently in understanding and analysing related problems critically and to determine the degree of achieving the subject learning outcomes.						
Student Study Effort Expected	Class contact:					
	▪ Lecture		33 Hrs.			
	▪ Tutorial/Mini-project discussion & presentation		6 Hrs.			
	Other student study effort:					
	▪ Self study/coursework		43 Hrs.			
	▪ Mini-project report preparation and presentation		24 Hrs.			
	Total student study effort		106 Hrs.			
Reading List and References	1. Azapagic A., Perdan S., Clift R. and Surrey G., Sustainable Development in Practice, John Wiley & Sons, Ltd., latest edition.					
	2. Burall P., Product Development and the Environment, The Design Council, latest edition.					
	3. Fuad-Luke A., EcoDesign: The Sourcebook, Chronicle Books, latest edition.					
	4. Ottman J.A. Green Marketing, NTC Business Books, latest edition.					
	5. William McDonough & Michael Braungart, Cradle to Cradle: Remaking the Way We Make Things, latest edition.					
	6. Ulrich, K.T. and Eppinger, S.D., Product Design and Development, McGraw-Hill, latest edition.					

Revised July 2016

Subject Description Form

Subject Code	ME42005
Subject Title	CAD/CAE Technologies for Product Development
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: AMA2111 Mathematics I Exclusion: ME42008 Computer-Aided Technology for Design
Objectives	To provide students with computer-aided design (CAD) and computer-aided engineering (CAE) technologies and the ability in using CAD and CAE software for product design and development.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Use CAD and CAE technologies to support product design activities, including geometry modeling, design solution modeling, analysis and evaluation, in different design process of the whole product design and development cycle. b. Understand data exchange standards and practices between CAD and CAE models and systems and their interoperability and associativity. c. Use CAD and CAE commercial software systems for product design and development in terms of geometry modeling, kinetics simulation, design solution analysis and evaluation. d. Optimize design solutions with the aid of CAD and CAE technologies.
Subject Synopsis/ Indicative Syllabus	<i>Computer-aided Design</i> - Geometric Models of Products • Wireframe model • Surface model • Solid Model - Geometry modeling technologies • Curve Modeling • Surface Modeling • Solid Modeling - Product kinetics modeling and simulation <i>Design Analysis and Evaluation</i> - Finite Element Modeling and Analysis • Basic concept of finite element method • Modeling techniques • Mesh types • Boundary constraints • Material and Properties • Symmetry in modeling and analysis - Mechanical and thermal stress analyses - Dynamic response - Product optimization in terms of product size, shape and material - Non-linear stress analysis

	CAD/CAE Integration - Data exchange standards: STL, STEP and IGES - Interoperability and associativity between CAD and CAE - Model defect and repairing Case Studies - CAD case studies - CAE case studies - CAD and CAE integration																																														
Teaching/Learning Methodology	Lectures will be given to explain the theories behind CAD and CAE and their applications. (Outcomes b, c and d) Tutorials will be used to teach the students on how to conduct product design, analysis and evaluation using state-of-the-art CAD and CAE software commercial software systems. Students will be given various assignments to learn how to represent and model the products from geometry perspective, how evaluate and analyze the design solutions from thermal, mechanical and physical perspectives and how to optimize the design solutions in terms of product size, shape and material. (Outcomes a, c and d) A mini-project will be given to students so that they will go through all the design phases in using computer-aided technologies to achieve design objectives. (Outcomes a to d) <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="4">Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>Lecture</td><td></td><td>√</td><td>√</td><td>√</td></tr><tr><td>Tutorial</td><td>√</td><td></td><td>√</td><td>√</td></tr><tr><td>Case study</td><td></td><td></td><td>√</td><td></td></tr><tr><td>Mini-project</td><td>√</td><td>√</td><td>√</td><td>√</td></tr></table>	Teaching/Learning Methodology	Outcomes				a	b	c	d	Lecture		√	√	√	Tutorial	√		√	√	Case study			√		Mini-project	√	√	√	√																	
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	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Overall Assessment: $0.50 \times \text{End of Subject Examination} + 0.50 \times \text{Continuous Assessment}$	
	Examination is adopted to assess students on the overall understanding and the ability of applying the concepts. It is supplemented by the tests, written and computer assignments which provide timely feedbacks to both lecturers and students on various topics of the syllabus. Written reports on various case studies and mini-project are used to assess the students' knowledge in the application of state-of-the-art CAD/CAE software to facilitate the product design and analysis process. Mini-project report and presentation assess the students' ability to assimilate the learnt knowledge for solving a more realistic, open-ended design problem systematically.	
Student Study Effort Expected	Class contact:	
	▪ Lecture	30 Hrs.
	▪ Tutorial	3 Hrs.
	▪ Guided study of CAD/CAE	6 Hrs.
	Other student study effort:	
	▪ Performing CAD/CAE in design (tutorial problems)	20 Hrs.
	▪ Performing modeling of design problems (case studies and mini-project)	24 Hrs.
	▪ Literature search and private study	23 Hrs.
	Total student study effort	106 Hrs.
Reading List and References	1. Michael E. Mortenson, Geometric Modeling, John Wiley & Sons, latest edition. 2. Kunwoo Lee, Principles of CAD/CAM/CAE System, Addison-Wesley Longman, latest edition. 3. Vince Adams and Abraham Askenazi, Building Better Products with Finite Element Analysis, Onword Press, latest edition. 4. J.Y.H. Fuh, Y.F. Zhang, A.Y.C. Nee, M.W. Fu, Computer-aided injection mold design and manufacture, Marcel Dekker, Inc, latest edition.	
Last Update	July 2024	

Subject Description Form

Subject Code	ME42007
Subject Title	Design for Product Safety and Reliability
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: ME22002 Integrated Product Development Fundamentals or ME32002 Engineering Design Fundamentals
Objectives	To provide students an overview of the product liability and legal aspects in launching of new consumer products and develop their understanding of the management strategy in achieving product safety.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Identify problems related to liability, safety and reliability for an existing product design and apply their knowledge in reliability engineering to devise a technically and economically feasible solution. - Apply knowledge of mathematics and engineering sciences via analytical and computational approaches to assess the risks of a product design and development project, and to assess the impacts of various key elements in achieving product safety. - Develop systematically a safer and more reliable design for an existing product via a group project and present in a professional manner their ideas using multimedia and written reports.
Subject Synopsis/ Indicative Syllabus	Product Reliability – Definition of product reliability, reliability programme plan, reliability requirements, parameters, modeling, prediction, test requirement, and design for reliability. Product Liability - Meaning of product liability. Definition of defective product. Product liability in Hong Kong. Product liability law in Hong Kong. Product liability law in other Jurisdictions. The Management of Design Risks - Management strategy in product safety. Reducing product design risks through design reviewing systems. Personal and environmental risk identification of the whole product life from manufacturing to end of services disposal. Product Safety Standards - The consumer Product Safety Acts. The safety standards used in different countries such as Underwriters Laboratories Inc. (UL) in USA, British Standards in United Kingdom and International Electro-technical Commission (IEC) in Europe. Overview of the application and testing procedures in obtaining product safety markings for new products. Planning, implementation and control in product test and assurance.

	Product Risk Identification Methods - Fault Tree Analysis (FTA). Failure Mode and Effect Analysis(FMEA). Hazard and Operability Study (HAZOP) and Hazard Analysis Critical Control Point (HACCP). The use of quantitative and statistical methods in assessing product risks and design optimisation.																																				
	Product Risk Management - Product Risk transfer through insurance and contract conditions.																																				
Teaching/Learning Methodology	1. Lectures give coverage and exposure and arouse interest. (Outcomes a to c)																																				
	2. Group discussions and tutorials help students consolidate lecture materials. (Outcomes a to c)																																				
	3. Assignments, through which students learn to compile, assimilate, assess and analyze. (Outcomes a to c)																																				
	4. Through thematic projects students would keep abreast of current product liability law and strategies for management of design risks. The presentation of reports allows students develop communication skills. (Outcomes a to c)																																				
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1. For continuous assessment evaluation, each student is required to submit a minimum of three reports. One of these reports is group-based and the other two are individual assignments. Besides assessing all the written assignments, students will be required to present the group and/or individual projects in class.																																					

	Class presentation and participation in discussions will be assessed. - To achieve the intended learning outcomes, it is considered that more emphasis on formative assessment would be appropriate as students' performance will be improved via written and verbal feedback. - Marked assignments provide feedback and reinforcement on learning key concepts and outcomes. - Through presentations/discussions, students will learn how to: - Work effectively with diverse group of people; - Persuasively explain in both oral and written form their product safety concepts; - Tackle diverse and unstructured questions; - Tell thoughts, feelings, ideas so that others may understand; - Supports and leads others in discussion. - The examination will be used to assess the knowledge acquired by the students to deal with product design risks in a strategic manner. It provides a reference of standards with which the learning outcomes are measured.	
Student Study Effort Expected	Class contact:	
	▪ Lecture and seminar	33 Hrs.
	▪ Tutorial and group discussion	6 Hrs.
	Other student study effort:	
	▪ Performing group project	25 Hrs.
	▪ Conducting case study and assignment	23 Hrs.
	▪ Literature search and private study	18 Hrs.
	Total student study effort	105 Hrs.
Reading List and References	- Abbot, Howard: Safer by design: a guide to the management and law of designing for product safety, Gower, latest edition. - Hammer, Willie: Product Safety management and engineering, American Society for Safety Engineers, latest edition. - The Law Reform Commission of Hong Kong: Report on Civil Liability for Unsafe Products, latest edition.	

Revised July 2018

Subject Description Form

Subject Code	ME42011
Subject Title	Fundamentals of Robotics
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: ME31002 Linear systems and control; or ME41004 Mechatronics and Control
Objectives	1. To provide students with the concepts and techniques for the design, modeling, analysis of robots. 2. To provide students with the fundamental knowledge of machine vision for robot guidance and automation.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Identify different types of robots and their applications in the industry. b. Construct the kinematics and dynamics equations of robots for motion analysis. c. Apply motion planning algorithms to generate the path for robots. d. Apply different machine vision and image processing algorithms to automate robots.
Subject Synopsis/ Indicative Syllabus	Robot Modelling. Degrees of freedom, coordinate frame and homogeneous transformation, Denavit-Hartenberg (DH) convention, forward and inverse kinematics, Jacobian matrix, singularity, robot dynamic models, state estimation, motion planning and obstacle avoidance. Computer Vision. Image formation, acquisition, histogram, edge and line detections, image enhancement, filtering, object recognition, stereo vision, camera modeling and calibration, deep neural networks in computer vision. Laboratory Work There is at least 6-hour laboratory sessions or an equivalent project. Typical Experiments are: 1. Programming and control of robots. 2. Path planning of mobile robots for collision avoidance. 3. Programming of image processing and computer vision.

Teaching/Learning Methodology	Lectures aim at providing students with an integrated knowledge required for understanding and analyzing different robots, including system modeling, trajectory planning and image processing (Outcomes a to d) Tutorials aim at enhancing students’ analytical and problem solving skills on robotics. Students will be able to solve real-world problems using the knowledge they acquired in the class. (Outcomes a to d) The project/experiments aims to have hand-on experience to automation of a robot system with vision or other functions. (Outcomes a to d) <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="4">Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>1. Lectures</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>2. Tutorials</td><td></td><td>√</td><td>√</td><td>√</td></tr><tr><td>3. Homework assignments</td><td></td><td>√</td><td>√</td><td>√</td></tr><tr><td>4. Project or experiments</td><td>√</td><td>√</td><td>√</td><td>√</td></tr></table>	Teaching/Learning Methodology	Outcomes				a	b	c	d	1. Lectures	√	√	√	√	2. Tutorials		√	√	√	3. Homework assignments		√	√	√	4. Project or experiments	√	√	√	√					
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Total	100%																																		

Student Study Effort Expected	Class contact:	
	▪ Lecture	30 Hrs.
	▪ Tutorial/Laboratory	9 Hrs.
	Other student study effort:	
	▪ Reading and review	36 Hrs.
	▪ Coursework (assignments, project)	40 Hrs.
	Total student study effort	115 Hrs.
Reading List and References	1. Introduction to Robotics: Analysis, Control, Applications, Saeed B. Niku, John Wiley and Sons, latest edition. 2. Modern Robotics: Mechanics, Planning, and Control, Kevin M. Lynch, Frank C. Park, latest edition. 3. Introduction to Autonomous Mobile Robots, Roland Siegwart, Illah Reza Nourbakhsh and Davide Scaramuzza, latest edition. 4. Introduction to Autonomous Robots: Mechanisms, Sensors, Actuators, and Algorithms, Nikolaus Correll, Bradley Hayes, Christoffer Heckman and Alessandro Roncone, latest edition. 5. R. C. Gonzalez and R. E. Woods, Digital Image Processing, Prentice Hall, latest edition. 6. Deep Learning, Ian Goodfellow, Yoshua Bengio and Aaron Courville, latest edition.	

Revised July 2021

Subject Description Form

Subject Code	ME42014
Subject Title	Networked Robots and IoT
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: ME35002 Sensors and Actuators for Mechatronics
Objectives	The subject aims to equip students with knowledge of: 1. Fundamentals of Internet of things. 2. Networks of robots, sensors, and systems. 3. Hands-on project.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. mathematically model different networks that can represent multi-robot systems and IoT. b. integrate and process multimodal sensor data across complex networks. c. design algorithms to coordinate multiple robotic systems over a network. d. develop robotic/sensing systems to solve different practical problems. e. work in teams and conduct experiments related to networked robots and IoT technologies.
Subject Synopsis/ Indicative Syllabus	Introduction to IoT. Fundamental concepts; Application domains; Technologies and systems used in IoT. Integration of robotics with IoT. Sensing and control over networks. Modelling of networks. Fundamentals of linear algebra; Introduction to graph theory; Edges and vertices; Topologies; Undirected and directed networks; Laplacian matrices and network dynamics. Consensus protocols. Agreement in undirected networks; Agreement in directed networks; Agreement and Markov chains; Edge agreement; Stability of consensus. Collective decision-making. Cooperative networked robots. Formation control with shapes and relative states; Networks of mobile robots; Collective object transport; Task allocation; Coverage problem. Robot swarms. Sensor networks. Distributed sensing and estimation; Distributed linear least squares; Distributed Kalman filtering; Implementation over networks. Local communication. Practical work. Introduction of hardware used for IoT and networked robotics. Conduct a hands-on project where students will apply different concepts studied during the course.

Teaching/Learning Methodology	1. Lectures aim at providing students with fundamental knowledge required for understanding and analysing different networked systems, including its mathematical models, controller design, and algorithms. (Outcomes a to e) 2. Tutorials aim at enhancing students’ analytical and problem-solving skills in networked robots and IoT. Students will be able to solve real-world problems using the knowledge they acquired in the class. (Outcomes b to e) 3. The experiments/project aim to provide hands-on experience for developing networked robots and IoT and reinforcing the acquired knowledge. (Outcomes a to e) <table><tr><th>Teaching/Learning Methodology</th><th colspan="5">Outcomes</th></tr><tr><th></th><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr><tr><td>1. Lectures</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>2. Tutorials</td><td></td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>4. Project or experiments</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td></tr></table>	Teaching/Learning Methodology	Outcomes						a	b	c	d	e	1. Lectures	√	√	√	√	√	2. Tutorials		√	√	√	√	4. Project or experiments	√	√	√	√	√										
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Student Study Effort Expected	Class contact: Lectures Tutorials Other student study effort: Course work Self-learning Total student study effort 33 Hrs. 6 Hrs. 40 Hrs. 36 Hrs. 115 Hrs.																																								

Reading List and References	• Lecture notes, articles, and handouts. • Graph Theoretic Methods in Multiagent Networks, by M. Mesbahi and M. Egerstedt. Latest edition. • Graph Theory You need Before Undergrad Research in 8 Undergrad Lectures, by M. Omar. Latest Edition. • Wireless Sensor Networks, by S. Chai, B. Zhang, et al. Latest Edition • Swarm Robotics: A Formal Approach, by H. Hamann, Latest Edition. • The Internet of Things, by S. Greengard. Latest Edition.
Develop Date	June 2024

Subject Description Form

Subject Code	ME43003
Subject Title	Product Testing Technology
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: ME33001 Mechanics of Materials
Objectives	To equip students with basic knowledge and universal standards of common product testing and examination technologies.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Apply knowledge of mathematics, engineering sciences and computing simulation to analyze and test a product design via analytical, experimental and computational approaches. - Understand and explain the effects of various important factors including materials, manufacturing processes, environmental and health issues, reliability and safety issues on product design and development. - Work effectively as a member and apply project management technique in the capacity of a team leader to complete a multi-disciplinary product testing project. - Appreciate the state-of-the-art product testing technologies and present a design project via written report. - Recognize the need to develop the ability of life-long learning.
Subject Synopsis/ Indicative Syllabus	<i>Purpose and Classification of Product Testing and Examination</i> - Damage and degradation of products, environmental attack, crack initiation, aging, fault in manufacturing process; classification of testing and examination methods. <i>Destructive Testing</i> - Tensile and shear strength tests; Drop tests for home appliances and toys; Impact and fracture toughness tests for plastics and metallic materials; Scratch and wear tests of surface coatings; Hardness test; Creep and durability tests for static and dynamic products. <i>Non-destructive Testing (NDT)</i> - Damage detection in products; embedded sensor technology; Wireless sensing technique; Ultrasonic spectroscopy and detection technique; Vibration and acoustic emission technique; Acousto-ultrasonic reproducibility; C-scan of composite products; Thermal wave imaging and full-field NDE; Microwave evaluation; Eddy current and Magnetic flux techniques. <i>Product Examination Techniques</i> - Surface morphology examination using optical technique, scanning electron microscopy (SEM) and atomic force microscopy (AFM); Chemical analysis using EDX and XRF; Structure examination using XRD. <i>Standards and Data Handling</i> - Design for inspection; Testing codes and standards; Data collection and analysis techniques. <i>Virtual Testing</i> - Product drop test simulations using CAE technique.

Teaching/Learning Methodology	1. The lectures are aimed at providing students with an integrated knowledge required for understanding and analyzing product testing technology and methodology. (Outcomes a and b). 2. The tutorials are aimed at enhancing the analytical skills of the students. Examples on the analysis of testing methods and testing results will be involved. So the students will be able to solve real-world problems using the knowledge they acquired in the class. (Outcomes a, b and e). 3. The experiments will provide the students with hands-on experience on the instrumentation and measurement. It also trains students in the analysis and presentation of experimental data. (Outcomes a and b). 4. The mini-project is aimed at enhancing the written and oral communication skills and team-work spirit of the students. The students are expected to apply the knowledge learnt in product testing technologies. The students are required to participate in the mini-project through literature survey, information search, discussions, report writing and presentation of results. Innovative thinking is encouraged. (Outcomes a, b, c, d and e). <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="5">Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr><tr><td>Lecture</td><td>√</td><td>√</td><td></td><td></td><td></td></tr><tr><td>Tutorial</td><td>√</td><td>√</td><td></td><td></td><td>√</td></tr><tr><td>Experiment</td><td>√</td><td>√</td><td></td><td></td><td></td></tr><tr><td>Mini-project</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td></tr></table>	Teaching/Learning Methodology	Outcomes					a	b	c	d	e	Lecture	√	√				Tutorial	√	√			√	Experiment	√	√				Mini-project	√	√	√	√	√												
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	1. The continuous assessment will comprise of four components: one test (20%), assignments (10%), project reports (10%) and oral presentation (10%). The test is aimed at assessing the interim knowledge gained by the student. The assignments are aimed at assisting the students in preparation for the tests and checking the progress of their study. The project report is aimed at assessing the capability of the student in analyzing and reporting experimental data, self-learning and problem-solving skills, and English writing capability. The oral presentation is aimed at assessing the student's communication and presentation skills. 2. The examination will be used to assess the knowledge acquired by the students for understanding and analyzing the product problems related to property testing and defect/motion detecting technologies.	
Student Study Effort Expected	Class contact:	
	▪ Lecture	30 Hrs.
	▪ Laboratory / Tutorial	9 Hrs.
	Other student study effort:	
	▪ Reviewing and Reading	26 Hrs.
	▪ Assignment / Laboratory Report	40 Hrs.
	Total student study effort	105 Hrs.
Reading List and References	1. Mechanical Testing, ASM International, ASM Handbook Volume 8, latest edition. 2. Sampling and analysis, Upper Saddle River, N.J.: Prentice Hall, latest edition. 3. Nondestructive testing of materials, Amsterdam; Washington, D.C.: IOS Press; Tokyo: Ohmsa, latest edition. 4. Practical non-destructive testing, Raj Baldev, New Delhi: Narosa Pub. House; Materials Park, Ohio: Distribution in North America only by ASM International, latest edition. 5. Encyclopedia of Materials Characterization, TA418.7.B73, latest edition.	

Revised July 2014

Subject Description Form

Subject Code	ME44002
Subject Title	Engine Technology
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: ME34002 Engineering Thermodynamics
Objectives	To provide students with the fundamental knowledge of engine technology, and its combustion-related emissions.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Understand and evaluate physical parameters of engine design and operating characteristics. - Apply the fundamental knowledge of solving air-standard and real air-fuel engine cycles. - Apply the fundamental knowledge of thermochemistry and fuels. - Understand the general principles of engine combustion, emissions controls and standards.
Subject Synopsis/ Indicative Syllabus	Introduction - Historical perspective of engines. Engine classifications. Terminology and abbreviations. Engine components. Basic engine cycles. Engine Design and Operating Characteristics - Engine parameters. Indicated work per cycle. Mean effective pressure. Brake torque and power. Dynamometers. Air-fuel and fuel-air ratios. Specific fuel consumption. Fuel efficiencies. Volumetric efficiency. Specific emissions and emission index. Relationships between performance parameters. Engine design and performance data. Noise abatement. Engine Cycles - Air-standard cycles. Otto Cycle. Diesel cycle. Dual cycle. Comparison of Otto, Diesel and Dual cycles. Real air-fuel engine cycles. Thermochemistry and Fuels - Thermochemistry. Gasoline, diesel and alternative fuels. Engine Combustion and Emissions - Spark ignition engine combustion, ignition and burning rate analysis. Compression ignition engine combustion, fuel injection, ignition delay. Engine emissions controls and standards. Introduction to Fuel Cell Technology – Working principle, types of fuel cells, cell components, polarization curve, transport phenomena; heat and water management.

Teaching/Learning Methodology	Lectures are used to deliver the fundamental knowledge in relation to internal combustion engines (outcomes a to d). Tutorials will be conducted to facilitate discussions of typical examples and coursework assignments (outcomes a to d). <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="4">Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>Lecture</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>Assignment/Tutorial</td><td>√</td><td>√</td><td>√</td><td>√</td></tr></table>						Teaching/Learning Methodology	Outcomes				a	b	c	d	Lecture	√	√	√	√	Assignment/Tutorial	√	√	√	√																	
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Reading List and References	1. C.R. Ferguson and A.T. Kirkpatrick, Internal Combustion Engines, John Wiley & Sons Inc., latest edition 2. W.W. Pulkrabek, Engineering Fundamentals of the Internal Combustion Engine, Prentice Hall, latest edition. 3. J.C. Guibet, Fuels and Engines- Technology, Energy and Environment, Vol. 1 & 2, Technip, Paris, latest edition.
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July 2023

Subject Description Form

Subject Code	ME44004
Subject Title	Heat and Mass Transfer
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: ME34002 Engineering Thermodynamics; and ME34004 Fluid Mechanics
Objectives	To provide students with the fundamental knowledge of heat and mass transfer.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Apply the fundamental knowledge of heat transfer mechanisms, namely conduction, convection and radiation. - Evaluate different types of heat exchangers. - Apply the numerical techniques in heat transfer applications. - Apply the fundamental knowledge of mass transfer.
Subject Synopsis/ Indicative Syllabus	Introduction - Conduction, convection and radiation. Fourier's law. Newton's law of cooling. Conduction - The plane wall. Insulation and thermal resistance. Radial systems. The overall heat transfer coefficient. Critical thickness of insulation. Heat-Source systems. Cylinder with heat sources. Heat transfer from extended surfaces. Unsteady conduction in slab or cylinder, Lumped-heat-capacity method. Forced and Free Convection - Governing equation for the boundary layer. Fluid and thermal boundary layer. The relation between fluid friction and heat transfer. Flow over a flat plate. Flow across cylinders/tubes and spheres. Heat transfer in laminar tube flow with constant temperature and constant heat flux. Heat transfer coefficients for free convection of plates and cylinders. Numerical Simulation - General differential equations for heat conduction. Energy balance method. Finite-difference solutions for differential equations of heat conduction. Explicit and implicit methods. Grid shape and size. Gauss-Seidel iteration. Accuracy and stability. Heat Exchanger - Heat exchanger types. The overall heat transfer coefficient. Heat exchanger analysis: Log mean temperature difference, parallel and counterflow heat exchangers. Heat exchanger analysis: The Effectiveness-NTU Method. Radiation - Black body and grey body. Absorptivity and emissivity. View factors. Irradiation and radiosity. Radiation exchange between surfaces and its network approach. Mass Transfer - Basic equations in mass transfer. Analogy between heat and mass transfer. Mass diffusion. Boundary conditions. Steady mass diffusion through a wall.

	Water vapour migration in buildings. Simultaneous heat and mass transfer.					
Teaching/Learning Methodology	Lectures are used to deliver the fundamental knowledge in relation to heat transfer and mass transfer (outcomes a to d).					
	Tutorials will be conducted to facilitate discussions of typical examples and coursework assignments (outcomes a to d).					
	Teaching/Learning Methodology	Outcomes				
		a	b	c	d	
	Lecture	√	√	√	√	
Assignment/Tutorial	√	√	√	√		
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)			
			a	b	c	d
	1. Assignment	30%	√	√	√	√
	2. Test	20%	√	√		
	3. Examination	50%	√	√	√	√
	Total	100%				
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:					
	Overall Assessment:					
	0.50 × End of Subject Examination + 0.50 × Continuous Assessment					
	1. The continuous assessment will comprise two components: assignments (30%) and tests (20%). The assignments are aimed at evaluating the progress of students study, assisting them in fulfilling the respective intended subject learning outcomes, and enhancing the integration of their knowledge learnt. The mid-term test(s) covers the first half of the subject material and provides useful feedback to both the lecturer and students on the learnt topics.					
2. The examination (50%) will be used to assess the knowledge acquired by the students for understanding and analyzing the problems critically and independently; as well as to determine the degree of achieving the intended subject learning outcomes.						

Student Study Effort Expected	Class contact:	
	▪ Lecture	33 Hrs.
	▪ Tutorial	6 Hrs.
	Other student study effort:	
	▪ Self-study/Coursework	67 Hrs.
	Total student study effort	106 Hrs.
Reading List and References	1. Y.A. Cengel and A.J. Ghajar, Heat and Mass Transfer: Fundamentals and Applications, McGraw-Hill, latest edition. 2. J.P. Holman, Heat Transfer, McGraw Hill, latest edition. 3. F.P. Incropera, D.P. Dewitt, T.L. Bergman and A.S. Lavine, Principles of Heat and Mass Transfer, John Wiley & Sons, Inc., latest edition.	

Revised July 2014

Subject Description Form

Subject Code	ME44008
Subject Title	Air Conditioning for Indoor Air Quality
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: ME34002 Engineering Thermodynamics or ME34003 Thermofluid Mechanics Exclusion: ME44001 Air Conditioning for Indoor Thermal and Environmental Quality
Objectives	To provide students with the fundamental knowledge of air conditioning for indoor air quality.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Appreciate and understand the concepts and components of air conditioning and refrigeration systems and applications. b. Apply the general knowledge of indoor thermal comfort and environmental health. c. Apply the knowledge of moist air properties and conditioning processes. d. Apply the knowledge of heating and cooling load required for a building. e. Apply the knowledge of refrigeration systems and cycles.
Subject Synopsis/ Indicative Syllabus	<i>Introduction of Air Conditioning and Refrigeration Systems and Applications</i> - Basic components of air conditioning and refrigeration systems. The complete air conditioning system. Central mechanical equipment. All-air systems, air-and-water systems, all-water systems. Unitary air conditioners. Heat pumps. Heat recovery systems. Thermal storage. <i>Moist Air Properties and Conditioning Processes</i> - Moist air and standard atmosphere. Fundamental parameters. Adiabatic saturation. Wet bulb temperature and the Psychrometric chart. Space air conditioning- design and off-design conditions. <i>Space Heating and Cooling Loads</i> - Outdoor and Indoor design conditions. Heat transmission in building structures. Infiltration. Heat losses from air ducts. Auxiliary heat sources. Supply air for space heating. Source media for space heating. Heat gain, cooling load and heat extraction rate. Solar radiation. Outside and interior surface heat balance. Zone air heat balance. Implementation of the heat balance method. <i>Refrigeration</i> - Refrigerants. Mechanical vapour-compression refrigeration cycles. Modifications to basic cycles. Reciprocating compressors. Cooling towers. <i>Indoor Thermal Comfort</i> - Physiological considerations. Thermal comfort indices and conditions. Hot and humid, and extreme cold environments. <i>Indoor Environmental Health</i> - Terminology and standards. Health sciences. The basic concerns of indoor air quality (IAQ). Prediction of indoor air quality model. Physical agents. Methods to control contaminants. Gas and particulate removal applications.

Teaching/Learning Methodology	1. The subject intends to equip students with fundamental knowledge of air conditioning for indoor thermal and environmental quality. Systematic lectures are required to achieve such foundation building coupled with assignments (outcomes a, b, c, d and e). 2. Tutorials are used to illustrate the application of fundamental knowledge to practical situations (outcomes a, b, c, d and e). It is intended to make use of these teaching/learning methodologies to achieve the intended subject learning outcomes as indicated in the following table: <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="5">Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr><tr><td>Lecture</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>Tutorial</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td></tr></table>	Teaching/Learning Methodology	Outcomes					a	b	c	d	e	Lecture	√	√	√	√	√	Tutorial	√	√	√	√	√																	
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Student Study Effort Expected	Class contact:	
	▪ Lecture	33 Hrs.
	▪ Tutorial	6 Hrs.
	Other student study effort:	
	▪ Coursework	33 Hrs.
	▪ Self-study/	33 Hrs.
	Total student study effort	105 Hrs.
Reading List and References	1. ASHRAE Handbooks on HVAC Systems and Equipment, Fundamentals, Refrigeration, and HVAC Applications, latest edition. 2. F.C. McQuiston, J.D. Parker and J.D. Spitler, Heating, Ventilating and Air Conditioning- Analysis and Design, John Wiley & Sons, Inc., latest edition. 3. W.T. Grondzik W.T.; J.S. Reynolds ; B. Stein; A.G. Kwok Mechanical and Electrical Equipment for Buildings, John Wiley & Sons, latest edition.	

July 2023

Subject Description Form

Subject Code	ME45001
Subject Title	Aerodynamics
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: ME34004 Fluid Mechanics
Objectives	To equip students with necessary knowledge of flow physics, analytical and numerical techniques for the prediction of forces acting on and performance analysis of aerodynamic bodies.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Formulate and solve problems relating to low-speed flow past two-dimensional airfoils and aerodynamic bodies by applying inviscid and incompressible flow theories. - Formulate and solve problems relating to downwash and induced drag phenomena for finite wings by applying the techniques derived from laws of vortex motion. - Formulate and solve problems relating to compressible flow through nozzles/diffusers and supersonic flow past aerodynamic bodies by applying one-dimensional compressible flow equations and knowledge of flow compressibility and wave phenomena in aerodynamics. - Analyze and interpret data obtained from experiments in incompressible and compressible aerodynamics. - Present effectively in completing written reports of laboratory work and the given task.
Subject Synopsis/ Indicative Syllabus	<i>Fundamental Principles and Equations</i> - Control volume concept for fluid. Continuity equation. Momentum equation. Energy equation. Substantial derivative. Angular velocity, vorticity and strain. Dimensional analysis. <i>Inviscid and Incompressible Flow</i> - Stream function and velocity potential. Potential flow. Laplace's equation and its elementary solutions. <i>Incompressible Flow over Two-Dimensional Airfoils</i> - Airfoil nomenclature and characteristics. The Kutta condition. Circulation and lift. Kelvin's circulation theorem and starting vortex. General thin airfoil theory. Symmetric and cambered airfoils. Aerodynamic Center. Panel method for arbitrary lifting bodies. Viscous airfoil drag. <i>Finite Wings</i> - Downwash and induced drag. Vortex system on finite wing. Law of vortex motion. Prandtl's lifting line theory. Lifting-surface theory and vortex lattice numerical method.

	Inviscid and Compressible Flow - Definition of compressibility and total conditions. Governing equations. One-dimensional flow: weak waves and plane normal shock waves. Two-dimensional supersonic flow: oblique shock and expansion waves. Flow through nozzles and diffusers. Compressible Flow over Airfoils - Velocity potential equation and its linearized form. Prandtl-Glauert compressibility correction. Critical Mach number. The sound barrier. Supersonic pressure coefficients. Application to supersonic airfoils.																																															
Teaching/Learning Methodology	Lectures are used to deliver the fundamental principles and equations of aerodynamics as well as solution techniques (outcomes a to d). Tutorials are used to illustrate the application of fundamental principles to practical engineering situations (outcomes a to c). Projects, in the form of design problems or case studies, are used to allow students to deepen their knowledge on a selected topic through search of information, analysis of data and report writing (outcomes a, b and e). Experiment(s) on evaluating the effects of configurations of an airfoil on its flow field and aerodynamic characteristics, either in laboratory or numerical setup, is (are) provided for bridging the knowledge of fluid mechanics with current subject. Students are exposed to proper use of knowledge taught and analysis skills on evaluating their experimental results (outcomes a, d and e). <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="5">Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr><tr><td>Lecture</td><td>√</td><td>√</td><td>√</td><td>√</td><td></td></tr><tr><td>Tutorial</td><td>√</td><td>√</td><td>√</td><td></td><td></td></tr><tr><td>Project/Laboratory</td><td>√</td><td>√</td><td></td><td></td><td>√</td></tr><tr><td>Experiment</td><td>√</td><td></td><td></td><td>√</td><td>√</td></tr></table>	Teaching/Learning Methodology	Outcomes					a	b	c	d	e	Lecture	√	√	√	√		Tutorial	√	√	√			Project/Laboratory	√	√			√	Experiment	√			√	√												
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	Examination is adopted to assess students on the overall understanding and the ability of applying the concepts. It is supplemented by continuous assessment including assignments projects and test(s), which provide timely feedback to both lecturers and students on various topics of the syllabus. Assigned homework and test are designed to enhance the students' learning of fundamental principles in aerodynamics. The projects provide students an opportunity to capitalize on the knowledge they learn for tackling practical aerodynamic problems arising from real practice. Written report and oral presentation on a specific project or case study is used to assess the students' knowledge in contemporary aircraft maintenance engineering.	
Student Study Effort Expected	Class contact:	
	▪ Lecture	33 Hrs.
	▪ Tutorial	6 Hrs.
	Other student study effort:	
	▪ Self-study	45 Hrs.
	▪ Homework assignment	12 Hrs.
	▪ Project/case study	12 Hrs.
	Total student study effort	108 Hrs.
Reading List and References	1. Anderson Jr., J. D., Fundamentals of Aerodynamics, McGraw-Hill, latest edition. 2. Houghton, E. L., and Carpenter, P. W., Aerodynamics for Engineering Students, Butterworth & Heinemann, latest edition. 3. Bertin, J. J. and Cummings, R. M., Aerodynamics for Engineers, Pearson Prentice-Hall, latest edition. 4. Anderson Jr., J. D., Aircraft Performance and Design, McGraw-Hill, latest edition.	

Revised July 2014

Subject Description Form

Subject Code	ME45002
Subject Title	Aircraft Systems
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: ME34004 Fluid Mechanics
Objectives	To develop students' fundamental knowledge and basic concepts on components and operating principles of essential mechanical and electrical systems in civil transport aircraft.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Demonstrate good understanding of the principles of flight control and various systems in civil transport aircraft; and formulate and solve problems relating to aircraft systems. - Identify the flight control and utility functions to be considered in the design of an aircraft hydraulic system. - Explain major electrical loads and the characteristics of modern aircraft electrical system. - Explain the need for cabin and avionics conditioning and outline recent advances in aircraft environmental control system design. - Explain the design philosophy and objectives of aircraft emergency systems.
Subject Synopsis/ Indicative Syllabus	<i>Flight Control Systems</i> - Principles of flight control. Primary and secondary flight controls. Flight phases. <i>Power Plant</i> - Fuel efficiency. Effect of specific thrust. Specific fuel consumption and flight speed. Engine cycle and performance. <i>Fuel Systems</i> - Characteristics of aircraft fuel systems. Fuel system components. Aircraft mass and payload. <i>Hydraulic Systems</i> - Flight control and utility functions. Emergency power sources. Landing-gear system. Braking and anti-skid. <i>Electrical systems</i> - Characteristics of civil aircraft electrical system. Electrical loads. Emergency power generation. <i>Pneumatic systems</i> - Pitot-static systems. Use of engine bleed air. Bleed air control. Thrust reversers. <i>Environmental Control Systems</i> - The need for cabin and equipment conditioning. Environmental control system design. Air distribution systems. Cabin pressurization. <i>Emergency Systems</i> - Warning systems. Fire detection and suppression. Emergency oxygen. Explosion suppression. Passenger evacuation.

Teaching/Learning Methodology	Lectures are used to deliver the fundamental knowledge in relation to various aircraft systems (outcomes a to e). Tutorials are used to illustrate the application of fundamental knowledge to practical situations (outcomes a to e). Industrial visits and special seminars delivered by invited industrial professionals are used to relate the concepts learnt on class to engineering practices. Students are expected to achieve better understanding of aircraft systems through these activities (outcomes a to e). <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="5">Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr><tr><td>Lecture</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>Tutorial</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>Industrial field visit and special seminar</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td></tr></table>	Teaching/Learning Methodology	Outcomes					a	b	c	d	e	Lecture	√	√	√	√	√	Tutorial	√	√	√	√	√	Industrial field visit and special seminar	√	√	√	√	√											
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Student Study Effort Expected	Class contact:	
	▪ Lecture	33 Hrs.
	▪ Tutorial	6 Hrs.
	Other student study effort:	
	▪ Course work	20 Hrs.
	▪ Self-study	45 Hrs.
	Total student study effort	104 Hrs.
Reading List and References	1. The Rolls-Royce Book of the Jet Engine, latest edition, Rolls-Royce Ltd, latest edition. 2. SAE Aerospace Information Report 5005, Aerospace – Commercial Aircraft Hydraulic Systems, latest edition. 3. I. Moir and A.G. Seabridge, Design and Development of Aircraft Systems – An Introduction, First Edition, AIAA Education Series, latest edition.	

Revised July 2014

Subject Description Form

Subject Code	ME46002
Subject Title	Numerical Methods for Engineers
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: AMA2111 Mathematics I
Objectives	To teach students numerical methods of solving typical engineering problems.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Formulate simple engineering problems with knowledge in data analysis and engineering mathematics. - Solve non-linear equations, simultaneous linear algebraic equations, eigenvalue problems, using numerical methods. - Perform numerical differentiation and integration and analyze the errors. - Know the contemporary issues, and understand the impact of artificial intelligence in a global and societal context. - Use MATLAB or other data analysis tools to compute the solutions of engineering problems using the appropriate numerical methods.
Subject Synopsis/ Indicative Syllabus	Introduction to Mathematical Modelling and Computational Methods – Importance of computational modelling in engineering. Data representation and errors. Applications of commercial software packages such as MATLAB. Functions and plotting using MATLAB. Computer Solution of Non-linear Equations - Bracketing Methods. Bisection Method. Open Methods. Newton-Raphson Method. Secant Method. Convergence of methods. Determination of multiple roots. Engineering applications. Simultaneous Linear Equations - Solving simultaneous linear equations by Matrix Inversion. Cramer's Rule. Gauss Elimination. Gauss-Jordan Elimination. LU decomposition method. Engineering applications and choice of methods. Eigenvalues Problems. Optimization - Unconstrained optimization. Multi-dimensional optimization. Unconstrained optimization. Curve Fitting and Data Analysis - Interpolation using splines. Linear Least-Squares Regression. Nonlinear Regression. Introduction to Machine Learning Algorithms. Numerical Differentiation and Integration - Taylor's series expansion. Finite differences for the first derivative and the second derivative. High-accuracy differentiation formulas. Trapezoidal rule. Simpson's rule. High-order Newton-Cotes formulas. Applications of numerical differentiation and integration in heat transfer, solid mechanics and fluid flow problems.

Teaching/Learning Methodology	Lectures are used to deliver the fundamental knowledge in relation to numerical methods. (Outcomes a - d) Tutorials will be conducted in small groups to facilitate discussions. (Outcomes a - d) Computational workshops provide hands-on experience in using software to solve numerical problems. (Outcomes b - e) <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="5">Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr><tr><td>Lecture</td><td>√</td><td>√</td><td>√</td><td>√</td><td></td></tr><tr><td>Tutorial</td><td>√</td><td>√</td><td>√</td><td>√</td><td></td></tr><tr><td>Computational workshop</td><td></td><td>√</td><td>√</td><td>√</td><td>√</td></tr></table>						Teaching/Learning Methodology	Outcomes					a	b	c	d	e	Lecture	√	√	√	√		Tutorial	√	√	√	√		Computational workshop		√	√	√	√											
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	▪ Private study	25 Hrs.
	Total student study effort	116 Hrs.
Reading List and References	1. S.C. Chapra and R.R. Canale, Numerical Methods for Engineers, McGraw-Hill, latest edition. 2. S.S. Rao, Applied Numerical Methods for Engineers and Scientists, Prentice-Hall, latest edition. 3. S.C. Chapra, Applied Numerical Methods with MATLAB for Engineers and Scientists, McGraw-Hill, latest edition. 4. D.M. Etter, Engineering Problem Solving with Matlab, Prentice-Hall, latest edition.	

Revised December 2021

Subject Description Form

Subject Code	ME46003
Subject Title	Numerical Fluid Mechanics and Heat Transfer
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: ME34003 Thermofluid Mechanics, ME34002 Engineering Thermodynamics or ME34004 Fluid Mechanics
Objectives	To equip students with numerical methods and computational techniques in analyzing fluid dynamics and heat transfer problems which are usually encountered in the design of thermofluid systems.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Understand and apply numerical differentiation method, and analyze the stability and errors involved. - Apply knowledge of thermofluid science/engineering to formulate numerical equations for solving steady-state/transient fluid mechanics or heat transfer problems, and apply appropriate mathematics methods for their evaluation. - Apply knowledge of mathematics and thermofluid science/engineering via computational approaches to analyze and predict the performance of thermofluid systems/products.
Subject Synopsis/ Indicative Syllabus	<i>Introduction to Numerical Methods for Product Analysis</i> – Mathematical modeling for fluid mechanics and heat transfer systems. Numerical approximations of scientific equations. Direct and iterative methods for solving simultaneous equations. Stability and major errors involved in numerical methods. <i>Numerical Differentiation</i> – Finite-differences for the first derivative and the second derivative. Finite-differences for partial differentiation. High-accuracy differentiation formulas. <i>Finite-Difference Methods in Solving Heat Transfer Problems</i> – Governing equations for heat transfer. Boundary conditions in heat conduction and heat convection. Steady-state and transient heat transfer problems. Dimensionless differential equations. Discretization. Explicit scheme finite-difference. Implicit scheme finite-difference. Direct and iterative mathematics methods. Analysis on solution stability and estimation of errors. <i>Finite-Difference Methods in Solving Fluid Dynamics Problems</i> – Classification of partial differential equations for fluid dynamics. Navier-

	Stokes equations. Grid types. Explicit and implicit scheme finite-difference formulations. Introduction to turbulence and its modeling. Introduction to computational approach – Introduction to commercial CFD software and their applications to solve fluid mechanics problems.																												
Teaching/Learning Methodology	1. The lectures are aimed at providing students with necessary background knowledge in related mathematical principles and computational approaches for analysis of thermofluid problems. (Outcomes a to c) 2. The tutorials and in-class exercises are aimed at enhancing the students’ skills in effectively using numerical and computational approaches to solve thermofluid problems. Thus, some tutorial classes will be held in the Computational Laboratory. (Outcomes a to c) 3. The homework assignments are to get students engaged with learning activities continuously and to provide them with self-assessment opportunities on their progress of learning. (Outcomes a to c) <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="3">Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th></tr><tr><td>Lecture</td><td>√</td><td>√</td><td>√</td></tr><tr><td>Tutorials/In-class exercises</td><td>√</td><td>√</td><td>√</td></tr><tr><td>Homework assignments</td><td>√</td><td>√</td><td>√</td></tr></table>	Teaching/Learning Methodology	Outcomes			a	b	c	Lecture	√	√	√	Tutorials/In-class exercises	√	√	√	Homework assignments	√	√	√									
Teaching/Learning Methodology	Outcomes																												
	a	b	c																										
Lecture	√	√	√																										
Tutorials/In-class exercises	√	√	√																										
Homework assignments	√	√	√																										
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="3">Intended subject learning outcomes to be assessed</th></tr><tr><th>a</th><th>b</th><th>c</th></tr><tr><td>1. Homework assignments/ In-class exercises</td><td>30%</td><td>√</td><td>√</td><td>√</td></tr><tr><td>2. Test</td><td>20%</td><td>√</td><td>√</td><td>√</td></tr><tr><td>3. End-of-semester Examination</td><td>50%</td><td>√</td><td>√</td><td>√</td></tr><tr><td>Total</td><td>100%</td><td>√</td><td>√</td><td>√</td></tr></table>	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed			a	b	c	1. Homework assignments/ In-class exercises	30%	√	√	√	2. Test	20%	√	√	√	3. End-of-semester Examination	50%	√	√	√	Total	100%	√	√	√
Specific assessment methods/tasks	% weighting			Intended subject learning outcomes to be assessed																									
		a	b	c																									
1. Homework assignments/ In-class exercises	30%	√	√	√																									
2. Test	20%	√	√	√																									
3. End-of-semester Examination	50%	√	√	√																									
Total	100%	√	√	√																									

Assessment Methods in Alignment with Intended Learning Outcomes	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Overall Assessment: $0.5 \times \text{Continuous Assessment} + 0.5 \times \text{Examination}$. - Homework assignments and in-class exercises are aimed at evaluating students' progress in study, and assisting them in fulfilling the respective subject learning outcomes. Homework assignments and in-class exercises should include analyses of thermofluid systems, case-study of problems encountered in thermofluid design, and applications of computational technique (including CFD software) to solve thermofluid problems. - Test and end-of-semester examination will be used to assess the degree of achieving the subject learning outcomes by individual student. Their understanding of numerical methods and ability to apply them to critically analyze thermofluid problems will be evaluated.	
Student Study Effort Expected	Class contact:	Time
	Lectures	26 Hrs.
	Tutorials/In-class Exercises	13 Hrs.
	Other student study effort:	Time
	Performing assignments including computational work	50 Hrs.
	Self-study	31 Hrs.
	Total student study effort	120 Hrs.
Reading List and References	- S.C. Chapra and R.R. Canale, Numerical Methods for Engineers, McGraw-Hill, latest edition - S.S. Rao, applied Numerical Methods for Engineers and Scientists, Prentice-Hall, latest edition A. Cengel Yunus, and J. Ghajar Afshin, Heat and Mass Transfer-Fundamentals and Applications, 4th edition in SI units, McGraw-Hill, 2011. H. K. Versteeg and W. Malalasekera, An Introduction to Computational Fluid Dynamics, 2nd edition, Pearson Prentice Hall 2007.	
Last Update	January 2025	

Intended Blank

Training Subjects

Subject Description Form

Subject Code	ME29004/IC2117
Subject Title	Applied Engineering Fundamentals
Credit Value	2 Training Credits
Level	2
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	This subject offers fundamental engineering techniques to students, project-based learning approach will be adopted in this subject aiming to enhance the problem-solving skills for all Year 1 students in the Engineering Faculty.
Intended Learning Outcomes	Upon completion of the subject, students will: a) Be able to demonstrate an understanding of engineering design b) Develop their problem-solving ability and global outlook c) Be able to demonstrate an understanding on typical engineering material properties and basic prototyping and joining processes
Subject Synopsis/ Indicative Syllabus	- Engineering Design and Prototyping Processes Engineering design concept, prototyping for design verification, digital modelling, measurement techniques, standard components. Basic Prototyping techniques; 3D Printing, Laser Cutting and standard components. Engineering material; polymers, metal, composite, surface treatment processes; painting, plating. Application of adhesives and mechanical parts in joint of material. - Fundamental Engineering Projects There will be practical workshops, project briefing and demonstration sessions for the project which aim at developing students' creativity and problem-solving skills through practical and hands-on tasks at a level commensurate with their first-year engineering backgrounds. Students will work in small groups under the guidance of teachers/instructors to design and implement an engineering solution to some given problems.
Teaching/ Learning Methodology	<u>Mini Lectures</u> The Mini lectures are aimed at providing students the basic knowledge and concept about engineering design and the common processing technique for prototyping and design verification <u>Workshop Tutorials</u> The workshop tutorials focus on deepening students' knowledge and enhancing their ability to apply concepts and skills to complete specific tasks. The practical works aim to reinforce learning by enabling students to explore the diverse topics covered in the course through active engagement in research, practice, questioning, and problem-solving in an integrated and hands-on manner.

	Fundamental Engineering Project For the fundamental engineering project, students will be required to design and develop an engineering solution to a given problem under the guidance of instructors. Students will be given opportunities to develop creativity, problem-solving skills and research for information from this project.				
Assessment Methods in Alignment with Intended Learning Outcomes (Note 4)	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)		
			a	b	c
	1. Assignments	80	✓	✓	✓
	2. Quizzes	20	✓		✓
	Total	100			
	Assignments are designed to enable students to periodically reflect upon and apply the knowledge acquired throughout the training. Quizzes are designed to facilitate students to review the breadth and depth of their understanding on specific topics.				
Student Study Effort Expected	Class Contact				
	▪ Mini lecture, In-class Assignments and project works		60 Hrs.		
	Other Student Study Effort		0 Hrs.		
	Total Student Study Effort		60 Hrs.		
Reading List and References	1) Training material, manual and articles published by Industrial Centre. 2) “Engineering fundamentals & problem solving” (2018) A. R. Eide, New York, NY: McGraw-Hill				

Subject Description Form

Subject Code	ME39002/IC348
Subject Title	Appreciation of Manufacturing Processes
Credit Value	3 Training Credits
Level	3
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	This subject aims at developing students' understanding on: • the principles and operations of common manufacturing processes, and • the properties and application of different materials.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a) demonstrate a holistic understanding on the working principle, capability and operation of common manufacturing processes. b) justify appropriate manufacturing processes for different materials and specific product requirements. c) select and use various common engineering materials for specific purpose. d) collaboratively execute an application-oriented project through group work and discussions and inspires oneself to learn continuously about the latest industrial practices and technologies.
Subject Synopsis/ Indicative Syllabus	1) Properties and uses of common engineering materials such as metal, alloy and polymers 2) Working principles and operation of • Common metal removing processes such as CNC machining; • Common production processes such as casting and moulding; • Common sheet metal parts fabrication processes such as forming and turret pressing; • Common joining and cutting processes such as welding and laser cutting; • Common surface finishing processes such as electro-plating;

	• Rapid prototyping and production technologies; • Manufacturing metrology and part inspection.																																		
Teaching/Learning Methodology	Min-lectures aim at providing students an understanding of the principles and application of common manufacturing technologies, properties and selection of common engineering materials. Hands-on activities will be used for students to appreciate the working principles, capability and operation procedures of common manufacturing processes. Group product assembly will be used to enable students to apply acquired practical knowledge and skills to produce a functional product, and to facilitate students in performing group collaboration and problem solving skills learning.																																		
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="4">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>1. Individual Assignments</td><td>60</td><td>√</td><td>√</td><td>√</td><td></td></tr><tr><td>2. Product Assembly</td><td>10</td><td></td><td></td><td></td><td>√</td></tr><tr><td>3. Individual Report</td><td>30</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>Total</td><td>100 %</td><td colspan="4"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: The Individual Assignments are aimed at assessing student’s practical ability in using various processes to produce the components for the product. The Product Assembly is aimed at assessing student’s group collaboration, organization, time management and problem solving capability. The Individual Report is aimed at assessing student’s appreciation, understanding and application of all the processes involved in the product.	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)				a	b	c	d	1. Individual Assignments	60	√	√	√		2. Product Assembly	10				√	3. Individual Report	30	√	√	√	√	Total	100 %				
Specific assessment methods/tasks	% weighting			Intended subject learning outcomes to be assessed (Please tick as appropriate)																															
		a	b	c	d																														
1. Individual Assignments	60	√	√	√																															
2. Product Assembly	10				√																														
3. Individual Report	30	√	√	√	√																														
Total	100 %																																		

Student Study Effort Expected	Class contact:	
	Min-lecture / Hands-on Practice / Product Assembly / Report Writing	90 Hrs.
	Other student study effort:	
	Nil	0 Hrs.
	Total student study effort	90 Hrs.
Reading List and References	1. Fundamental of machining processes: Conventional and nonconventional processes, Hassan El Hofy, CRC, 2006 2. Reading Materials published by the Industrial Centre	

Subject Description Form

Subject Code	ME39003/IC382
Subject Title	Multidisciplinary Manufacturing Project
Credit Value	3 Training Credits
Level	3
Pre-requisite/ Co-requisite/ Exclusion	ME39002/IC348 or EIE2901/IC2114 or AAE3103/IC381
Objectives	The subject provides opportunity for students to work in a multidisciplinary project team to accomplish realistic engineering goals. Through the project, students will apply and integrate the engineering knowledge and practical skills acquired from prior engineering subjects and industrial trainings. Students will also be able to analyse engineering problems from multiple perspectives, and synthesize a solution from ideas contributed by teammates of multiple disciplines.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - apply engineering knowledge in carrying out an industrial project starting from problem definition, design, manufacturing, down to assembly, testing and evaluation; - select and use appropriate technology building blocks, components and manufacturing processes to develop a solution to meet given specifications and constraints; - Work collaboratively and effectively in a multidisciplinary team to accomplish mutual project goals; and - Communicate effectively in a multidisciplinary project team.
Contribution of the Subject to the Attainment of Outcomes of EIE Programmes	<u>Category A: Professional/academic knowledge and skills</u> - Design systems, components and processes to meet given specifications and constraints. - Use modern engineering/IT tools appropriate to EIE practice. <u>Category B: Attributes for all-roundedness</u> - Work with others collaboratively in a multi-disciplinary team and have a knowledge of leadership

Subject Description Form

Subject Synopsis/ Indicative Syllabus	Students will be divided into groups to design and manufacture an engineering product that satisfy an existing demand in IC or a certain customer from the industry. Throughout the project, students will encounter situations that reinforce the following skills: 1) Project specification: Identification of client needs and wants; Identification of resource constraints such as time, manpower, equipment, budget; Formulation of project plan. 2) Engineering design: Selection of design methodology; collaborative design; Make-or-buy decisions; Design prototyping; Testing and simulation. 3) Product manufacturing: Material procurement; Component machining; PCB fabrication; Programming; Assembly and fine-tuning. 4) Project collaboration: Determination of project stages and milestones; CAD and PDM; Leadership and Collaborative decision making; Tolerances and fits; Project documentations.
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Subject Description Form

Learning Methodology	Students will be divided into groups of 5-8 to design and manufacture an engineering product. Each project group will be formed by students from two or more engineering streams. The project topics will be provided by the subject supervisor team. Topics will be either initiated by supervisors or by commercial clients. All topics shall demand two or more skillsets including Mechanics, Electronics, and IT. Typical topics include: automated production equipment, mobility products, robotic toys, airframe structures, cabin installations, aircraft maintenance tools, jigs and gauges, <i>etc.</i> The subject is divided into two stages: • Design Stage During this period, the project team, under the guidance of the supervisors and clients, have to discover, understand and analyze the requirement of the project; and apply their knowledge to design a solution. Furthermore, students are required to search and track down parts and components with suppliers to obtain materials for the following manufacturing stage. • Manufacturing stage During this period, the project team will fabricate, test, and debug the product they designed. The supervisors will guide and monitor the groups on personal commitment, cooperation and coordination among team members. Regular group tutorials in the form of student-centred project meeting will be arranged between project group and respective supervisors.
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Subject Description Form

Assessment Methods in Alignment with Intended Learning Outcomes

Assessment Methods	Weighting (%)	Intended Learning Outcomes Assessed			
		a	b	c	d
1. Quality of final product	30	✓	✓		
2. Report	20	✓	✓	✓	✓
3. Presentation and demonstration	20			✓	✓
4. Reflective Journal	30	✓	✓	✓	✓
Total	100				

Group assessment components

Quality of final product will be assessed by the supervisor team during demonstration. The assessment is to determine how well the group's solution meets with client's requirement in terms of completeness and functionality. The assessment also determines how well the group has carried out the manufacturing in terms of accuracy and craftsmanship. This addresses the intended learning outcomes (a) & (b).

Report submitted at the end of project will be summative evidence of how well the group applied knowledge and made decisions collectively. Compulsory report chapters include: Technical description of final design; Justification of technology building blocks used; Critical review on project execution; and Record of internal communications. This addresses the intended learning outcomes (a), (b), (c) & (d).

Individual assessment components

Oral presentation and demonstration in an exhibition booth setting allow individual members to demonstrate their ability in presenting engineering contents clearly and logically. Through Q&A session supervisors can also determine the effectiveness of individual members' effort toward the final product outcomes. This addresses the intended learning outcomes (c) & (d).

Individual reflective journal serves as summative evidence of how well the student has functioned in the group and embrace the multidisciplinary collaboration concept. Compulsory journal contents include: Technical description of design and manufacturing tasks performed; Critical review of technical ideas proposed and adapted; Critical review on personal performance in the project execution and the collaboration experience. This addresses the intended learning outcomes (a), (b), (c) & (d).



Subject Description Form

Student Study Effort Expected	Class Contact	
	▪ Project works	78 Hrs.
	▪ Tutorial	12 Hrs.
	Other Study Effort	0 Hrs.
	Total Study Effort	90 Hrs.
Reading List and References	1. H. Damith ; S. David, Foundations of Robotics A Multidisciplinary Approach with Python and ROS, Springer, 2022 2. M. Habib, J. Davim, Engineering Creative Design in Robotics and Mechatronics. IGI Globa, 2013 3. P. Harpum, 'Design Management', in <i>Engineering Project Management</i>, 3rd ed., N. Smith, Ed. Oxford: Blackwell, 2008, pp. 234-254. 4. Alur, Rajeev. Principles of Cyber-physical Systems. Cambridge, Massachusetts: MIT, 2015. 5. Valvano, Jonathan W. Introduction to ARM Cortex-M Microcontrollers. 5th ed. , Jonathan W. Valvano, 2017	

ME39003/IC382: Multidisciplinary Manufacturing Project

Subject Description Form

Subject Code	ME39004/IC390
Subject Title	Applied Engineering Essentials: Python, Safety Practices, and Advanced CAD modelling
Credit Value	2 Training Credits
Level	3
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	This subject offers a wide spectrum of fundamental engineering practice that are essential for a professional engineer. This subject includes Advanced CAD modelling CAD, Safety, Mechanism Design Practice and Scientific Computing Languages and it applications that aims at providing the technical skills to the students.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a) Interpret basic occupational health and industrial safety requirements for engineering practice; b) Apply scientific computing software for computing in science and engineering including visualization and programming; c) Apply advanced modeling techniques in CAD to effectively communicate technical information and documentation in accordance with engineering standards.

Subject Synopsis/ Indicative Syllabus	Students are required to take 3 modules: <u>1) Industrial Safety</u> Safety Management: Overview, essential elements of safety management, safety training, accident management, and emergency procedures. Safety Law: F&IU Ordinance and principal regulations, OSH Ordinance and principal regulations. Occupational Hygiene and Environmental Safety: Noise hazard and control; dust hazard and control; ergonomics of manual handling. Safety Technology: Mechanical lifting, fire prevention, dangerous substances and chemical safety, machinery hazards and guarding, electrical safety, first aid, job safety analysis, fault tree analysis, and personal protective equipment. <u>2) AI application with Python</u> Problems solving with Python: Functions and Python packages to solve engineering problem (i.e. plot displacement diagram). AI application: Develop applications that leverage AI techniques for data manipulation and visualization, including the use of libraries for machine learning, data processing, and graphical user interfaces. <u>3) Computer Aided Design</u> Assembly Modeling: Learn to create and manage assemblies using both bottom-up approaches, explore various mate types to define component relationships, understand the management of subassemblies for complex projects. Advanced Part Modeling: Utilize advanced features such as lofts, sweeps, and patterns for complex geometry creation, implement design tables for efficient parametric modeling, and apply optimization strategies to enhance part performance and manufacturability. Surface Modeling: Gain an understanding of surface modeling fundamentals and applications in product design and explore methods to convert surface models into solid bodies for integration into assemblies.
Learning Methodology	The learning and teaching methods include lectures, workshop tutorials, and practical works. The lectures are aimed at providing students with an overall and concrete background knowledge required for understanding key issues in engineering communication, use of standard engineering components and systems, and importance of industrial safety. The workshop tutorials are aimed at enhancing students' in-depth knowledge and ability in applying the knowledge and skills to complete specific tasks. The practical works aim at facilitating students to review the diverse topics covered in this course and perform active learning with research, practice, questioning, and problem solving in a unified activity.

Assessment Methods in Alignment with Intended Learning Outcomes					
	Assessment Methods	Weighting (%)	Intended Learning Outcomes Assessed		
			a	b	c
	1. Assignment	80	✓	✓	✓
	2. Test	20		✓	
	Total	100			
	Assignments in the form of system configuration or fabrication tasks will be used to assess how well students understand the working principle, capabilities, and operation of the aviation systems and processes. Students’ skill-level will be evaluated by the artifacts they produced, while their engineering judgment and critical thinking be evaluated by individually filled task worksheets.				
Test is designed to facilitate students to review the breadth and depth of their understanding on specific topics.					
Student Study Effort Expected	Class Contact				
	Min-lecture /Hands-on Practice/ Product Assembly /Report Writing			60 Hrs	
	Other Study Effort			0 Hrs.	
	Total Study Effort			60 Hrs.	
Reading List and References	Reference Software List:				
	1. SolidWorks from Dassault Systèmes Solidworks Corp.				
	2. Python from Python Software Foundation				
	Reference Standards and Handbooks:				
	1. BS EN ISO 128 – Technical product documentation. General principles of representation				
Reading List and References	Reference Books:				
	1. Joshi, P. (2017). <i>Artificial intelligence with python</i> . Packt Publishing Ltd.				
	2. Training material, manual and articles published by Industrial Centre.				